

Zscaler for Users - Administrator

EDU-200

Hands-on Lab



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Zscaler For Users - Administrator (EDU-200) Hands-On Lab Guide

Dec 2025, Rev.1.0

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About the Zscaler for Users - Administrator (EDU-200) Hands-on Lab

Welcome to the Zscaler for Users – Administrator (EDU-200) Hands-on Lab. Throughout the hands-on lab, you will practice skills that you learned during the Zscaler for Users - Administrator (EDU-200) e-learning. By the end of the hands-on lab, you will be able to:

- Configure connectivity to the Zero Trust Exchange.
- Configure access control policies to allow or deny traffic through the Zero Trust Exchange.
- Use the administration interfaces to interpret and analyze traffic patterns.

Lab Description and Scenario

For these lab exercises, consider the scenario where your organization (Labtenant) has deployed the **Zscaler Zero Trust Exchange** to protect against Internet security threats, control Internet access, and provide secure access to private resources. Your organization is also interested in the end user experience, which can be reported on with the Zero Trust Exchange.

You will deploy the **Zscaler Client Connector (ZCC)** application to end user devices to control Internet access, provide access to private applications, and to probe, benchmark, and measure the digital experiences for each user within the organization.

To support the rollout, you are tasked to:

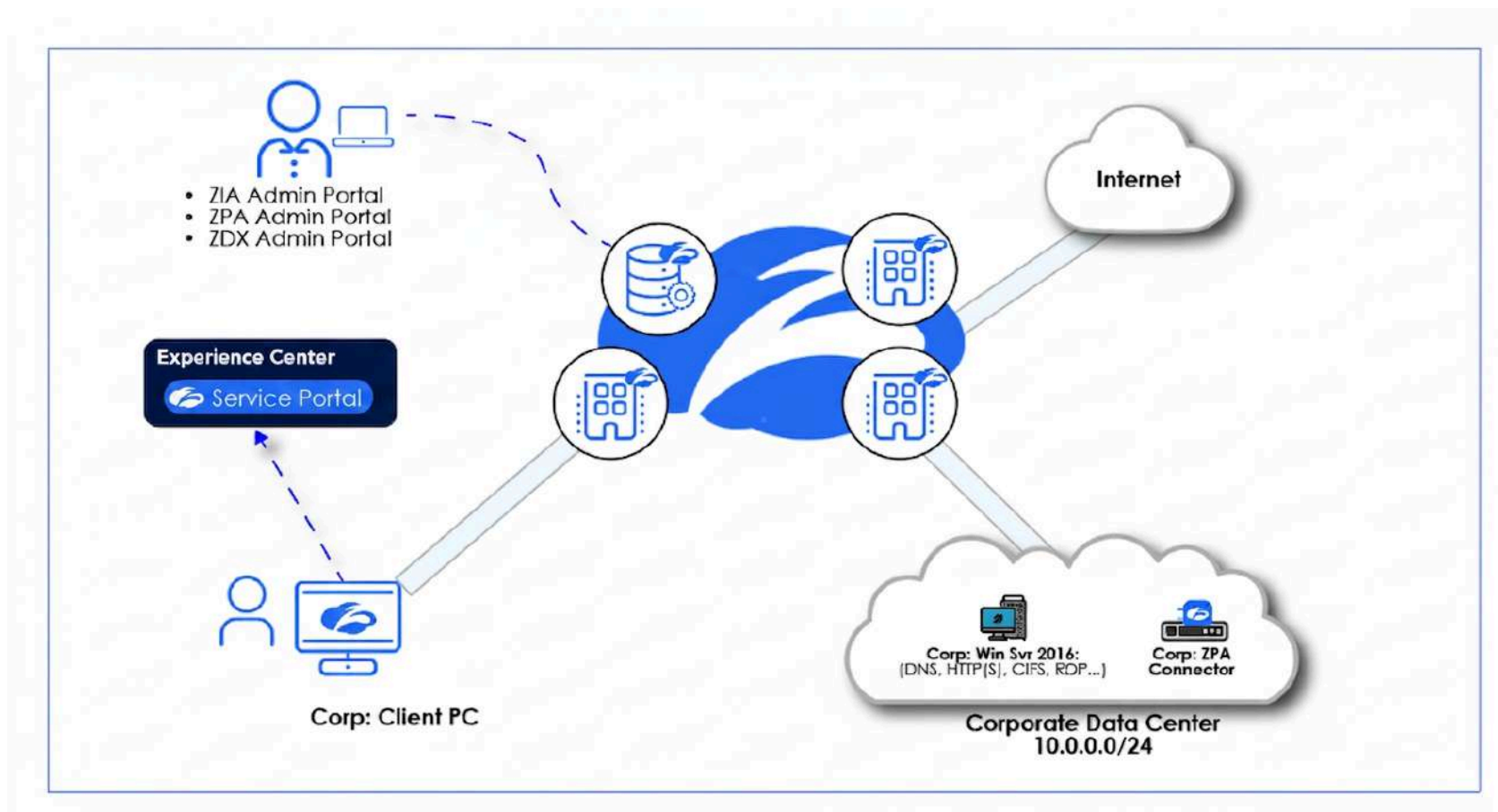
- Connect users to the Zero Trust Exchange by deploying **ZCC** to Windows client devices.
- Configure SSL inspection.
- Deploy **Zscaler Digital Experience (ZDX)** reporting and create an initial set of **ZDX** applications and probes.
- Configure user access to both Internet and private applications.
- Verify Cyberthreat Protection services.
- Verify Data Protection services.

Lab Setup

You will have access to **Zscaler Zero Trust Exchange (Service Portal (formerly ZIdentity)/ZIA/ZPA/ZDX)** and a **Skytap 'Pod'**.

For your lab exercises, you will configure the **Service Portal** (Formerly **ZIdentity**), **ZIA**, **ZPA**, and **ZDX** services and explore the end user experience. You will use a Windows 10 Client PC in Skytap to connect to the **Zero Trust Exchange**.

Lab Topology



- Windows Client PC
 - Name - **Corp: Client PC**
 - The Client PC is joined to Active Directory pattraining.safemarch.com.
 - Username | Password: **student** | **Admin-123!**

- Windows Server
 - Name - **Corp: Win Svr**
 - Provides local directory services through Active Directory.
 - Hosts the Active Directory domain pattraining.safemarch.com, to which the **Corp: Client PC** is joined.
 - Runs HTTP/HTTPS Intranet applications and file services.
 - Functions as the DNS server for hosts in the data center.
 - Username | Password: **administrator** | **Admin-123!**
- App Connectors
 - Names - **Corp: App Connector**
 - It's a Red Hat Enterprise 9.4 VM, pre-deployed in the network to simulate the results of having previously instantiated the OVA/OVF file to deploy the VM with the App Connector RPM already installed.
 - Username | Password: **admin** | **zscaler**

Lab Credentials

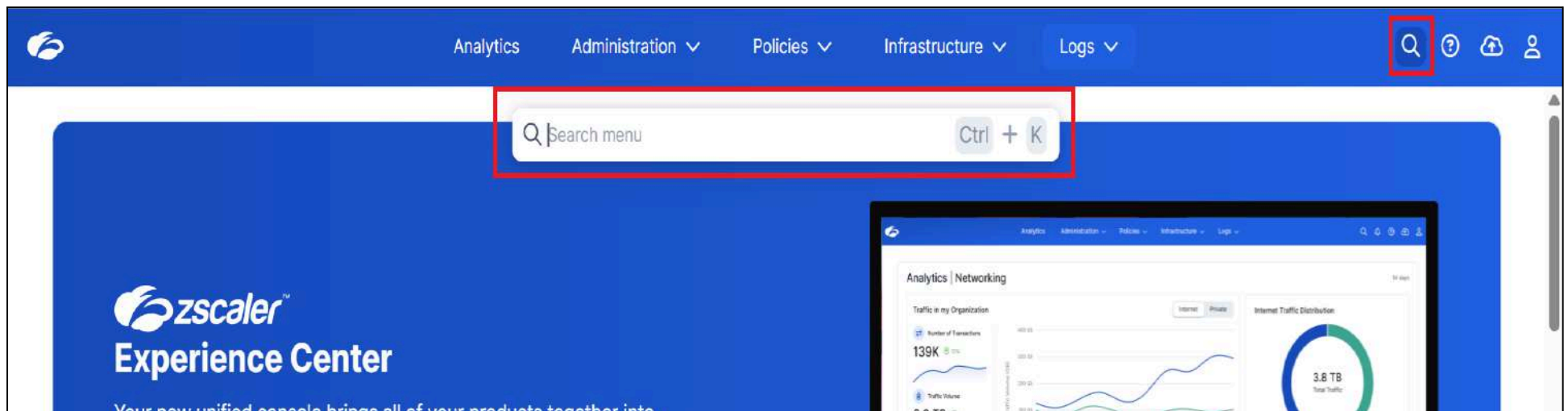
Virtual Machine Name	User Name	Password	IP Address
Corp: Client PC	student	Admin-123!	192.168.1.1
Corp: Win Svr 2016	administrator	Admin-123!	10.0.0.0/24
Corp: App Connector	admin	zscaler	10.0.0.4
User Information	User Name	Password	
	Joe	Admin-123!	
	Bev	Admin-123!	

Lab Access Information

For this course, each student is assigned a unique student ID with a dedicated virtual environment (hosted in Skytap) and corresponding Student FQDN along with a set of admin and test users accounts for Zscaler. User credentials and portal addresses are all contained in the lab access instructions provided to you for your assigned training pod. These include:

- Your **Student ID** which is a unique identifier for your pod (e.g., zs9901)
- Your **Student FQDN** which is a unique domain for your pod (e.g., zs9901.labtenant.com)
- Your admin login credentials for access to the Zscaler Portals
 - The login name will be in the format **student@<Student FQDN>** where **<Student FQDN>** is the domain assigned to your pod. For example, student@zsxxxx.labtenant.com.
 - The password is unique for your session and is set as the same password on all the configured admin and user accounts for the pod

Note: To navigate to any specific feature you can use the search option available in the Experience Center tenant.



Lab 1: Connect to the Virtual Lab Environment

Scenario: As the Zscaler administrator, you need to establish a functional, pre-configured environment (the Virtual Lab) that mirrors Lab tenant's potential production setup to validate the platform and access its core management console (Experience Center) before you can begin configuration tasks.

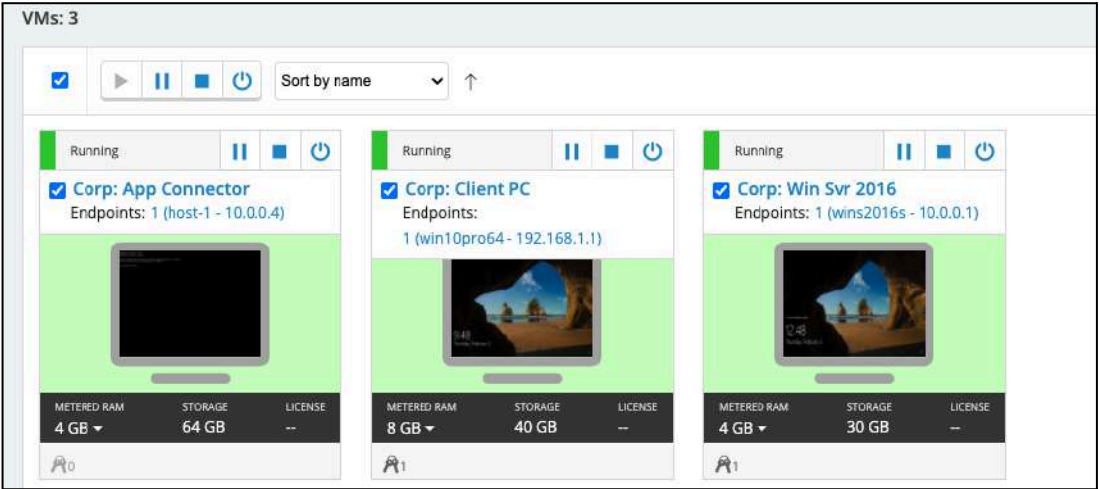
Objectives: By the end of this lab, you will

- **Connect to Skytap** and your lab virtual machines (VMs).
- **Familiarize** yourself **with** the login options and **clipboard copy/paste functions**.

Approximate Time: 10 minutes

Task 1.1: Verify Lab Access

- 1. Start and connect to your Skytap pod:
 - a. Click the **Skytap Portal URL** provided in your access instructions.
 - b. Ensure the checkbox is checked to **select all VMs**.
 - c. If machines have not already started as displayed in the image below, Click the **Play** button. All the VMs will launch.

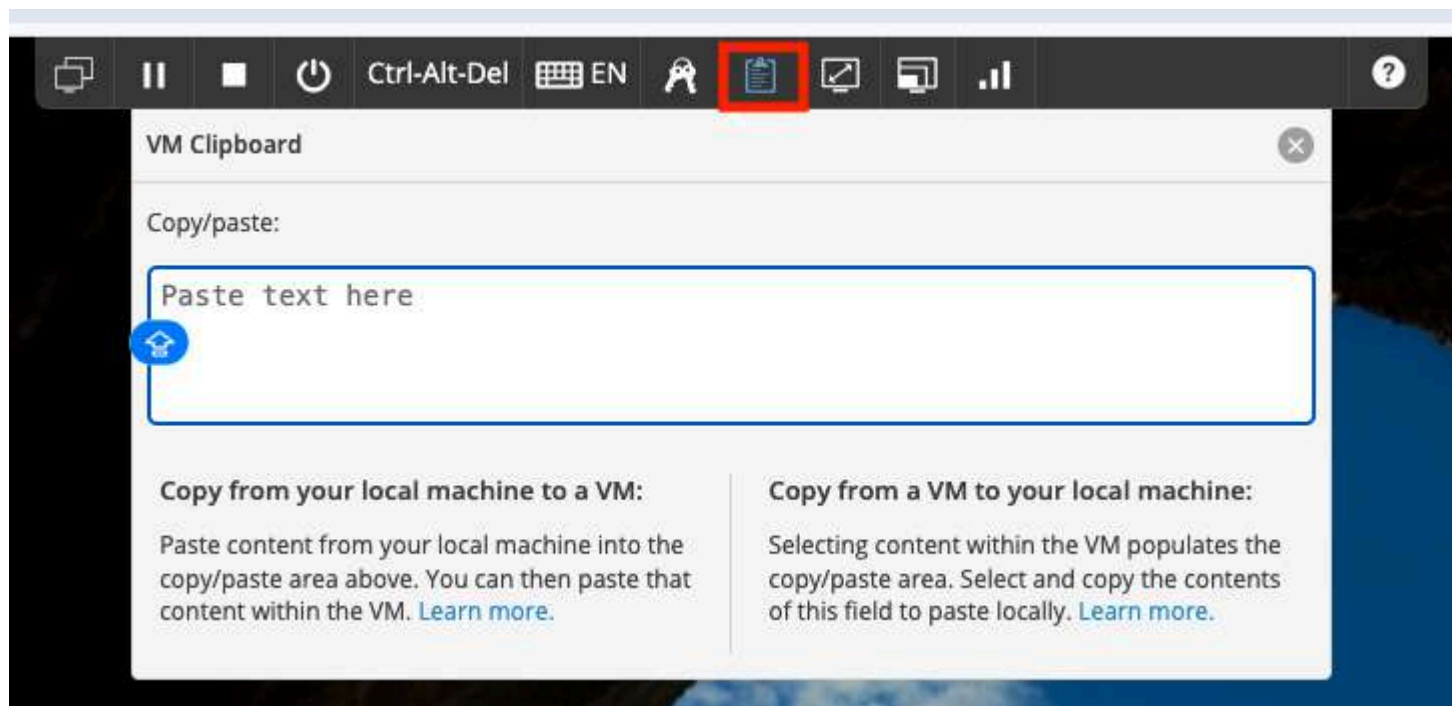


Note: VMs in the environment are set to start in stages, so the whole process will take a few minutes.

Task 1.2: Use the Lab Clipboard

During these labs, you will frequently copy commands, URLs, and configuration values from the lab guide into the virtual machines.

1. Use the clipboard icon in the lab interface:
 - a. Copy text from your host to the VM clipboard.
 - i. Click the **clipboard icon**.
 - ii. Paste text into the **Copy/paste:** field.



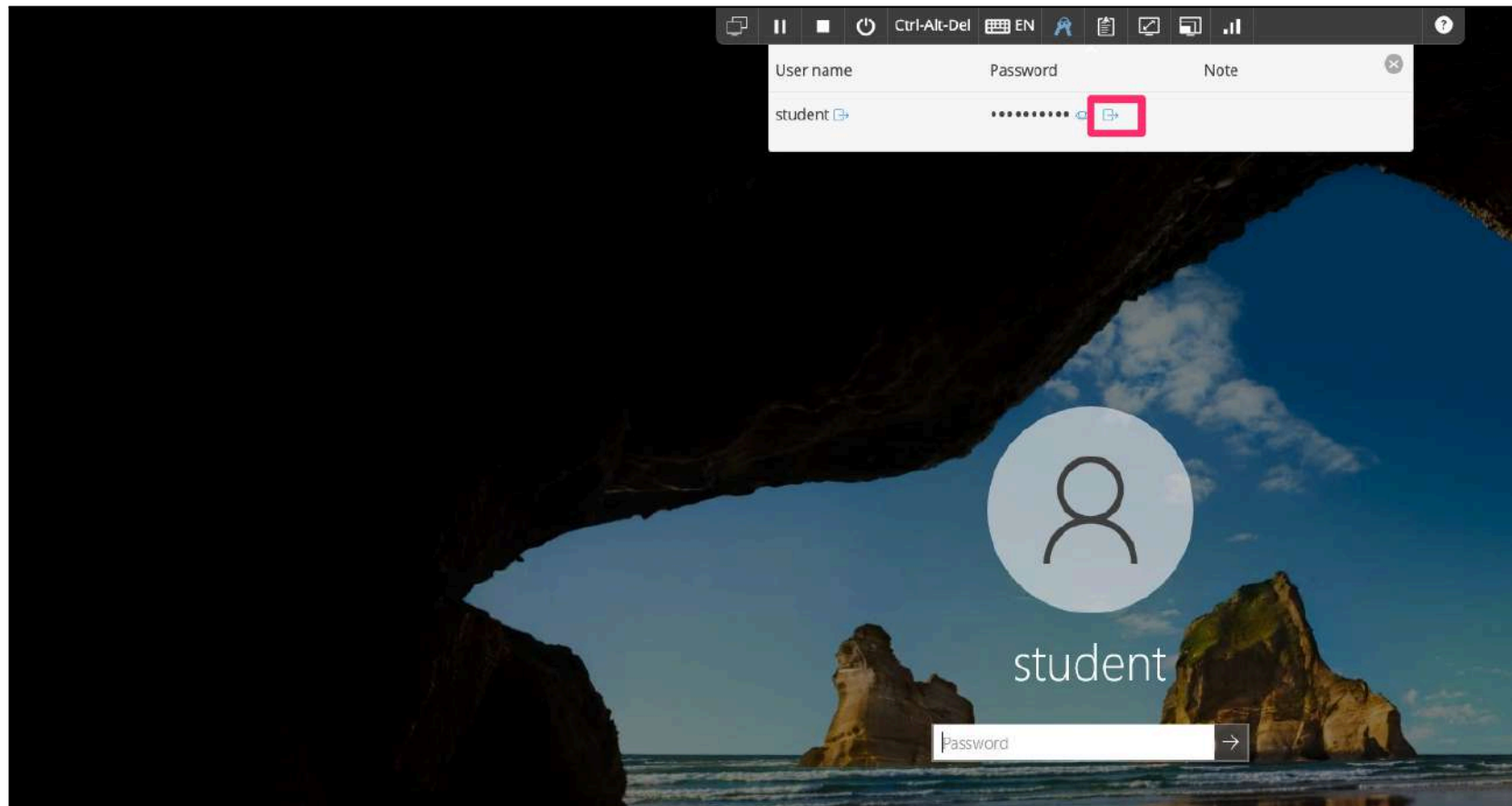
- b. Paste in the VM:
 - i. In the VM, use **Ctrl+V** (or **right-click > Paste**) to paste the copied text.

Note: Even if you are on a Mac, the virtual machine is running Windows, so they must use Ctrl+V (not Cmd+V) inside the VM.

Task 1.3: Sign In to the Experience Center Landing Page

Note: Throughout this lab guide, **<Student FQDN>** represents the domain assigned to your tenant. For example, if your student admin username is **student@zs9901.labtenant.com**, then your **Student FQDN** is **zs9901.labtenant.com**

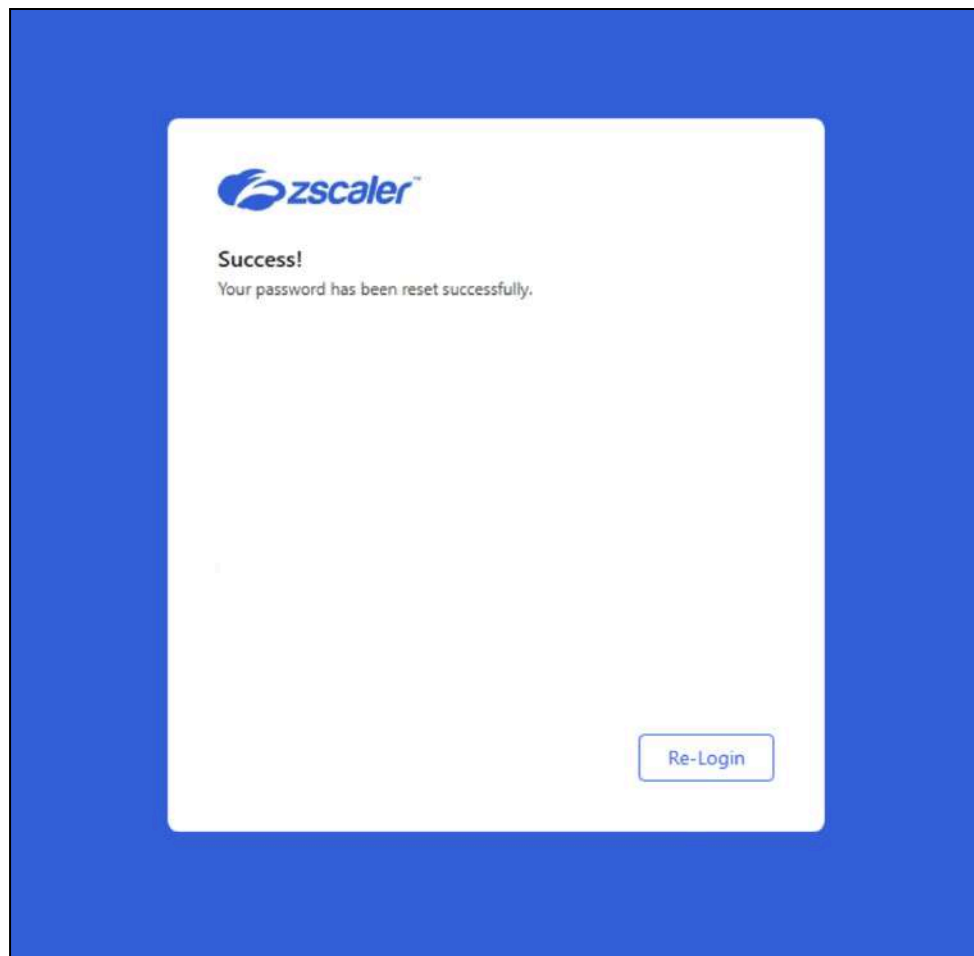
1. Login to the **Corp: Client PC** as the local Student User:
 - a. Launch the **Corp: Client PC** from the Skytap portal.
 - b. Click the **Ctrl-Alt-Del** icon in the Skytap Toolbar.
 - c. Click the **Credential (keys)** icon in the Skytap Toolbar.
 - d. Click the **Insert Credentials** icon next to the password field for the **student** username.



2. In a browser window, open Experience Center Landing Page URL “<https://console.zscaler.com/>”

Note: You will be redirected to the **Service Portal** (formerly ZIdentity) login page to validate your identity.

- a. Authenticate with your student admin account:
 - i. Username: **student@<Student FQDN>**
 - ii. Password: as provided in the Lab Access Instructions
 - iii. Click on **Sign in**.
- b. You will be prompted to change your password upon successful authentication.



Lab 2: Configure User and Administrator Access Entitlements

Scenario: You've secured the Zscaler console and now need to **establish a clean foundation for who can sign in and what they can access in Labtenant.com. Two new users—Joe (Support/helpdesk) and Bev (IT/admin)—must be onboarded** with the right departments and groups so they inherit the correct entitlements. The goal is to **ensure Joe has limited, read-only access appropriate for Tier 1 support**, while **Bev has full administrative access**, and to **validate both users' experiences** by logging in and confirming what each can and cannot do.

Objectives: By the end of this lab, you will be able to:

- **Create required org structure** (departments, groups)
- **Provision users** and place them **correctly so policies and entitlements apply**
- **Configure entitlements and confirm they're applied to the right identity/group**
- **Validate access** is correct **via role-based UI restrictions and service status**

Approximate Time: 40 minutes

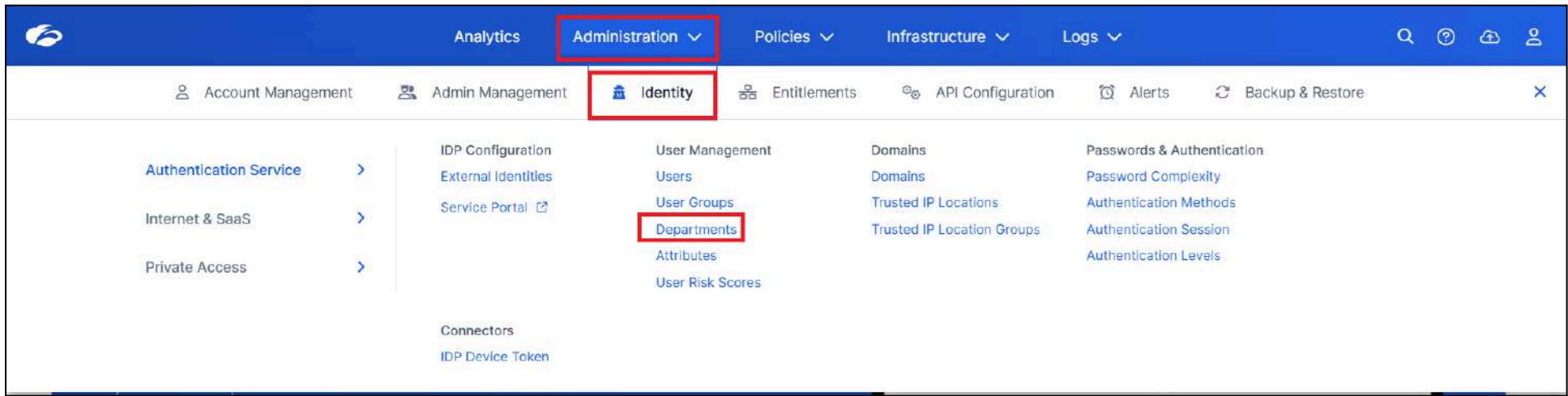
Note: The table below summarizes the user, department, user group, and permissions on service and administrative entitlements, which you will configure in later tasks.

User	Task	Department	User Group	Service Entitlement	Administrative Entitlement
Joe	2.2	Support	Helpdesk Level 1		ZIA(Read-only), ZDX(Read-only)
	2.3	Support	Helpdesk Level 1	ZIA	
Bev	2.2	IT	Zscaler Global Administrators(built-in group)		Service Portal (formerly ZIdentity) , Workflow Automation, ZIA,ZPA,ZDX,ZCC,Zscaler Cloud and Branch connector, Business Insights

Task 2.1: Create Departments and User Groups

In this task, you will create the **Support** and **IT** departments, then create users **Joe** and **Bev**, assigning one user to each department.

- 1. In the **Zscaler Admin Console**, select the **Administration** tab.
 - a. Go to **Identity > Departments**



- 2. Create the Support department.
 - a. Select **Add Department**.
 - b. In the **Add Department** section, enter the following values:
 - i. Department Name: **Support**
 - ii. Description: **Support Department**

Add Department

DEPARTMENT

Department Name

Support

Description

Support Department

SaveCancel

c. Click **Save**.

3. Create the IT department.

a. Select Add Department and enter the following values:

- i. Department Name: **IT**
- ii. Description: **IT Department**

b. Select **Save**.

Edit Department

DEPARTMENT

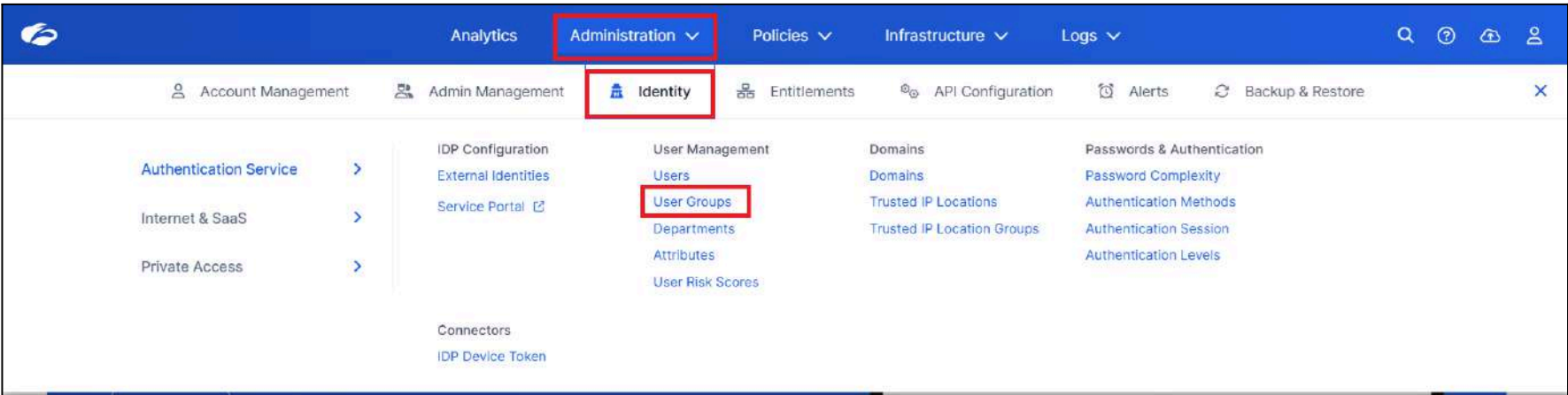
Department Name

IT

Description

IT Department

- 4. Confirm that **IT** and **Support** appear in the **Departments** section.
- 5. Create the Helpdesk L1 user group.
 - a. In the **Experience Center Dashboard**, select the **Administration** tab.
 - b. Go to **Identity > User Groups**.
 - c. Select on **Add User Group**,
 - d. In the Add Group pane, enter the following value:
 - i. User Group Name: **Helpdesk L1**
 - e. Select **Save**.



Add Group

INFORMATION

User Group Name

Helpdesk L1

Assign Users

Select

Source

UI

Description

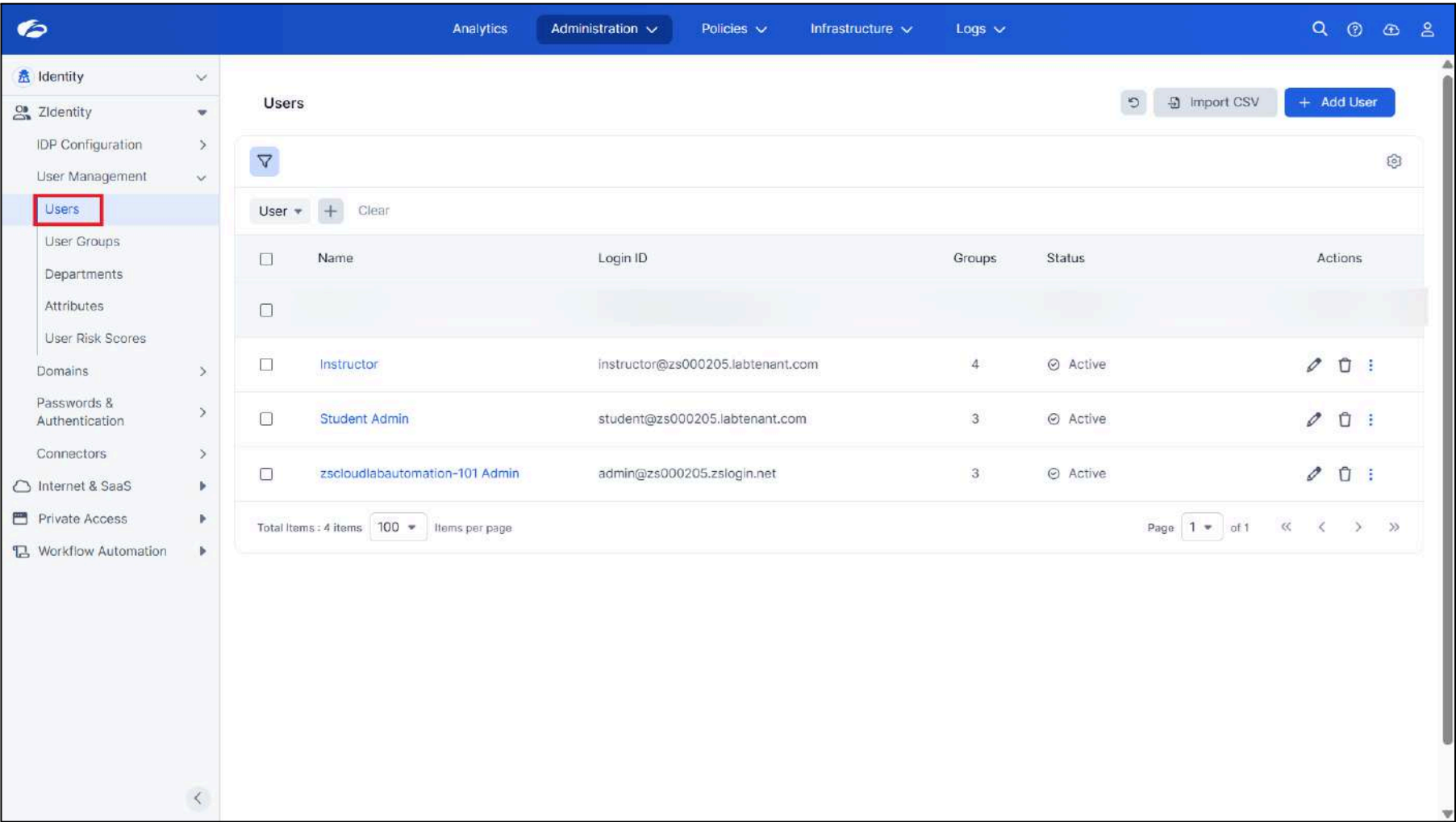
Save

Cancel

Task 2.2: Create Users

In this task, you will create users **Joe** and **Bev** and map them to the departments and user groups you created in the previous task.

- 1. Create User Joe:
 - a. From the **left navigation panel**, go to **Users**.



- b. Click **Add User** button  at the top right corner and set the following:

- Login ID: **joe@<Student FQDN>**
- Same As Login ID: **Enabled**

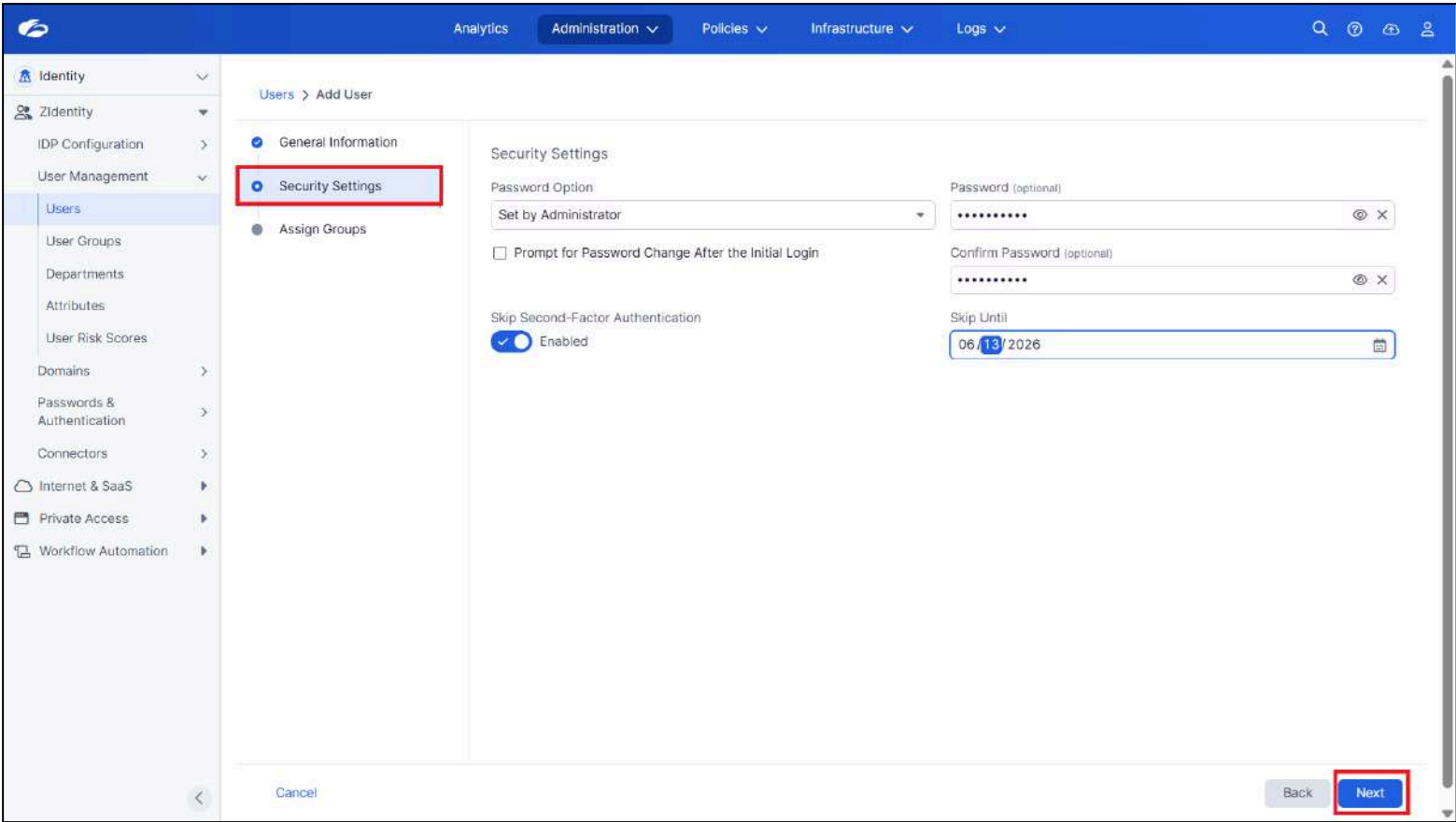
- Primary Email: **joe@<Student FQDN>**
- First Name: **Joe**
- Last Name: **Web**
- Department: **Support**

c. Click **Next**.

The screenshot shows the Zscaler Administration console interface. The left sidebar contains a navigation menu with options like Identity, ZIdentity, IDP Configuration, User Management, Users, User Groups, Departments, Attributes, User Risk Scores, Domains, Passwords & Authentication, Connectors, Internet & SaaS, Private Access, and Workflow Automation. The main content area is titled 'Users > Add User' and features a tabbed interface with 'General Information' (selected and highlighted with a red box), 'Security Settings', and 'Assign Groups'. The 'General Information' tab contains several input fields: 'Login ID' (joe@z...labtenant.com), 'Primary Email' (joe@zsc...labtenant.com), 'Secondary Email (optional)' (e.g., johndoe2@gmail.com), 'Last Name' (Web), 'Department' (Support), 'First Name' (Joe), 'Display Name' (Joe Web), and 'Manager (optional)' (Select). There are also toggle switches for 'Status' and 'Same As Login ID', both of which are turned on. At the bottom of the form, there is a 'Cancel' button and a 'Next' button, which is highlighted with a red box.

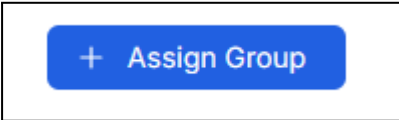
d. In the **Security Settings** tab and configure the following:

- Password option : **Set By Administrator**
- Password: **Admin-123!**
- Confirm Password: **Admin-123!**
- Prompt for Password change After the Initial Log in: **Unchecked**
- Multi-Factor Bypass Skip Second Factor Authentication: **Enabled**
- Skip until: Set as far out as allowed.

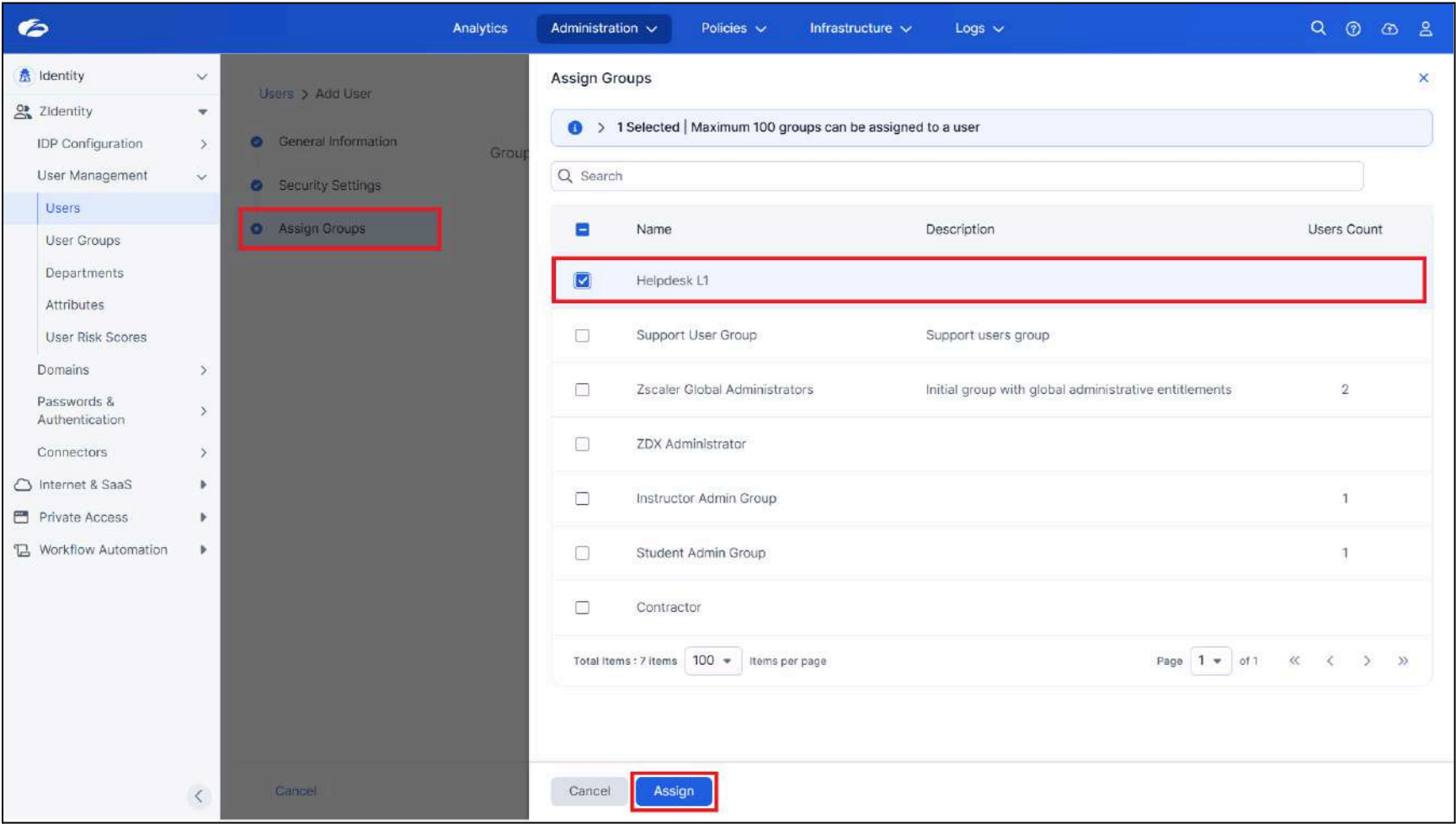


Note: For lab purposes, the second factor of authentication is skipped. Zscaler recommends enabling Multi-Factor Authentication (MFA) for all users.

- e. Click **Next**.
- f. In the **Assign Groups** tab:
 - i. Click the **+ Assign Group** button.



- ii. Ensure the checkbox is selected for the group **Helpdesk L1**.
 - iii. Click **Assign**.



- g. Click **Save**.

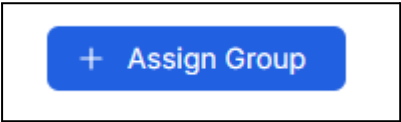
2. Create User Bev:

- a. Click **Add User** and set the following:
 - i. Login ID: **bev@<Student FQDN>**
 - ii. Same As Login ID: **Enabled**
 - iii. Primary Email: **bev@<Student FQDN>**
 - iv. First Name: **Bev**
 - v. Last Name: **Carey**
 - vi. Department: **IT**

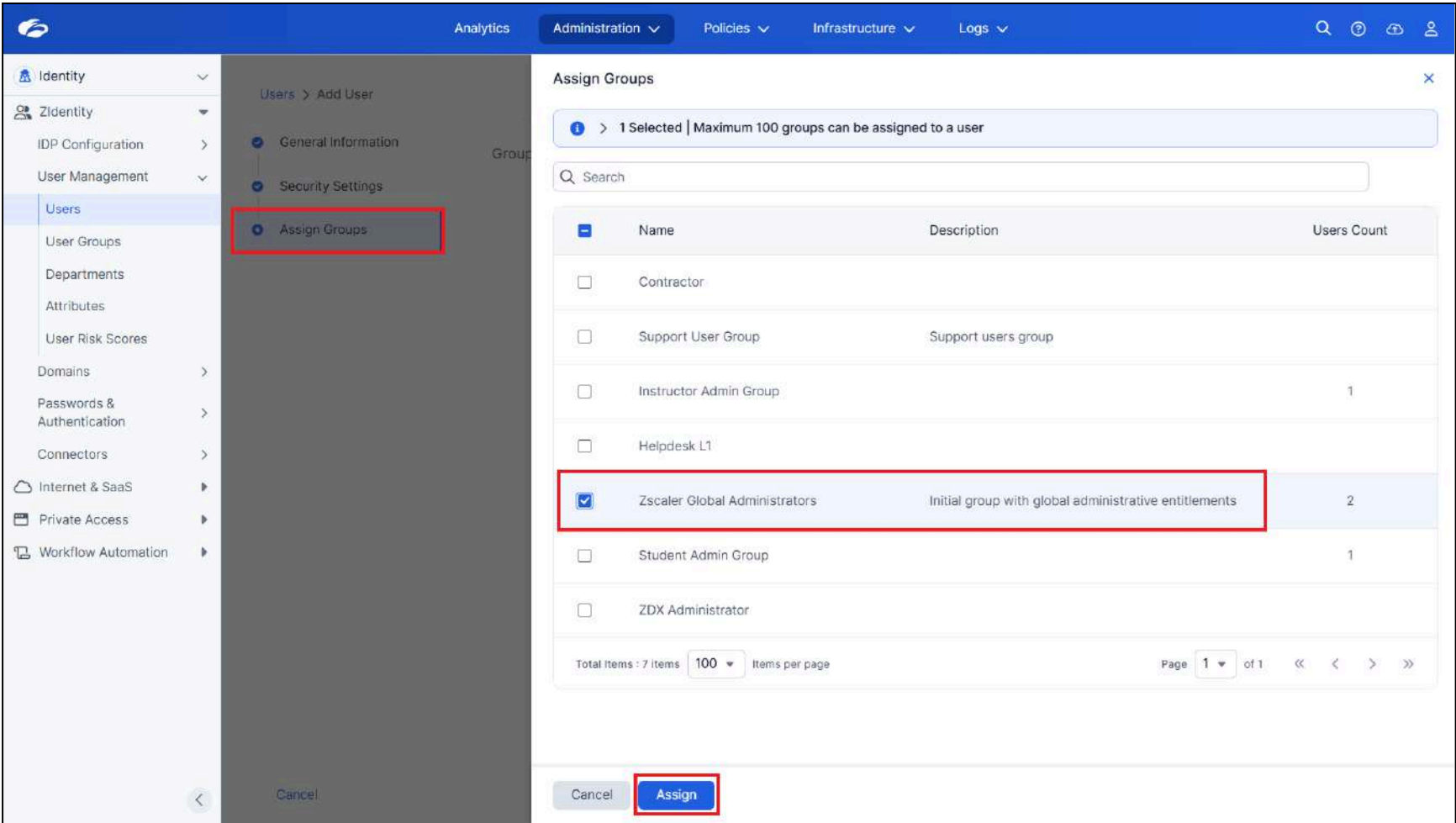
- b. Click on **Security Settings** tab and configure the following:
 - i. Password option : **Set By Administrator**
 - ii. Password: **Admin-123!**
 - iii. Confirm Password: **Admin-123!**
 - iv. Prompt for Password change After the Initial Log in: **Unchecked**
 - v. Multi-Factor Bypass Skip Second Factor Authentication: **Enabled**
 - vi. Skip until: Set as far out as allowed.

- c. Click **Next**.

- d. In the **Assign Groups** tab:
 - i. Click the **+ Assign Group** button.



- ii. Ensure the checkbox is selected for the group **Zscaler Global Administrators**.
 - iii. Click **Assign**.



- e. Click **Save**.

Task 2.3: Add Administrative and Service Entitlements

In this task, you will **assign administrative entitlements** and **service entitlements** to users **Joe** and **Bev**.

- 1. Assign Administrative Entitlement for ZIA (Helpdesk L1):
 - a. From the **Experience Center Dashboard**, navigate to **Administration > Admin Management > Role Based Access Control > Administrative Entitlements**.
 - b. Click on **Zscaler Internet Access**.



Administrative Entitlements	
Name	Cloud Name
 Business Insights	zsccloud.net
 Workflow Automation	app.us1.zsworkflow.net
 ZIdentity Administration	---
 Zscaler Client Connector	zsccloud.net
 Zscaler Cloud and Branch Connector	zsccloud.net
 Zscaler Digital Experience	zsccloud.net
 Zscaler Internet Access	zsccloud.net
 Zscaler Private Access	zpatwo.net

- c. In the **Zscaler Internet Access - Administrative** pane click **Assign Groups**.
- d. Select check box for **Helpdesk L1 Group**.
- e. From the **Role** drop-down menu, select **Executive Insights App** role.
- f. Click **Next** and **Assign**.

Note: For simplicity, this lab uses the predefined **Executive Insights App** role. It conveniently provides read-only access to the Zscaler portal, which is the intended permission level for this user.

In a production environment, you can **create a custom role** for this purpose as needed under **Administration > Admin Management > Role Based Access Control > Roles**.

Admin Management

Administrator Management

Role Based Access Control

Roles

Entitlements

Administrative Entitlements

Audit Logs

← Zscaler Internet Access - Administrative

Cloud Name: ZSCLOUD | Organization Name: zsccloudlabautomation-101

User Groups (4) | Users

+ Assign Groups

Search...

	Name	Role	Users	Actions
☐	Contractor	Super Admin	Users	
☐	Instructor Admin Group	Super Admin	Users	
☐	Student Admin Group	Super Admin	Users	
☐	Zscaler Global Administrators	Super Admin	Users	

AnalyticsAdministrationPoliciesInfrastructureLogs

Admin Management

Administrator Management

Role Based Access Control

Roles

Entitlements

Administrative Entitlements

Audit Logs

← Assign Groups

Select Groups & Roles

Summary

Set same role for all selected groups

Search...

	Name	Users	Role
☐	Contractor	Users	Select
☒	Helpdesk L1	Users	Executive Insights App
☐	Instructor Admin Group	Users	
☐	Student Admin Group	Users	
☐	Zscaler Emergency Access	Users	

Search...

Executive Insights App

Super Admin

Showing 6 of 6

CancelNext

- 2. Assign Administrative Entitlement for ZDX (Helpdesk L1):
 - a. Select the **Back** option **next to Zscaler Internet Access - Administrative**

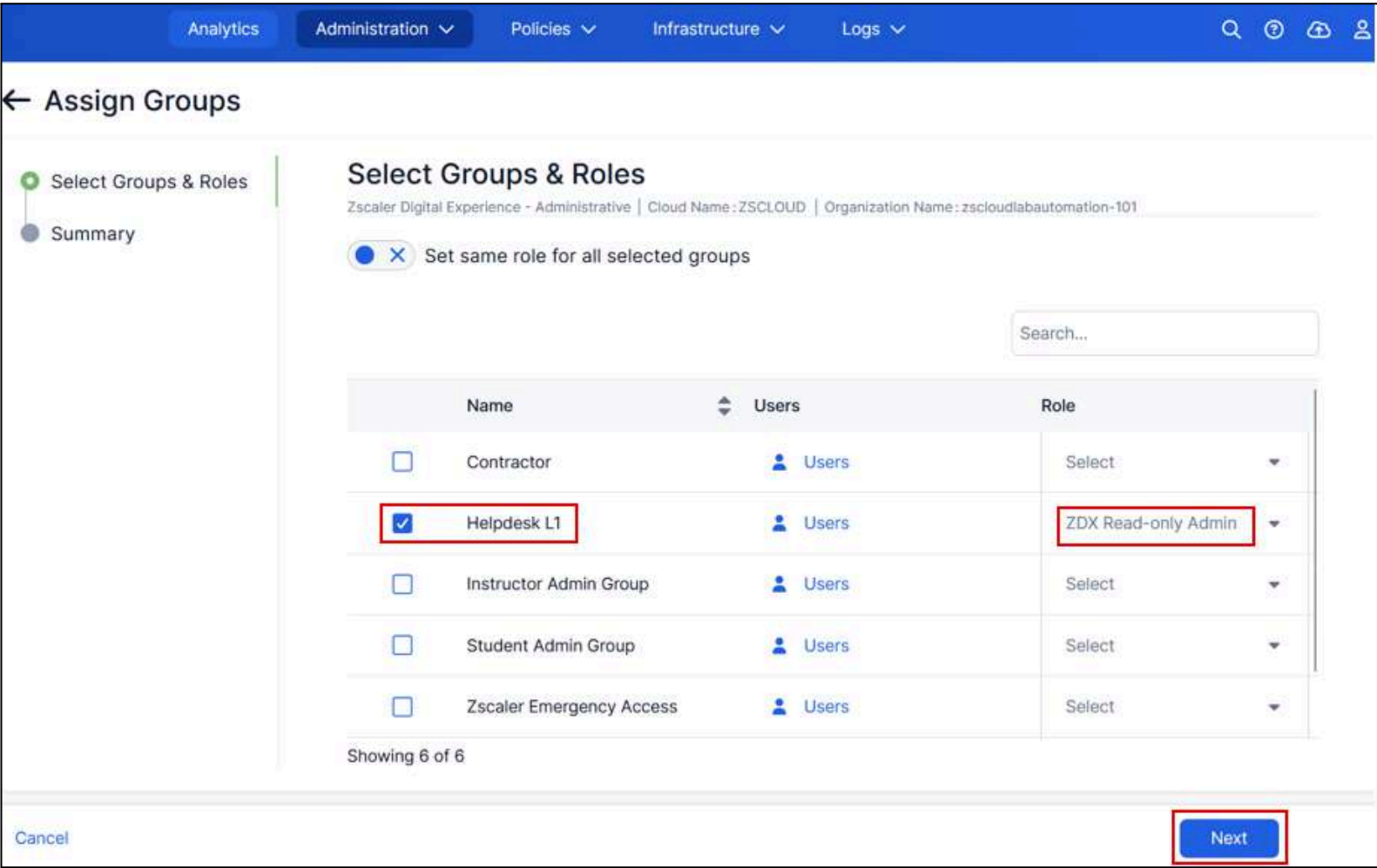


- b. Click **Zscaler Digital Experience**.

Administrative Entitlements		
Name		Cloud Name
	Business Insights	zsccloud.net
	Workflow Automation	app.us1.zsworkflow.net
	ZIdentity Administration	---
	Zscaler Client Connector	zsccloud.net
	Zscaler Cloud and Branch Connector	zsccloud.net
	Zscaler Digital Experience	zsccloud.net
	Zscaler Internet Access	zsccloud.net
	Zscaler Private Access	zpatwo.net

- c. In the **Zscaler Digital Experience - Administrative** pane:
 - i. Click **Assign Groups**.
 - ii. Select the check box for **Helpdesk L1 Group**.

- iii. From the **Role** drop-down menu, select **ZDX Read-only Admin**.
- d. Click **Next** and **Assign**



Note: The ZDX Read-only Administrator role is a pre-built role.

- 3. Assign Service Entitlement for ZIA (Helpdesk L1):
 - a. From the **Experience Center Dashboard**, navigate to **Administration > Entitlements > End User Entitlements > Service Entitlements**.
 - b. Click on **Zscaler Internet Access**.

Service Entitlements

Name	Cloud Name	Organization Name
 Zscaler Digital Experience	zscloud.net	zscloudlabautomation-101
 Zscaler Internet Access	zscloud.net	zscloudlabautomation-101
 Zscaler Private Access	zpatwo.net	Zscaler Internal Employee Tenant Reque:

- c. In the **Zscaler Internet Access - Service** pane:
- i. Click **Assign Groups**.

ii. Select check box for **Helpdesk L1 Group**.
- d. Click **Next** and **Assign**.

← Assign Groups

Select Groups

Summary

Select Groups

Zscaler Internet Access - Services | Cloud Name: ZSCLOUD | Organization Name: zscloudlabautomation-101

Search...

	Name	Users
☐	Contractor	 Users
☐	Dynamic Group Administrators	 Users
☐	Dynamic Group Guest Domains	 Users
☐	Dynamic Group Registered Domains	 Users
☒	Helpdesk L1	 Users
☐	Instructor Admin Group	 Users
☐	Student Admin Group	 Users
☐	Support User Group	 Users

Showing 10 of 10

Cancel

Next

4. Verify the entitlements for user Joe:
- a. From the **Experience Center Dashboard**, navigate to **Administration > Identity > Users**.

i. Click on the user **Joe**.

Users

↶

Import CSV

+ Add User

▼

User + Clear

☐	Name	Login ID	Groups	Status	Actions
☐	Bev Carey	bev@zs5005.safemarch.com	3	🟢 Active	
☐	HVAC	hvac@zs5005.safemarch.com	3	🟢 Active	
☐	Instructor	instructor@zs5005.safemarch.com	3	🟢 Active	
☐	Joe Web	joe@zs5005.safemarch.com	3	🟢 Active	

b. In the Joe user's **pane**:

- i. Navigate to the **Entitlements** tab.

ii. Verify that both the administrative and service entitlements are assigned to user **Joe**.

Joe Web

Actions ▾ ×

General Information

Security Settings

Groups

Entitlements (3)

Administrative Entitlements (2)

Service	Roles	Cloud Name	Organization Name
Zscaler Digital Experience	ZDX Read-only Admin	zscloud.net	Internal Safemarch - zs5005
Zscaler Internet Access	Executive Insights App	zscloud.net	Internal Safemarch - zs5005

Total Items : 2

Service Entitlements (1)

Service	Cloud Name	Organization Name
Zscaler Internet Access	zscloud.net	Internal Safemarch - zs5005

Total Items : 1

c. Click the **X** button to close the **User** pane.

Users

User

+

 Clear

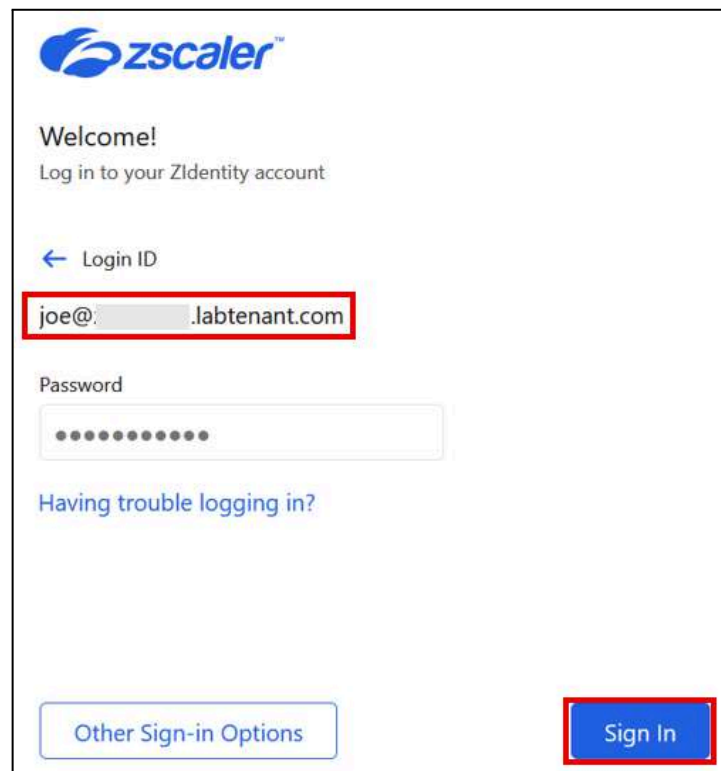
	Name	Login ID	Groups	Status
☐	Bev Carey	bev@zs5005.safemarch.com	3	🕒 Active

Note: You can verify the entitlements for user Bev, as you did for user Joe. From the **Experience Center Dashboard**, click the **Administration > Identity > Users**. Click on the user Bev and review the Entitlements tab.

Task 2.4: Validate Entitlement Access for Users

In this task, you will log in to the **Experience Center** to validate administrative entitlements and access for users **Joe** and **Bev**.

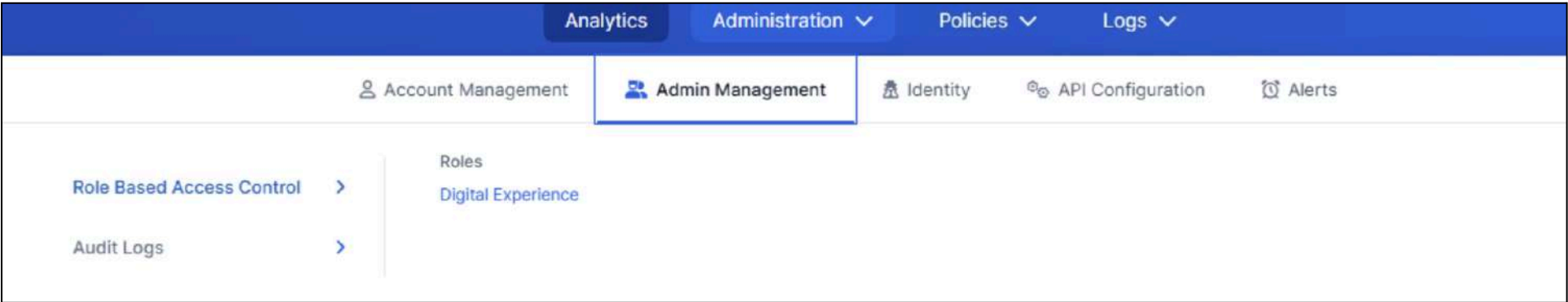
1. Validate administrative entitlements for Joe
 - a. On the **Corp Client PC**, Open a **new incognito window** in Chrome browser.
 - b. Navigate to **Experience Center Landing Page**
 - i. "<https://console.zscaler.com/>"
 - c. Authenticate with user Joe's account:
 - i. Username: **joe@<Student FQDN>**
 - ii. Password: **Admin-123!**
 - d. Click **Sign in**.



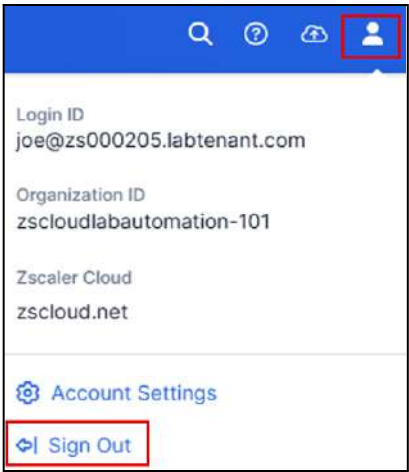
The screenshot displays the Zscaler login interface. At the top is the Zscaler logo. Below it, the text 'Welcome!' and 'Log in to your ZIdentity account' is shown. A 'Login ID' field is present, with a red box highlighting the text 'joe@.labtenant.com'. Below the login ID field is a 'Password' field, also with a red box highlighting the masked text '.....'. A link 'Having trouble logging in?' is located below the password field. At the bottom of the form, there are two buttons: 'Other Sign-in Options' and 'Sign In'. The 'Sign In' button is highlighted with a red box.

- e. In the **Experience Center Dashboard**, observe that the **Administration** and **Policy** tabs are restrictive compared to that of our **Student** user.

- f. Confirm that Joe cannot access **Admin Management**, **Sign-On Policies**, or other administrative configuration areas that were previously available to the **student@<FQDN>** user.
- g. Confirm that Joe can only view existing **Digital Experience** data and **cannot create** or **modify** objects



- h. Click on the **Account settings Icon** on the top right corner of the dashboard to **Sign out**.



Note: The Helpdesk Level 1 Group has read-only privileges for ZIA and ZDX. User Joe inherits this read-only access, which removes access to core administrative areas (such as Policies and Admin Management) and limits him to view-only analytics and experience data.

- 2. Validate administrative entitlements for Bev:
 - a. In the same **incognito window**, navigate to **Experience Center Landing Page**:
 - i. [“https://console.zscaler.com/”](https://console.zscaler.com/)

b. Authenticate with user **Bev's** account:

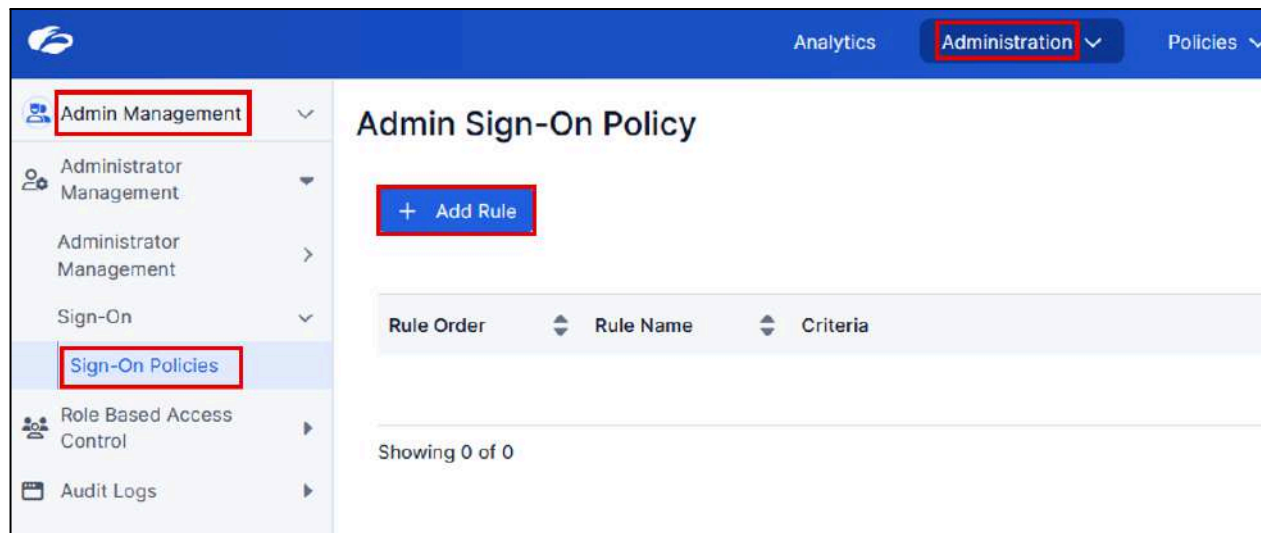
i. Username: **Bev@<Student FQDN>**

ii. Password: **Admin-123!**

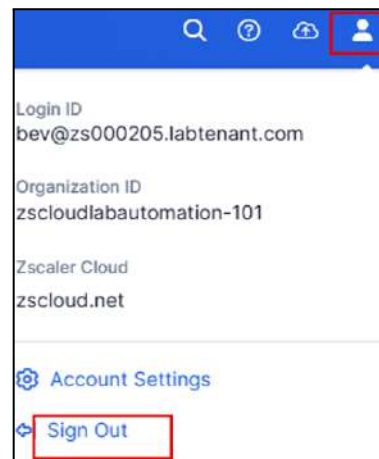
c. Click **Sign in**.

d. In the **Experience Center Dashboard**, Navigate to **Administration > Admin Management > Sign-On policies**.

e. Confirm that in the **Admin Sign-On Policy** window, user **Bev** is allowed to **Add Rule** (button is enabled) because they are part of the **Zscaler Global Administrators** group.



f. Click on the **Account settings Icon** on the top right corner of the dashboard to **Sign out**.



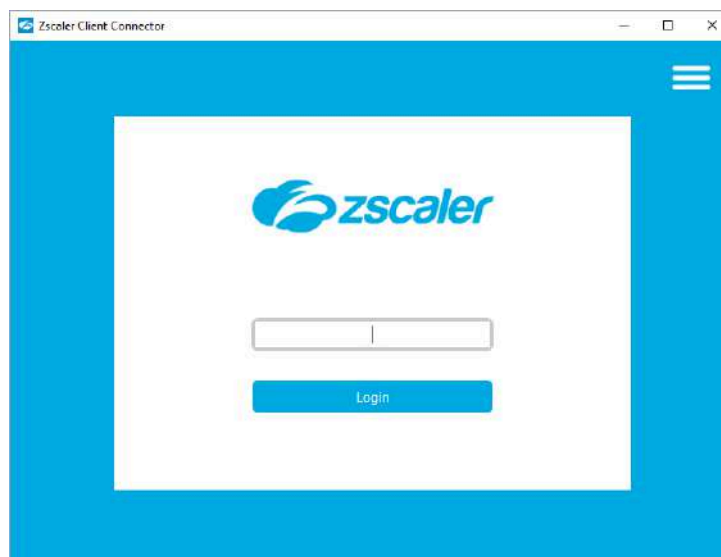
Note: The *Add Rule* button is only enabled for users with sufficient administrative privileges. As a member of the *Zscaler Global Administrators* group, Bev has full administrative access, including managing sign-on policies.

3. Validate service entitlement for Joe via Zscaler Client Connector:

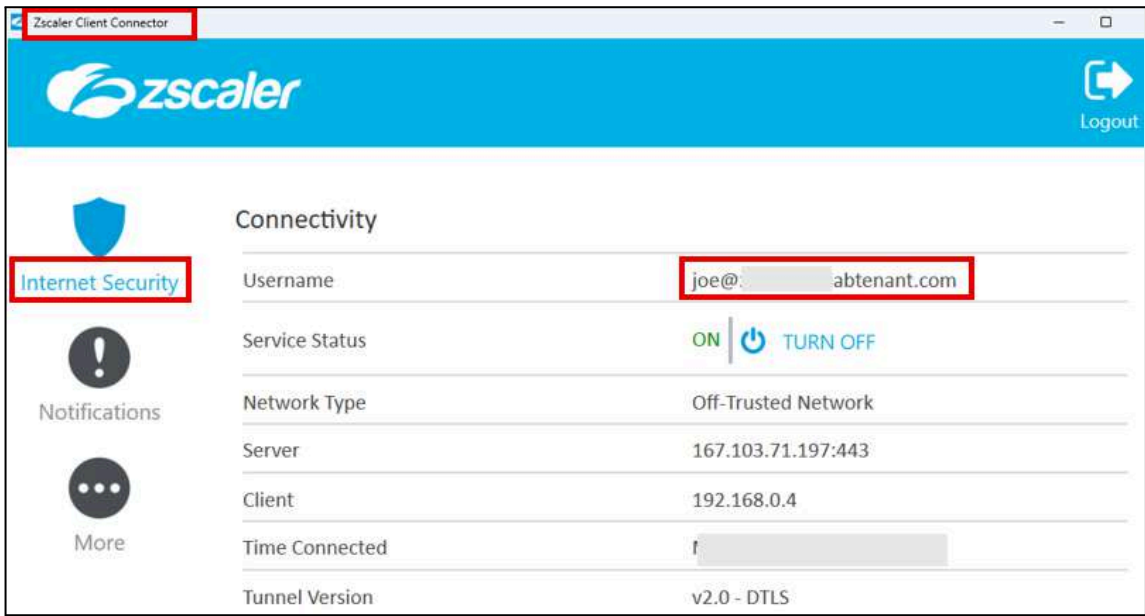
- a. On the **Corp: Client PC**, in the Windows taskbar, (bottom-right corner), click the **Show hidden icons arrow** ^.
- b. Select the **Zscaler Client Connector** icon, then click **Open Zscaler**.



- c. From the **Zscaler Client Connector login page**, authenticate with user **Joe's** account:
 - i. Username: **joe@<Student FQDN>**
 - ii. Password: Admin-123!
- d. Click **Sign in**.



- e. Verify **Joe's** service entitlement access for **Zscaler Internet Access**:
- f. In the **Zscaler Client Connector** console, verify that **Zscaler Internet Access** is **ON** for user Joe.



- g. After verifying Joe's service entitlement, **log out of Zscaler Client Connector**. You will log in again later with a different account.

Lab 3: Forward Traffic with the Zscaler Client Connector

Scenario: As the Zscaler administrator, your goal is to now configure the **Zscaler Client Connector (ZCC)** to securely forward all user traffic, regardless of protocol, to Zscaler for inspection and protection. By doing so, you will enable firewall functionality, secure internet access, and ensure multi-platform compatibility for users both on and off trusted networks.

Objectives: By the end of this lab you will:

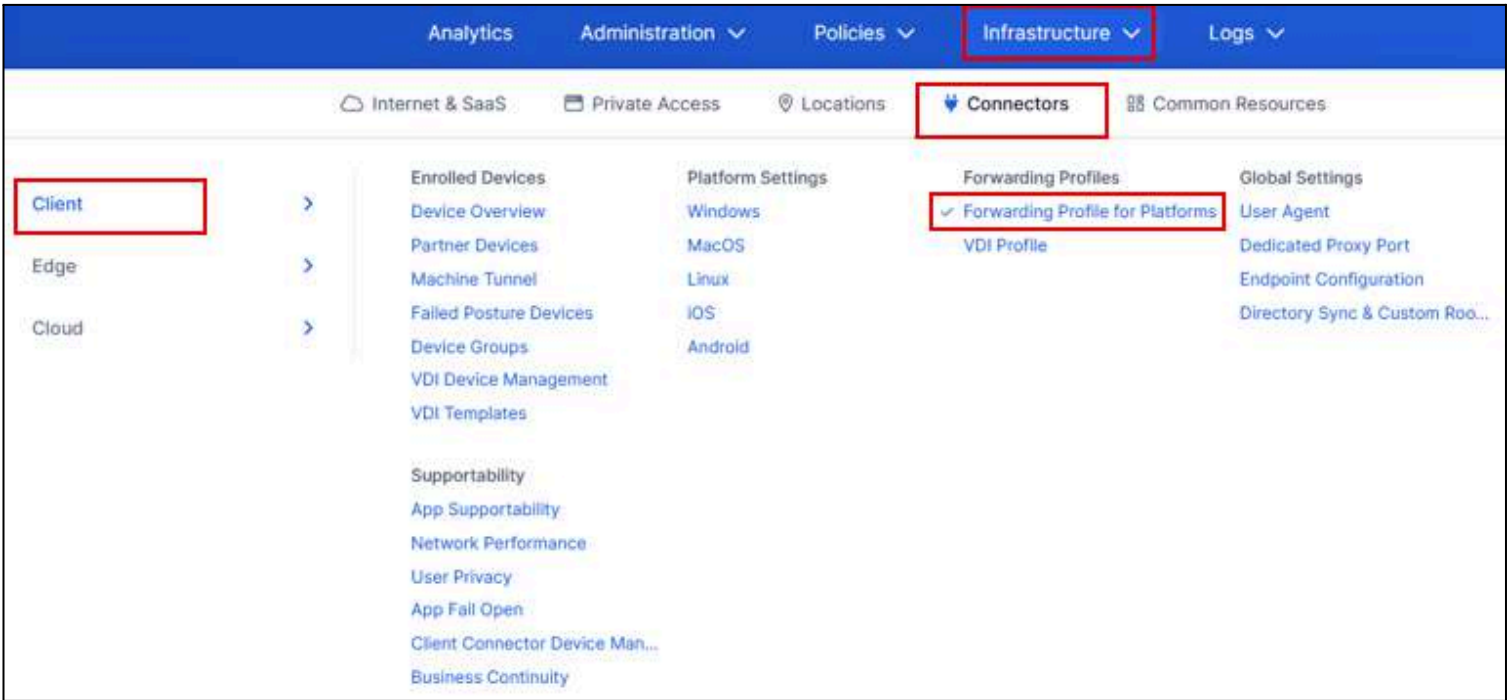
- **Configure forwarding options** to send all user traffic securely to Zscaler.
- **Setup forwarding profiles** and platform settings for Windows devices.
- **Confirm** features, such as **Private Access**, **Internet Security**, and **Digital Experience**, are enabled and functioning.
- **Confirm** the **Corp: Client PC traffic is forwarded** to Zscaler and verify entitlements and settings.

Approximate Time: 20 minutes

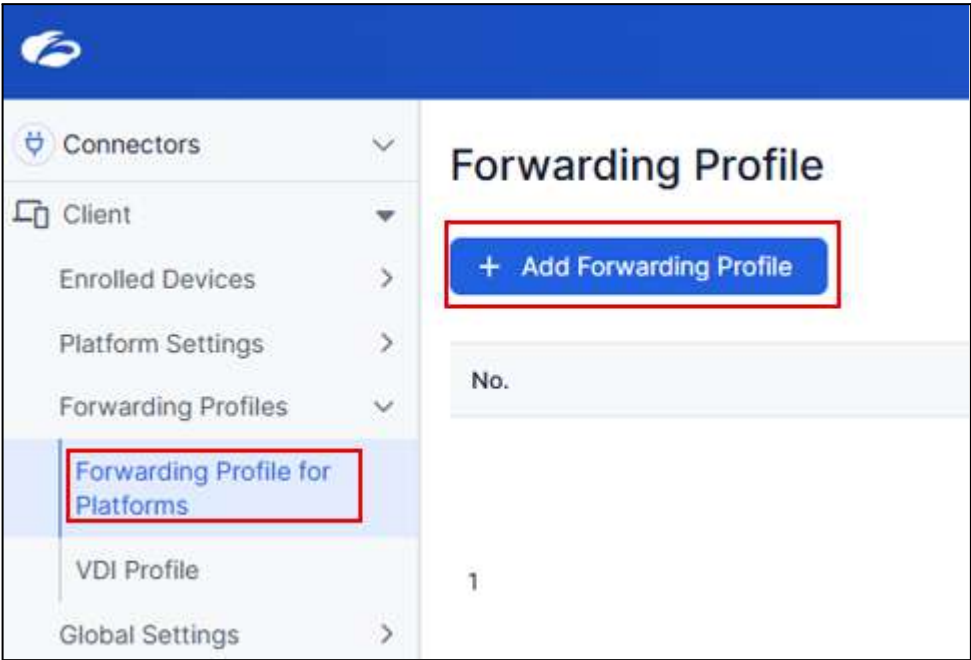
Task 3.1: Configure Traffic Forwarding Options

In this task, you will configure traffic forwarding so that all users send traffic, regardless of protocol to Zscaler.

- 1. Login to the Experience center:
 - a. Navigate to `https://console.zscaler.com` and authenticate with the student account.
 - i. Username: **student@<Student FQDN>**
 - ii. Password: The password you configured at initial login in Lab 1.
 - b. Click **Sign in**.
- 2. Create Forwarding Profile:
 - a. Navigate to **Infrastructure > Connectors > Client > Forwarding Profile for Platforms**.



- b. Click **Add Forwarding Profile**.



- c. Name the profile: **HandsOnLab**.
- d. Scroll down. In the **Windows Driver Selection** section, select **Packet Filter Based**.

Note: If Zscaler Client Connector version 4.8 or higher is installed on Windows, the application will default to the packet filter-based driver, ignoring the Route-Based configuration option

- e. In the **FORWARDING PROFILE ACTION FOR ZIA** section:
 - i. For **On-Trusted Network**, select **Tunnel**.
 - ii. For the **Tunnel version selection**, use **Z-Tunnel 2.0**.

Note: Tunnel 2.0 is necessary to intercept all traffic to tunnel to ZIA for Firewall functionality.

Add Forwarding Profile

PROFILE DEFINITION

Profile Name ⓘ

HandsOnLab

TRUSTED NETWORK CRITERIA

Add Condition ⓘ

Select

Add Condition

TRUSTED NETWORK EVALUATION

Evaluate Trusted NW against All Default Route Adapters ⓘ

v. 3.8.0+

v. 3.9.0+

Enable Split VPN-Trusted Network ⓘ

v. 3.8.0+

v. 4.1.0+

Skip Trusted Criteria match for VPN Adapters ⓘ

v. 3.9.0+

v. 3.9.0+

WINDOWS DRIVER SELECTION

Tunnel Driver Type ⓘ

Route-Based

Packet Filter-Based

FORWARDING PROFILE ACTION FOR ZIA

On-Trusted Network ⓘ

Tunnel

Tunnel with Local Proxy

Enforce Proxy

None

Tunnel version selection ⓘ

v. 2.0.0+

v. 2.0.0+

Z-Tunnel 1.0

Z-Tunnel 2.0

- f. For **VPN Trusted Network** and **Off-Trusted Network**, click the box next to **Same as “On-Trusted Network”**
- g. In the **FORWARDING PROFILE ACTION FOR ZPA** section,
 - i. For **On-Trusted Network**, select **Tunnel**.
 - ii. For **VPN Trusted Network** and **Off-Trusted Network**, select **Same as “On-Trusted Network”**
- h. Click **Save**.

VPN-Trusted Network ⓘ

☒ Same as "On-Trusted Network"

Tunnel

Tunnel with Local Proxy

Enforce Proxy

None

Off-Trusted Network ⓘ

☒ Same as "On-Trusted Network"

Tunnel

Tunnel with Local Proxy

Enforce Proxy

None

FORWARDING PROFILE ACTION FOR ZPA

On-Trusted Network

Tunnel

None

VPN-Trusted Network

☒ Same as "On-Trusted Network"

Tunnel

None

Off-Trusted Network

☒ Same as "On-Trusted Network"

Tunnel

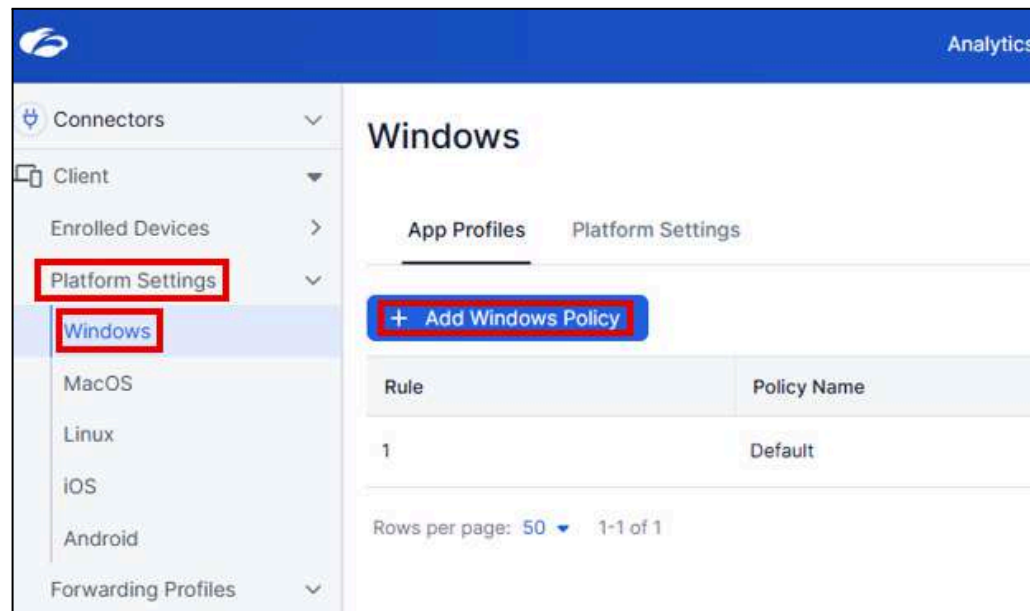
None

Save

Cancel

3. Add Windows App Profile :

- a. Use the left navigation panel, to navigate to **Platform Settings > Windows**.
- b. **Click Add Windows Policy** under the **App Profile** section.



- c. Configure the following settings (leave any unspecified fields at their default settings):
 - i. Name: **HandsOnLab**
 - ii. Rule Order: **1**
 - iii. Status: **Enable**
 - iv. Forwarding Profile: select the **HandsOnLab** profile created earlier.
 - v. Install Zscaler SSL Certificate: **On**
 - vi. User Groups: **Select All**.
- d. Click **Add**.

Add Windows Policy

Search

GENERAL

NameHandsOnLab

Rule Order

1

Status

Disabled

Enabled

Forwarding Profile

HandsOnLab

ZIA Posture Profile

Select ZIA Posture Profile

Install Zscaler SSL Certificate

Install WFP Driver

GROUPS

User Groups

All Selected

Users

None

Follow Global Settings for Partner Login

TRAFFIC STEERING

PAC and Proxy

App and IP Bypass

Disaster Recovery

DNS

Advanced

PAC CONFIGURATION

Add

Cancel

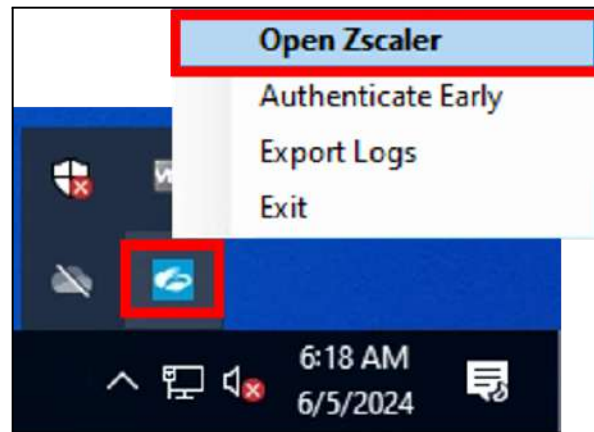
© 2025 Zscaler, Inc. All rights reserved. | ZSCALER CONFIDENTIAL INFORMATION

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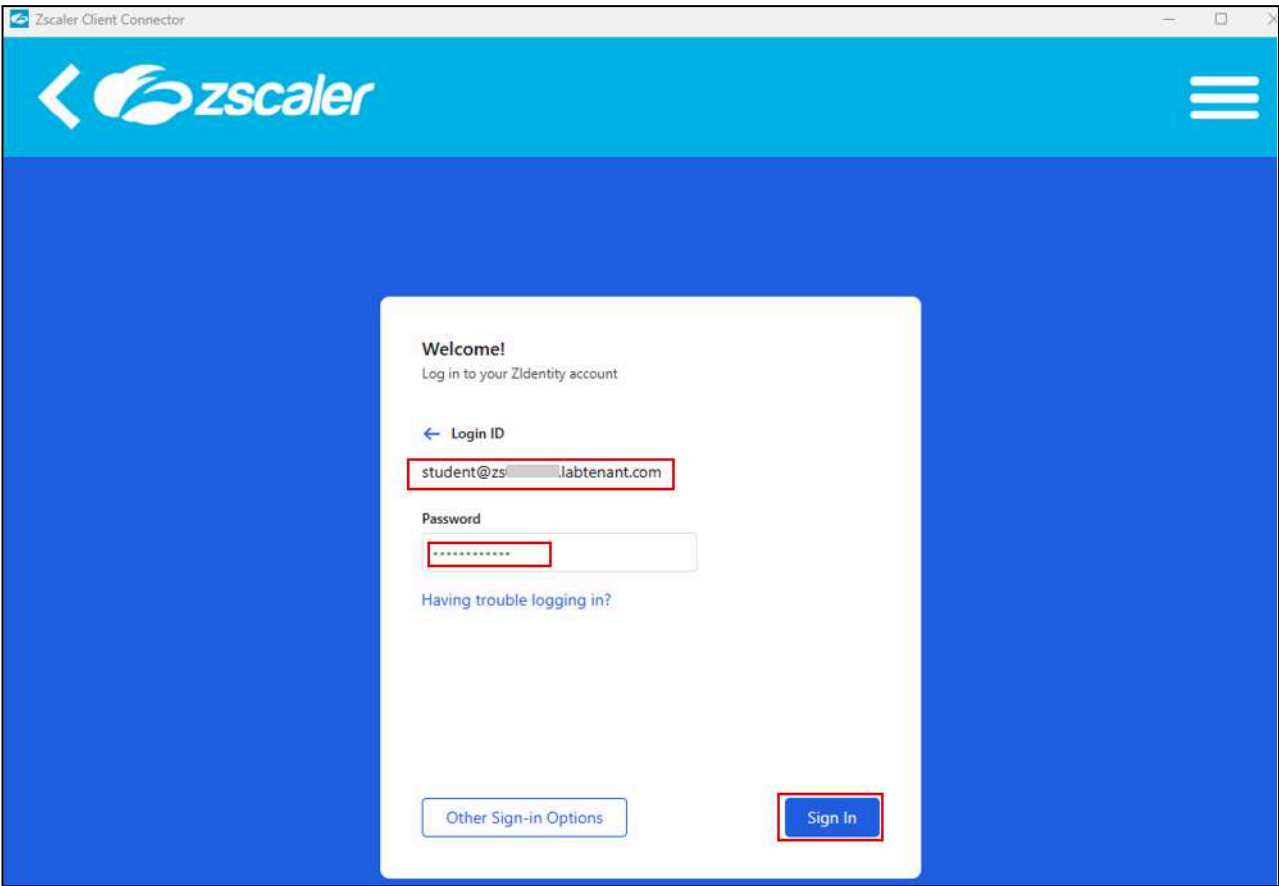
Task 3.2: Sign In to the Zscaler Client Connector and Verify Protection

In this task, you will log in to the **Zscaler Client Connector** to confirm that the **Corp: Client PC** is now protected by the app.

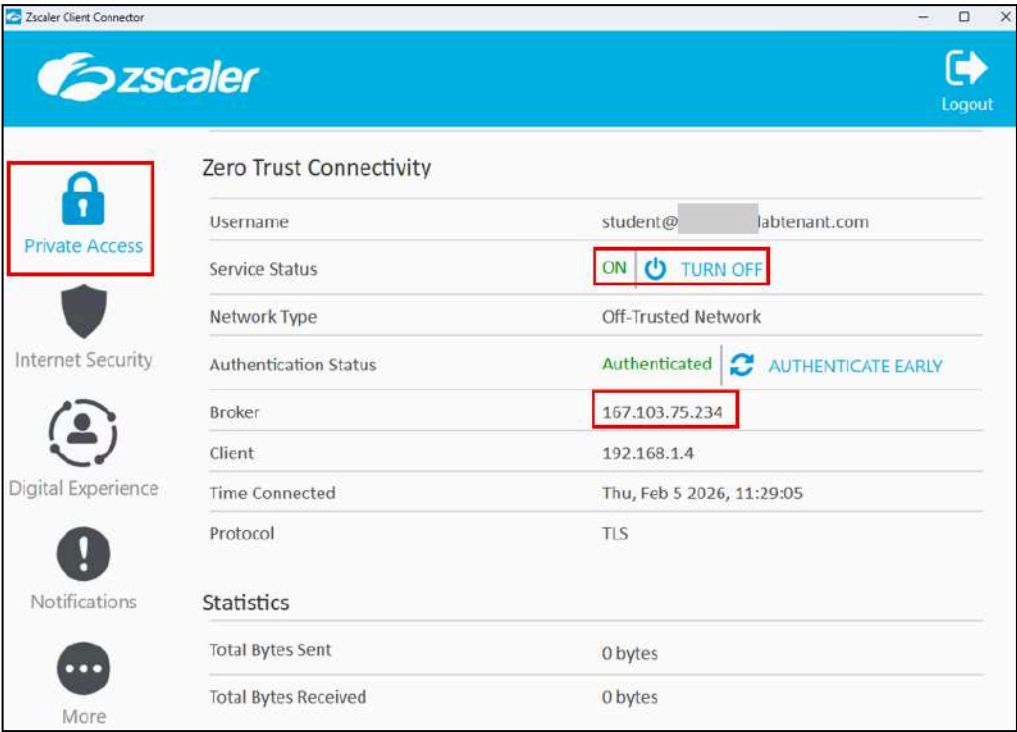
1. Open the Zscaler Client Connector:
 - a. On the **Corp: Client PC**, in the **Windows Task Bar**, (bottom-right corner), click the Show hidden icons option.
 - b. Select the **Zscaler Client Connector** icon and click **Open Zscaler**.



- c. In the **Zscaler Client Connector console**, Login using the student credentials:
 - i. Username: **student@<Student FQDN>**
 - ii. Password: **as set in Lab 1.**
 - d. Click **Sign in**



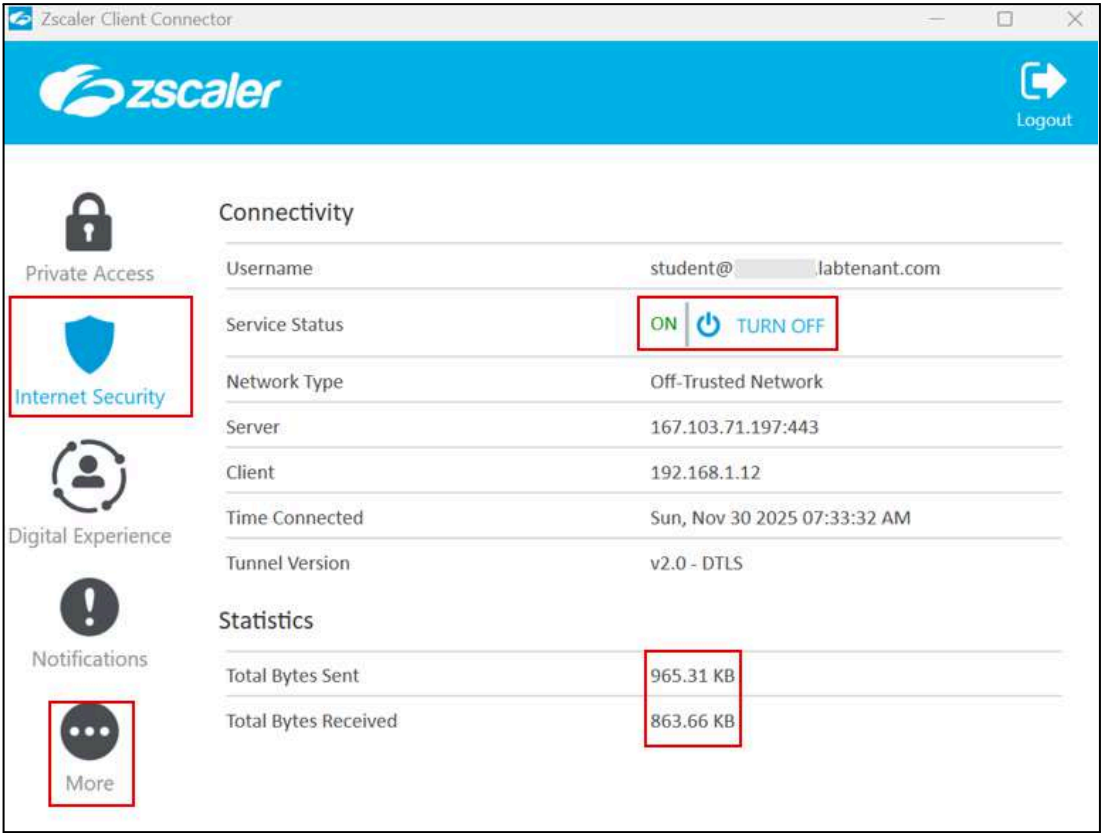
2. Verify Private Access Settings:
- a. On the **Private Access** page, confirm the following:
 - i. Service Status: **ON**.
 - ii. **Client** and **Broker IP** addresses: populated



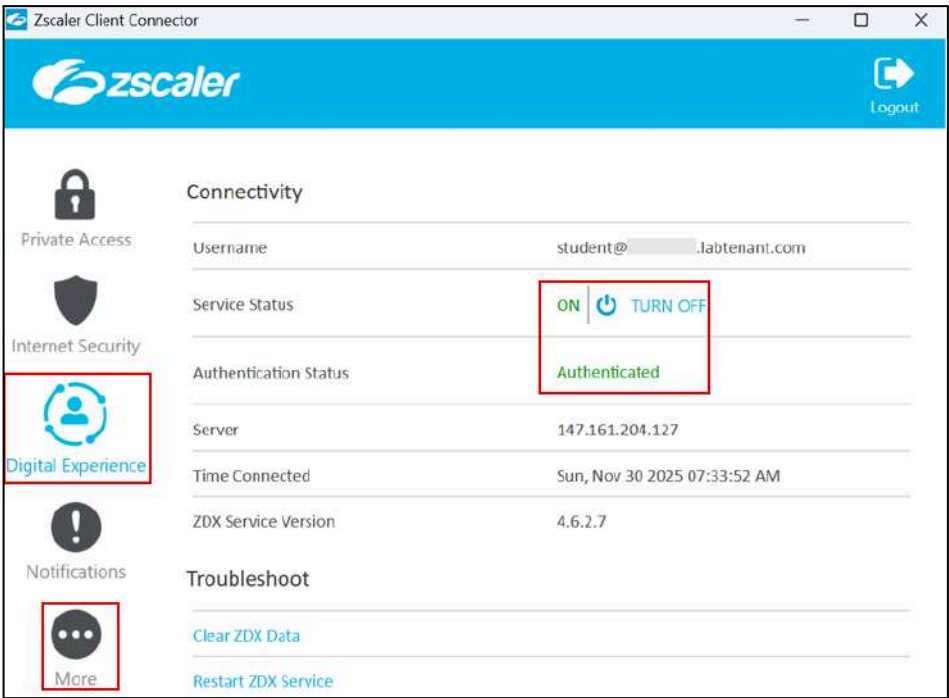
Note: If you do not see **ZPA** or **ZDX** in Zscaler Client Connector, your user may not have the correct service entitlements. To verify entitlements for the Student Admin user, navigate to **Administration > Users** in the portal, **select the Student Admin user account**, and **click the Edit icon**. In the Edit User pane, **go to the Entitlements section** and **confirm** that **ZPA** and **ZDX** are assigned to the user. The Broker IP address highlighted in the screenshot is an example and may not match the IP address you see in your own Zscaler Client Connector.

3. Verify Internet Security Settings:

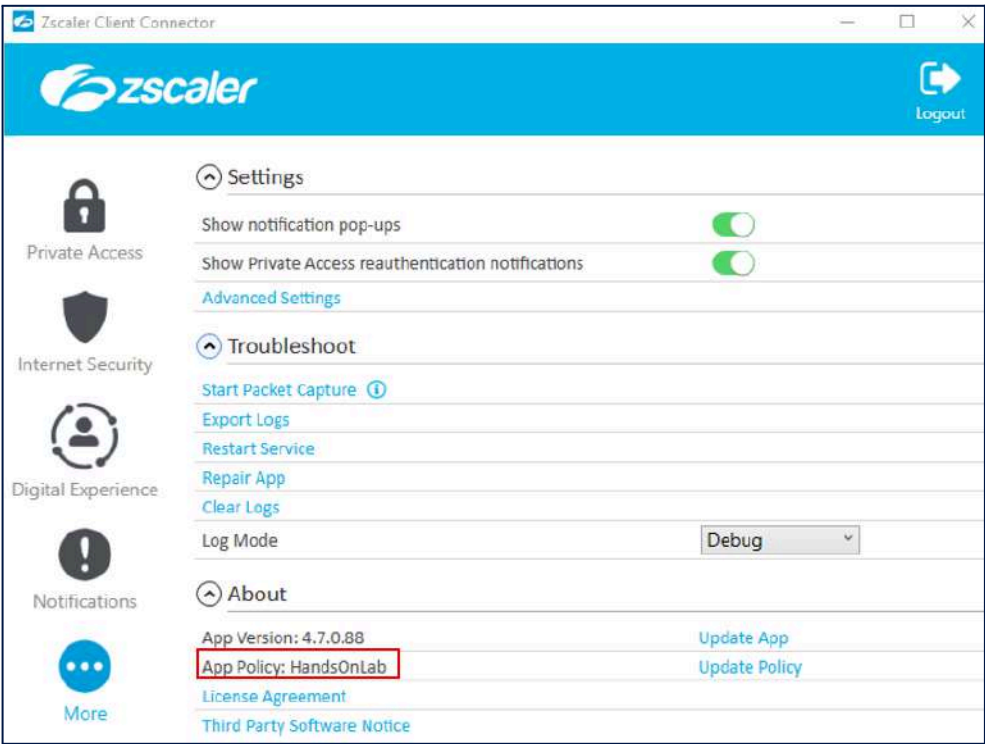
- a. On the **Internet Security** page, confirm the following:
 - i. Service Status: **ON**.
 - ii. Username: Displays correctly.
 - iii. **Total Bytes Sent / Received:** These values increment as you load web pages.



3. Verify Digital Experience Settings:
- a. On the **Digital Experience** page, confirm that:
 - i. Service Status : **ON**.
 - ii. Authentication Status : **Authenticated**.
 - iii. Username: Displays correctly.



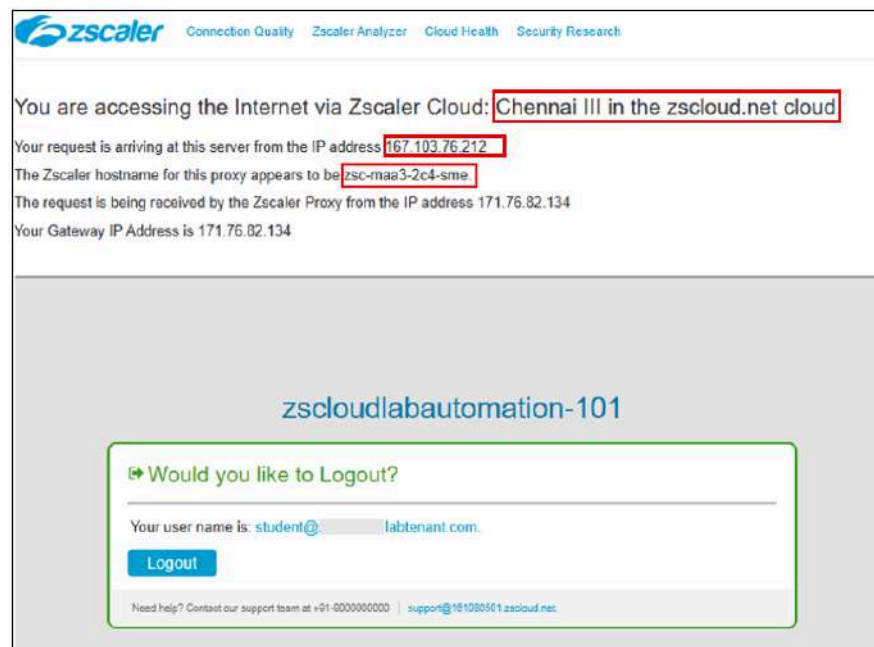
- 4. Verify Applied App Policy:
 - a. On the **More** page, confirm that:
 - i. Appy Policy: **HandOnLab**



Note: If the updated policy, **HandsOnLab** is not reflected on the Zscaler Client Connector (ZCC), click Update Policy in the ZCC under the **More** tab to force a policy refresh. Wait a few seconds and then re-verify.

5. Verify Traffic Forwarding:

- a. Navigate to <https://ip.zscaler.com>
- b. Confirm that you are accessing the Internet via a service edge geographically close to the user.



Note: The Service Edge location shown in results will typically reflect a location that is geographically close to the user.

Lab 4: Configure Firewall Filtering Policy

Scenario: As the **Zscaler administrator**, you are tasked with **blocking access** to the external application **WhatsApp** to prevent organizational users from using it. At the same time, you need to ensure that **SSH and ICMP traffic is enabled** for all other applications.

Objectives: By the end of this lab, you will:

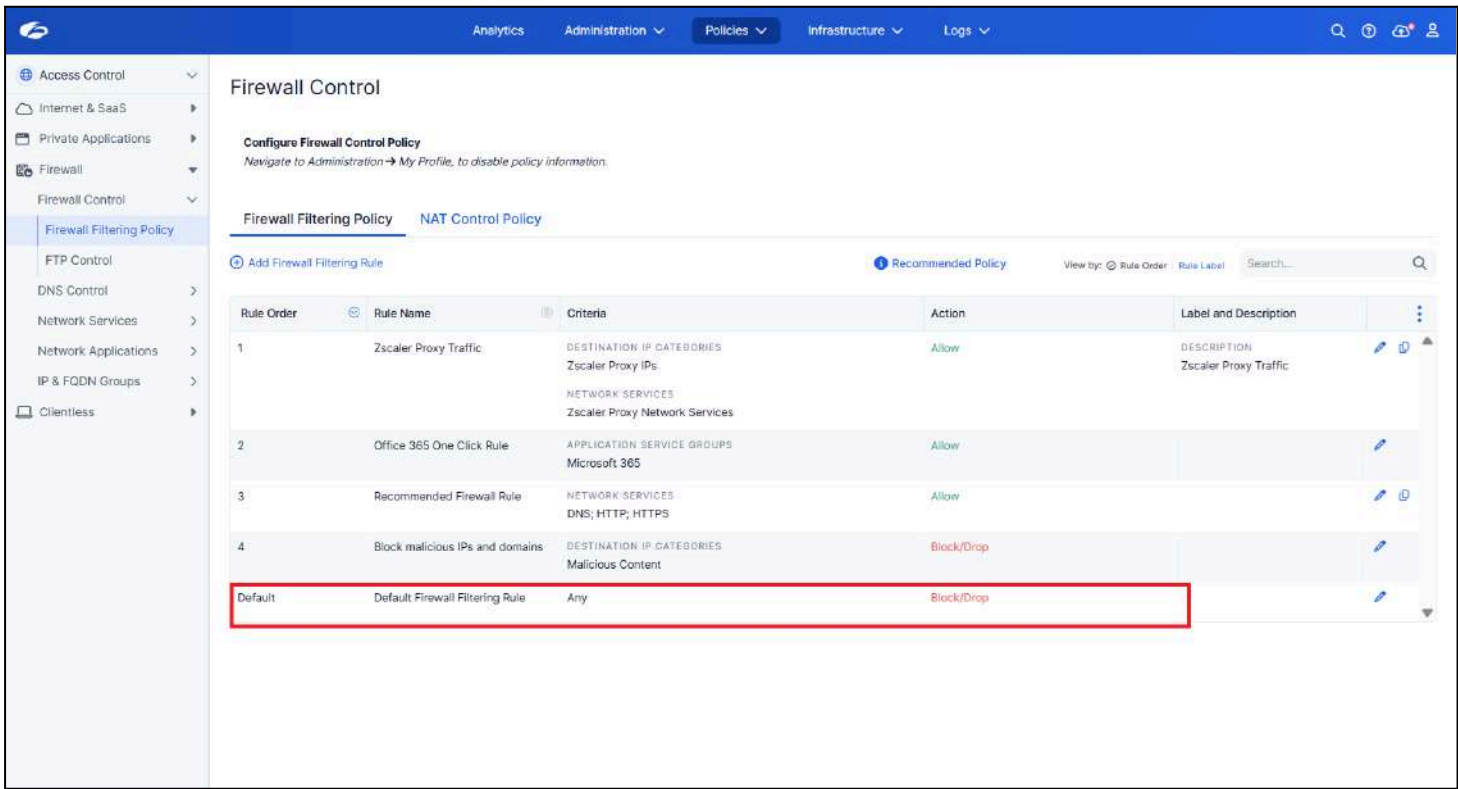
- **Verify** the **default firewall filtering policy** to confirm it blocks all non-web traffic.
- **Test default firewall rules** that block ICMP and SSH traffic.
- **Create** and **configure firewall filtering rules** to **allow** outbound ICMP and SSH traffic.
- **Create** and **configure firewall filtering rules** to **block** access to specific external applications like WhatsApp.
- **Verify firewall filtering policies** using practical, real-world testing scenarios.

Approximate Time: 25 minutes

Task 4.1: Test Non-Web Traffic with the Default Firewall Block

With **Z-Tunnel 2.0**, all traffic for all ports and protocols is sent to Zscaler for inspection. In this task, you will generate non-web traffic from the **Corp: Client PC** to verify that it is **blocked** by the **default Block/Drop rule**.

- 1. Verify the Default Firewall Policy:
 - a. From the **Experience Center Portal**, go to **Policies > Access Control > Firewall > Firewall Filtering Policy**.
 - b. Confirm the following:
 - i. The **Default Rule** is set to **Block/Drop**.
 - ii. There are no other rules that would allow non-web traffic (refer to the example image).



Note: Rules such as the Office 365 One Click Rule, Zscaler Proxy Traffic, and Recommended Firewall Rule explicitly allow specific traffic required for the proper functioning of relevant features. **ICMP and SSH traffic in this lab exercise will not match any of these rules** and will therefore fall to the **Default Rule** until explicitly allowed in subsequent steps.

2. Test the default firewall policies on the **Corp: Client PC VM**:

a. Generate non-web traffic via **Zscaler Client Connector**.

- i. Open a **command prompt** and type: `ping www.google.com`

Note: This should fail with the request timing out since the outbound ICMP packets are dropped by the Default Firewall rule.

b. SSH Traffic Test

- i. Open **puTTY**. (double-click the desktop icon)
- ii. in the Host Name field , enter: `github.com`
- iii. Click **Open**

Note: This should fail. Normally, the session would prompt for a username and password. However, since the outbound TCP connection is dropped by the Default Firewall Rule, the session will display a PuTTY Fatal Error after timing out (~20 seconds).

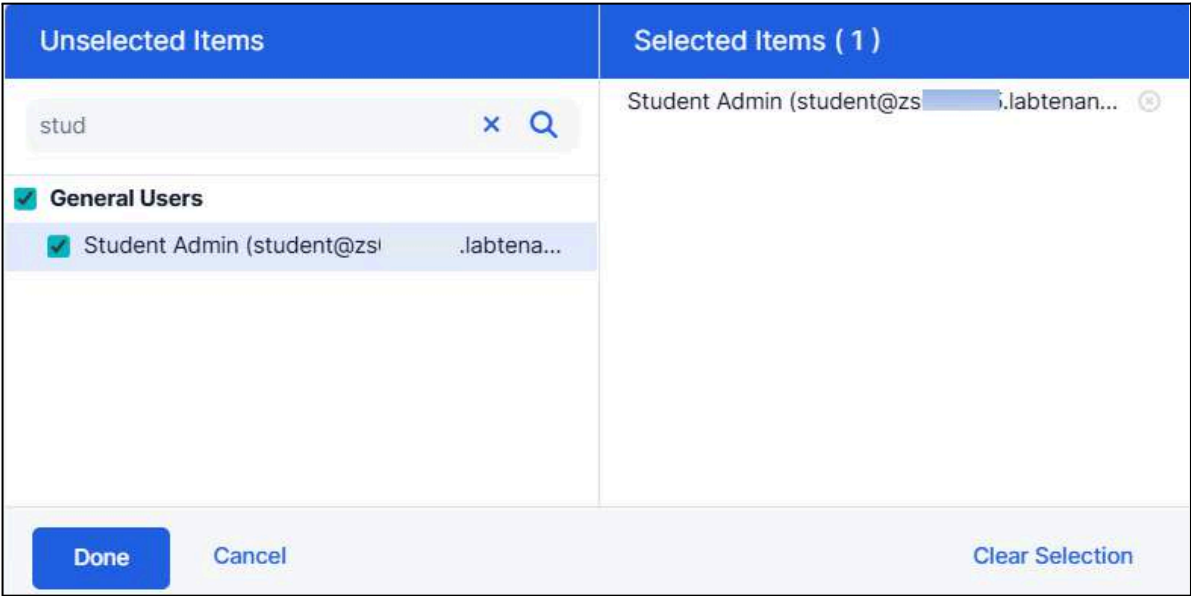
3. End the Unsuccessful Sessions:

- a. Close the command prompt
- b. Exit puTTY by closing the application.

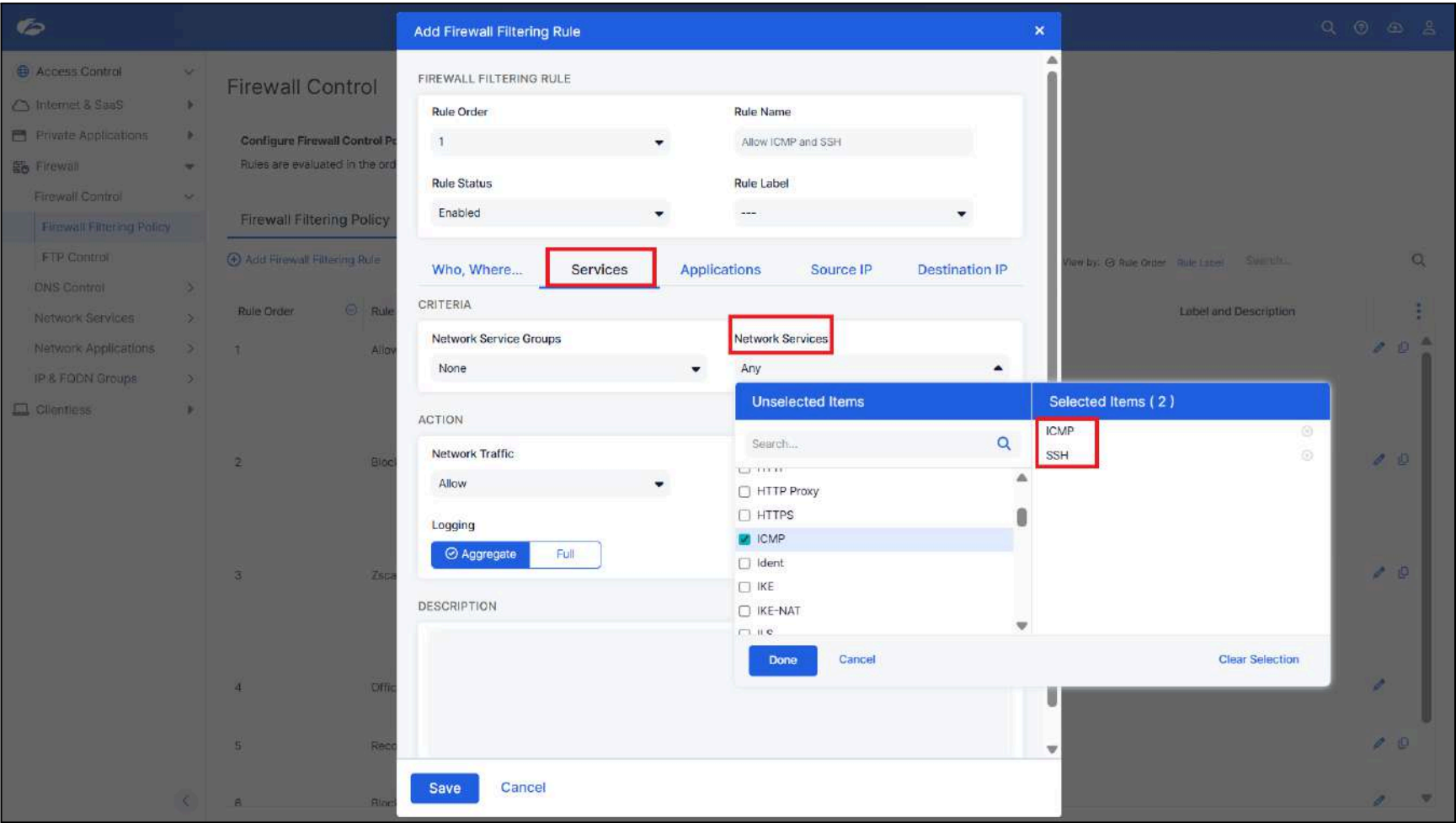
Task 4.2: Configure Firewall Policy to Allow Outbound SSH and ICMP Traffic

ICMP and SSH traffic blocked in the previous task now needs to be permitted to pass. In this task, you will configure the firewall policies to allow this traffic.

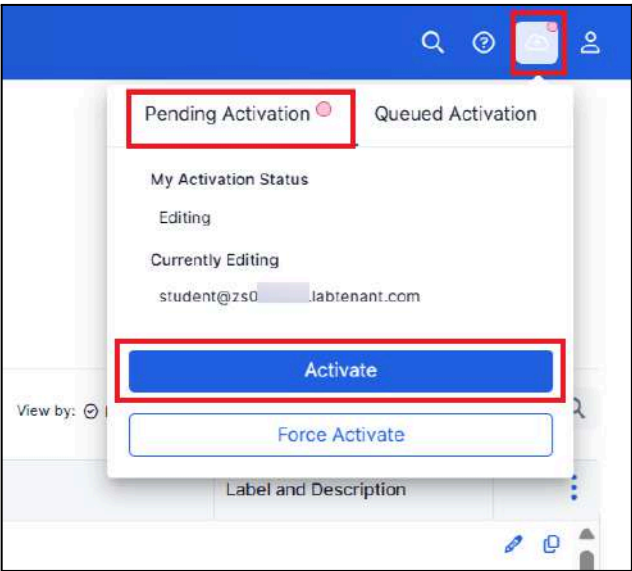
1. Create a Firewall Filtering Rule to allow ICMP and SSH:
 - a. From the **Experience Center Portal** go to **Policies > Access Control > Firewall > Firewall Filtering Policy**.
 - b. Click **Add Firewall Filtering Rule**.
 - c. Configure the rule:
 - i. Rule Order: **1**
 - ii. Rule Status: **Enabled** (default)
 - iii. Rule Name: **allow-outbound-icmp_ssh-student-any**
 - **[action]-[direction]-[services]-[source-scope]-[destination-scope]**
 - d. On the Who, Where, Criteria tab select your Student user account.

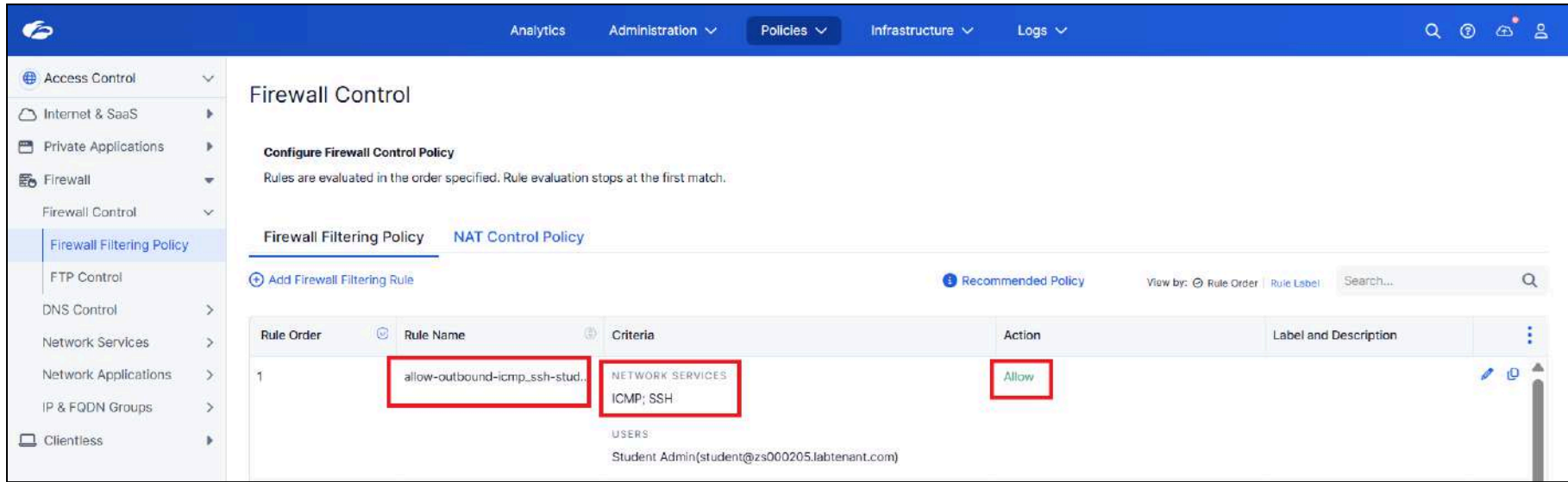


- e. Click on the **Services** tab.
- f. Under the **Criteria** section select the **Network Services** drop-down.
- g. **Search** for and **select ICMP, SSH**
- h. Click **Done**



- i. Under the **ACTION** section ensure that Network Traffic is set to **Allow**.
- j. Click **Save** and **Activate** the changes from the **Activation** icon (cloud with arrow))in the top-right corner..





Test

ICMP traffic with the new rule:

- k. On the **Corp: Client PC**, open a command prompt and execute: `ping www.google.com`

Note: This should now show a response since the outbound ICMP packets match the Allow ICMP_SSH rule and pass through the firewall.

2. Test SSH traffic with the new rule:

- a. On the **Corp: Client PC** open PuTTY. (Double click on the Desktop icon to launch PuTTY)
- b. Type **Github.com** in the Host Name and click **Open**

Note: PuTTY should now attempt to establish a TCP connection and prompt for login and you will likely be prompted to accept the fingerprint of the external server, indicating that the SSH packets match the Allow ICMP and SSH rule and pass through the firewall.

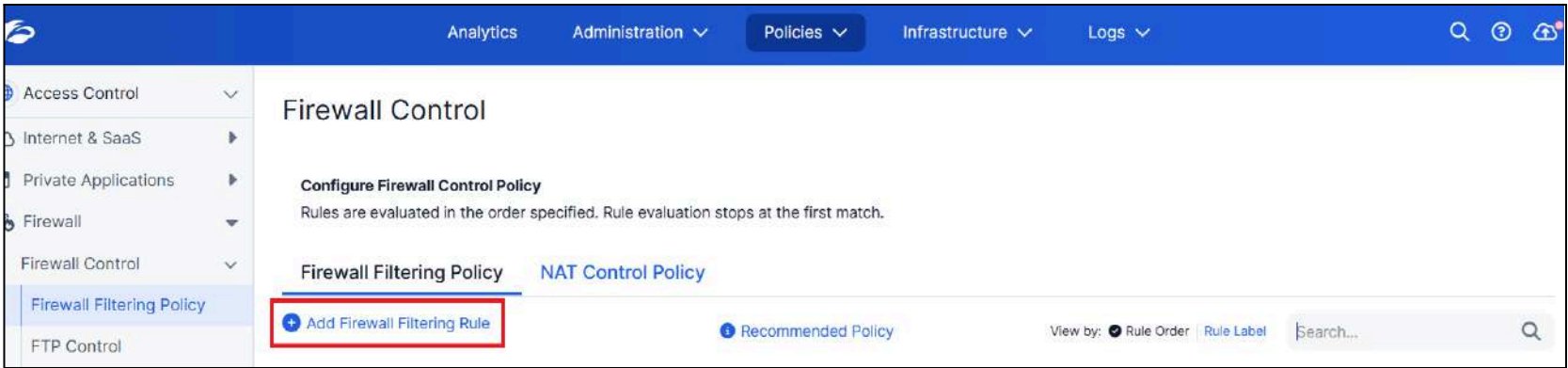
3. Close the sessions:

- a. **Close** the **Command Prompt** window.
- b. **Exit** PuTTY application.

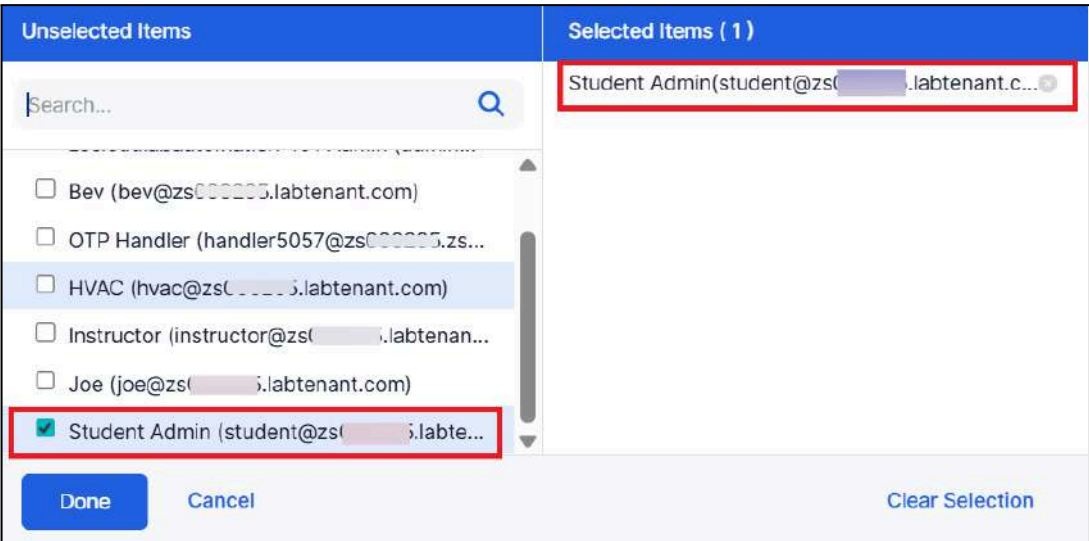
Task 4.3: Configure a Firewall Filtering Policy to Restrict Applications

In this task you will create and activate a Firewall Filtering Policy to block access to the external application WhatsApp.

- 1. Create a Firewall Filtering Rule
 - a. From the **Firewall Control** menu. Or, **Policies > Access Control > Firewall > Firewall Filtering Policy**
 - b. Click **Add Firewall Filtering Rule** to create a rule.

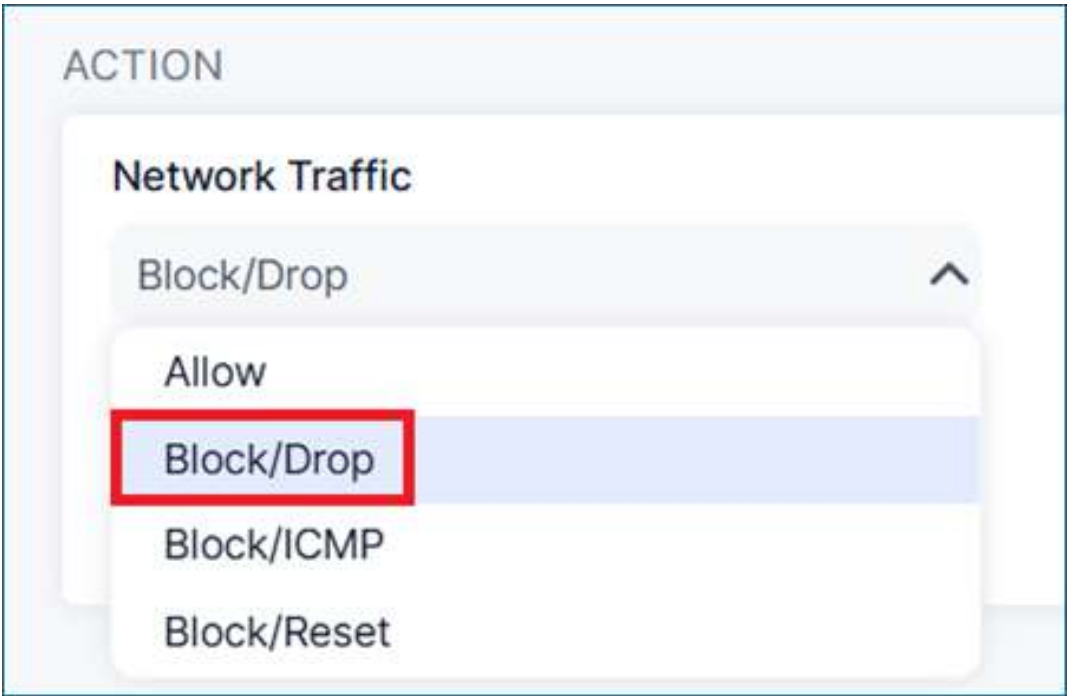


- c. In the **Add Firewall Filtering Rule** window input the below details:
 - i. Rule Name: **block_drop_all_whatsapp_student_any**
 - ii. Rule Order: **2**
 - iii. Rule Status: **Enabled** (default)
- d. Under the **Who, Where & When** tab
 - i. User: Select the **Student Admin**



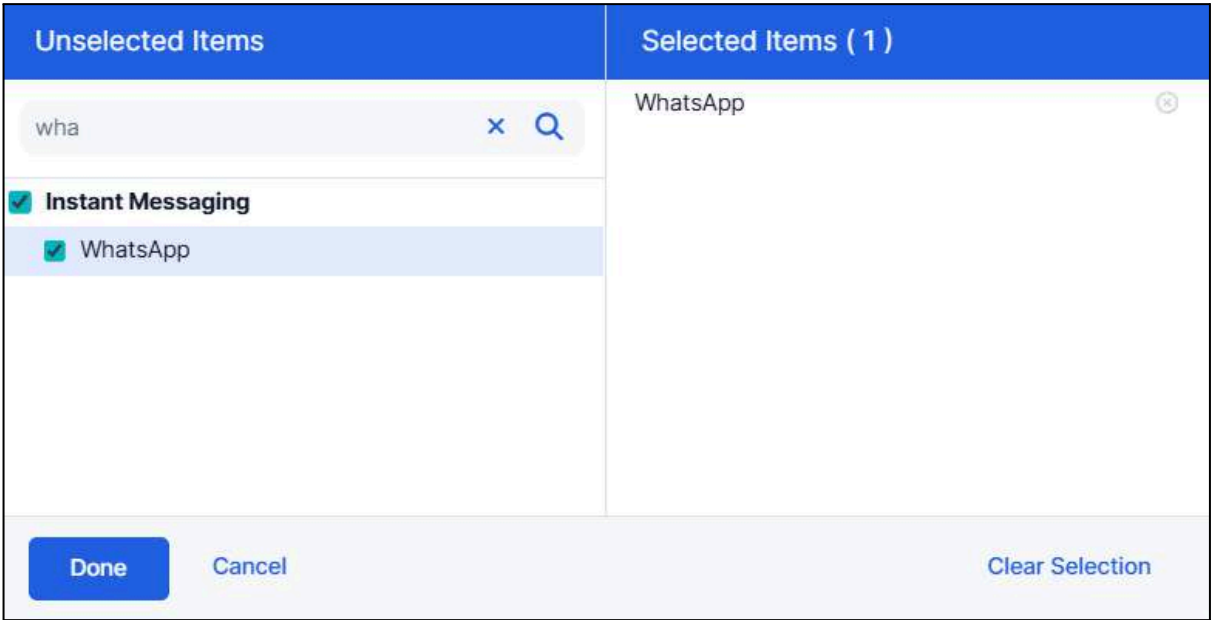
e. **Action**

- i. Network Traffic: **Block/Drop**

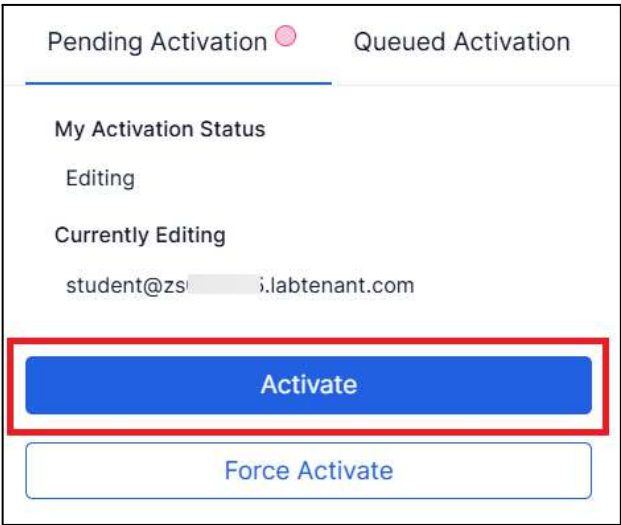


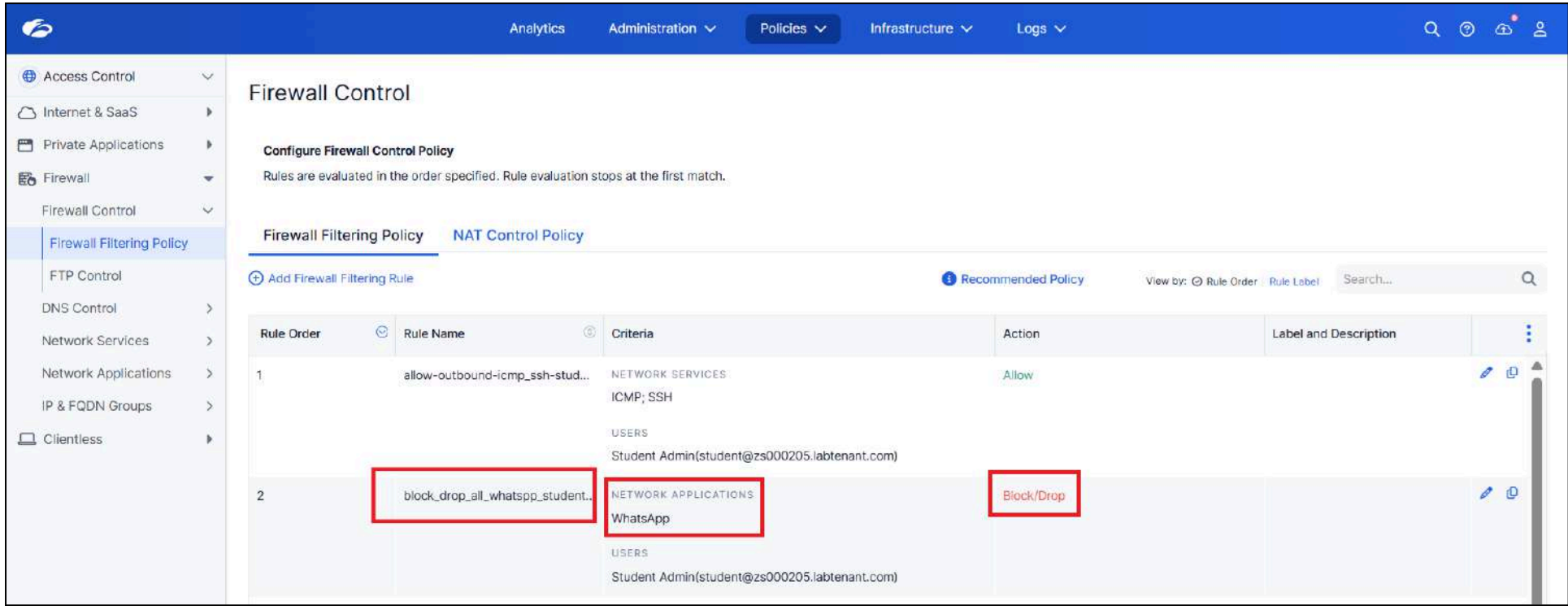
f. Under the **Applications** tab

- i. Network Applications: **Search** for and **select WhatsApp**



- g. Click **Done** and **Save** to save the rule.
- h. Activate changes from the **Activation** icon (Cloud with arrow) at the top-right of the screen.

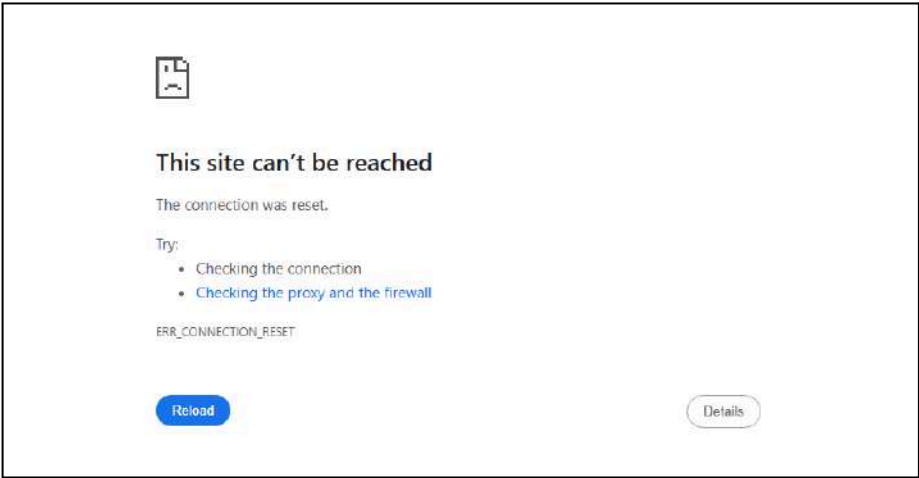




2. **Test** the Firewall Rule

- i. Launch a web browser and navigate to `https://web.whatsapp.com`

Note: Access to `web.whatsapp.com` should be restricted.



Note: While URL Filtering Policies can block access to web.whatsapp.com, this method may not restrict users who have the WhatsApp application installed on their system. By using a Firewall Filtering Policy, you can block access at the network layer, ensuring complete enforcement of access controls.

Lab 5: Configure SSL Inspection Policies

Scenario: As the Zscaler administrator, you are tasked with **enabling SSL/TLS inspection** to **monitor and enforce policies** on users' **encrypted web traffic**. Since most modern threats are hidden within HTTPS traffic, enabling **SSL inspection provides visibility into encrypted sessions** and allows policies to be enforced effectively. You need to **configure SSL inspection for all destinations and define exemptions for trusted or privacy-sensitive categories**.

Objectives: By the end of this lab, you will:

- **Enable and configure SSL inspection policies** for all encrypted traffic.
- **Verify Zscaler's certificate installation** on the **Windows Corp: Client PC**.
- **Analyze** and resolve **certificate pinning errors**.
- **Analyze SSL inspection logs** and **identify inspection or bypass events**.

Note: The Zscaler Root CA Certificate is installed on Windows Client PCs when the Zscaler Client Connector is installed. **For Windows devices, the Root CA must be installed in both:**

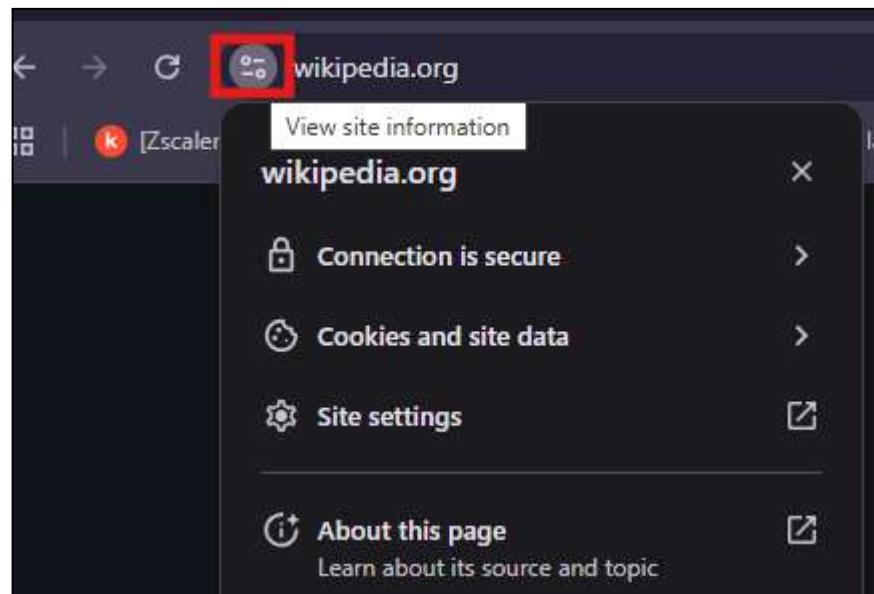
- a. The System Certificate Store (Trusted Root Certification Authorities).
 - b. The Firefox Certificate Store (based on the setting in the App Profile).
-

Approximate Time: 30 minutes

Task 5.1: Enable SSL Inspection for All Destinations

In this task, you will enable SSL inspection for all encrypted traffic.

1. Verify the current SSL Inspection status:
 - a. On the **Corp: Client PC**, open a Chrome browser window in **Incognito Mode**.
 - i. Navigate to: `www.wikipedia.org`
 - b. Click the **site information (filter)** icon in the URL field.

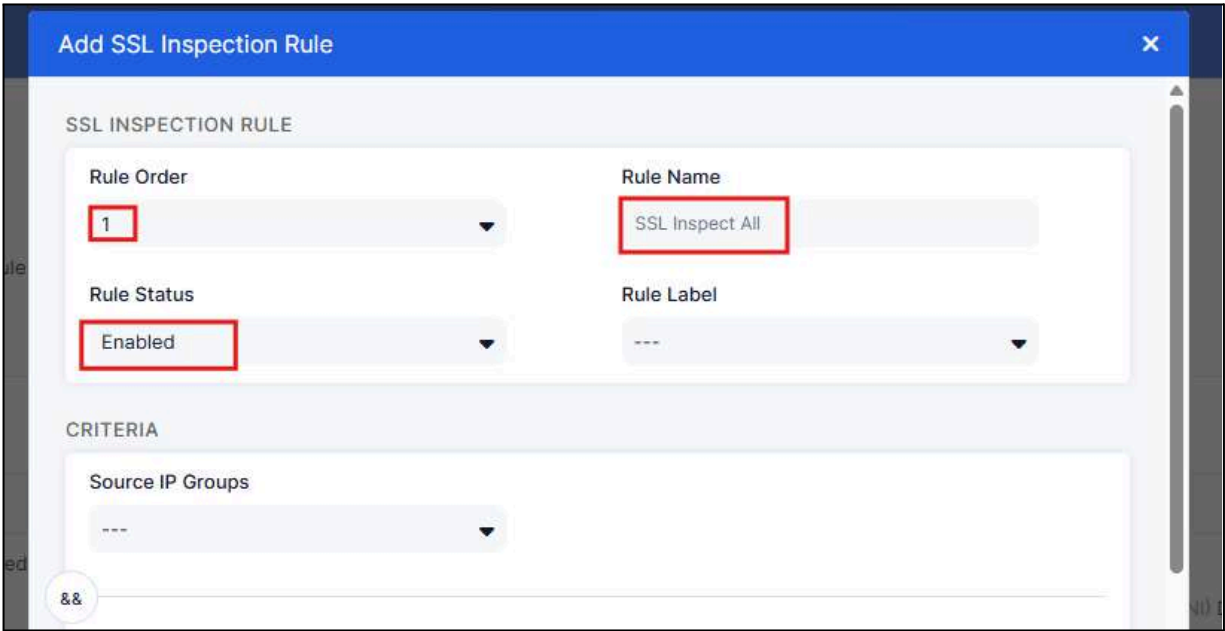
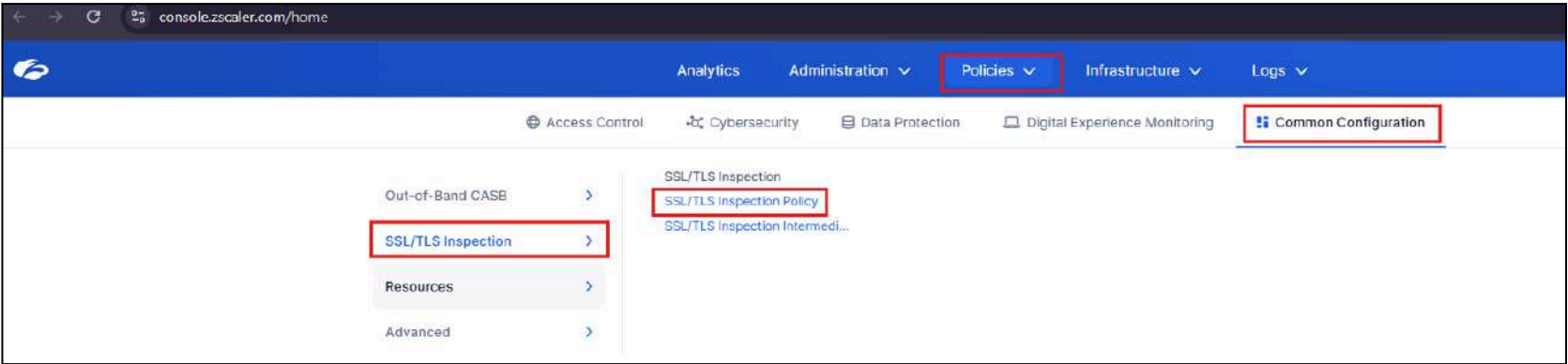


- c. Click **Connection is secure** > **Certificate is valid**.

Note: You will see that you have received the certificate from the destination Web server. You can identify the site information icon by its appearance. It resembles three horizontal lines connected by dots. In some browsers it can also be a padlock icon. This icon is typically displayed to the left of the URL in the browser's address bar.



- 2. Enable SSL Inspection:
 - a. Back in the **Zscaler Experience Center** window, navigate to the **Policies > Common Configuration > SSL/TLS Inspection > SSL/TLS Inspection Policy**.
 - b. On the **SSL Inspection Policy** tab, click **+Add SSL Inspection Rule**.
 - c. Configure the rule as follows:
 - i. Rule Order: **1**.
 - ii. Rule Name: **SSL Inspect All**



d. Scroll down to the **ACTION** section and select **Inspect** under **Action**.

ACTION

Action

Inspect

Do Not Inspect

Block

Override Default Intermediate CA Certificate

Yes

No

Intermediate CA Certificate

Zscaler Intermediate CA Certificate

Untrusted Server Certificates

Allow

Pass Through

Block

Block No Server Name Indication (SNI)

OCSP Revocation Check

Block Undecryptable Traffic

Minimum Client TLS Version

TLS 1.0

Minimum Server TLS Version

TLS 1.0

Enable HTTP/2

DESCRIPTION

Save

Cancel

e. Click **Save** and **Activate**.

Pending Activation

Queued Activation

My Activation Status

Editing

Currently Editing

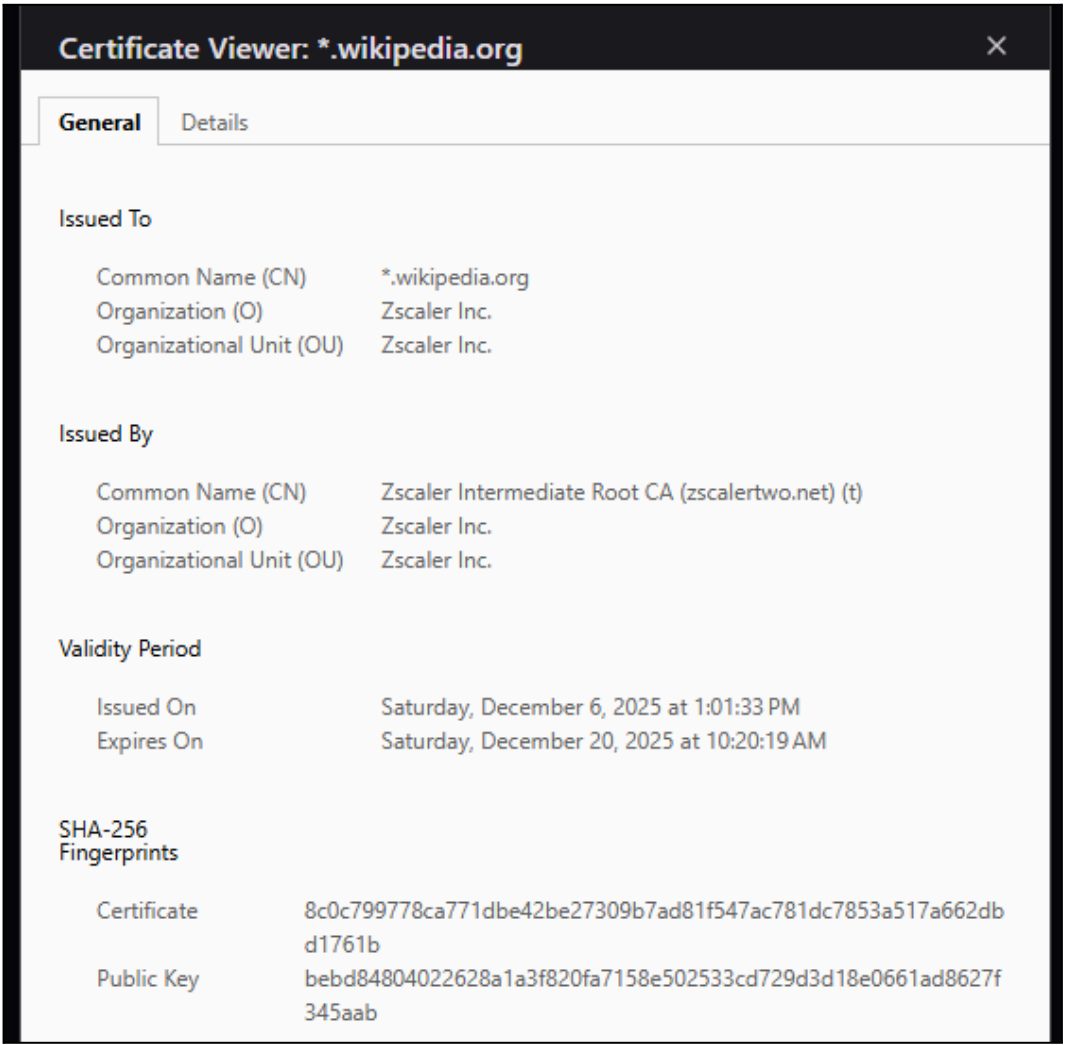
instructor@zs000198.labtenant.com

Activate

Force Activate

- 3. Verify Inspection status:
 - a. Close the **Incognito** browser or refresh it, (to ensure the new policy is applied)
 - b. Re-open a browser window in **Incognito Mode**, and visit `www.wikipedia.org`
 - c. Click the **site information (filter)** icon in the URL field > **Connection is secure** > **Certificate is valid**.

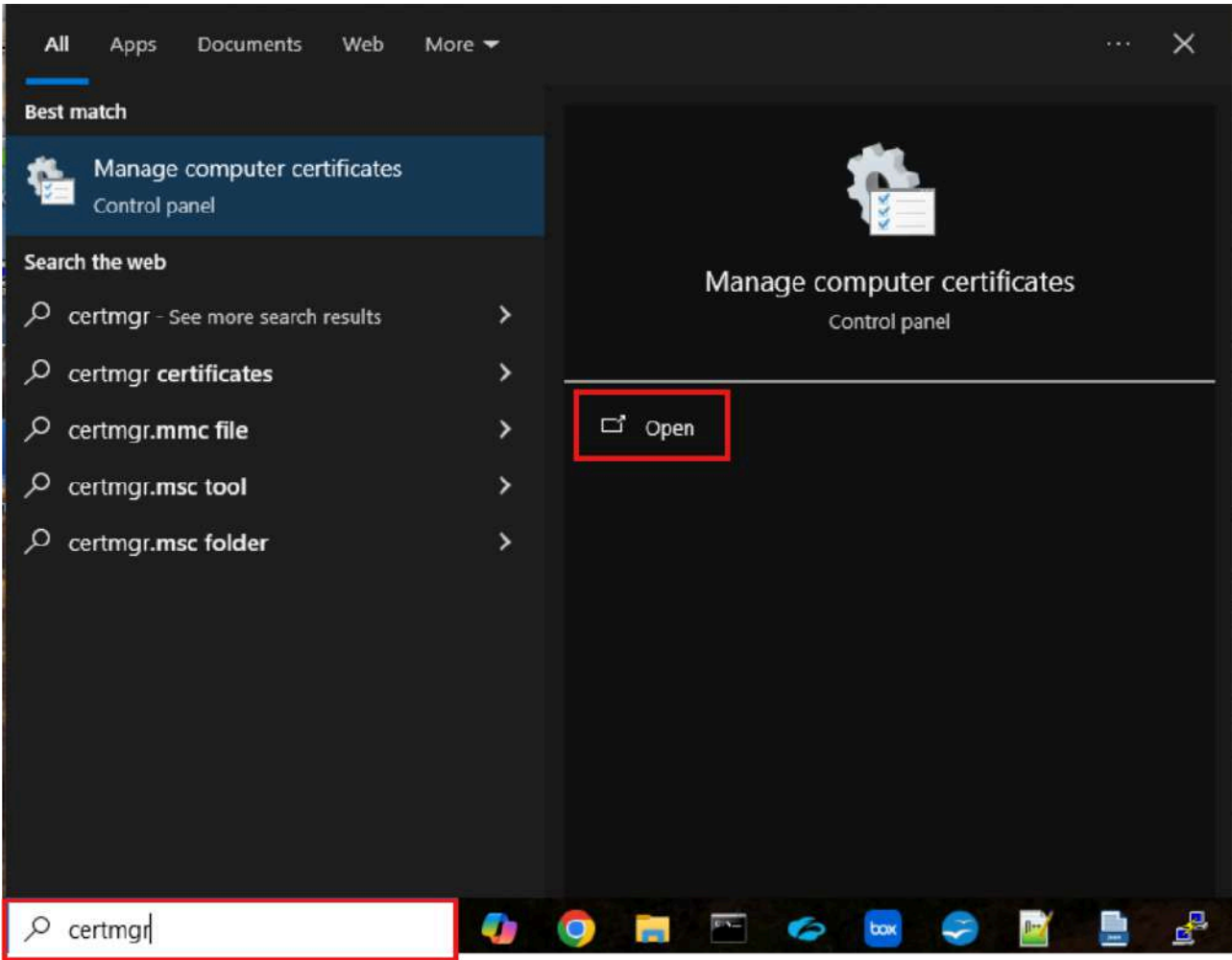
Note: You will see that you have received a server certificate signed by Zscaler.



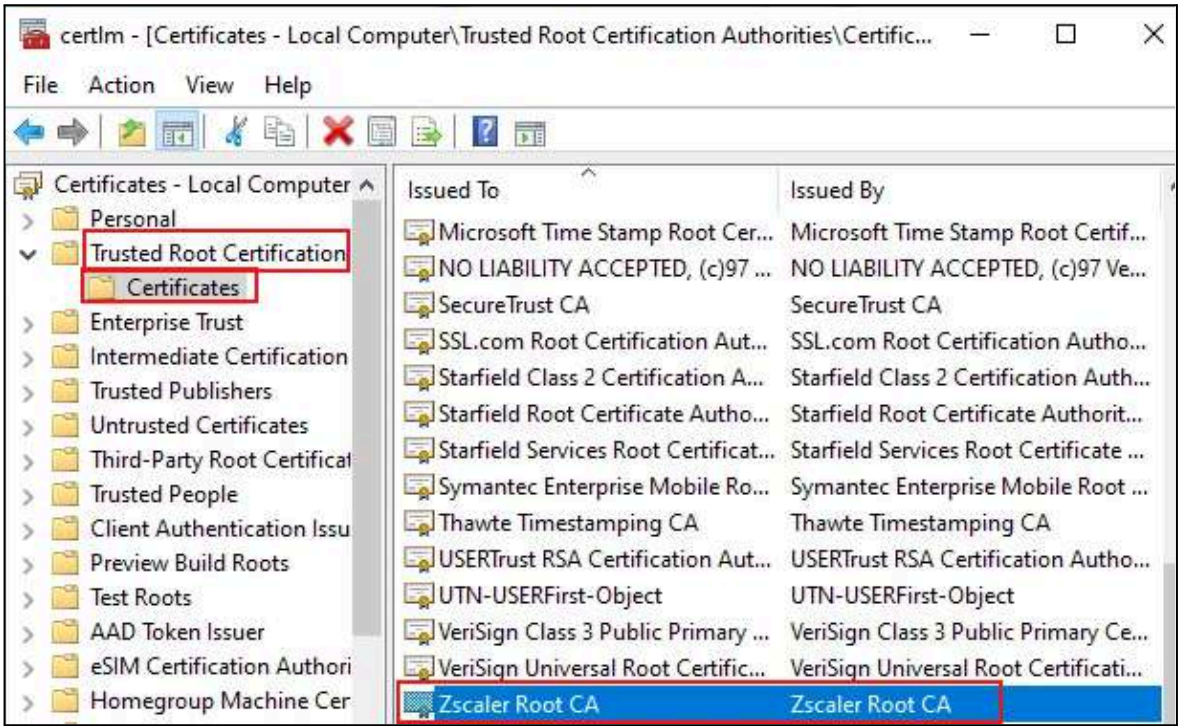
Task 5.2: Verify Zscaler CA Installation [Optional]

In this task, you will **verify** whether the **Zscaler Root CA** is properly **installed** on the **Corp: Client PC**.

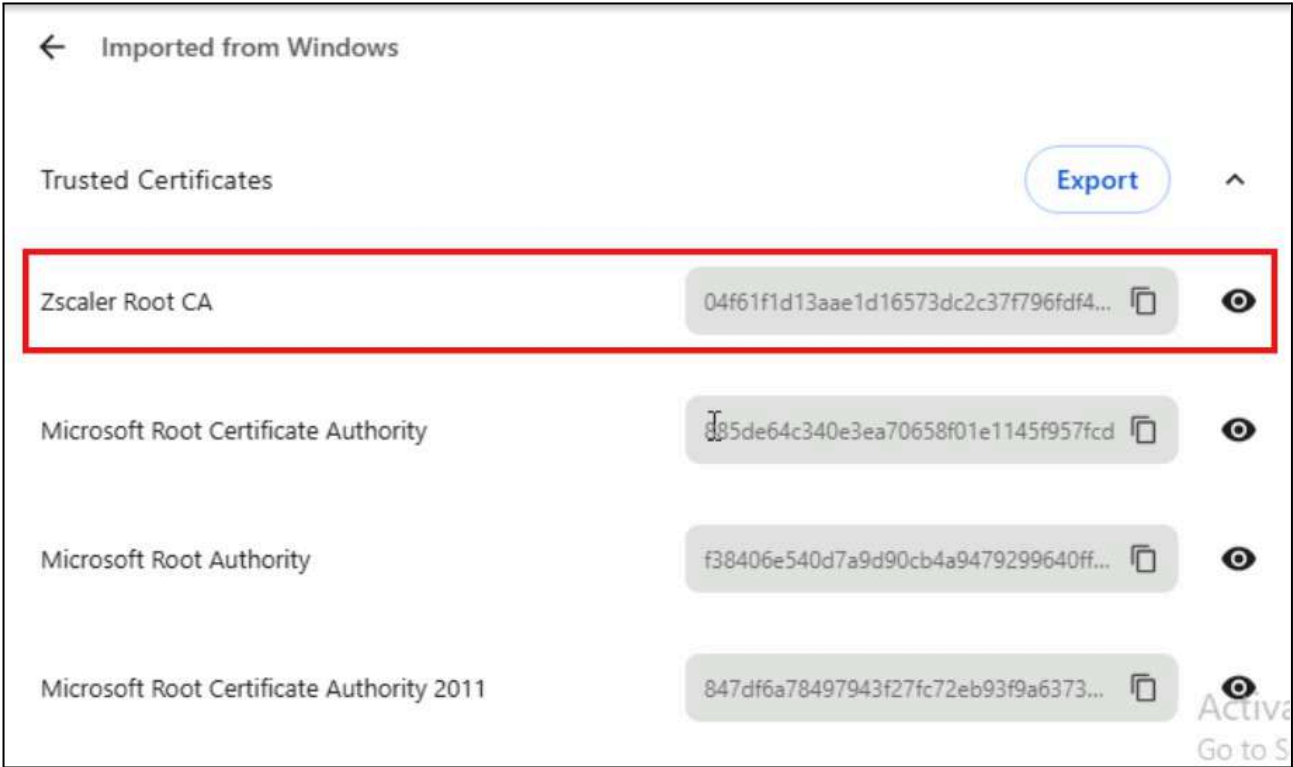
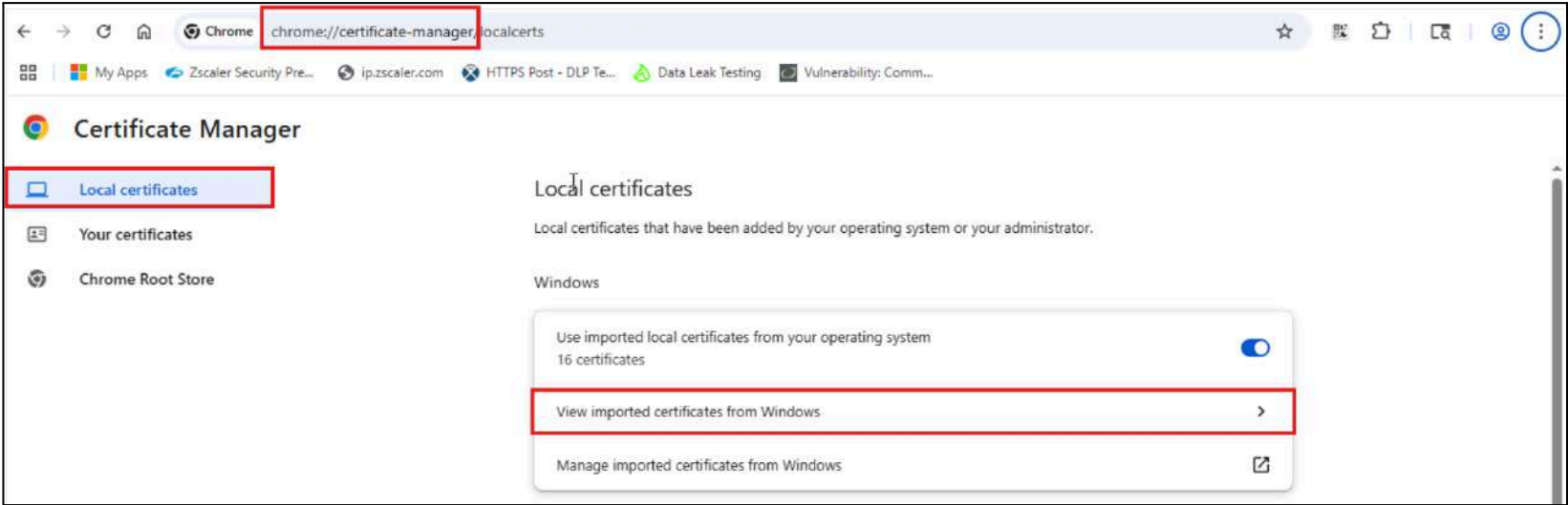
- 1. Open the certmgr tool:
 - a. Type **certmgr** in the Windows search bar or run it in the command prompt.



- 2. Navigate to:
 - a. **Trusted Root Certification Authorities > Certificates.**
- 3. Confirm the following:
 - a. The Zscaler Root CA Certificate is listed.
 - b. The certificate is not expired.



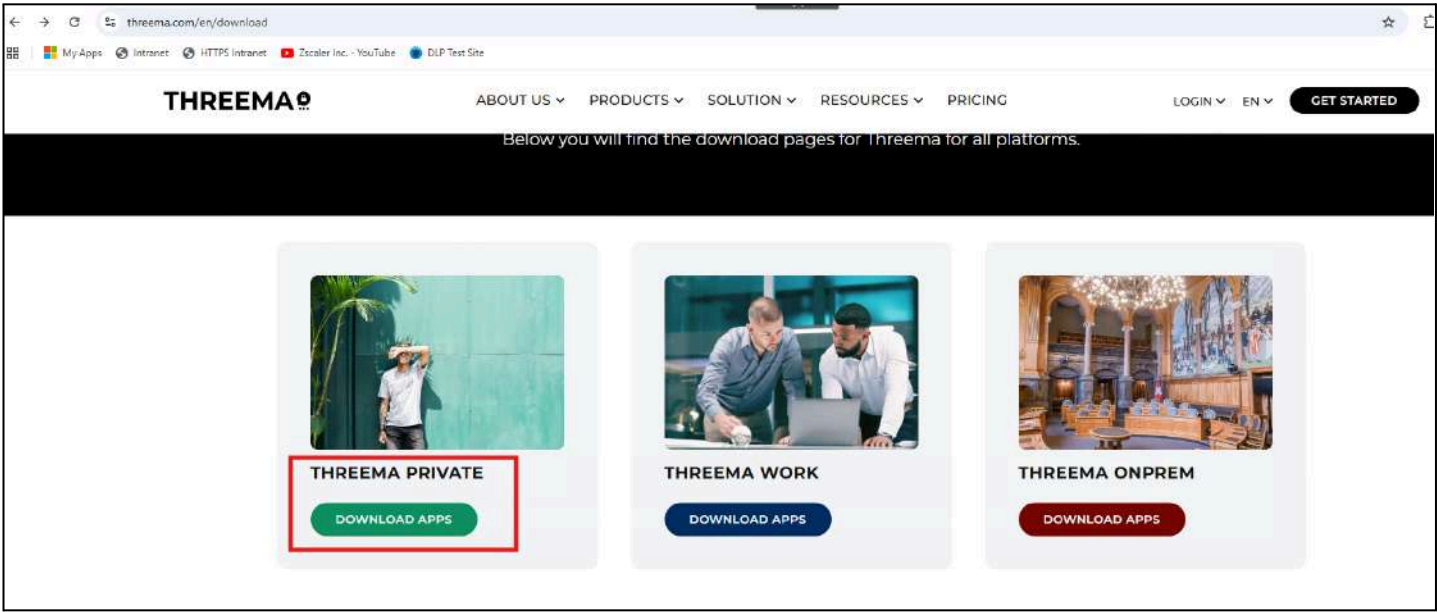
- 4. Verify via Google Chrome:
 - a. Open a browser window and type: `chrome://certificate-manager/` in the address bar.
 - b. Click on **Local certificates.**
 - c. Select on **View imported certificates from windows.**



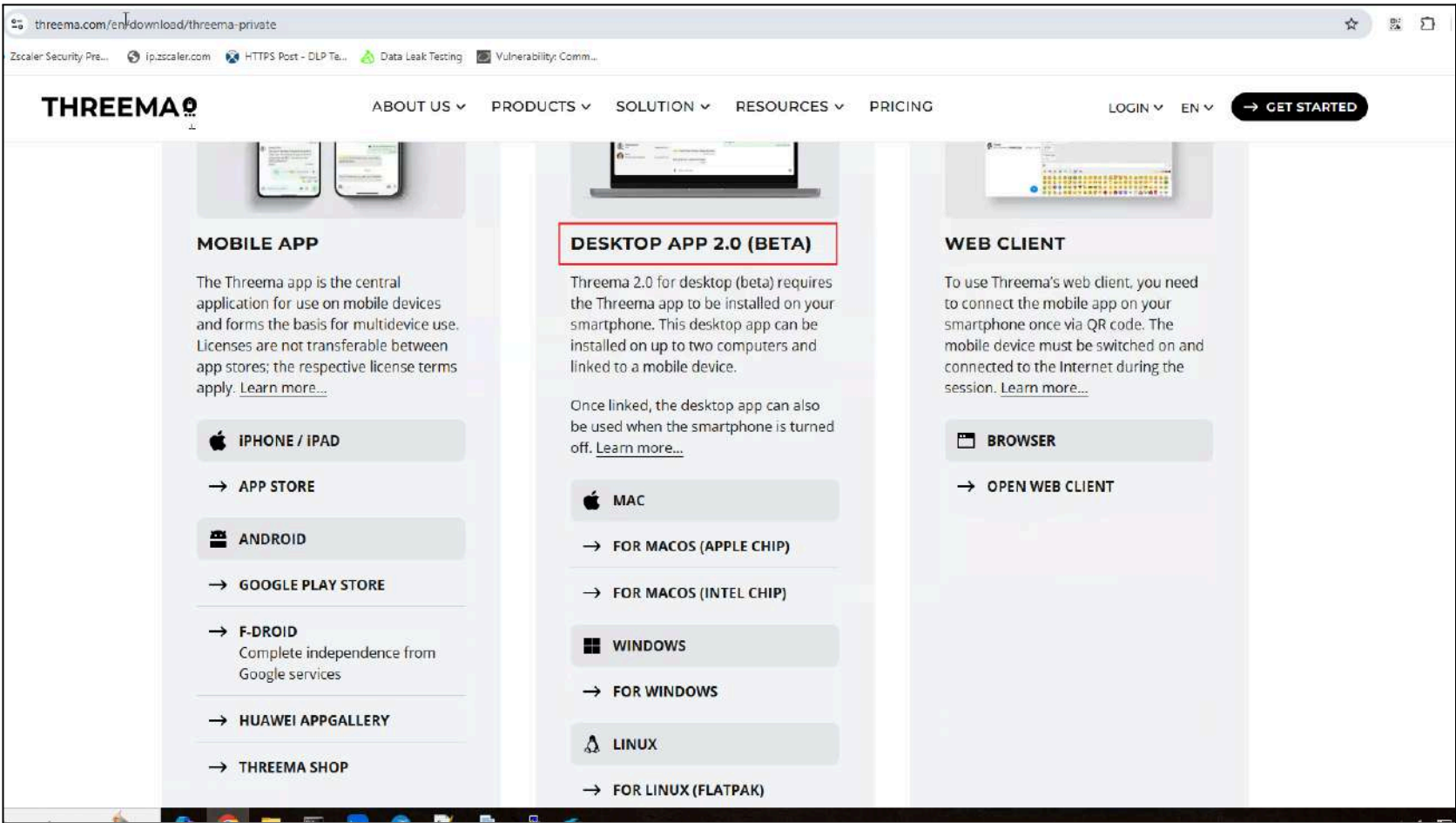
Task 5.3: Review Certificate Pinning Errors

Certificate pinning is a client-side mechanism used to secure applications by verifying that the SSL certificate presented by a backend server matches a trusted certificate hardcoded into the application. While this approach helps protect against man-in-the-middle (MITM) attacks, it can also lead to SSL connection failures when the pinned certificate does not match intercepting proxies, such as Zscaler's SSL inspection certificates. **In this task, you will learn two common methods to resolve certificate pinning errors in applications.**

1. Recreate Certificate Pinning Error:
 - a. Open a **Chrome Browser** on **Corp:Client PC**.
 - b. Download **Threema Private Desktop** for **Windows**.
 - i. Go to <https://threema.com/en/download>
 - ii. Click on the **Download Apps** of the **Threema Private**



iii. Click on download for **Windows** and an msix file will get downloaded



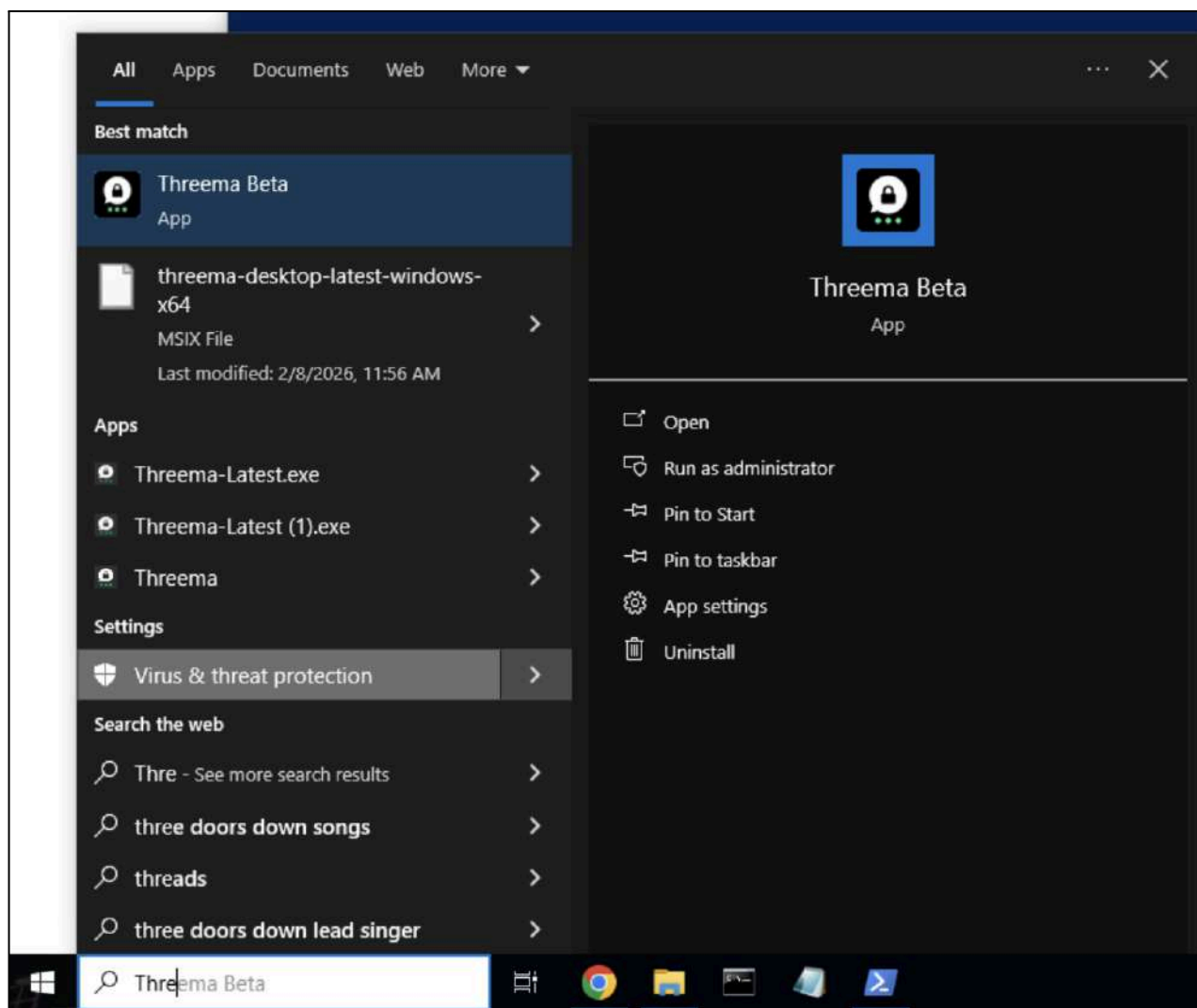
c. Open the msix File:

i. From the **Start** menu Open **PowerShell** and enter:

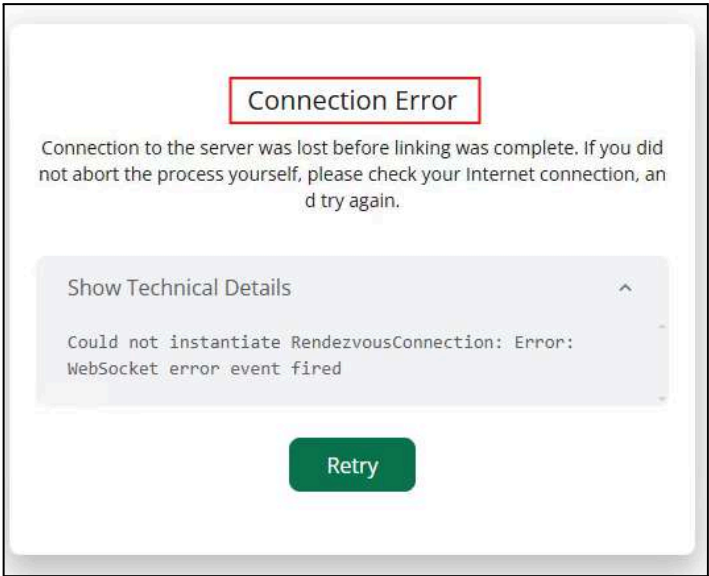
- Add-AppxPackage -Path .\Downloads\threema-desktop-latest-windows-x64.msix

```
PS C:\Users\Student.PATRaining> Add-AppxPackage -Path .\Downloads\threema-desktop-latest-windows-x64.msix
PS C:\Users\Student.PATRaining> █
```

ii. From the Start menu, open Threema Beta.

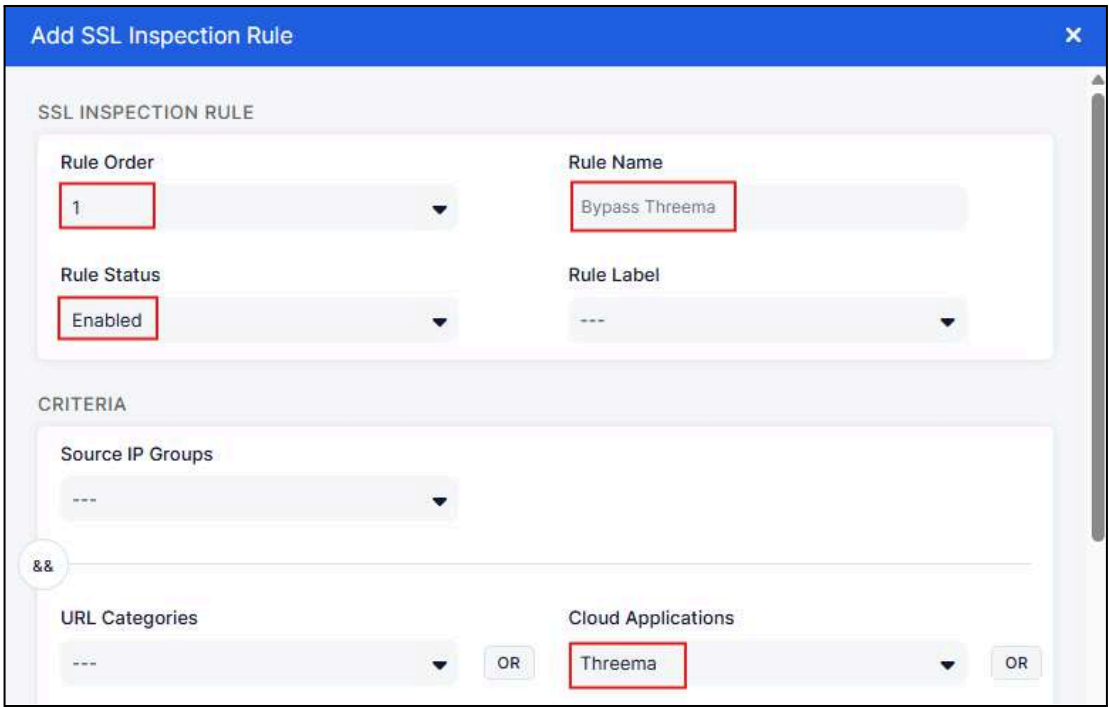


d. Observe the error message displayed during installation.

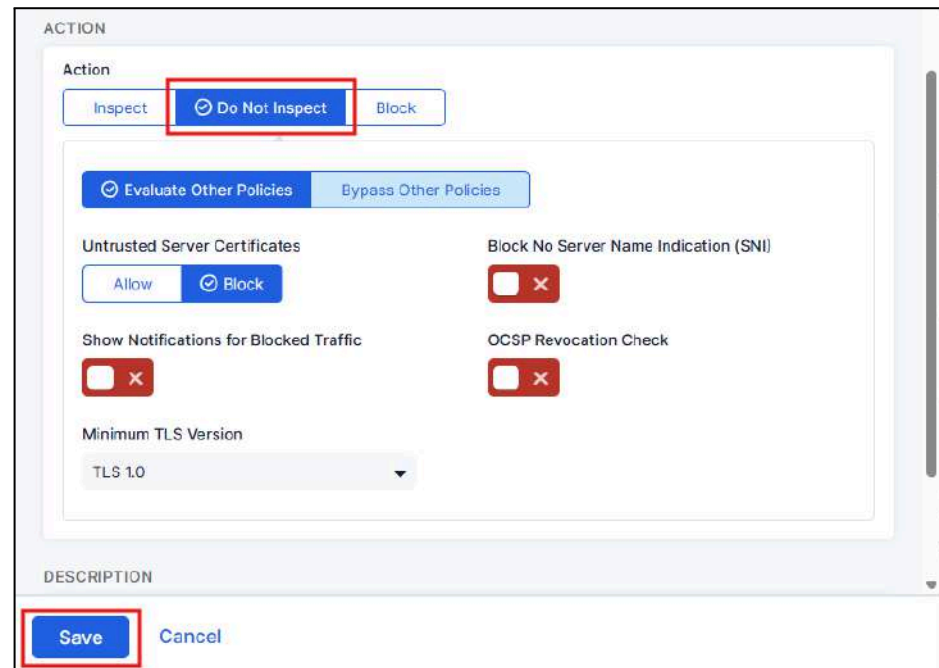


2. Configure an SSL Bypass Rule.

- a. From the **SSL Inspection page**, click on **Add SSL Inspection Rule** and configure the following:
 - i. Rule Order: **1**
 - ii. Rule Name: **Bypass Threema**
 - iii. Cloud Applications: Click on **Instant Messaging** and Select **Threema** from the list. Click **Done**.

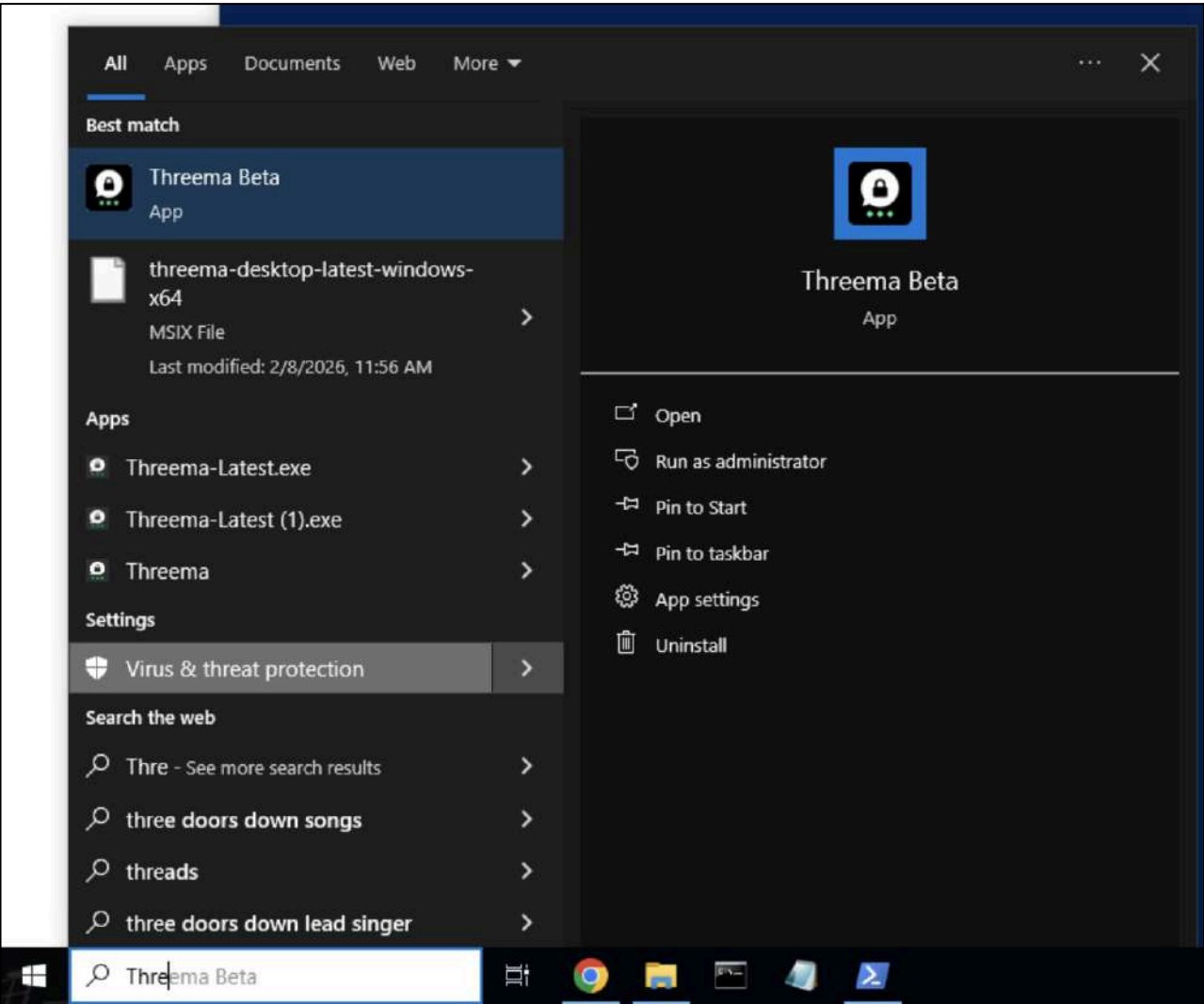


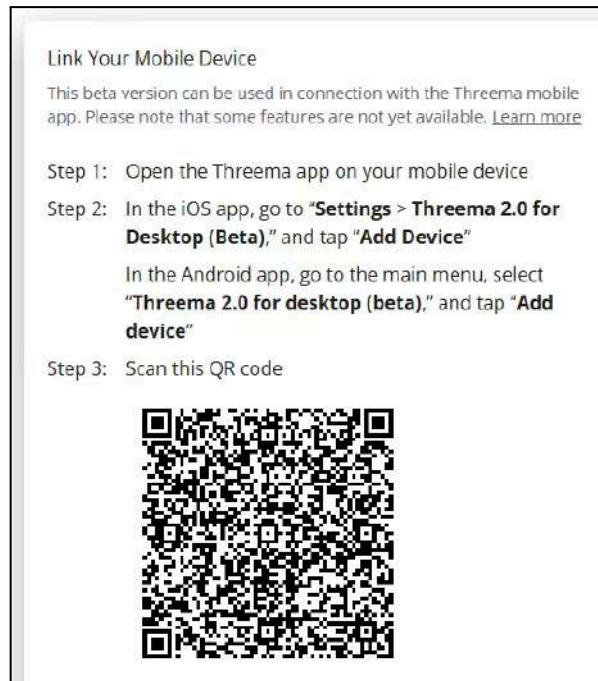
- iv. Scroll down. Under Action, Select **Do Not Inspect**.
- v. Click **Save** and **Activate** via the **Activation** icon in the top-right corner.



3. Reattempt Installation of Threema:

- a. Right-click the **Windows Taskbar** and select **Task Manager**.
 - i. Find **Threema** (if there) in the list of running applications.
 - ii. Click **End Task**.
- b. Reopen the Downloads folder.
 - i. From the Start menu, open Threema Beta.





4. Analyze SSL Inspection and Bypass Logs

- a. From the **Experience Center**, navigate to **Logs > Web Insights > Logs**.
- b. Configure the following:
 - i. Timeframe: **Last 1 hour** (you can also select other timeframes, such as Last 2 hours or Current Day)
 - ii. Add filters: **SSL Inspected** and **Bypassed Transaction**.
- c. Click **Run Query**.

Note: Allow time for logs to populate (typically ~5 minutes). Use the available filters and operators to explore differences between SSL inspection and bypass events.

Insights

Internet & SaaS

Web Insights

Mobile Insights

Firewall Insights

DNS Insights

Tunnel Insights

SaaS Security Insights

Endpoint DLP Insights

Email DLP Insights

Private Applications

Privileged Remote Access

Branch & Cloud Connectors

Insights

Logs

Timeframe

Last 1 Hour: 1/5/2026 1:43:46 PM - 1/5/2026 2:43:46 PM

Select Filters

Clear Filters

SSL Inspected

Bypassed Transaction

Add Filter

Download (.csv)

Display

Number of Records Displayed

1K

5K

10K

25K

Run Query

Insights Logs

Jan 05, 2026 01:45:58 PM - Jan 05, 2026 02:29:55 PM

15 Log Records Found

No.	Event Time	User	Policy Action	Location	Cloud Applic...	URL	URL
2	Monday, January 05, 2026 1:58:39 PM	stud...	Dropped due to failed client SSL handshake	Road War...	SIGNAL	updates2.signal...	Online
3	Monday, January 05, 2026 2:13:25 PM	stud...	Allowed	Road War...	None	self.events.data...	Profes...
4	Monday, January 05, 2026 2:18:43 PM	stud...	Allowed	Road War...	None	ctldl.windowsup...	Corpor...
5	Monday, January 05, 2026 2:18:43 PM	stud...	Allowed	Road War...	None	ctldl.windowsup...	Corpor...
6	Monday, January 05, 2026 2:18:43 PM	stud...	Allowed	Road War...	None	ctldl.windowsup...	Corpor...
7	Monday, January 05, 2026 2:20:31 PM	stud...	Allowed	Road War...	Comodoca	crl.comodoca.co...	Other I...
8	Monday, January 05, 2026 2:20:31 PM	stud...	Allowed	Road War...	Sectigo	ocsp.sectigo.co...	Corpor...
9	Monday, January 05, 2026 2:20:32 PM	stud...	Allowed	Road War...	Sectigo	ocsp.sectigo.co...	Corpor...
10	Monday, January 05, 2026 2:20:42 PM	stud...	Dropped due to failed client SSL handshake	Road War...	Threema	rendezvous-4.thr...	Interne...
11	Monday, January 05, 2026 2:24:19 PM	stud...	Allowed	Road War...	Common Offi...	client.wns.windo...	Other I...
12	Monday, January 05, 2026 2:25:12 PM	stud...	Allowed	Road War...	Microsoft Log...	login.microsoft...	Profes...
13	Monday, January 05, 2026 2:28:15 PM	stud...	Dropped due to failed client SSL handshake	Road War...	Threema	rendezvous-5.thr...	Interne...
14	Monday, January 05, 2026 2:28:46 PM	stud...	Dropped due to failed client SSL handshake	Road War...	SIGNAL	updates2.signal...	Online
15	Monday, January 05, 2026 2:29:55 PM	stud...	Allowed	Road War...	Whatsapp Web	www.whatsapp.c...	Online

Note: The Insights page allows you to analyze transactions processed by Zscaler through customizable charts and logs. It also provides filters and operators to drill down into specific insights for detailed analysis.

Lab 6: Configure Content Filtering & Access Control

Scenario: As the Zscaler administrator, your goal is to improve users' browsing experience by ensuring they only access safe and appropriate content. You will explore how **Zscaler Internet Access (ZIA)** enforces content filtering and access control to protect against threats such as phishing, malware, and data leakage. You will also verify existing threat protection configurations, analyze risk reports, and test policy behavior to assess how ZIA enforces organizational baselines for URL filtering, content categories, and cloud application controls.

Objectives: By the end of this lab, you will:

- **Explore ZIA's threat protection configurations** and analyze risk reporting.
- **Configure and validate a URL Filtering Policy** to block risky websites (e.g., gambling, adult, and shopping sites).
- **Test** the effect of **URL Filtering** control on end-user experience.
- **Create and verify a Cloud App Control policy** to allow access to specific apps (e.g., YouTube) while restricting others in the same category.

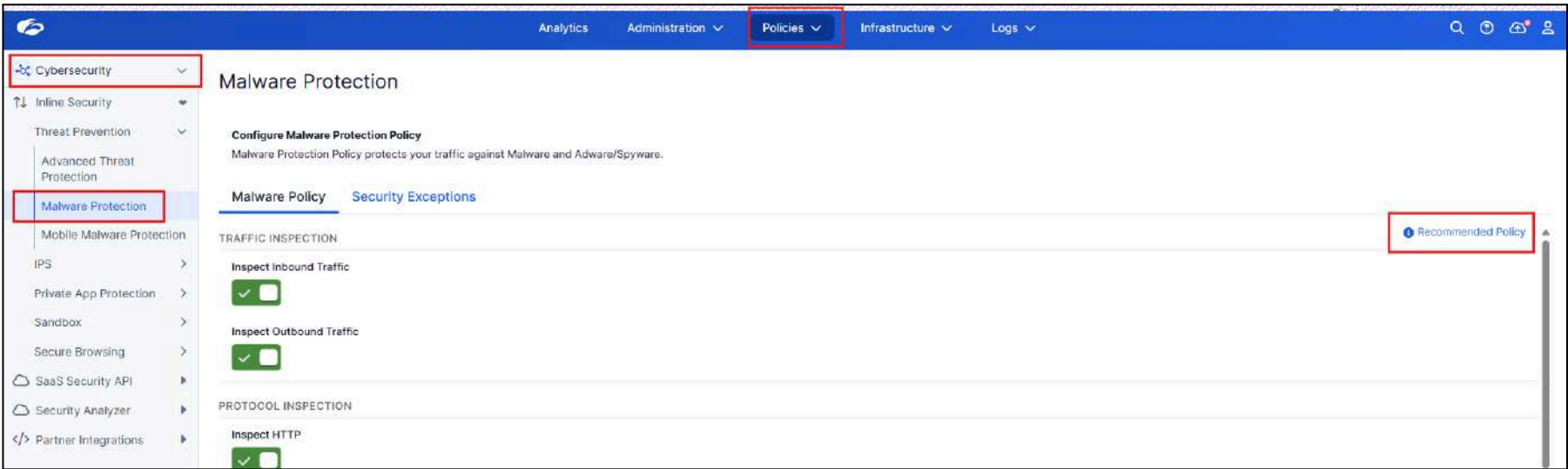
Approximate Time: 25 minutes

Task 6.1: View Threat Protection Reports

In this task, you will review current threat protection reports in the **ZIA Admin Portal**, examining Malware Protection, Advanced Threat Protection, and the Page Risk Score Index.

- 1. View Malware Protection Settings:
 - a. Go to **Policies > Cybersecurity > Malware Protection**.
 - b. Compare the current settings with the **Recommended Policy** settings and
 - c. Note **Malware Protections** are **enabled**.

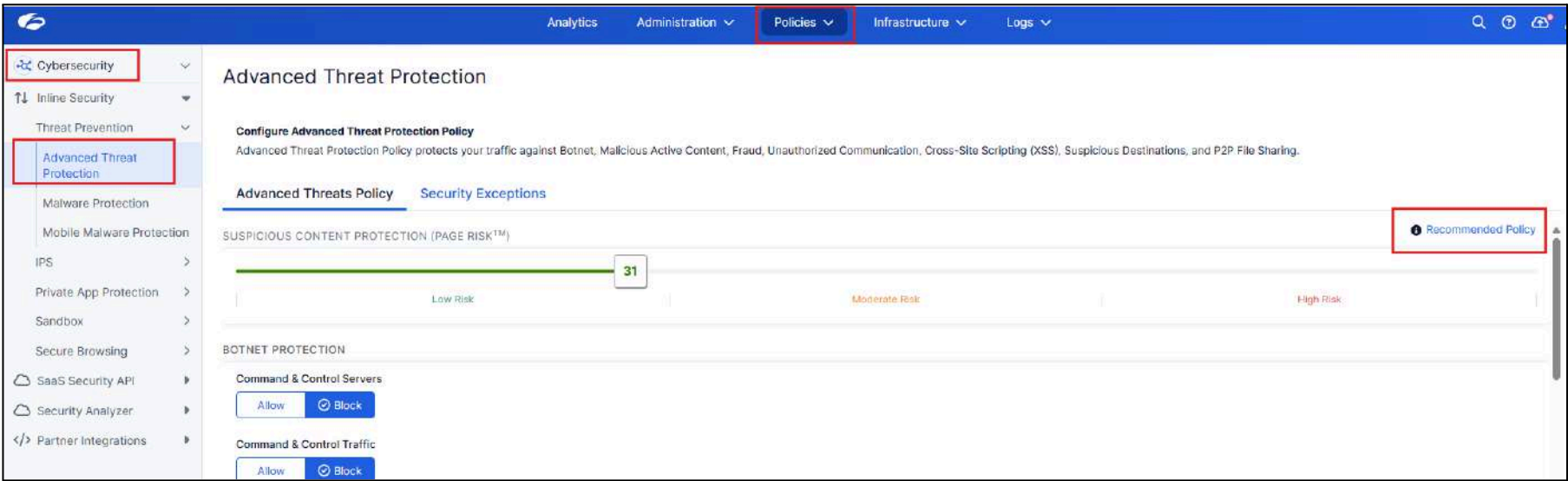
Note: Malware protection defends against spyware, botnets, malicious active content, and more.



- 2. View Advanced Threat Protection Settings:
 - a. Use the **left navigation panel** to go to **Advanced Threat Protection**. Or, **Policies > Cybersecurity > Advanced Threat Protection**

- b. Compare the current settings with the **Recommended Policy settings**, and note which **Advanced Threat Protections** are **enabled**.

Note: Advanced protections include settings against Command and Control Traffic, Phishing, Cryptomining, Tunneling, Cross-Site Scripting (XSS), and more.

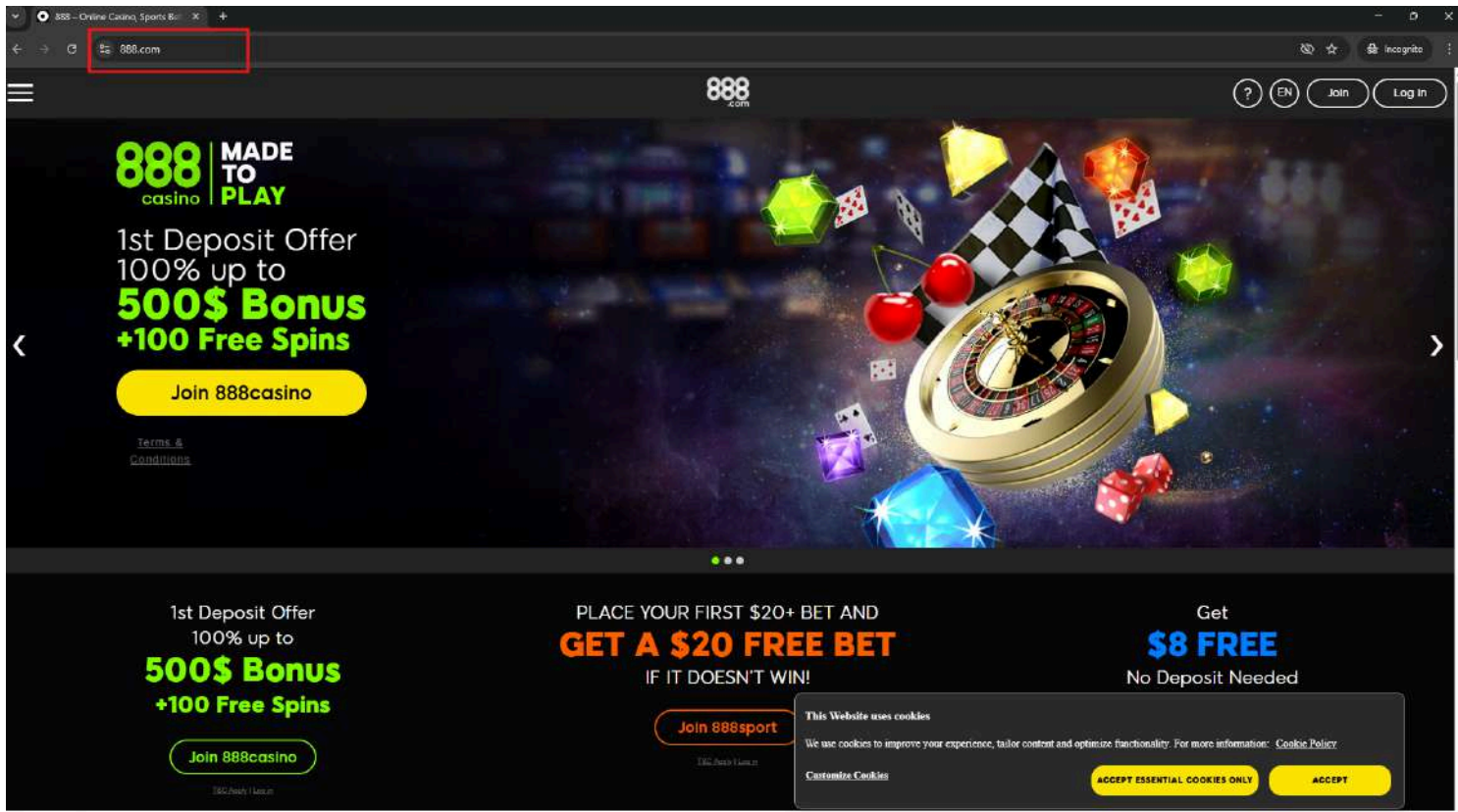


- 3. Review the Page Risk Score Index (**SUSPICIOUS CONTENT PROTECTION (PAGE RISK™)**), this calculates the score upon user access, analyzing the page's threat level dynamically.

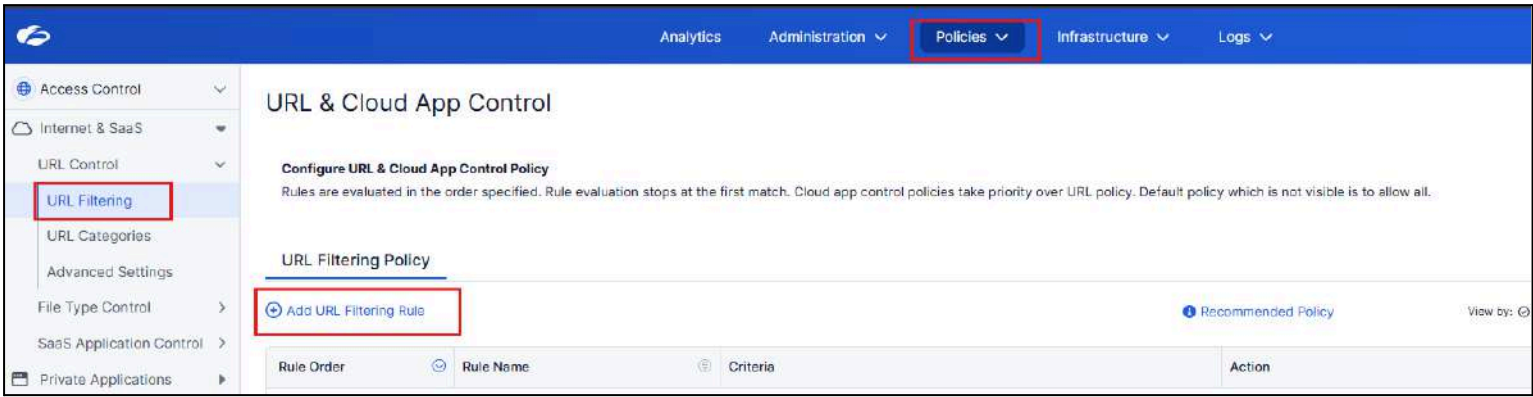
Task 6.2: Configure Content Filtering Controls

In this task, you will create a **URL Filtering Policy** to block risky URL categories such as gambling, adult content, and video streaming. You will also verify how content filtering controls work in ZIA to manage user access.

1. Test High-Risk URL Access (Before Configuration):
 - a. Open a browser on the **Corp: Client PC** and attempt to visit a risky or unsafe website, such as 888.com (a gambling site).
 - b. Verify that the website is currently accessible.

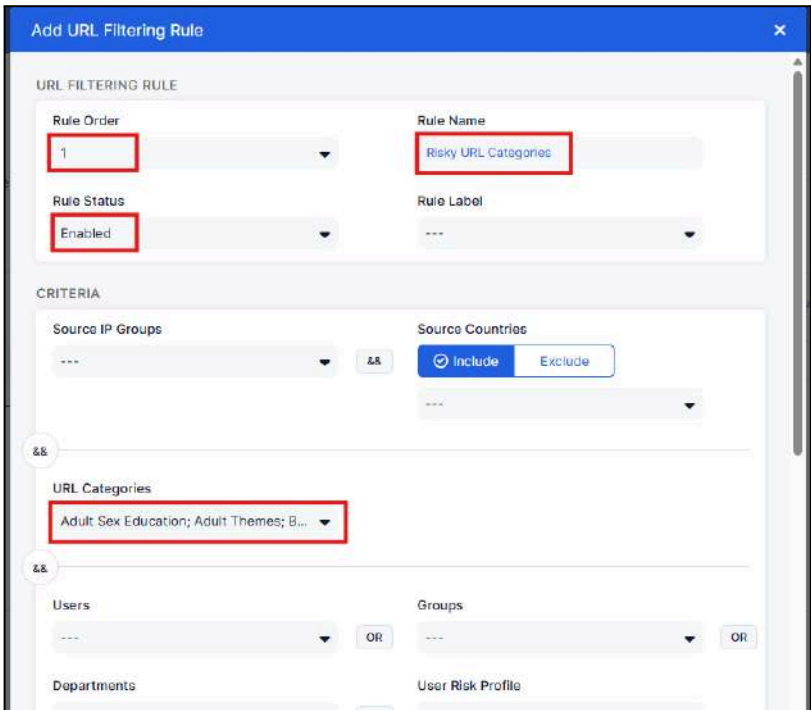


1. Create a URL Filtering Policy:
 - a. Go to **Policies > Access Control > Internet and SaaS > URL Control > Url Filtering**
 - b. On the **URL Filtering Policy tab**, click **Add URL Filtering Rule**



c. Configure the rule as follows:

- i. Rule Order: **1**
- ii. Rule Name: **Risky URL Categories.**
- iii. URL Categories:
 - **Adult Materials**
 - **Shopping and Auctions**
 - **Gambling**
 - **Video Streaming**



- iv. Request Methods and Protocols: Ensure all options are selected.
- v. **Action:** Select **Block**.

d. Click **Save** and **Activate**.

ACTION

Web Traffic

Allow

Caution

☒ Block

Isolate

Allow Override

☐

Redirect URL

Enter Text

DESCRIPTION

Save

Cancel

Task 6.3: Test End-User Experience with URL Filtering

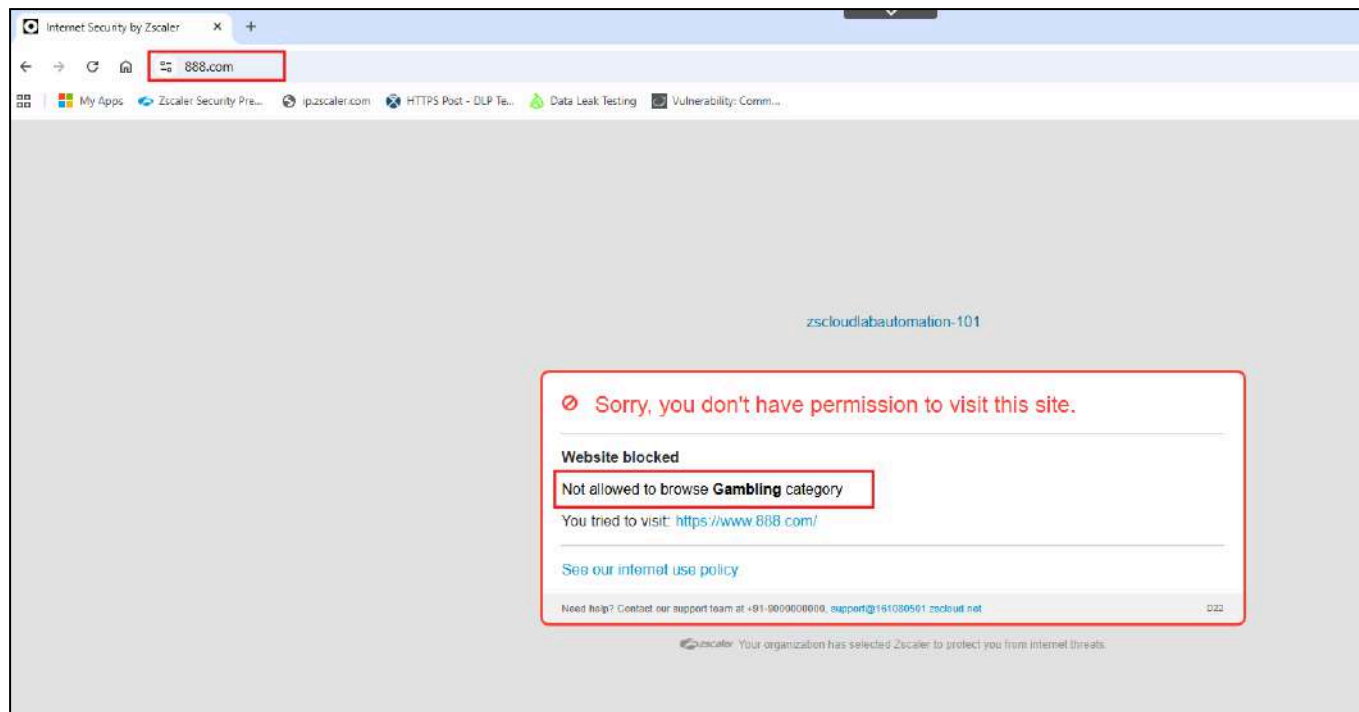
In this task, you'll test user access to various websites after applying the configured **URL Filtering Policy**.

1. Test Blocked URLs:

a. Open a browser and attempt to visit a previously accessible site, such as:

- i. 888.com (**Gambling Category**)
- ii. ebay.com (**Shopping and Auctions Category**)

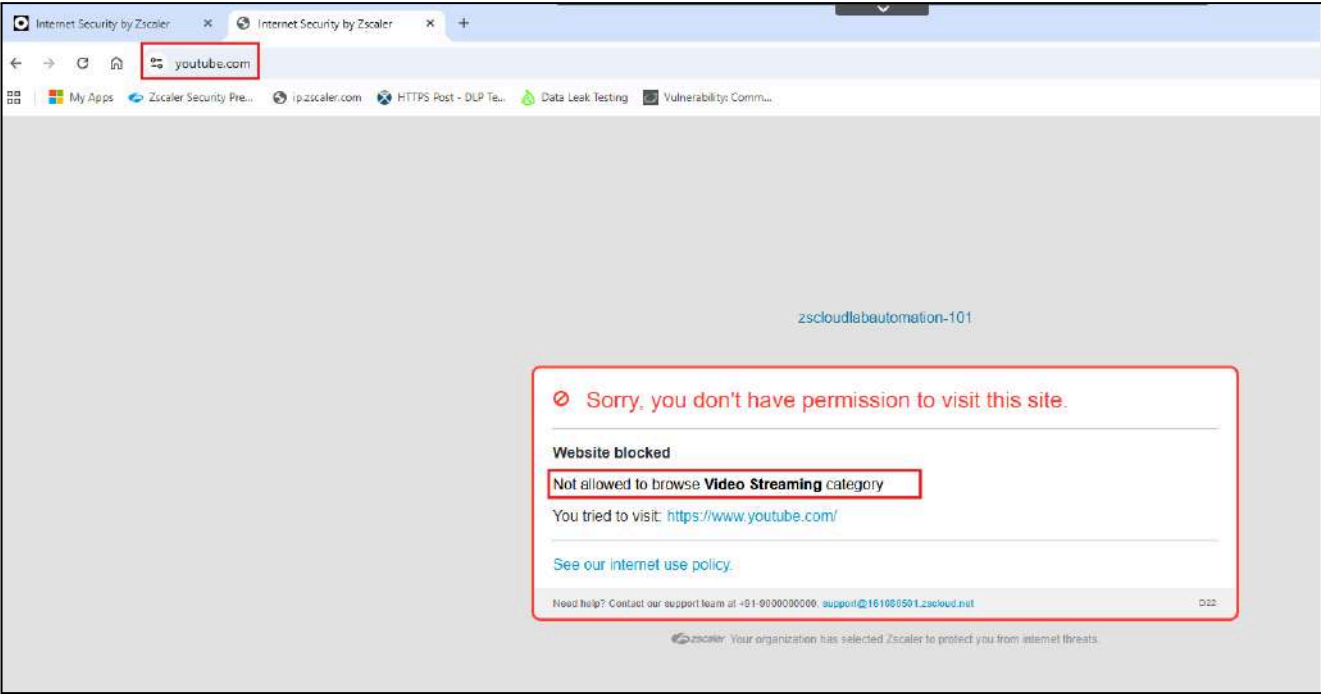
b. Verify that these sites are now blocked based on the configured **URL Filtering Policy**.



2. Test Video Streaming Site Access:

a. Attempt to access **Video Streaming** websites like YouTube.com Or Netflix.com.

b. Verify that these websites are also blocked under the configured rule.



Note: If testing does not work, try using Incognito Mode in the browser.

- 3. If the testing fails, verify URL Filtering Policy Status:
 - a. Navigate to **URL Filtering Policy** and click Edit to confirm that the **Rule Status** is set to **Enabled**.

Task 6.4: Configure Cloud App Control

The **Cloud App Control Policy** provides granular control over how users access specific cloud applications and websites. In this task, you will create a rule to **allow access to YouTube** while **restricting other entertainment and recreation websites**.

1. Add a Cloud App Control Policy:
 - a. Go to **Policies > Access Control > Internet & SaaS > SaaS Application Control > Policies** or use the **left navigation panel** to go to SaaS Application Control > Policies
 - b. Click the **Add** dropdown and select **Streaming Media**.
2. Create the Cloud App Control Rule:
 - a. Rule Order: **1**
 - b. Rule Name: **Allow YouTube**
 - c. Next, Under the **Criteria** section, click the drop-down for **Cloud Applications** and select **YouTube**, then click **Done**.

Add Streaming Media Rule

CLOUD APP CONTROL RULE

Rule Order	Rule Name
1	Allow YouTube
Rule Status	Rule Label
Enabled	---

CRITERIA

Cloud Applications	Cloud Application Instances
YouTube	None
Cloud Application Risk Profile	Users

- a. Action – Application Access: **Allow**
- b. Add Description: **Allow only YouTube**
- c. Click **Save** and **Activate**.

ACTION

Viewing/Listening

☒ Allow

Caution

Block

Isolate

Uploading

☒ Allow

Block

Daily Bandwidth Quota (MB)

Enter Text

Daily Time Quota (min)

Enter Text

Cascade to URL Filtering

☐ X

DESCRIPTION

Allow only YouTube

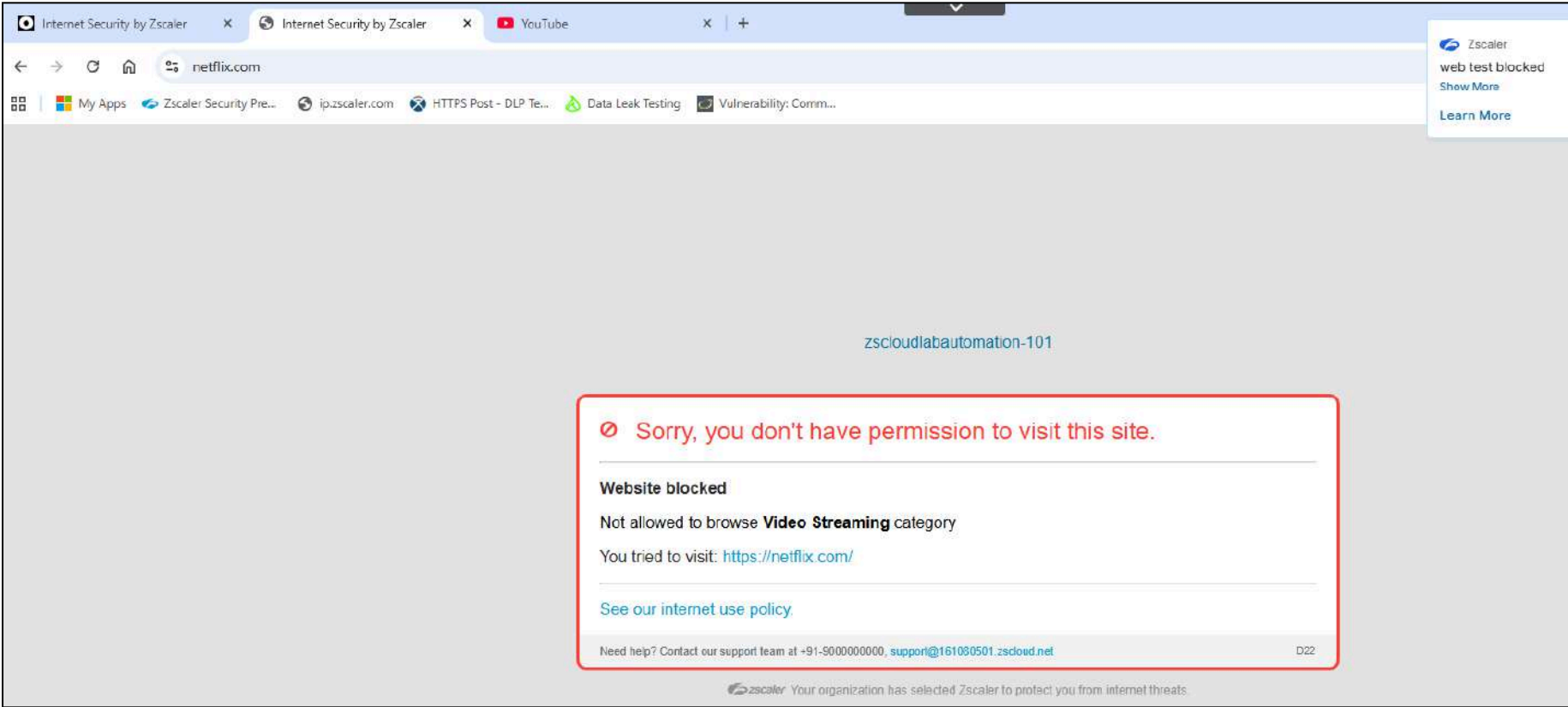
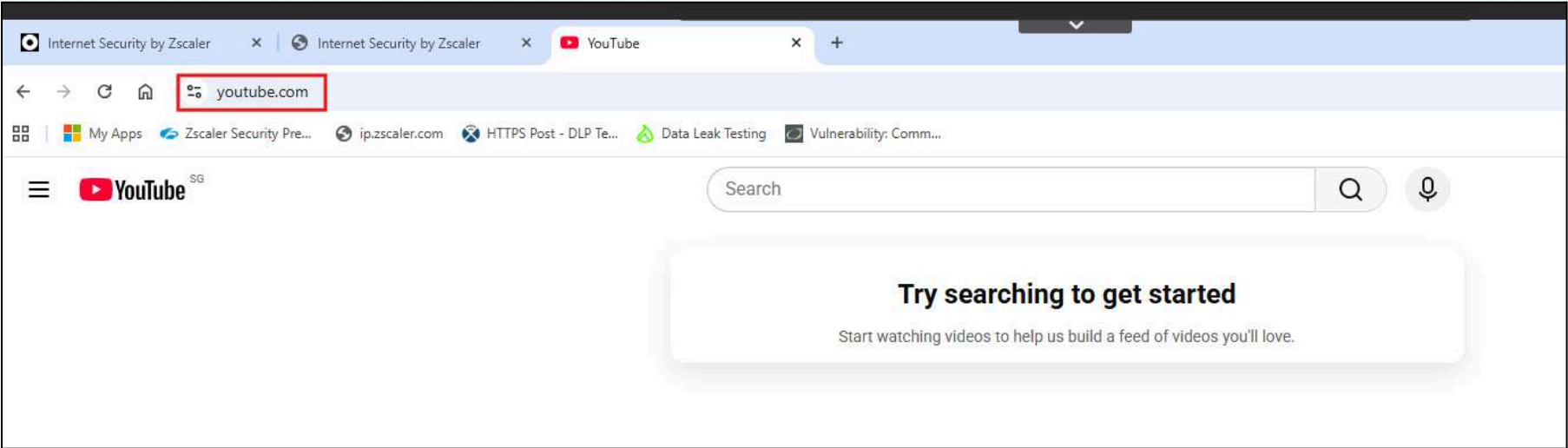
Save

Cancel

STREAMING MEDIA				
1	Allow YouTube	APPLICATIONS YouTube	Allow Viewing/Listening, Uploading	DESCRIPTION Allow only YouTube

2. Verify Cloud App Control Policy:
- a. Ensure the new rule appears in the **Streaming Media Rules List**.
 - b. Confirm on the **Corp: Client PC** that you can **access YouTube.com**, but access to other streaming websites like **Netflix.com** is **blocked**.

Note: SaaS Application policies are enforced with higher priority with respect to URL category based policies



Lab 7: Configure DNS Security

Scenario: Safemarch aims to protect its users by blocking access to malicious domains and enforcing acceptable use policies that prohibit access to gambling websites. As a Zscaler administrator, you will configure **DNS security controls**, enforce these policies, and validate them using the **DNS Insight logs**.

Objectives: By the end of this lab, you will:

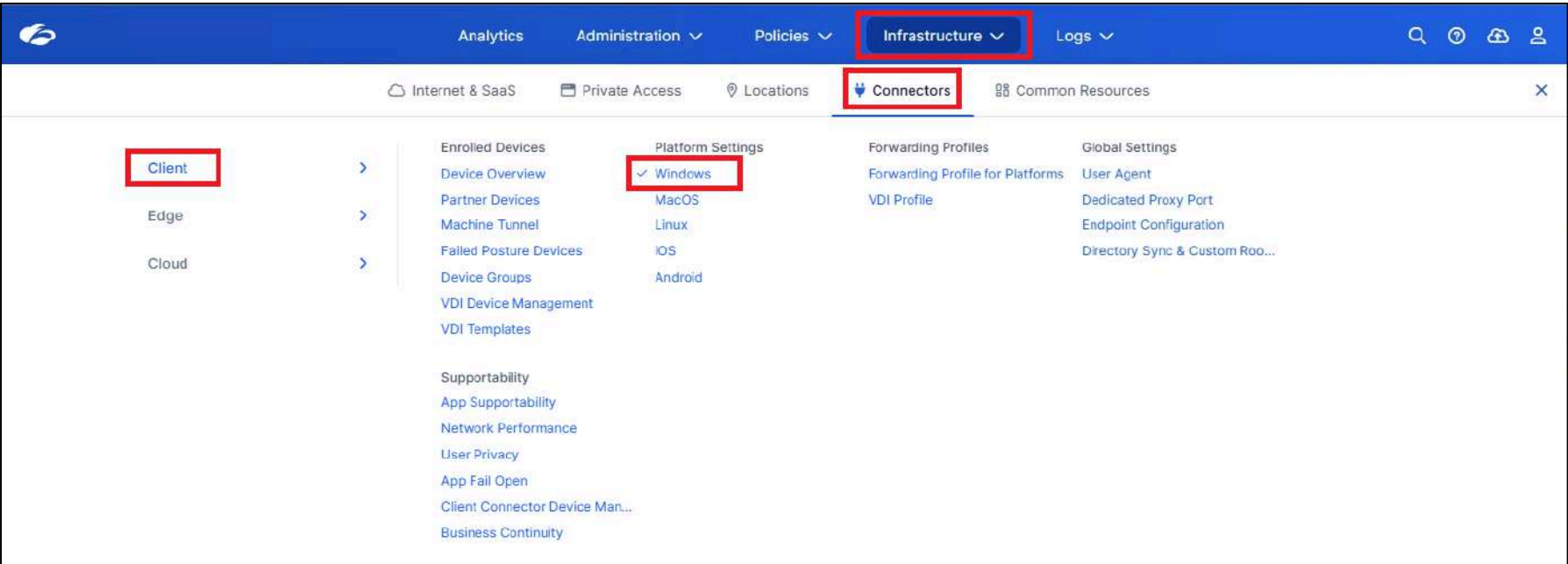
- **Configure** an **App Profile** to ensure that all DNS traffic flows through the Zero Trust Exchange.
- **Create and test** a **DNS Control Policy** to block gambling sites and ensure legitimate domains resolve correctly.
- **Verify** the **DNS Control Policy enforcement** using **DNS Insight Logs** to confirm that gambling-related DNS requests are being blocked.

Approximate Time: 20 minutes

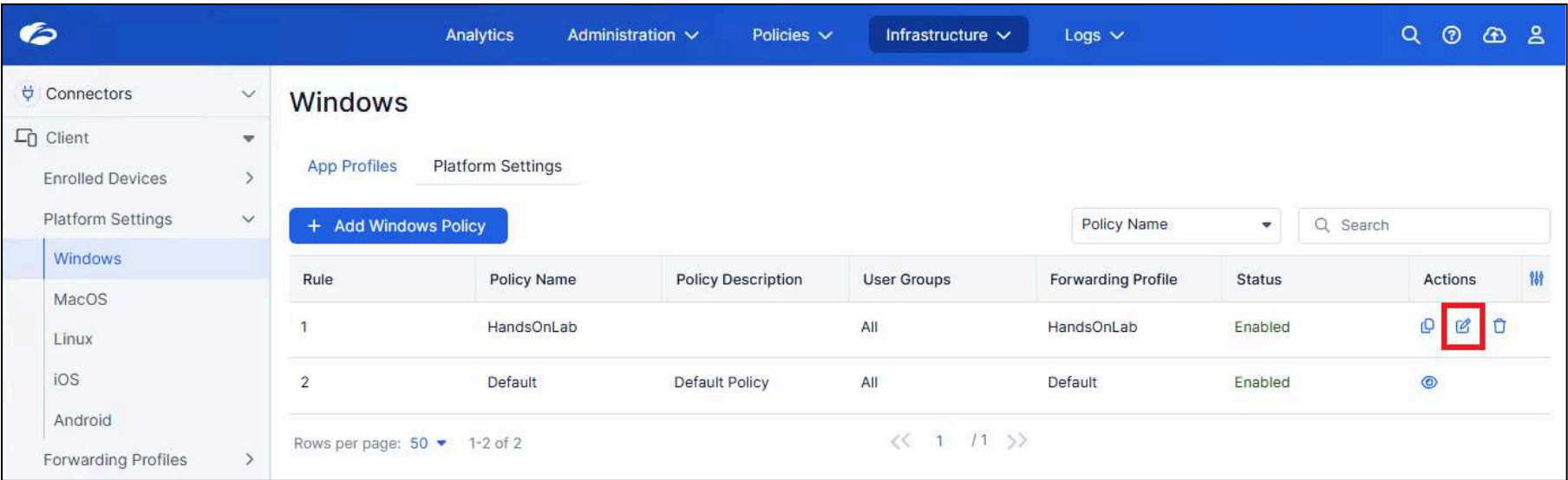
Task 7.1: Configure App Profile

In this task, you will update the **App Profile** to steer all DNS traffic through Zscaler for inspection. This ensures no DNS requests bypass monitoring.

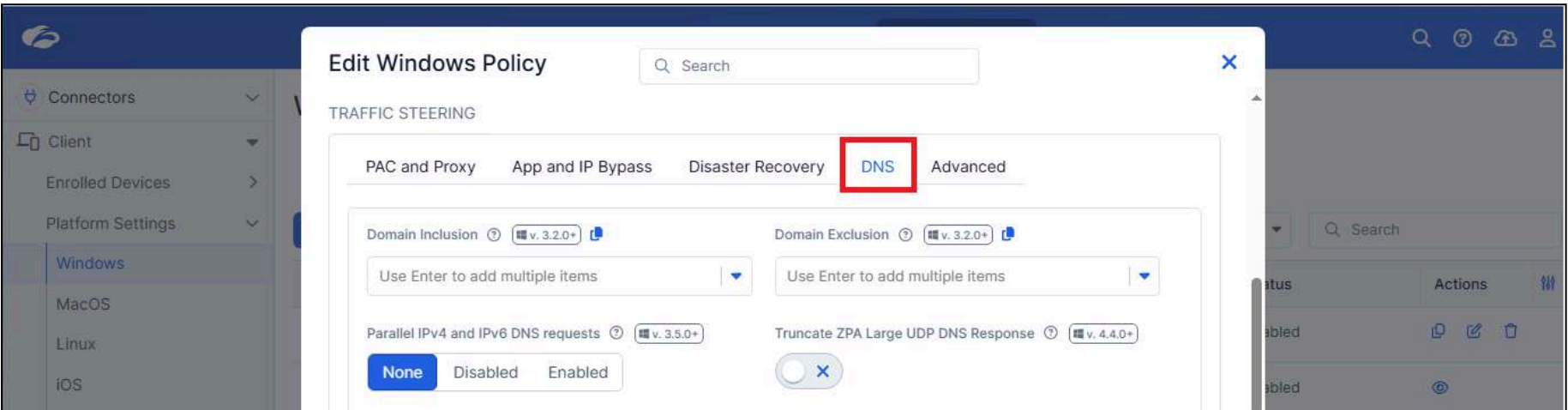
- 1. Edit HandsOnLab Windows App Profile:
 - a. From the **Experience Center**, navigate to **Infrastructure > Connectors > Client > Platform Settings > Windows**



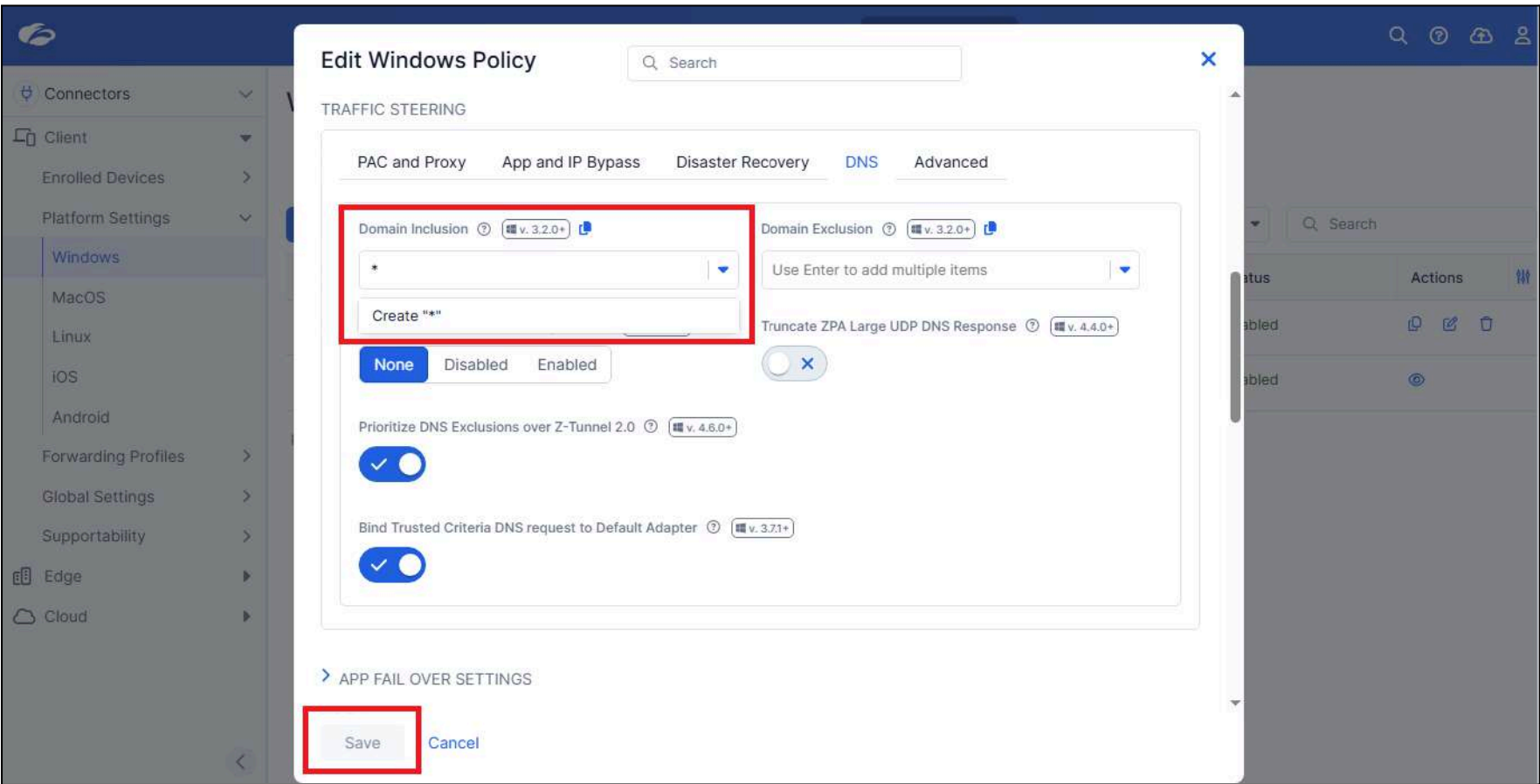
- b. Click on the **Edit** icon in the **HandsOnLab** policy page.



- c. Under Traffic Steering, Click on the **DNS** tab.
- d. In the Domain Inclusion text box, enter the wildcard *.

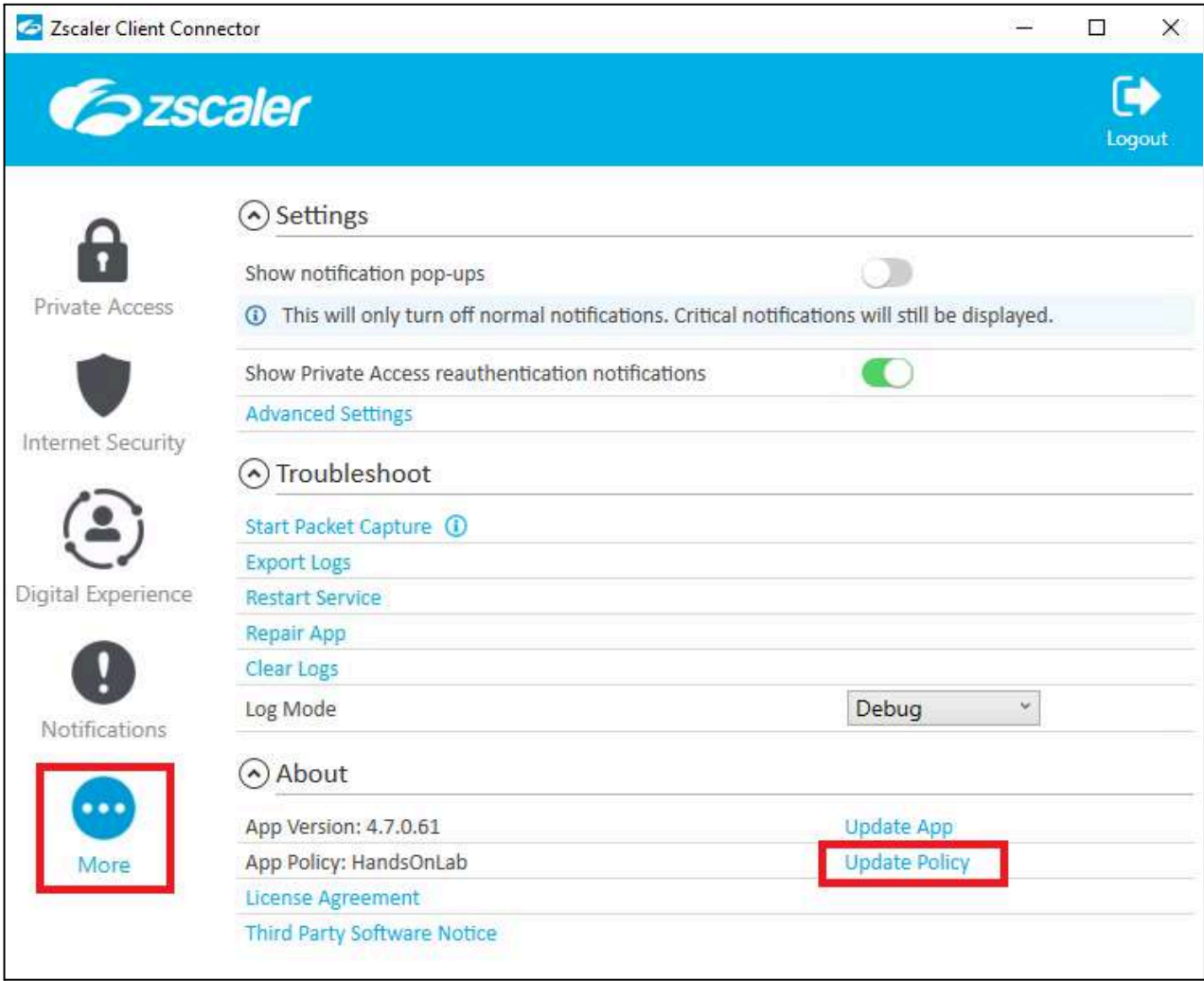


- e. Click on **Create “*”** and **Save**.



2. Save and update the policy:

- a. Go to **Zscaler Client Connector (ZCC)** on the **Corp: Client PC** > Select **More** and click **Update Policy**.

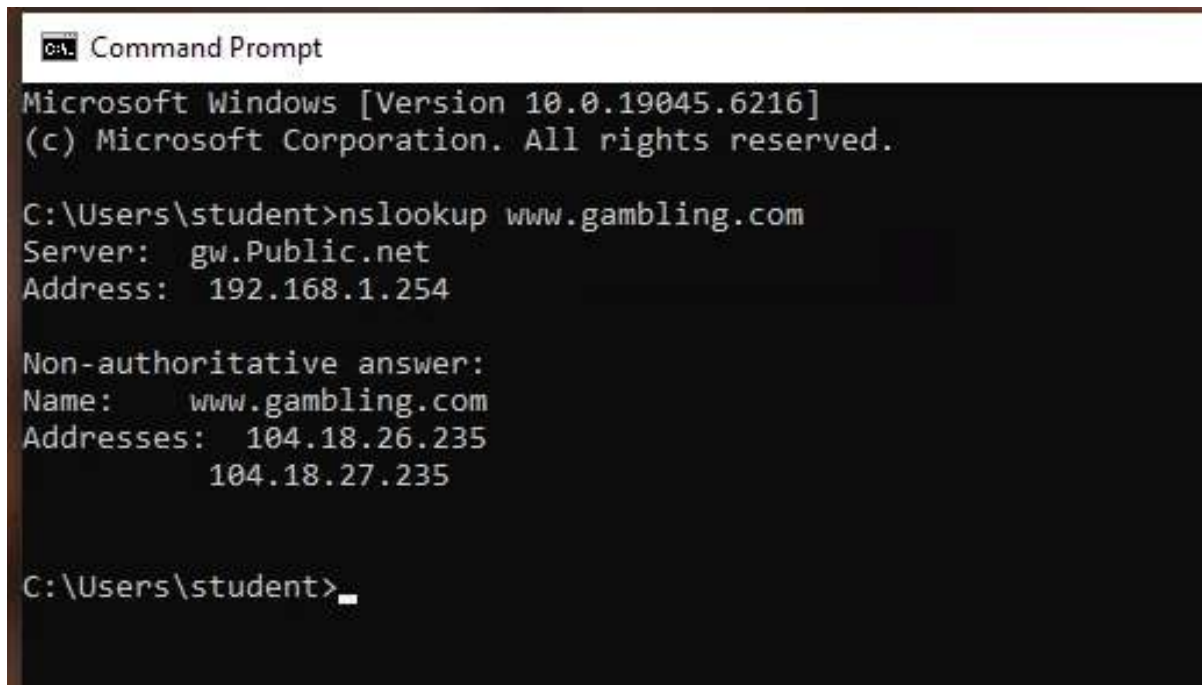


Task 7.2: Configure a DNS Control Policy

In this task, you will create and test a DNS Control rule that blocks gambling domains while allowing legitimate traffic.

1. Test DNS Resolution Before Configuration:
 - a. Open command prompt on the Corp: Client PC.
 - i. Execute: `nslookup www.gambling.com`

Note: You will observe that the DNS request resolves successfully because no policy has been configured yet.



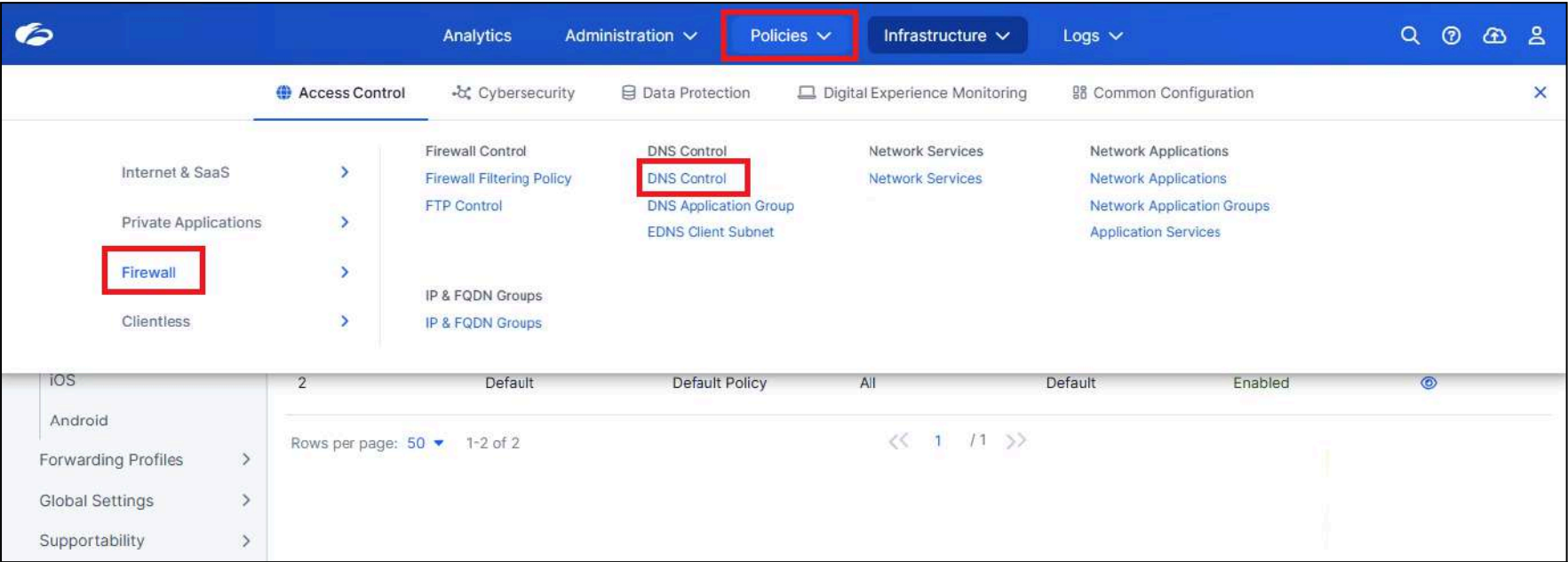
```
Command Prompt
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>nslookup www.gambling.com
Server: gw.Public.net
Address: 192.168.1.254

Non-authoritative answer:
Name: www.gambling.com
Addresses: 104.18.26.235
           104.18.27.235

C:\Users\student>
```

2. Create a DNS Control Policy:
 - a. From the **Experience Center**, navigate to **Policies > Firewall > DNS Control**.



- b. View the existing predefined **DNS Control Rules** which are based on best practice recommendations.
- c. Click **Add DNS Filtering Rule**.



3. Define the Rule Attributes:
- a. Rule order: **1**
 - b. Rule Name: **Block Gambling**
 - c. Rule Status: **Enabled**

Add DNS Filtering Rule

DNS FILTERING RULE

Rule Order

1

Rule Name

Block Gambling

Rule Status

Enabled

Rule Label

4. Define the DNS Application Parameters:
- a. Click on the **DNS Application** tab.

b. Under **Request** Categories, select **Gambling**. Click **Done**.

c. Under **Response** Categories, select **Gambling**. Click **Done**.

Add DNS Filtering Rule

Who, Where, & Wh...

Source IP

Destination/Resol...

DNS Application

CRITERIA

DNS Tunnels & Network Apps

Any

OR

DNS Application Group

Any

AND

Resolved IP-Based Countries

Any

OR

Request Categories

Gambling

Response Categories

Gambling

- d. Scroll down, under **Action** select **Network Traffic** to **Block**.

Add DNS Filtering Rule

ACTION

Network Traffic

Block

Logging

Full

NOTIFICATION

Client Connector EUN

Enable

Disable

Web EUN

Enable

Disable

DESCRIPTION

Save

Cancel

e. Click **Save** and **Activate** the changes from the **Activation** icon (cloud with arrow))in the top-right corner..

Search

Help

Activation

User

Pending Activation

Queued Activation

My Activation Status

Editing

Currently Editing

student@zso...labtenant.com

Activate

Force Activate

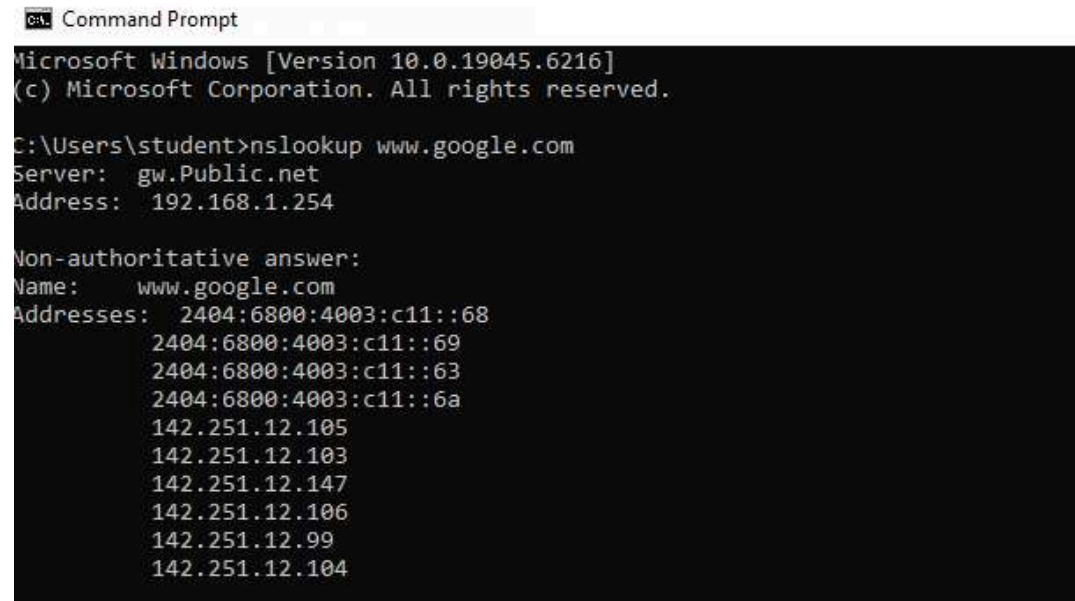
View by:

Label and Description

5. Test the New DNS Settings:

a. In the Command Prompt window:

b. Open the Command Prompt window and test the following: `nslookup www.google.com`



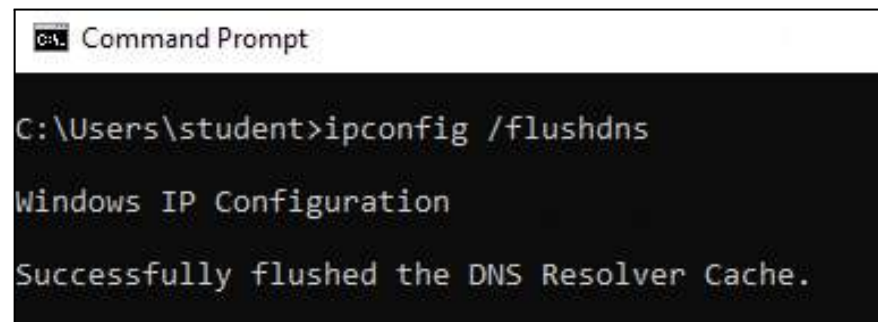
```
Command Prompt
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\student>nslookup www.google.com
Server: gw.Public.net
Address: 192.168.1.254

Non-authoritative answer:
Name: www.google.com
Addresses: 2404:6800:4003:c11::68
          2404:6800:4003:c11::69
          2404:6800:4003:c11::63
          2404:6800:4003:c11::6a
          142.251.12.105
          142.251.12.103
          142.251.12.147
          142.251.12.106
          142.251.12.99
          142.251.12.104
```

Note: Expected Result: The DNS request resolves successfully.

c. Type: `ipconfig /flushdns`



```
Command Prompt

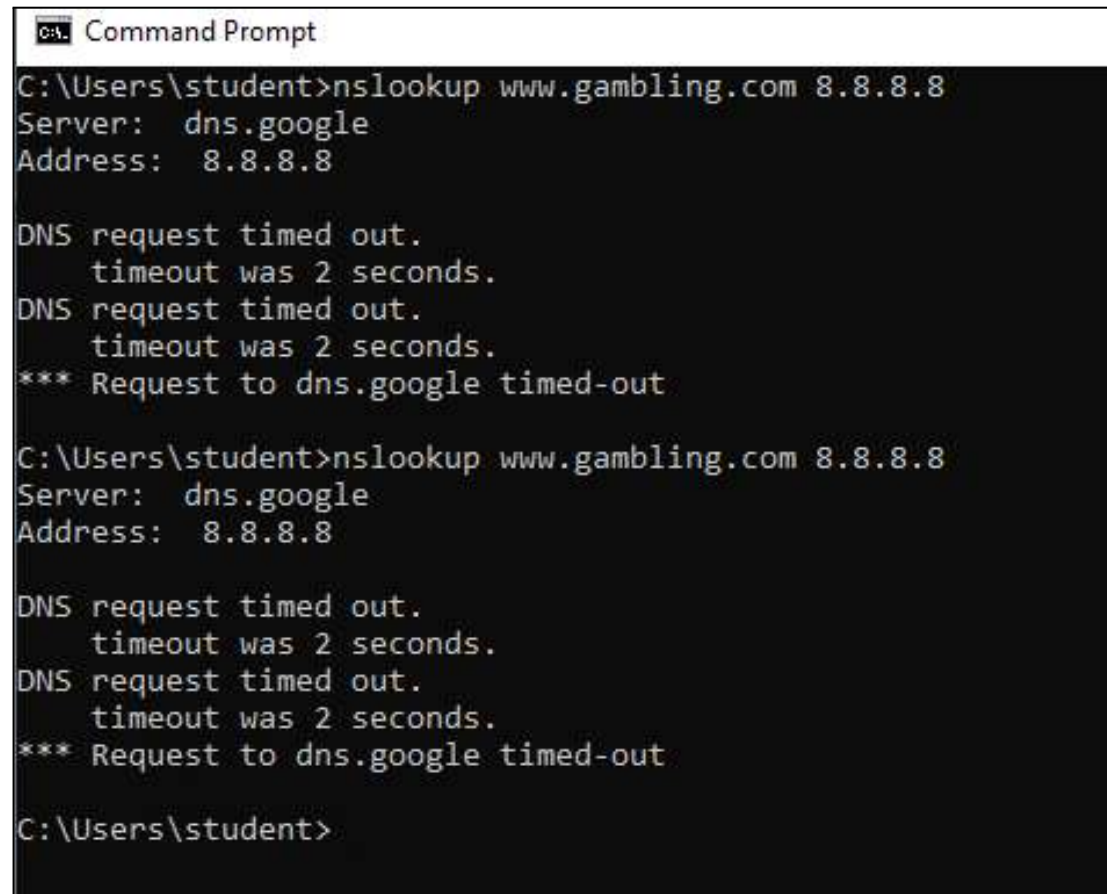
C:\Users\student>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.
```

d. Type: `nslookup www.gambling.com 8.8.8.8`

Note: Expected Result: The DNS request times out, confirming that gambling domains are blocked.



```
Command Prompt
C:\Users\student>nslookup www.gambling.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to dns.google timed-out

C:\Users\student>nslookup www.gambling.com 8.8.8.8
Server:  dns.google
Address:  8.8.8.8

DNS request timed out.
    timeout was 2 seconds.
DNS request timed out.
    timeout was 2 seconds.
*** Request to dns.google timed-out

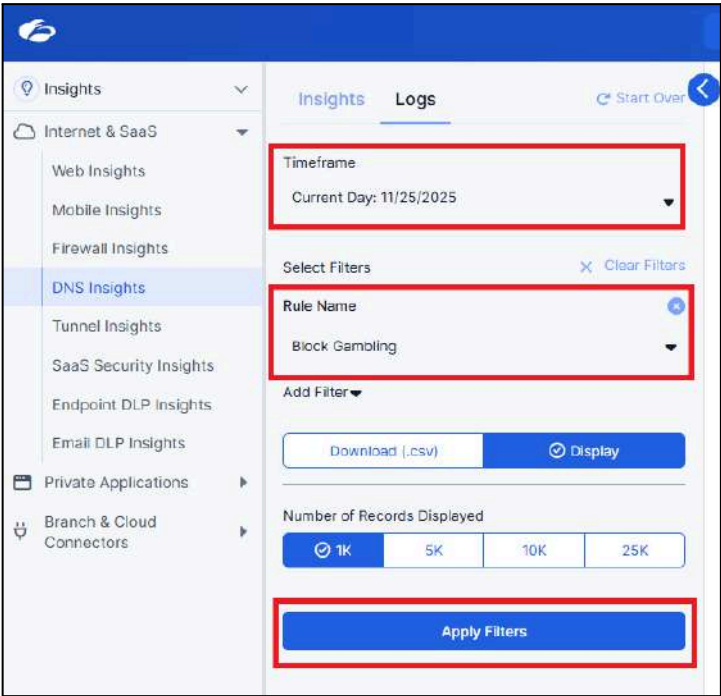
C:\Users\student>
```

Task 7.3: View DNS Insight Logs

In this task, you will validate that the **DNS Control Policy** has been applied and is blocking gambling domains as expected by reviewing DNS Insight Logs.

1. Access DNS Insight logs:
 - a. Navigate to **Logs > Internet & SaaS > DNS Insights > Logs** in the **Experience Center** landing page.
2. Configure log settings:
 - a. Set the **Timeframe** as **Current Day**.
 - b. Click **Add Filter** and add the filters:
 - i. **Rule Name:** Block Gambling
 - c. Click **Apply Filters**.

Note: Allow approximately 15 minutes for logs to become available. You can proceed to the next steps and return to review the log data once populated.



3. Verify Block Gambling activity:
- a. Confirm that DNS requests for gambling-related domains show hits on the Block Gambling **DNS Control rule** and that the action reflects **blocking** behavior

Insights Logs										
Nov 27, 2025 06:14:59 AM - Nov 27, 2025 06:15:19 AM										
4 Log Records Found										
No.	Event Time	User	Location	Client IP	Requested Domain	Server IP	Request Catego...	DNS Response Type	Request	
1	Thursday, November 27, 2025 6:1...	student@zs0002051...	Road Warrior	192.168.1.1	www.gambling.com	185.48.212.88	Gambling	Response type has Zscaler custo...	Block	
2	Thursday, November 27, 2025 6:1...	student@zs0002051...	Road Warrior	192.168.1.1	www.gambling.com	185.48.212.88	Gambling	Response type has Zscaler custo...	Block	
3	Thursday, November 27, 2025 6:1...	student@zs0002051...	Road Warrior	192.168.1.1	www.gambling.com	185.48.212.88	Gambling	Response type has Zscaler custo...	Block	
4	Thursday, November 27, 2025 6:1...	student@zs0002051...	Road Warrior	192.168.1.1	www.gambling.com	185.48.212.88	Gambling	Response type has Zscaler custo...	Block	

Lab 8: Enforce Policy with Unified DLP across Multi-Channel

Scenario: In this lab you will configure and test policies to protect sensitive data in motion and at rest, ensuring compliance with privacy regulations such as PCI-DSS (Payment Card Industry - Data Security Standard). Using Zscalers Unified DLP policies, you will secure multi-channel data by analyzing endpoint and SaaS application security logs

Objectives: By the end of this lab you will:

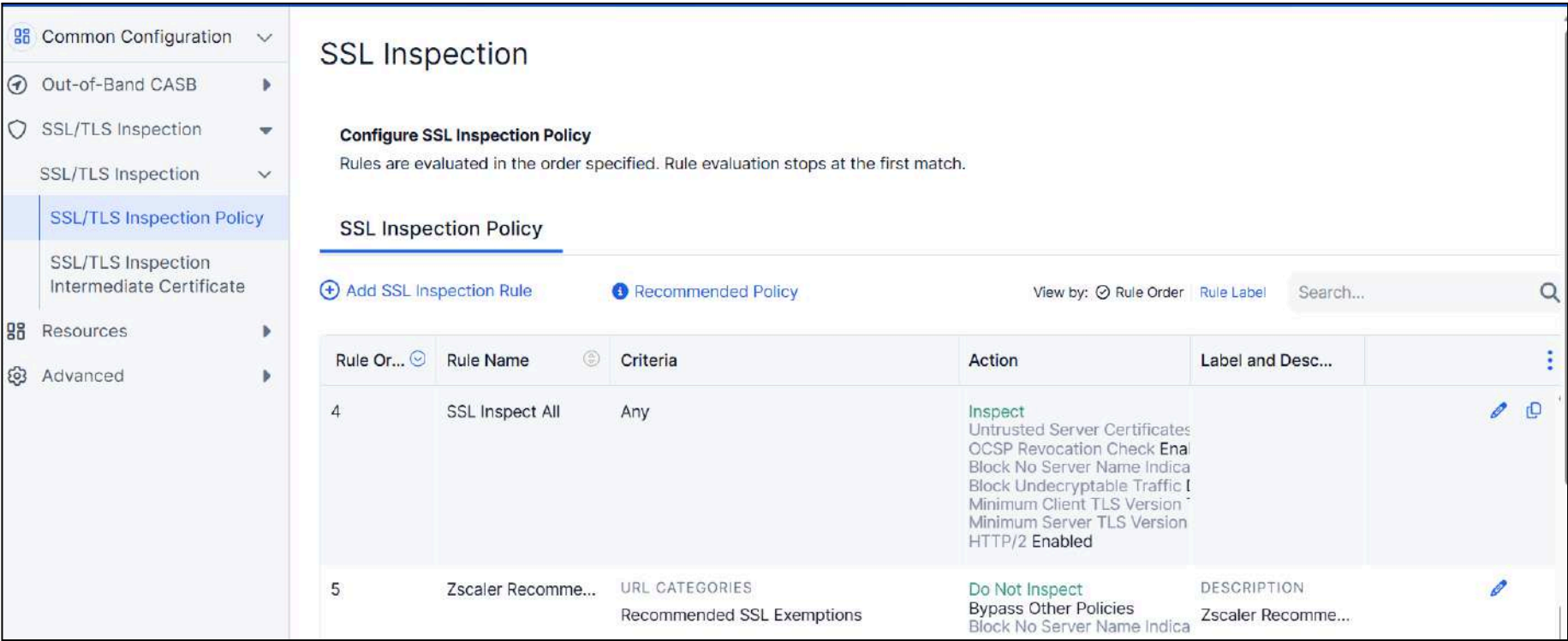
- **Verify and enable SSL/TLS inspection** for data traffic visibility and control.
- **Configure DLP Dictionaries, DLP Engines, and DLP Rules** to detect and block sensitive data, such as credit card information.
- **Test policy enforcement** to validate that DLP rules block sensitive information in data transfers.

Approximate Time: 15 minutes

Task 8.1: Protect PII Information for Data in Motion

In this task, you will create and test **PII (Personally Identifiable Information) policies** that safe-guard data-in-motion, such as credit card numbers embedded into unsearchable PDFs or other documents.

- 1. Verify SSL Inspection:
 - a. Go to **Policies > Common Configuration > SSL/TLS Inspection > SSL/TLS Inspection Policy**
 - i. Search for SSL Inspect All, which you created earlier.
 - ii. Verify that its **status** is set to **Enabled**.



- 2. Create a DLP Engine:
 - a. Navigate to **Policies> Data Protection > Common Resources > Dictionaries and Engines**.
 - b. In the DLP Dictionaries tab:
 - i. Search for the **Credit Cards** dictionary.
 - ii. Click the **Edit** button to explore its configuration.

c. You can explore the dictionary configuration

Edit DLP Dictionary

DLP DICTIONARY

Name

Credit Cards

Dictionary Type

Predefined

Confidence Score Threshold

Medium

BIN Number

Include

Exclude

Enter numbers separated by enter...

Add Items

Description

Detect leakage of credit card information

Save

Cancel

d. In the DLP Dictionaries & Engines, go to the DLP Engine tab.

e. Click **Add DLP Engines** and configure the engine as follows:

Data Protection

Policy

Inline Controls

Data Loss Prevention

SaaS Security

Out-of-Band CASB

SaaS Security Posture Management

DLP Dictionaries & Engines

DLP Dictionaries

DLP Engines

+ Add DLP Engine

No.	Name	Dictionary	Description
1	CCPA - High Volume	((Social Security Numbers (US) > 2...	

i. Name: **Sensitive Data**

ii. Expression: **All**

iii. Select a Predefined Dictionary: **Credit cards**

- iv. Violations Thresholds: **1** (empty field)
- v. Description: **Credit Card Number and Sensitive Data Violation**

g. Click **Save** and **Activate** the changes.

27 DLP ENGINE

28 Name

29 Sensitive Data

30 ENGINE BUILDER

31 EXPRESSION

32 ALL

33 Credit Cards > 1

34 ADD

35 Expression Preview

36 ((Credit Cards > 1))

37

38 DESCRIPTION

39 Credit Card Number and sensitive data violation

Save Cancel

3. Create a DLP Rule:

- a. Navigate to **Policies > Data Protection > Policy > Data Loss Prevention**
- b. Add the rule. Click **+Add DLP Rule** and configure the following:
 - i. Rule Order: **1**
 - ii. Severity: **High**
 - iii. Rule Name: **Data Violation Rule**

- iv. Rule Status: **Enabled**
- v. Content Matching : **Select DLP Engine**
- vi. DLP Engines: **Sensitive Data**
- vii. Leave all other options in **Criteria** and **DLP Incident Receiver** as **default**.

Add DLP Rule

DLP RULE

Rule Order

1

Severity

High

Rule Name

Data Violation Rule

Rule Status

Enabled

Rule Label

CRITERIA

Content Matching

Select DLP Engines

None

DLP Engines

Sensitive Data

Source IP Groups

Any

Save

Cancel

- viii. Action : Data Traffic: **Block**
- ix. End User Notification: **Show**

Add DLP Rule

DLP INCIDENT RECEIVER

Incident Receiver

ICAP

Zscaler Incident Receiver

Cloud-to-Cloud Forwarding

Zscaler Incident Receiver

None

ACTION

Data Traffic

Allow

Block

NOTIFICATION

Auditor Type

Hosted

External

End User Notification

Show

Hide

Auditor

None

Notification Message

Default

Note: You can optionally add a custom notification message by selecting the Notification Message drop-down, clicking the + (plus) icon, naming it CreditCard Block, and customizing the Allow and Block messages as needed.

- i. Click Save and **Activate** the changes from the **Activation** icon (cloud with arrow) in the top-right corner.

Add Custom Message

117 of 500 characters left

ADDITIONAL INFORMATION

Select the details to include in the notification

☒ File Name

☒ Destination

☒ Rule Name

☒ URL Category

☐ URL

☐ DLP Engines

PREVIEW

Allow Message:

Zscaler

Monitor

Posting content to the requested website violates your organization's policy. This activity was monitored by your organization.

Show more

Learn more

Save

Cancel

Data Protection

Policy

Inline Controls

Data Loss Prevention

SaaS Security

Out-of-Band CASB

SaaS Security Posture Management

Endpoint Controls

Outbound Email Controls

Common Resources

Data Loss Prevention

Configure Data Loss Prevention Policy

Rules are evaluated based on the configuration you select on the DLP Advanced Settings page (Administration > DLP Advanced Settings).

+Add DLP Rule

View by: Rule Order | Rule Label

Search...

Rule ...	Rule Name	Criteria	Action	Label and De...	
1	Data Violation Rule	DLP ENGINES Sensitive Data PROTOCOLS WebSocket SSL; WebSocket; Native FTP; HTTPS; HTTP	Block		

4. Test the Policy as a User:

a. Verify ZIA Activation:

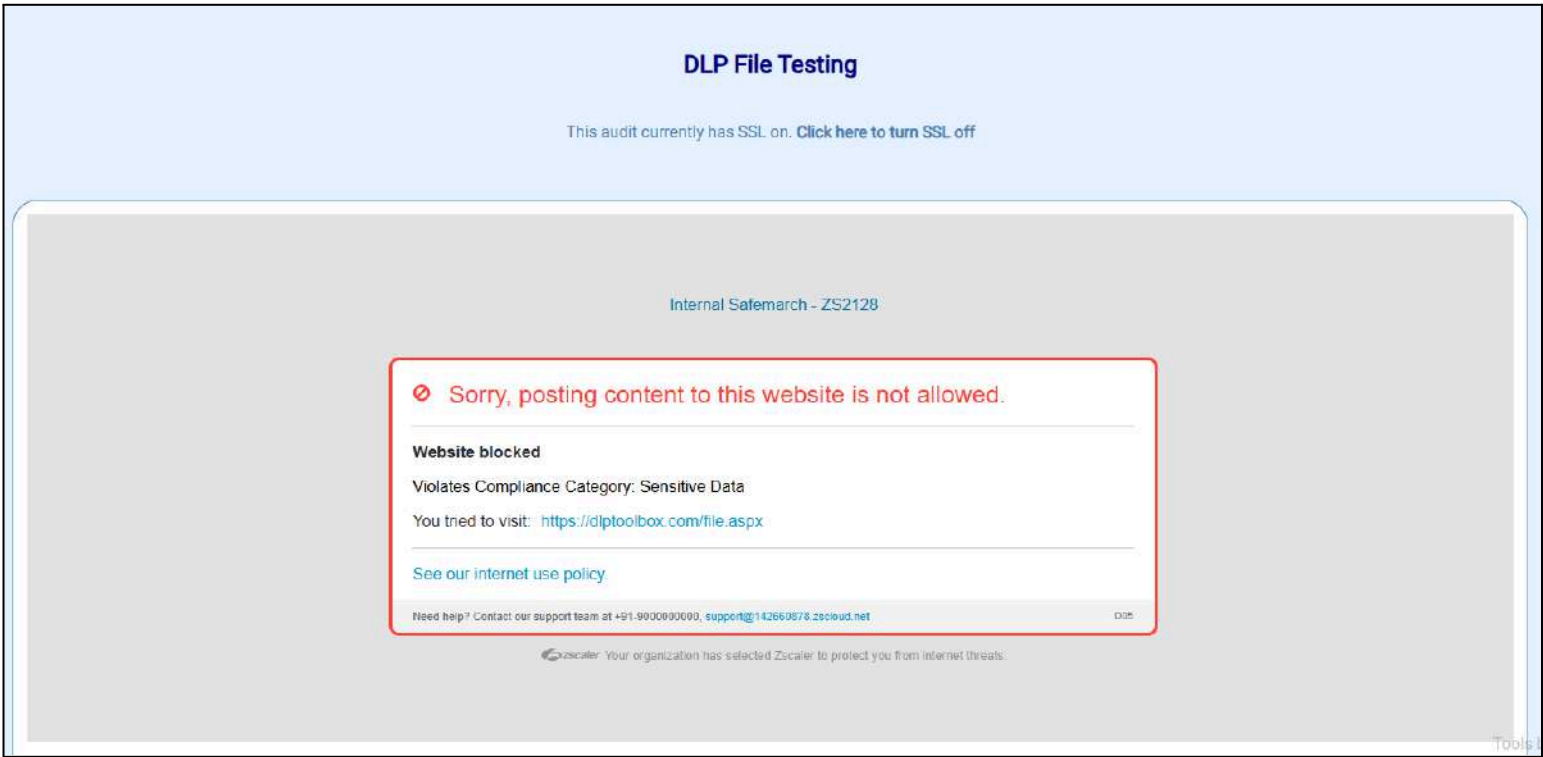
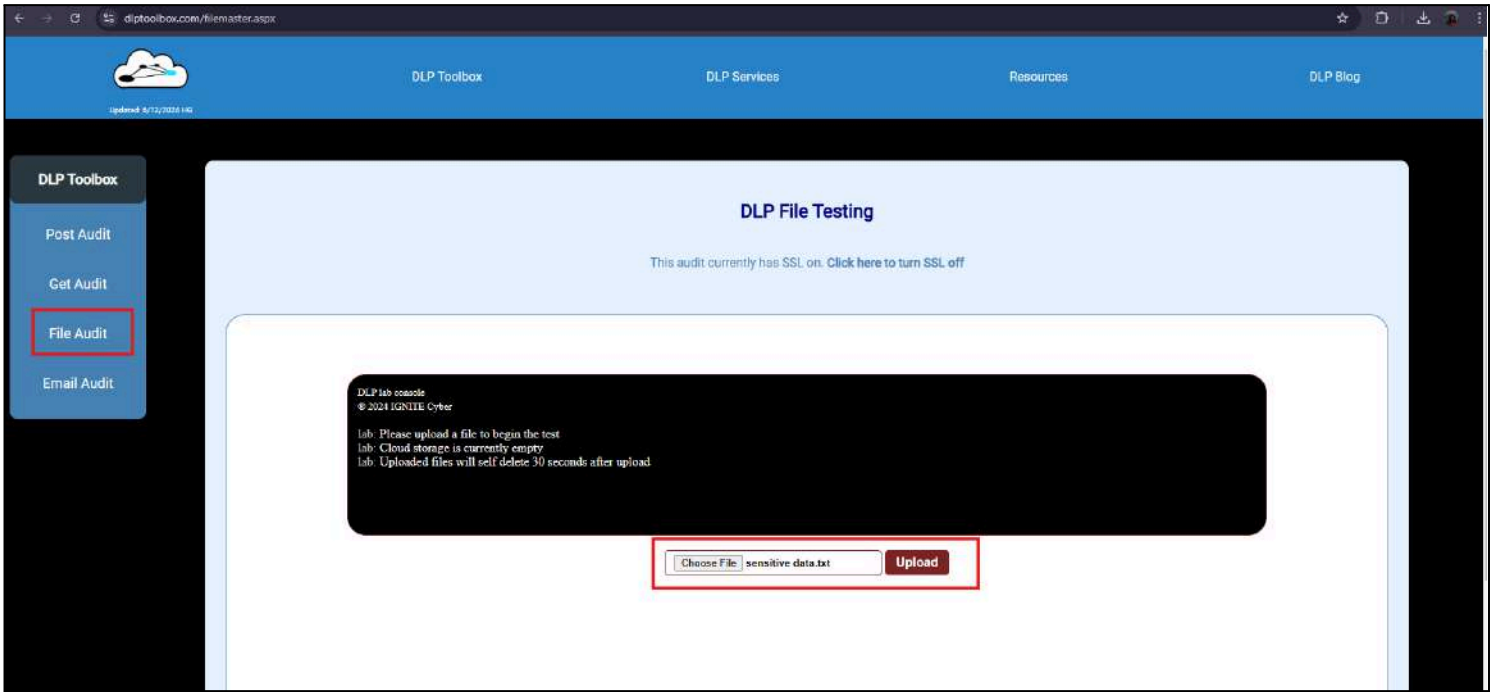
- i. On the **Zscaler Client Connector** ensure that the status for **Internet Security** is set to **enabled**.

b. Test Blocking Sensitive Data:

- i. Open a browser window on the **Client Corp PC**.
- ii. Navigate to: `https://dlptoolbox.com/`
- iii. From the left navigation pane, select **File Audit**.
- iv. Click **Choose file** and locate the file "**Sensitive Data.txt**" in the **DLP Test Files** folder on the **Desktop**.
- v. Click **Upload**

c. Verify Blocking Behavior:

- i. Confirm that a **Block Notification** is displayed during the upload attempt, indicating that the DLP Rule has blocked sensitive data from being transmitted.



Lab 9: Provision ZPA Infrastructure

Scenario: Safemarch Corporation is preparing to launch internal applications and requires secure, policy-driven connectivity without exposing those applications to the internet. **As a Zscaler administrator**, you will deploy and **provision ZPA App Connectors** across data centers and cloud platforms. These App connectors provide encrypted connectivity through the **Zero Trust Exchange (ZTE)** while keeping internal apps protected.

Objectives: By the end of this lab you will:

- **Deploy and provision** an **App Connector** using a provisioning key and certificate.
- **Activate** the **App Connector**, enabling secure communication with the ZPA infrastructure.
- **Validate** the **connection** and ensure the App Connector is functioning correctly.

Approximate Time: 25 minutes

Task 9.1: Provision an App Connector

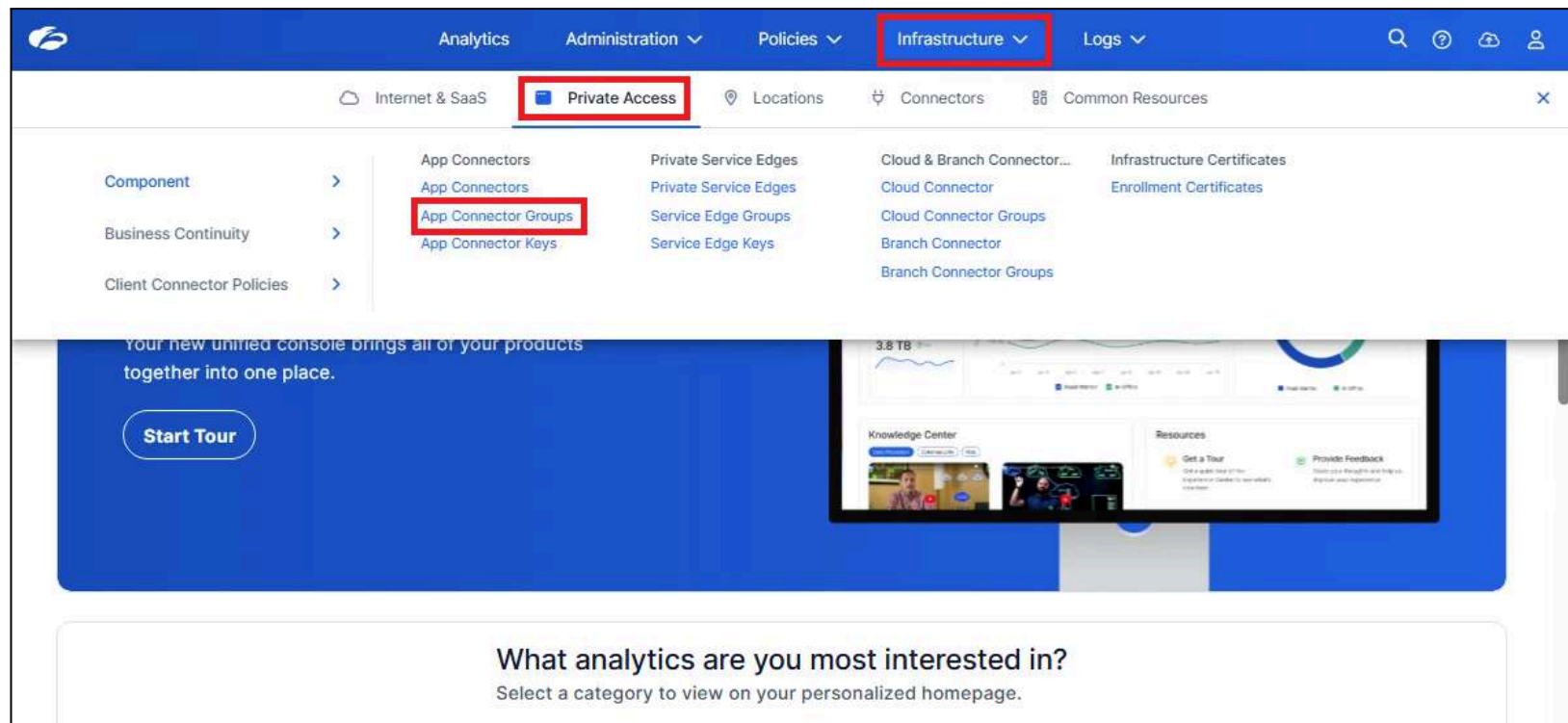
In this task, you will configure the **App Connector** by creating a provisioning key and registering it in the ZTE.

Note: The App Connector has been pre-installed with appropriate configurations for network, DNS, and NTP settings. In this task, you will activate the App Connector by providing the provisioning key created in the Experience Center Portal.

1. Access the ZPA Admin Portal:
 - a. On the **Corp: Win Svr 2016 VM**, log in with these credentials:
 - i. **Username: PATRAINING\Administrator**
 - ii. **Password: Admin-123!**
 - b. Navigate to **Experience Center Landing Page**
 - i. "<https://console.zscaler.com/>"
 - c. Authenticate with the **Student** account.

Note: Use the Windows 2016 server to access the Experience Center Portal, as you'll need to upload the provisioning key through WinSCP.

2. Configure the App Connector:
 - a. From the **Experience Center**, navigate to **Infrastructure > Private Access > App Connector Groups**.



b. From **Add App Connector Group**.

- i. Click **Add App Connector Group**
- ii. Name: **OnPrem**
- iii. Status: **Enabled**
- iv. Description: **Optional**
- v. Signing Certificate: **Connector**
- vi. DNS Settings (IPv4 and IPv6): **Leave as default**
- vii. Location: **NYC, NY, USA.**
- viii. Click **Next**.
- ix. Version Profile Configuration: **Leave as default**
- x. App Connector Software Update Schedule: **Sundays at 00:30.**
- xi. Click **Save**.

Private Access

Component

App Connectors

App Connectors

App Connector Groups

App Connector Keys

Private Service Edges

Cloud & Branch

Connector Resources

Infrastructure Certificates

Business Continuity

Client Connector Policies

Analytics

Administration

Policies

Infrastructure

Logs

Q

?

🔒

👤

← Add App Connector Group

DetailsUpdate Settings

Name

OnPrem

Description

Please enter the description here...

DNS Resolution Option

IPv4IPv6IPv4 and IPv6

Disaster Recovery

EnabledDisabled

Additional Features

Privileged Remote Access

EnabledDisabled

Hosting Detection

AutomaticManual

App Connector Location

New York, NY, USA

Status

EnabledDisabled

Signing Certificate

Connector

TCP Quick Acknowledgement

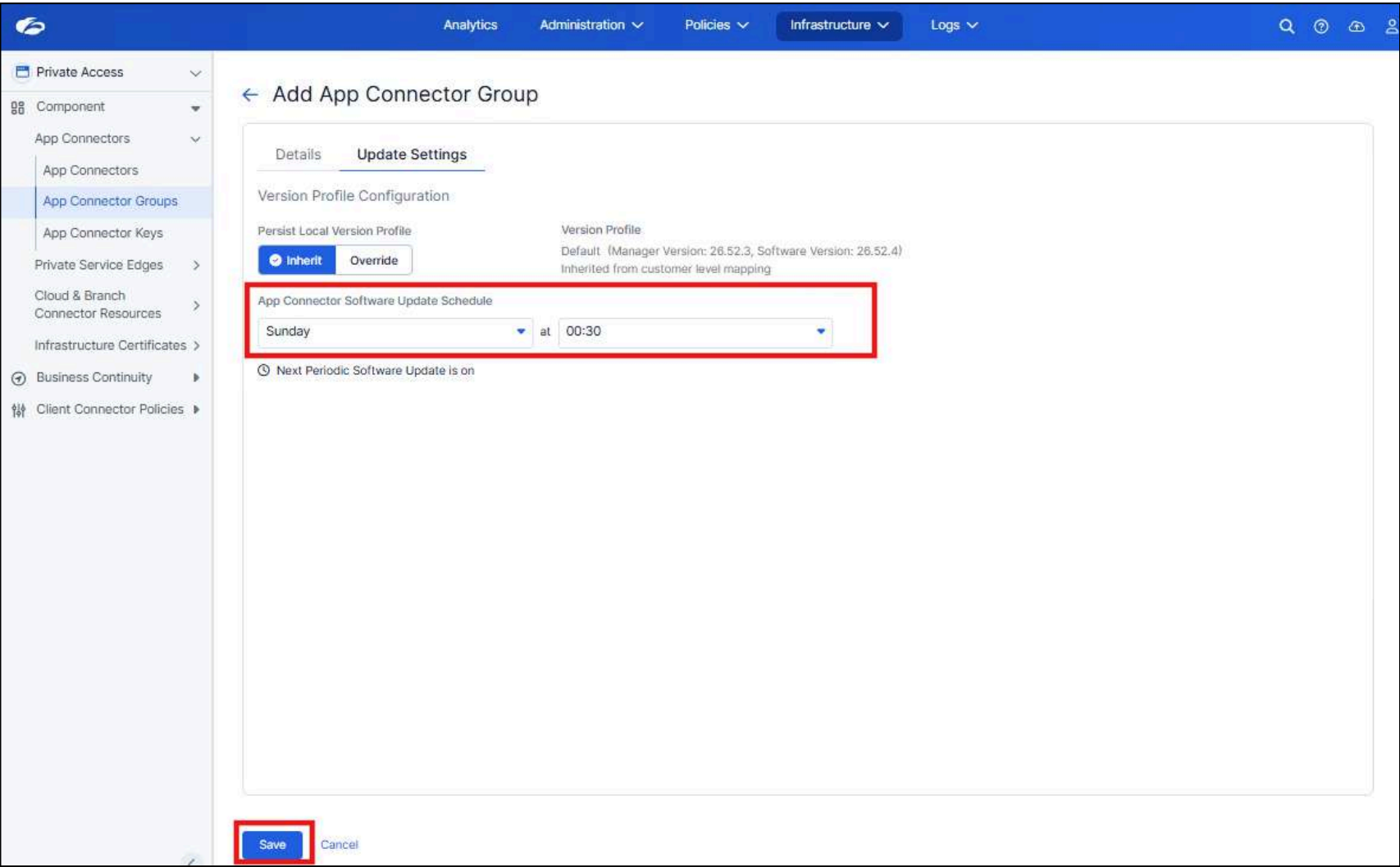
EnabledDisabled

Disable AppProtection

YesNo

Next

Cancel



- c. From **App Connector Keys**:
 - i. Click Add **App Connector Keys**
 - ii. Group: **OnPrem**
 - iii. Name: **OnPrem**
 - iv. Maximum Reuse Count: **4**
 - v. Signing Certificate: **Connector**
 - vi. Click **Save**.

Add App Connector Provisioning Key

View or Export Provisioning Key After Creation ☒

Group: OnPrem

Name: OnPrem

Maximum Reuse of Provisioning Key: 4

Signing Certificate: Connector

Save Cancel

⚠ Zscaler recommends setting the Maximum Reuse of Provisioning Key option to the number of App Connectors that will be immediately deployed, to reduce the possibility of compromise.

- 3. Save the Provisioning Key:
 - a. Copy the provisioning key:
 - i. Select **App Connector Provision Keys**. Click on the **Copy** icon.

App Connector Keys

Filters Applied Hide Filters

Name Maximum # of App Connectors Provisioning Key Utilization Count Reset

Name	Maximum # of App Connect...	Provisioning Key Utilization ...	App Connector Group	Provisioning Key	Actions
> OnPrem	4	0	OnPrem	4 api.private.zscaler.com...	

Rows per page: 20 1-1 of 1

- b. **Right-click on** an empty area of the **Desktop**.
- c. Click **New > Text Document**
- d. Name the new file: **provision_key**

Note: Remove any **.txt** extension if it appears, so the file name is exactly **provision_key** with no suffix.

- e. Double-click the **provision_key** file to **open it in Notepad**.
- f. **Paste** the **OnPrem App Connector provisioning key** (copied earlier from the portal) into the file.
- g. Click **File > Save** and close Notepad.

Task 9.2: Activate the App Connector

In this task you will activate the App Connector and ensure secure traffic flow to ZPA.

1. Configure the App Connector VM (**Corp: APP Connector**):

a. Log in to **Corp: APP Connector** with credentials:

i. Username: `admin`

ii. Password: `zscaler`

b. Start the **SSH Daemon** so you can connect with WinSCP:

i. `sudo systemctl start sshd`

ii. password: `zscaler`

c. Verify that SSH is running:

i. `sudo systemctl status sshd`

ii. Verify that status is **active (running)**.

2. Transfer the Provisioning Key:

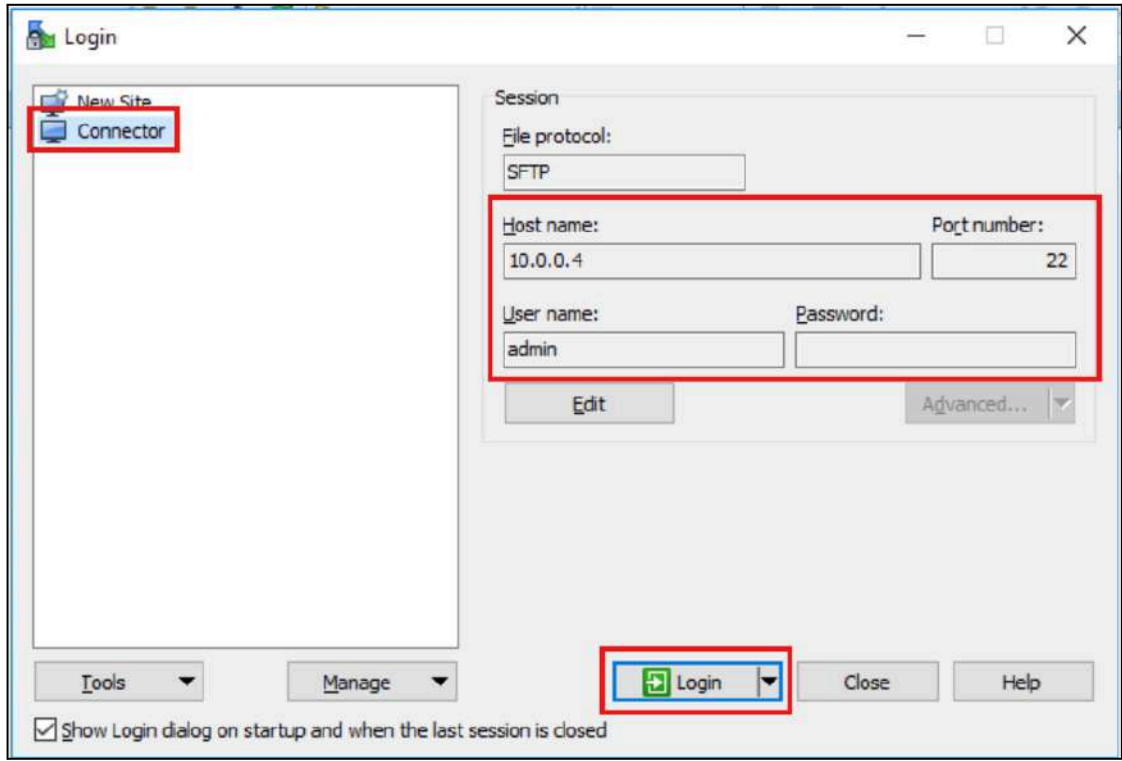
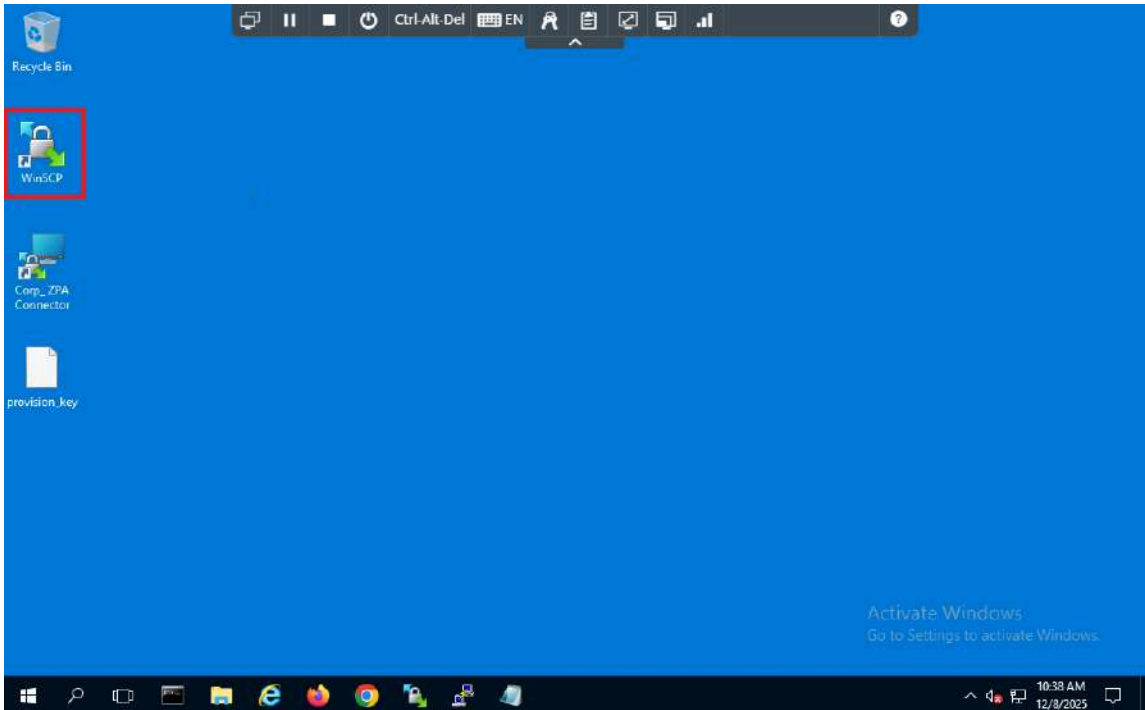
a. On **Corp: Win Svr 2016**, open **WinSCP** and configure the session as follows:

i. IP Address: `10.0.0.4`

ii. Port `22`

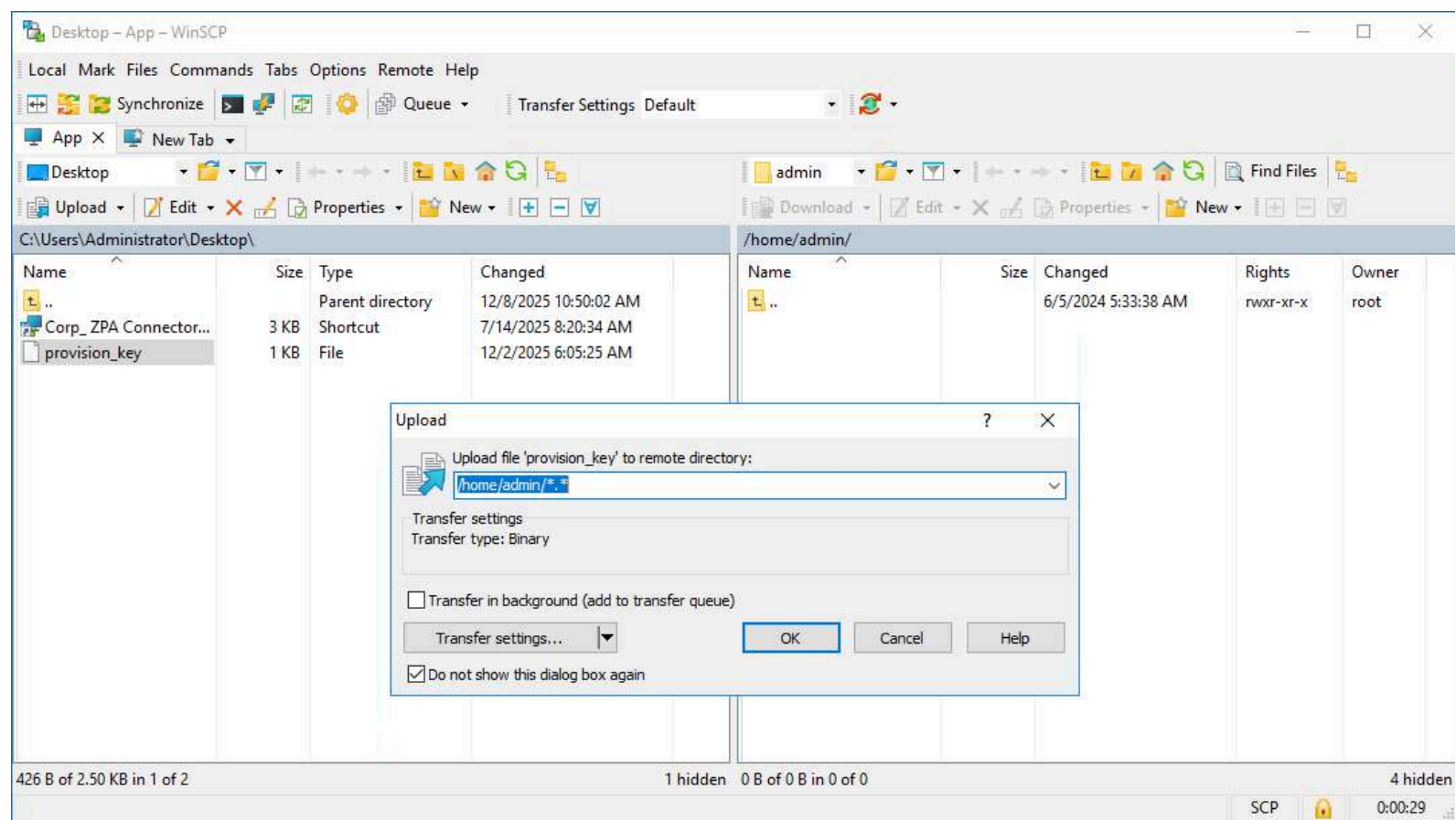
iii. Username: `admin`

iv. Password: `zscaler`



- b. In WinSCP, navigate to the key file:
 - i. Go to **C:\Users\Administrator\Documents***** and click **Browse**.
 - ii. Select **Desktop** and click **OK**. Confirm Desktop is selected before proceeding to the next step.

iii. Right-click on the **provision_key** and select **Upload**, or **drag and drop** from Left to right side panel.



c. Verify that the file is uploaded and close **WinSCP**.

3. Activate the App Connector

a. You should see a prompt similar to: `admin@zpa-connector ~]$`

Note: The ~ indicates you are in your home directory, which is /home/admin.

b. After you upload provision_key from WinSCP, it will be in the admin home directory by default.

c. Verify your current directory: `pwd`

i. Expected output: /home/admin

d. List files: `ls`

- i. Expected output: `provision_key`
- e. Activate the App Connector:
 - i. Stop the connector service: `sudo systemctl stop zpa-connector`
 - ii. Copy the provisioning key to the directory: `sudo cp provision_key /opt/zscaler/var/`

Note: Ensure the trailing slash (/) after var. Absence can cause service startup failures.

```
[admin@zpa-connector ~]$ pwd
/home/admin
[admin@zpa-connector ~]$ ls
provision_key
[admin@zpa-connector ~]$ sudo cp provision_key /opt/zscaler/var/
[sudo] password for admin:
[admin@zpa-connector ~]$ _
```

- iii. Confirm the file: `sudo ls /opt/zscaler/var/`

```
[admin@zpa-connector ~]$ sudo ls /opt/zscaler/var/
admin-passwd-message  image.bin  instance_id.bin  instance_id.crypt  provision_key  updater.version
[admin@zpa-connector ~]$
```

- iv. Restart the service: `sudo systemctl start zpa-connector`
- v. Enable the service to start on boot: `sudo systemctl enable zpa-connector`
- f. Verify that the service is running:
 - i. `sudo systemctl status zpa-connector`

Note: Expected Result: The App Connector is active and connected to the ZPA infrastructure. You should see the .pem key and certificate files as further confirmation.

4. Optional recovery (**Only if the connector is status fails or is red**):

- a. Stop the service: `sudo systemctl stop zpa-connector`
- b. Clear the `var` directory contents: `sudo find /opt/zscaler/var -mindepth 1 -delete`
- c. Ensure `provision_key` still exists in `/home/admin` (re-upload via **WinSCP on Win Svr 2016** if necessary): `ls`
- d. Re-copy the provisioning key: `sudo cp provision_key /opt/zscaler/var/`
- e. Start the connector service: `sudo systemctl start zpa-connector`
- f. Check status: `sudo systemctl status zpa-connector`

```

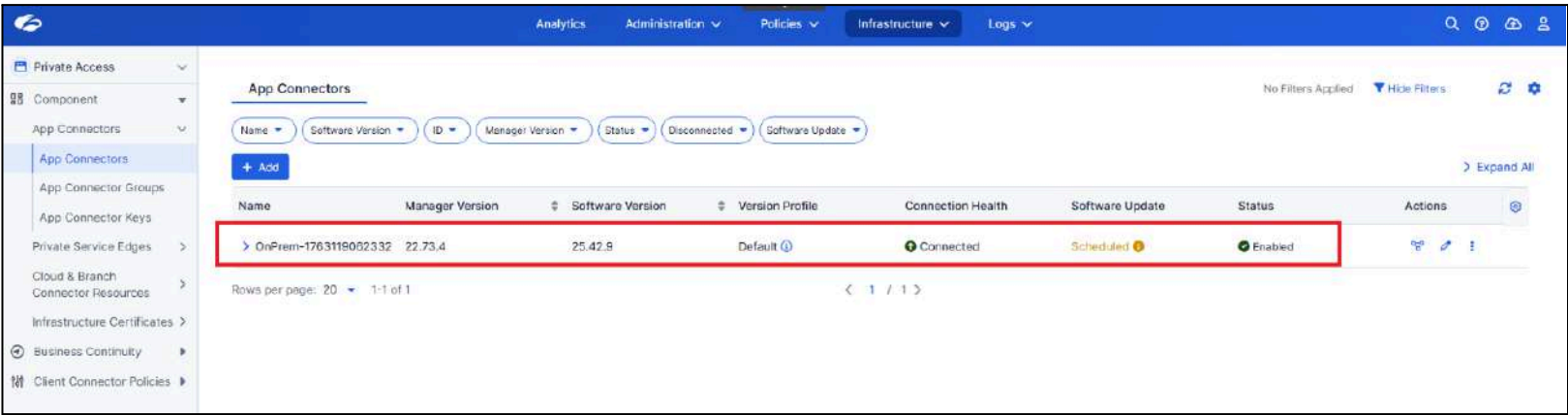
admin@zpa-connector ~$
admin@zpa-connector ~$ systemctl status zpa-connector
● zpa-connector.service - Zscaler Private Access Connector
   Loaded: loaded (/usr/lib/systemd/system/zpa-connector.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2018-02-27 02:07:20 PST; 14min ago
     Main PID: 1152 (zpa-connector)
    CGroup: /system.slice/zpa-connector.service
            └─1152 /opt/zscaler/bin/zpa-connector
               11735 zpa-connector-child

Feb 27 02:20:24 zpa-connector zpa-connector-child[11735]: Time skew: - 0.005375s
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: ----- Connector Status:ID=144123139134062689;Name=Training-1;Ver=17.01.2 -----
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Certificate will expire in 374 days, 14 hours, 32 minutes, 42 seconds
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Control connection state: fobh_connection connected, f10.0.0.41:50038:broker1.nyc3.prod.zpat...2541:443
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: RPC Messages: BrkReq = 0, BrkReqAck = 0, BindReq = 0, BindReqAck = 0, AppReqDisc = 0, AppReqAck = 0
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Broker data connection count = 0, backed_off connections = 0
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Data Transfer: Total ToBroker = 0 bytes, Total FromBroker = 0 bytes
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Mtunnels: Total Created = 0, Total Freed = 0, Current Active = 0, Alloc = 0, Free_q_cnt = 0
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Registered apps count = 0, alive app = 0, passive_health = 0, service_count = 0, target_count = 0
Feb 27 02:21:24 zpa-connector zpa-connector-child[11735]: Time skew: - 0.005768s
Hint: Some lines were ellipsized, use -l to show in full.
admin@zpa-connector ~$ psal

```

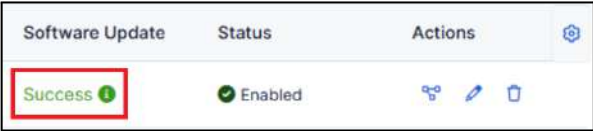
5. Verify in the ZPA Admin Portal

- a. In the Experience Center Landing Page (**Corp: Win Svr 2016**), navigate to:
 - i. **Infrastructure > Private Access > App Connectors**
- b. Confirm the following:
 - i. The App Connector is listed.



ii. Update Status field:

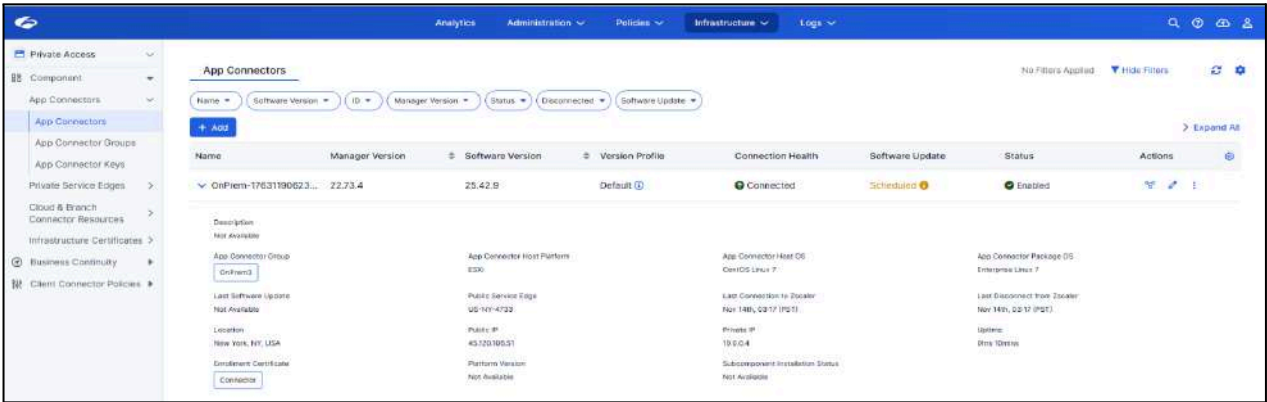
- **Success:** Indicates the App Connector is running the latest software.
- **Scheduled:** Indicates that an update will occur at the next update interval.



Note: There is no need to wait for the App Connector to update, continue with the lab and check back in a few minutes.

c. Click the name of the App Connector to expand details, including:

i. Host OS, PSE, Location, IP Addresses, and Uptime.



Lab 10: Add a Corporate Application

Scenario: Safemarch Corporation needs secure, scalable access to multiple internal applications that share a common domain pattern. To simplify access management and enforce reliable Zero Trust policies, you will configure a **wildcard application segment** for general access and a **dedicated application segment** for more granular user and application access. You will also implement ICMP access for troubleshooting.

Objectives: By the end of this lab, you will:

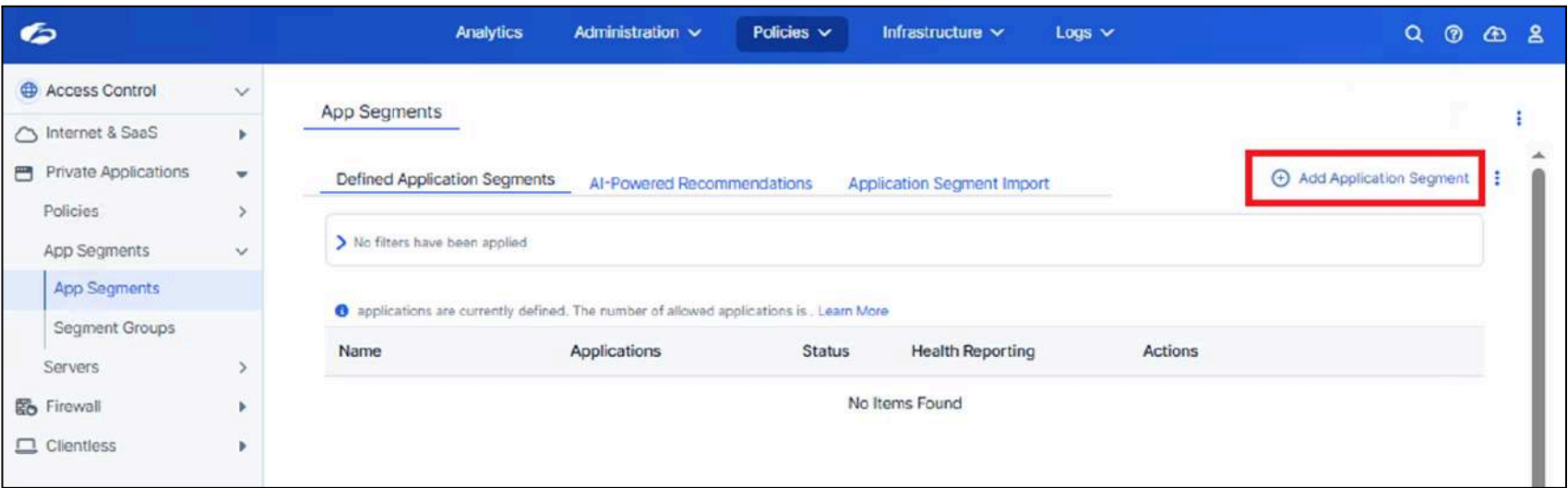
- **Configure a wildcard application segment** for internal applications (e.g., *.patraining.safemarch.com)
- **Configure user-specific access** to defined internal applications.
- **Enable ICMP access for troubleshooting** in the ZPA Admin Portal

Approximate Time: 35 minutes

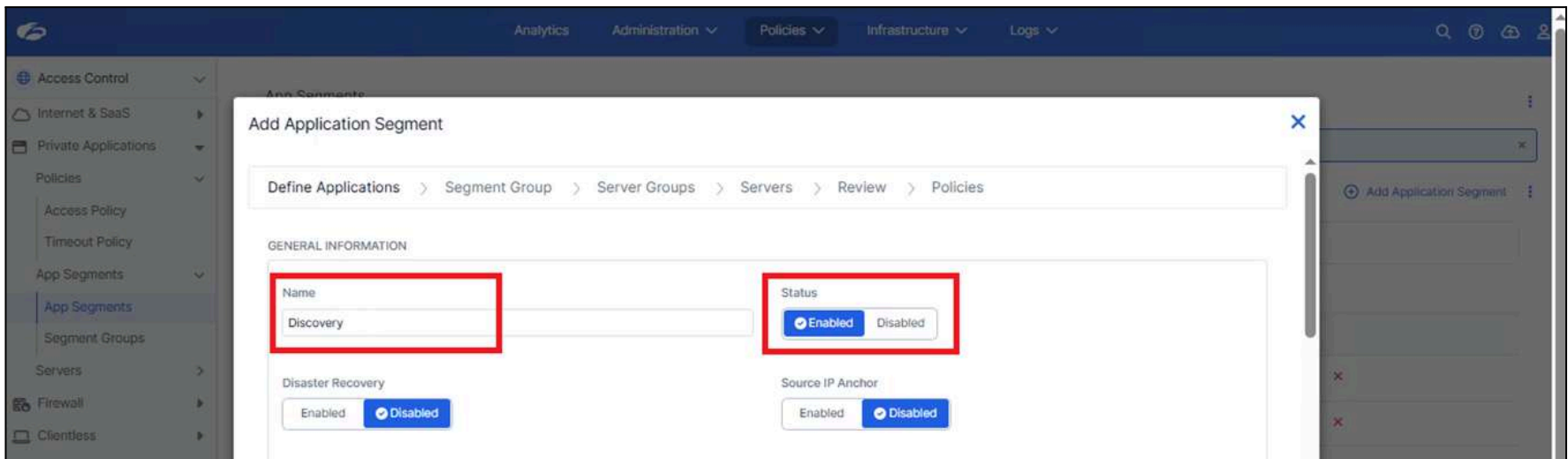
Task 10.1: Configure a Wildcard Application Segment for Internal Applications

In this task, you will configure a wildcard application segment to discover internal applications matching a domain pattern and allow general access for all users.

- 1. Add a Wildcard Application Segment
 - a. From the Experience Center, navigate to **Policies > Access Control > Private Applications > App Segments**.
 - b. Click the **+Add Application Segment** link to create a new segment.



- c. At the **Define Applications** step of the wizard > **General Information**:
 - i. Name: **Discovery**
 - ii. Status: **Enabled**
 - iii. Source IP Anchor: **Disabled**
 - iv. Add a suitable description.



d. Applications:

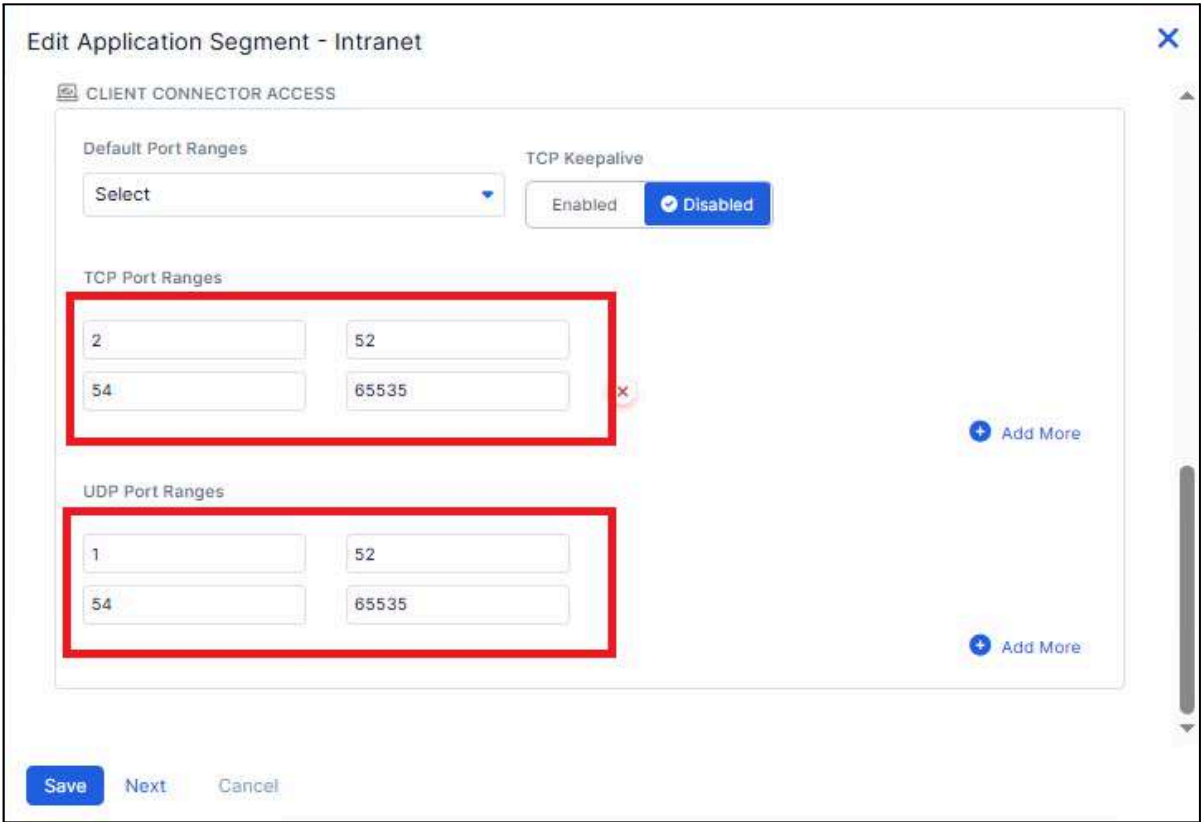
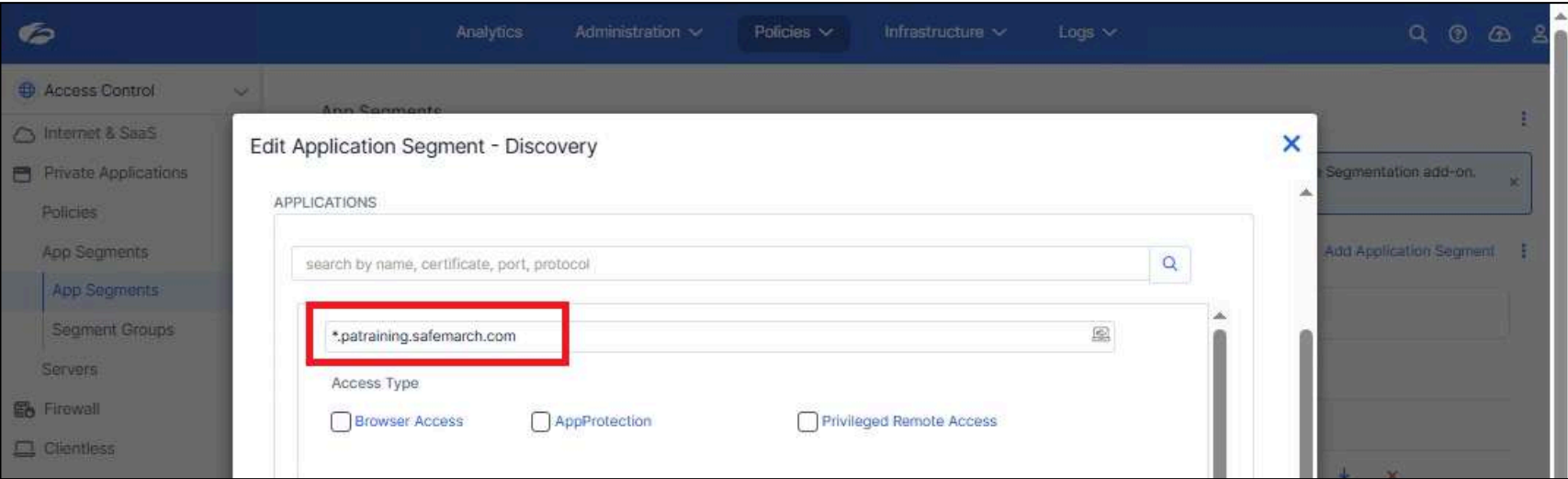
- i. In the Enter a domain or IP address field, specify: *.patraining.safemarch.com

e. Client Connector Access:

- i. Define the Port Ranges for **TCP** and **UDP**
- ii. TCP Port Ranges: **2-52** and **54-65535**
- iii. UDP Port Ranges: **1-52** and **54-65535**
- iv. TCP Keepalive: **Set to Disabled.**

f. Common Configuration:

- i. Leave **defaults** as is.



g. At the Additional Configuration step:

- i. Leave the following options unchanged:
 - Double Encryption: **Disabled**
 - Bypass: **Use Client Forwarding Policy**

- ICMP Access: Disabled

ii. Click **Next**.

Add Application Segment

COMMON CONFIGURATION

Health Reporting

Continuous

On Access

None

Client Connector can receive CNAME

Enabled

Disabled

App Connector Selection Method

Closer to Application

Closer to User

ADDITIONAL CONFIGURATION

Double Encryption

Enabled

Disabled

ICMP Access

Enabled

Disabled

Bypass during Reauthentication

Enabled

Disabled

Bypass

Use Client Forwarding Policy

Next

Previous

Skip

Cancel

Note: The ICMP Access option is typically disabled unless ICMP connectivity is required for application troubleshooting or testing.

h. At the Segment Group step:

- i. Click **Add Segment Group**.
- ii. Name: **Discovered Apps**

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iii. Status: **Enabled**

iv. Click **Next**.

Add Application Segment

Define Applications > Segment Group > Server Groups > Servers > Review > Policies

Select Segment Group Add Segment Group

Name
Discovered Apps

Description

Status
Enabled Disabled

Next Previous Skip Cancel

i. Add **Server Group**

i. Name: **CorpServers**

ii. Status: **Enabled**

iii. Set Dynamic Server Discovery: **Enabled**.

iv. App Connector Groups: **OnPrem**.

v. Click **Next**.

Add Application Segment

Define Applications > Segment Group > Server Groups > Servers > Review > Policies

This is a recommended configuration. You can however skip this for Client To Client applications.

Select Server Group

Add Server Group

Name

CorpServers

Description

Status

Enabled

Disabled

Dynamic Server Discovery

On

Off

App Connector Groups

OnPrem

Next

Previous

Skip

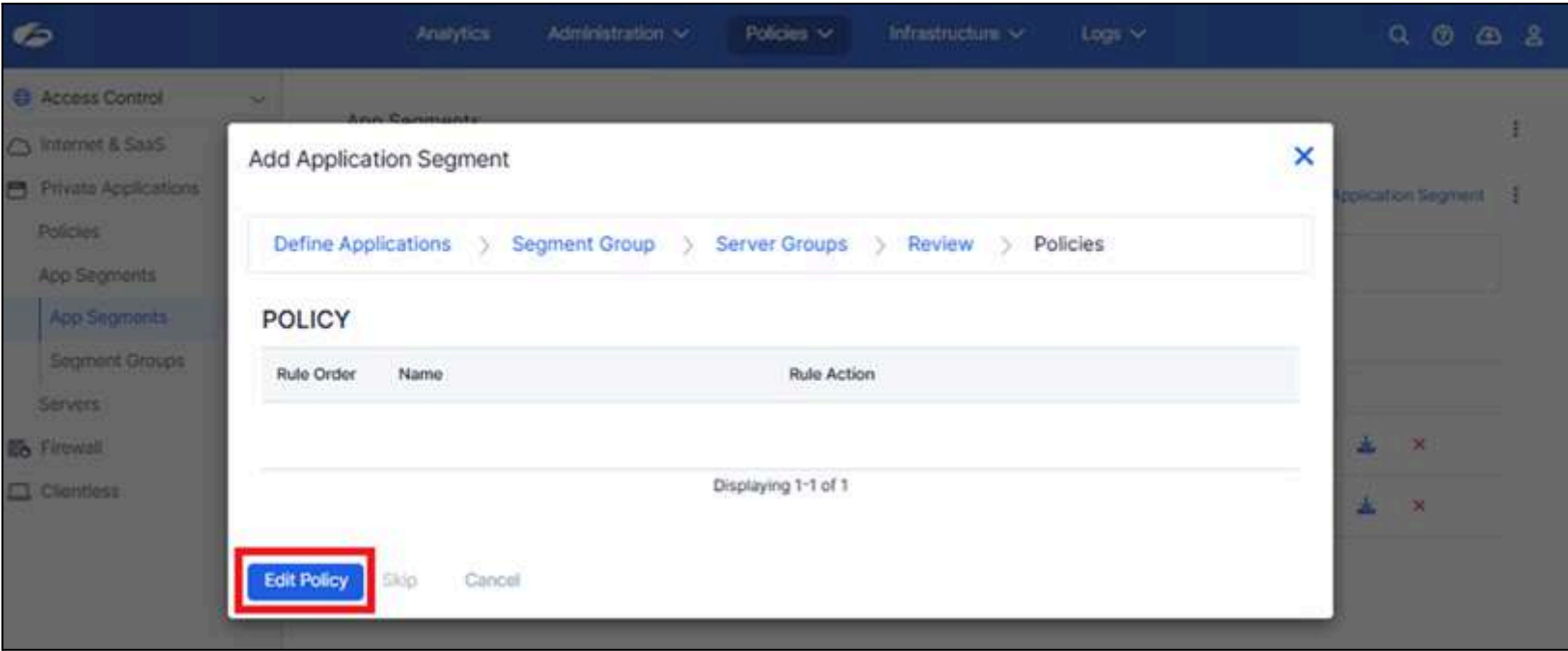
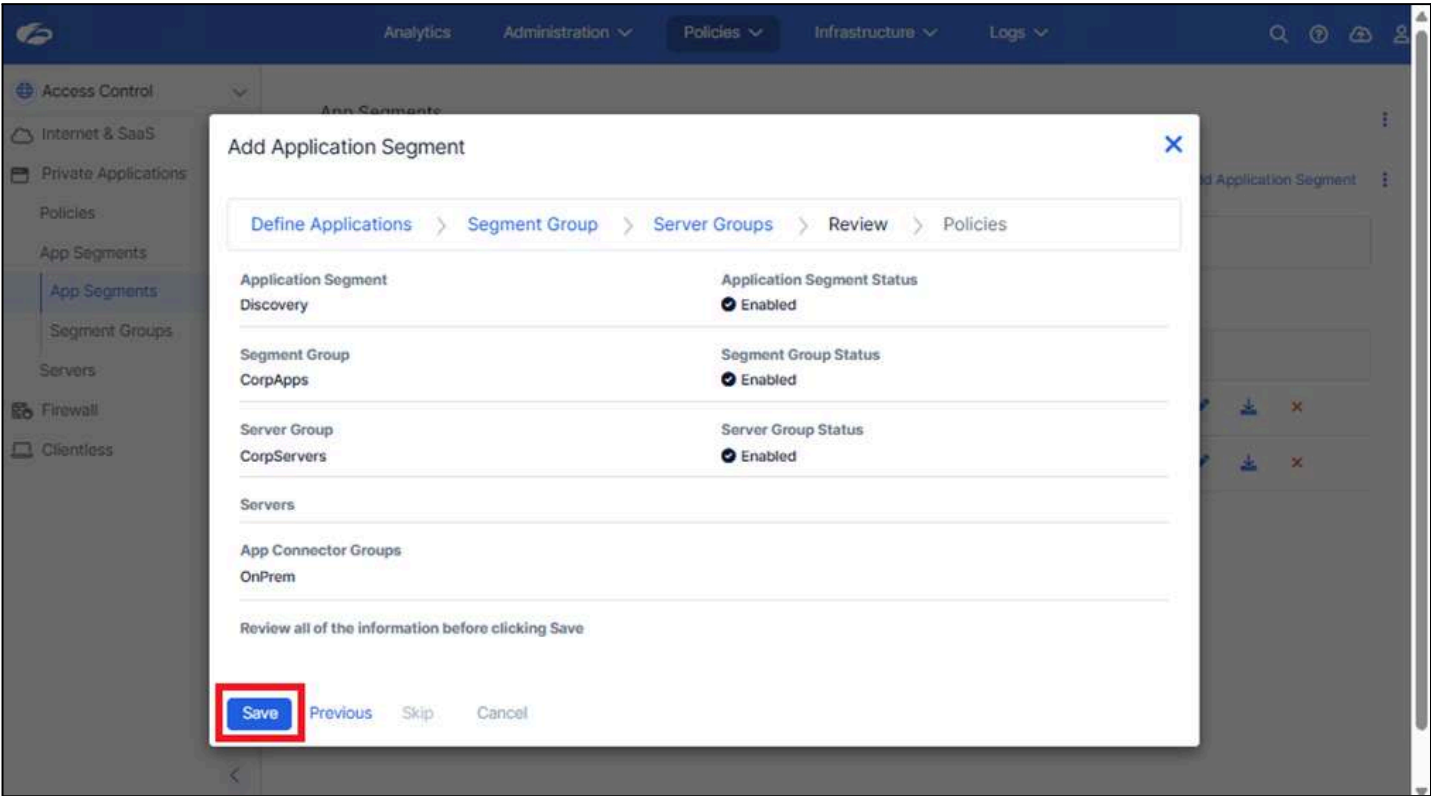
Cancel

2. At the **Review** step:

a. Confirm the configurations and click **Save**.

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- 3. Configure Access Policies:
 - a. Go to the **Policies > Access Policy** page in the ZPA Admin Portal.

- b. Configure the new rule:
- i. On the **Access Policy** page, click **Add**.
 - Name: **Discovered Apps**
 - Rule Action: **Allow Access**
 - Leave App Connector Selection Method to: **All App Connector Groups for the Application**.
 - Click **Add Criteria**, select **Applications**, and choose **Discovered Apps Segment Group**.
 - Do not add additional criteria.
 - Click **Save**.

The screenshot shows the 'Add Access Policy' dialog box. The 'Name' field is highlighted with a red box and contains the text 'Discovered Apps'. Below it is a 'Description' field. The 'ACTION' section has a red box around the 'Rule Action' field, which has three buttons: 'Allow Access' (selected), 'Block Access', and 'Conditional Access'. The 'TRAFFIC STEERING' section has a dropdown menu for 'App Connector Selection Method' set to 'All App Connector groups for the application'. The 'CRITERIA' section is empty with an 'Add Criteria' button. At the bottom are 'Save' and 'Cancel' buttons.

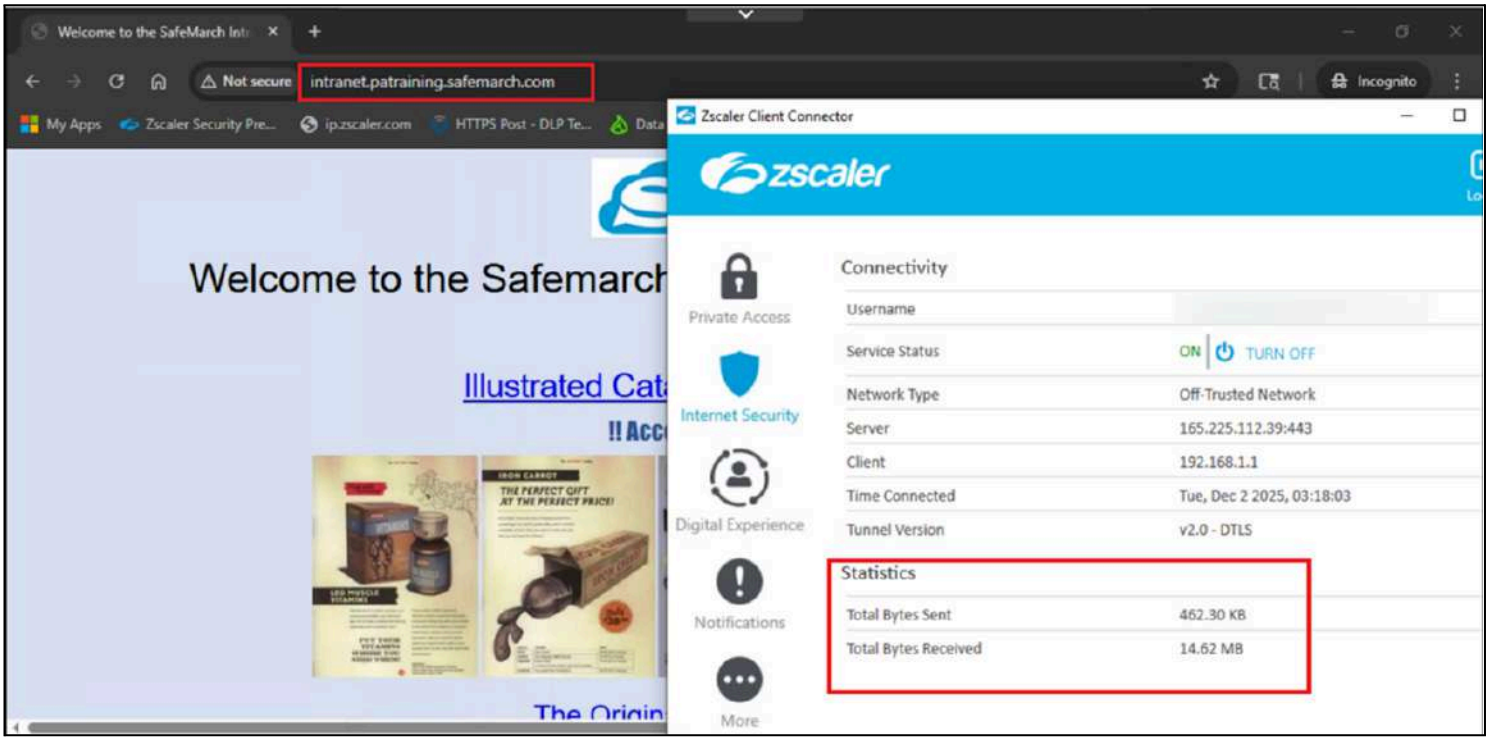
The screenshot shows the Zscaler policy configuration interface. On the right, a sidebar lists various categories: Applications, Branch Connector Groups, Chrome Enterprise Browser, Client Connector Posture Profiles, Client Connector Trusted Networks, Client Types, Cloud Connector Groups, Country Codes, Locations, Machine Groups, Platforms, Risk Scores, and User and Session Attributes. The 'Applications' category is highlighted with a red box. In the main area, under the 'TRAFFIC STEERING' section, the 'App Connector Selection Method' dropdown is set to 'All App Connector groups for the application' and is also highlighted with a red box. The 'ACTION' section at the top shows 'Allow Access' as the selected rule action. The 'CRITERIA' section at the bottom is currently empty.

This screenshot focuses on the 'CRITERIA' section of the Zscaler policy configuration. It shows two dropdown menus separated by an 'OR' operator. The first dropdown, 'Application Segments', has the placeholder text 'Select one or more application segments'. The second dropdown, 'Segment Groups', is set to 'Discovered Apps' and is highlighted with a red box. At the bottom left, the 'Save' button is highlighted with a red box, while the 'Cancel' button is not. An 'Add Criteria' button is visible in the top right corner of the criteria section.

10. Test the Access Policy:

- a. On the **Corp: Client PC**, open a **web browser in a window**
- b. Open the **Zscaler Client Connector** adjacent to it, placing the window next to the browser.
- c. Load `http://intranet.patrainig.safemarch.com` (a bookmark is provided)
- d. Confirm that the Intranet page loads correctly.

- e. Refresh the page and verify that the **Total Bytes Sent / Received** counters in the Zscaler Client Connector increment.

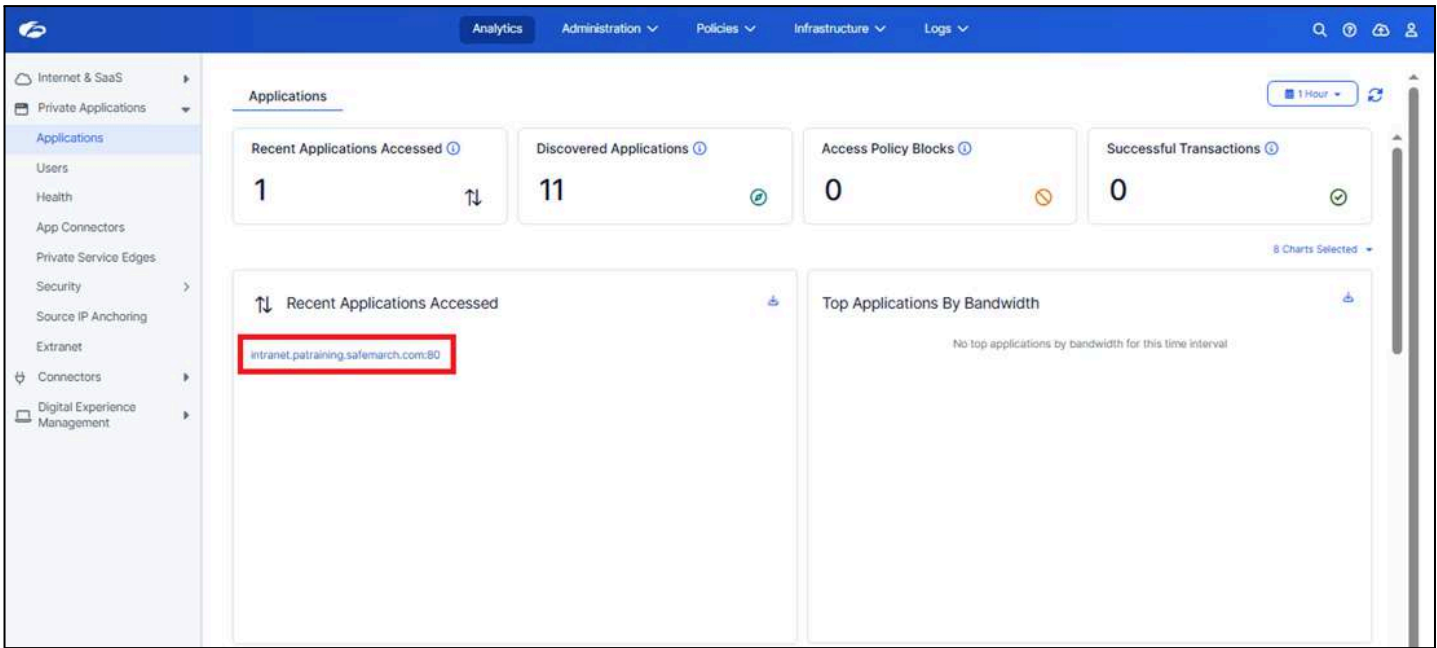
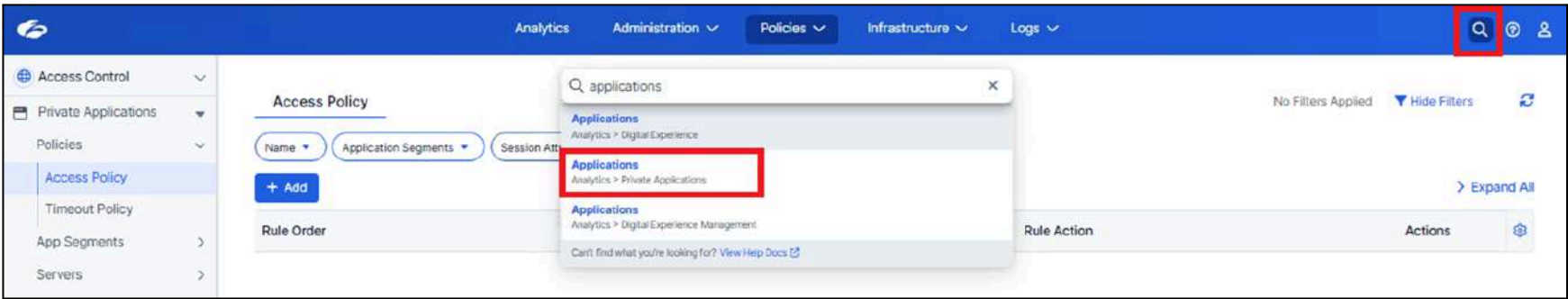


Task 10.2: Configure User-Specific Access to an Intranet Application

In this task, you will create a **dedicated application segment** to provide more granular access control that limits intranet access to the "Student" user.

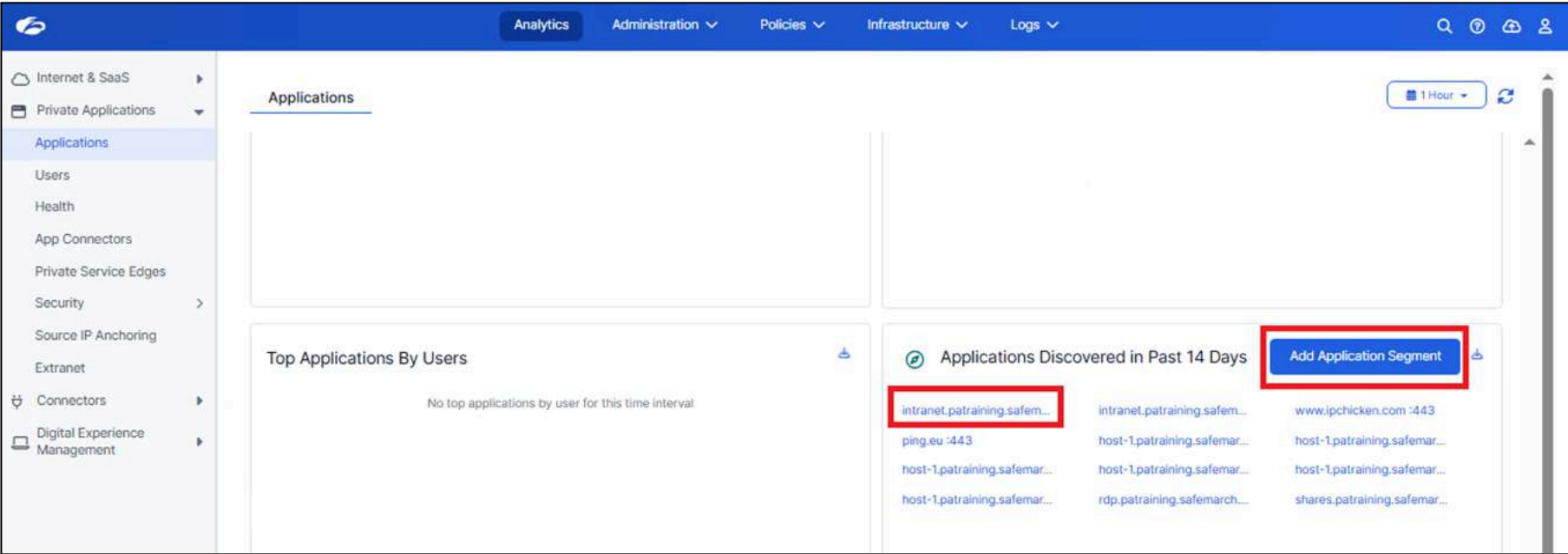
1. Review Recently Accessed Private Applications

- a. From the **Experience Center**, Go to the **search bar** and type **applications** to navigate into **Analytics > Private Applications**.
- b. Scroll to the **Applications Discovered in the past 14 days** section to identify recently accessed applications.

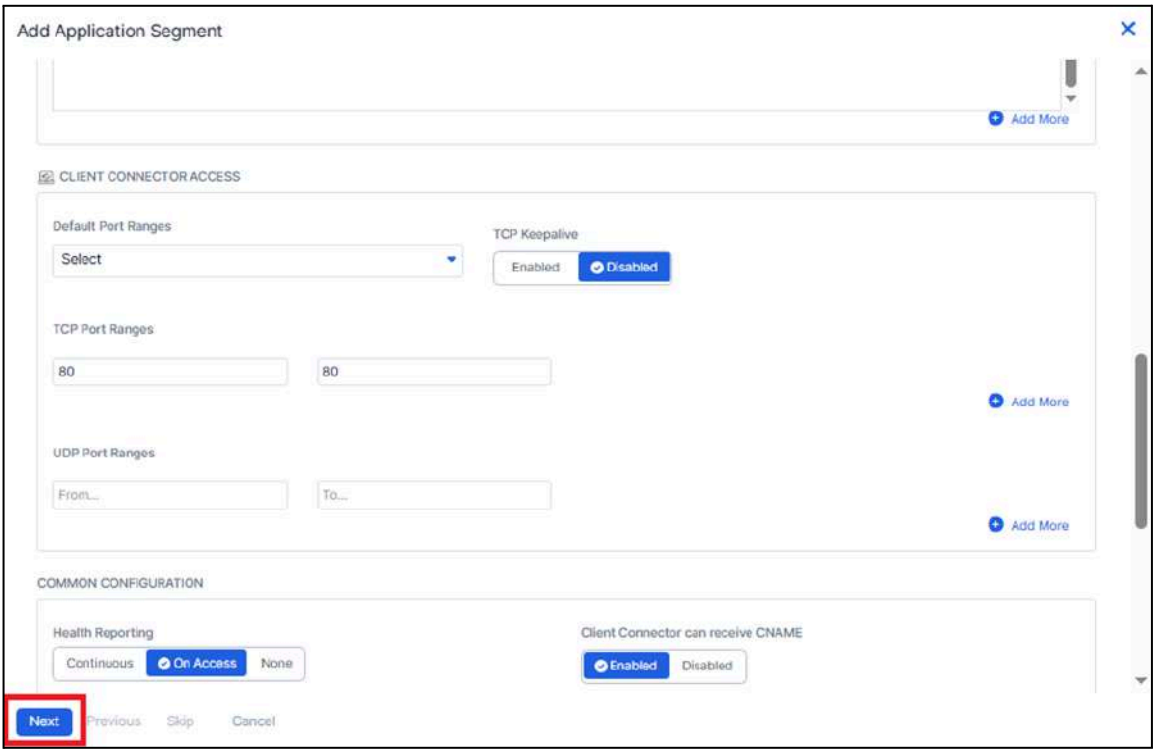
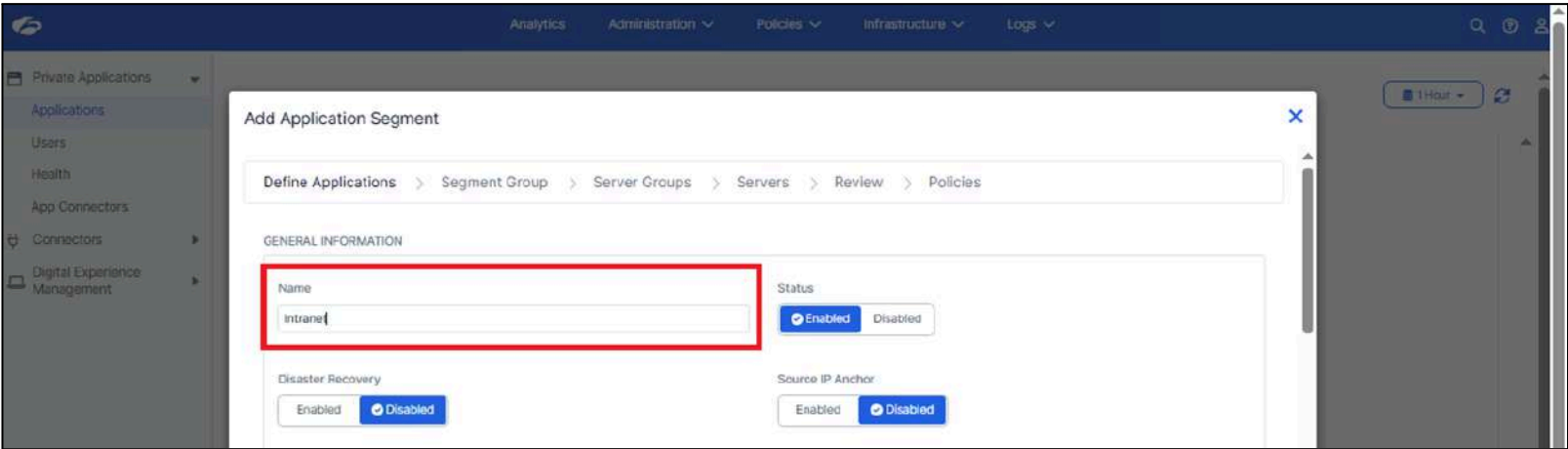


2. Configure the Intranet Application Segment

- a. Click **Add Application Segment**, select the application `intranet.pattraining.safemarch.com`, and click **Define Selected Applications**.
 - i. Alternatively, navigate to **Policies > App Segment**.



- b. At the Define Applications step:
 - i. Name: **Intranet**
 - ii. Status: **Enabled**
 - iii. Source IP Anchor: **Disabled**
 - iv. Add a suitable description.
 - v. Leave other settings as **default**.
 - vi. Click **Next**.



- c. Add Segment Group as follows:
- i. Name: CorpApps
 - ii. Status: Enabled
 - iii. Click **Next.**

Add Application Segment

Define Applications > Segment Group > Server Groups > Servers > Review > Policies

Select Segment Group

Add Segment Group

Name

CorpApps

Description

Status

Enabled

Disabled

Next

Previous

Skip

Cancel

- d. Add Server Group as follows:
- i. Name: **CorpServers** (Selected from the existing wildcard application segment)
 - ii. Click **Next**.
- e. Click **Save**.

Add Application Segment

Define Applications > Segment Group > Server Groups > Servers > Review > Policies

This is a recommended configuration. You can however skip this for Client To Client applications.

Select Server Group

Add Server Group

Server Group Type

App Connector

Extranet

Server Group

CorpServers

Next

Previous

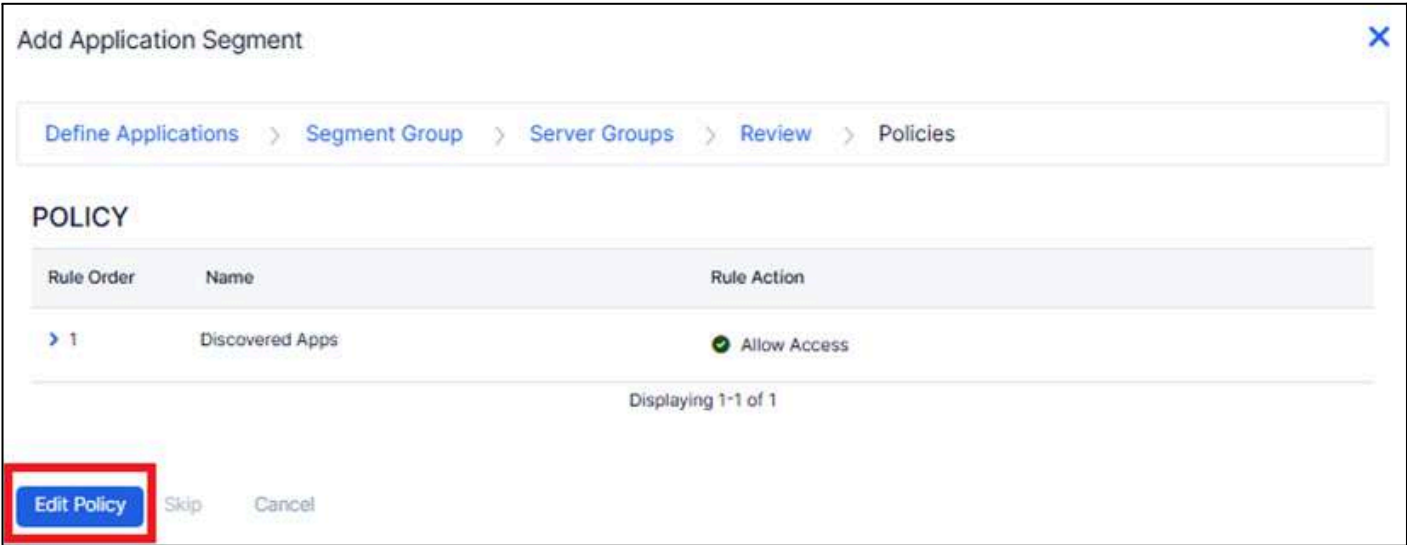
Skip

Cancel

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Note: After saving the Application Segment, you will be redirected to the Edit Policy page. From there, you can add a policy for the Discovery App Segment. Alternatively, you can navigate directly to the Policies > Access Policy page from the left-hand menu.



3. Add Access Policy

a. Navigate to **Policies > Access Policy**.

- i. Click **Add** to configure a new policy and set the following:
 - Rule Name: **Allow Intranet**
 - Rule Action: **Allow Access**
 - Keep the App Connector Selection Method: All App Connector Groups.

Add Access Policy

Name

Allow Intranet

Description

ACTION

Rule Action

Allow Access

Block Access

Conditional Access

TRAFFIC STEERING

App Connector Selection Method

All App Connector groups for the application

CRITERIA

+

Add Criteria

Save

Cancel

- b. Define the Criteria:
- i. Click **Add Criteria**.
 - ii. Select **Applications**, and choose the **Intranet Application Segment** that you just created.
 - iii. Add additional criteria for **User and Session Attributes**:
 - Click the **+** icon to add **User Attributes**.
 - Select **loginname** and choose the user: **Student@<FQDN>**.
 - iv. Click **Save**.

ACTION

Rule Action

Allow Access

Block Access

Conditional Access

TRAFFIC STEERING

App Connector Selection Method

All App Connector groups for the application

CRITERIA

Add Criteria

Applications

Branch Connector Groups

Chrome Enterprise Browser

Client Connector Posture Profiles

Client Connector Trusted Networks

Client Types

Cloud Connector Groups

Country Codes

Locations

Machine Groups

Platforms

Risk Scores

User and Session Attributes

CRITERIA

Application Segments

Intranet

OR

Segment Groups

Select one or more segment groups

Save

Cancel

Add Criteria

Edit Access Policy

Description

ACTION

Rule Action

Allow Access

Block Access

Conditional Access

TRAFFIC STEERING

Connector Selection Method

All App Connector groups and Extranet Resource...

CRITERIA

Application Segments

Intranet

OR

Connector Selection

Save

Cancel

Branch Connector Groups

Chrome Enterprise Browser

Client Connector Posture Profiles

Client Connector Trusted Networks

Client Types

Cloud Connector Groups

Country Codes

Locations

Machine Groups

Platforms

Risk Scores

User and Session Attributes

Add Criteria

Edit Access Policy

TRAFFIC STEERING

Connector Selection Method

All App Connector groups and Extranet Resource...

CRITERIA

Application Segments

Intranet

OR

Segment Groups

Select one or more segment groups

AND

User and Session Attributes

IDP: ZSlogin_Hosted_USER_72059248074078720

Select user or session criteria

Select User and Session Criteria

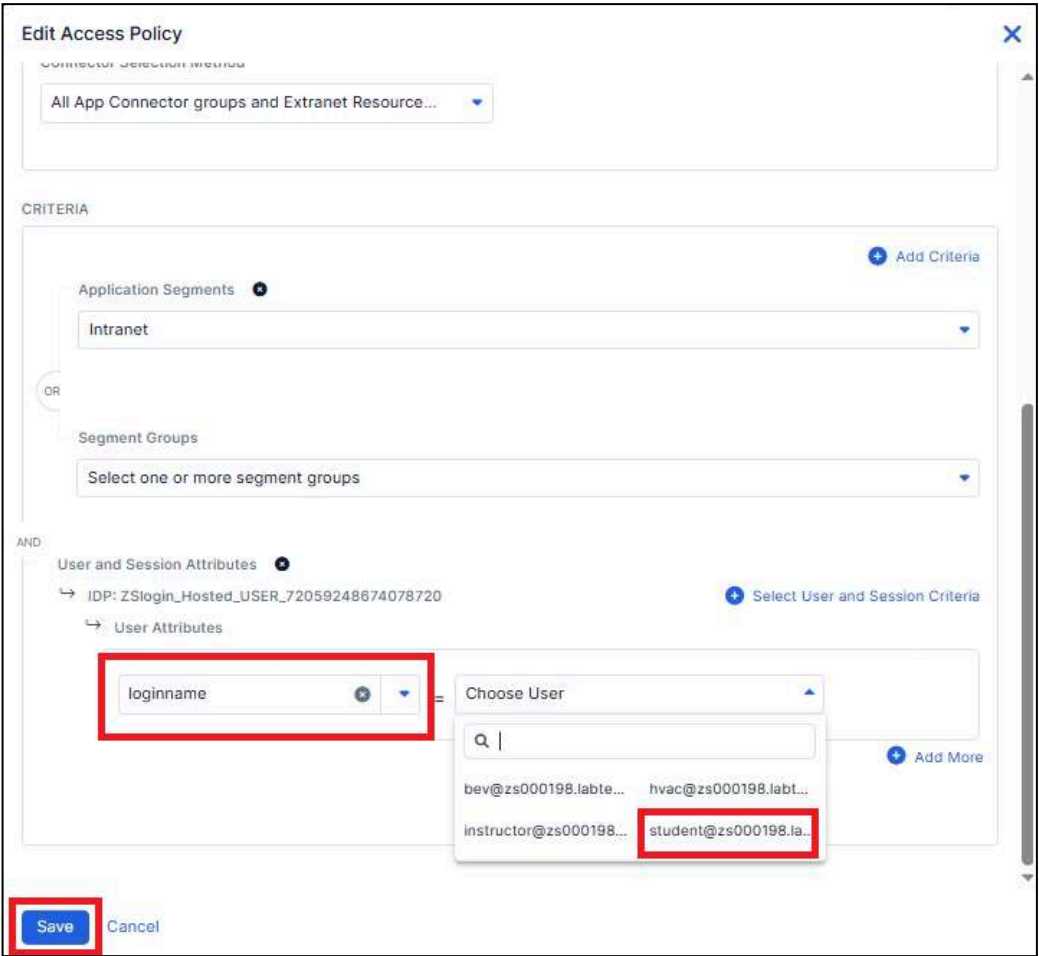
Session Attributes

User Attributes

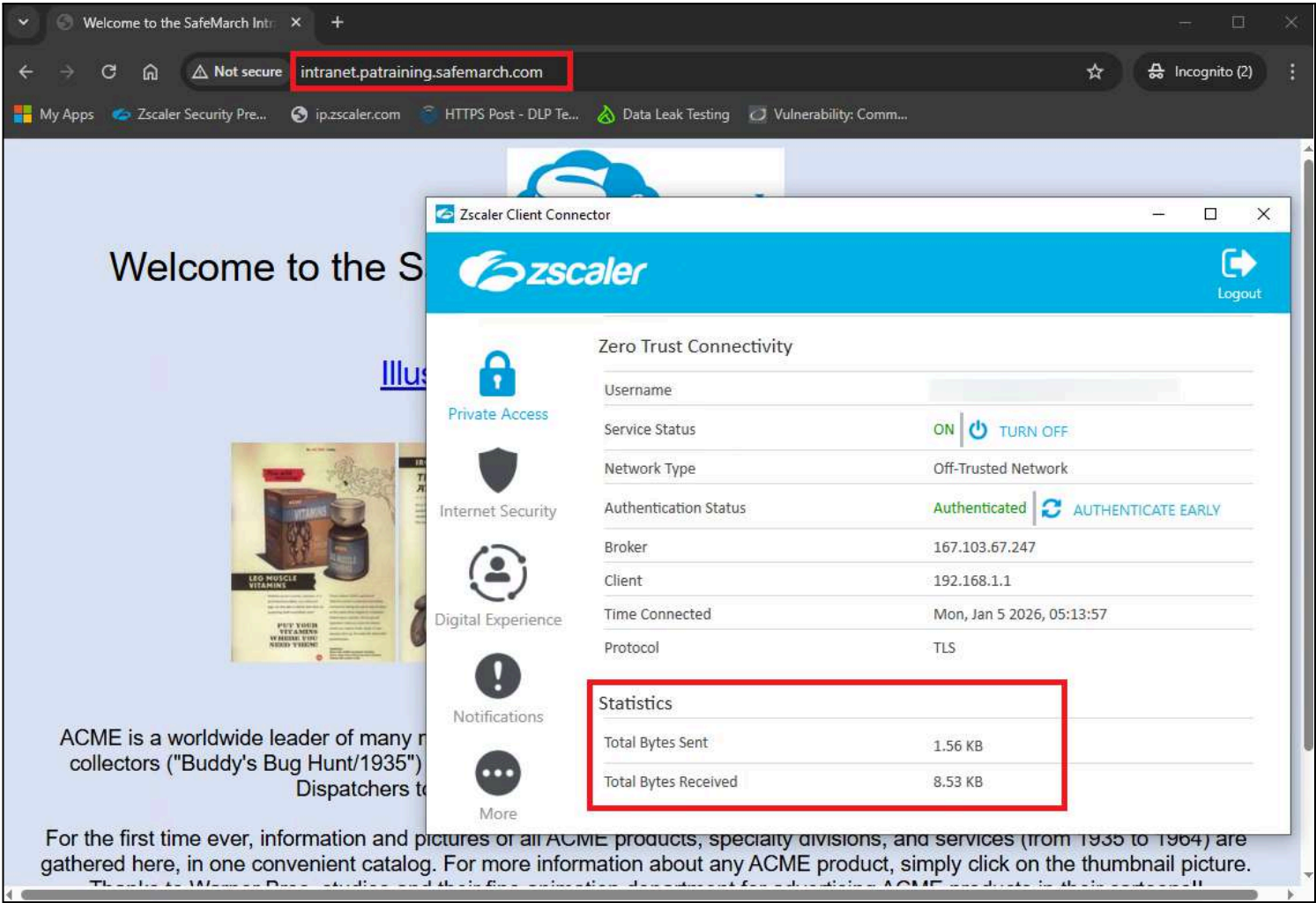
Group Attributes

Save

Cancel



4. Test the Policy:
- a. Open a web browser on the Corp: Client PC and the Zscaler Client Connector side-by-side
 - b. Navigate to: <http://intranet.pattraining.safemarch.com> using the provided bookmark.
 - c. Ensure that the intranet page loads successfully for the **Student** User.
 - d. Refresh the page and confirm that the **Total Bytes Sent/Received** counters in **Zscaler Client Connector** increment. On the **Corp: Client PC**, open a web browser in a window and open the Zscaler Client Connector adjacent to it, so you can see the traffic counters as you load web pages.

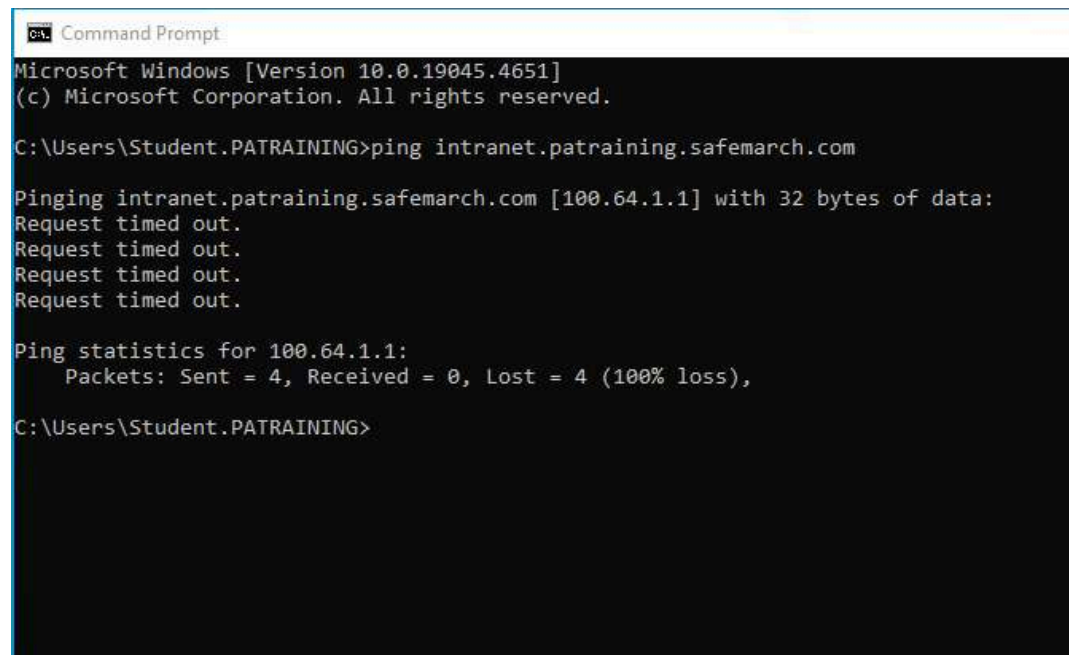


Task 10.3: Enable ICMP Access

In this task, you will enable ICMP access on an application segment so you can use ping to troubleshoot reachability to an internal application.

1. Verify ICMP Accessibility:

- a. On the **Corp: Client PC**, open **Command Prompt** and enter: `ping intranet.patrainig.safemarch.com`



```
Command Prompt
Microsoft Windows [Version 10.0.19045.4651]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student.PATRAINING>ping intranet.patrainig.safemarch.com

Pinging intranet.patrainig.safemarch.com [100.64.1.1] with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

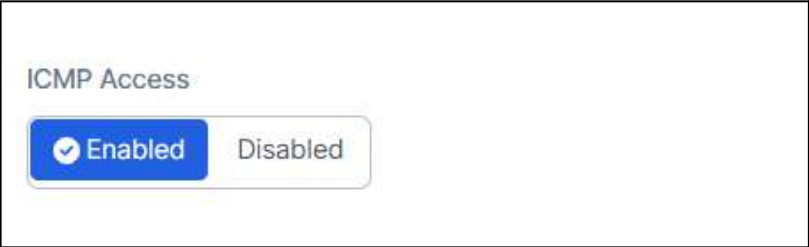
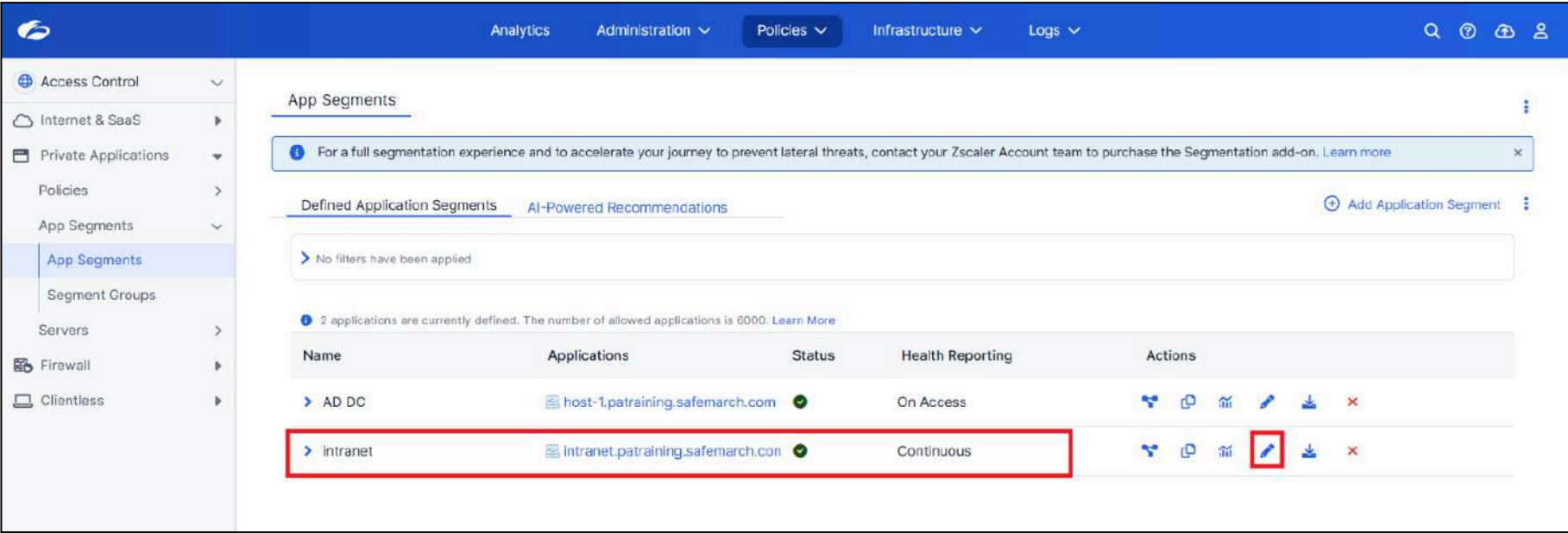
Ping statistics for 100.64.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\Student.PATRAINING>
```

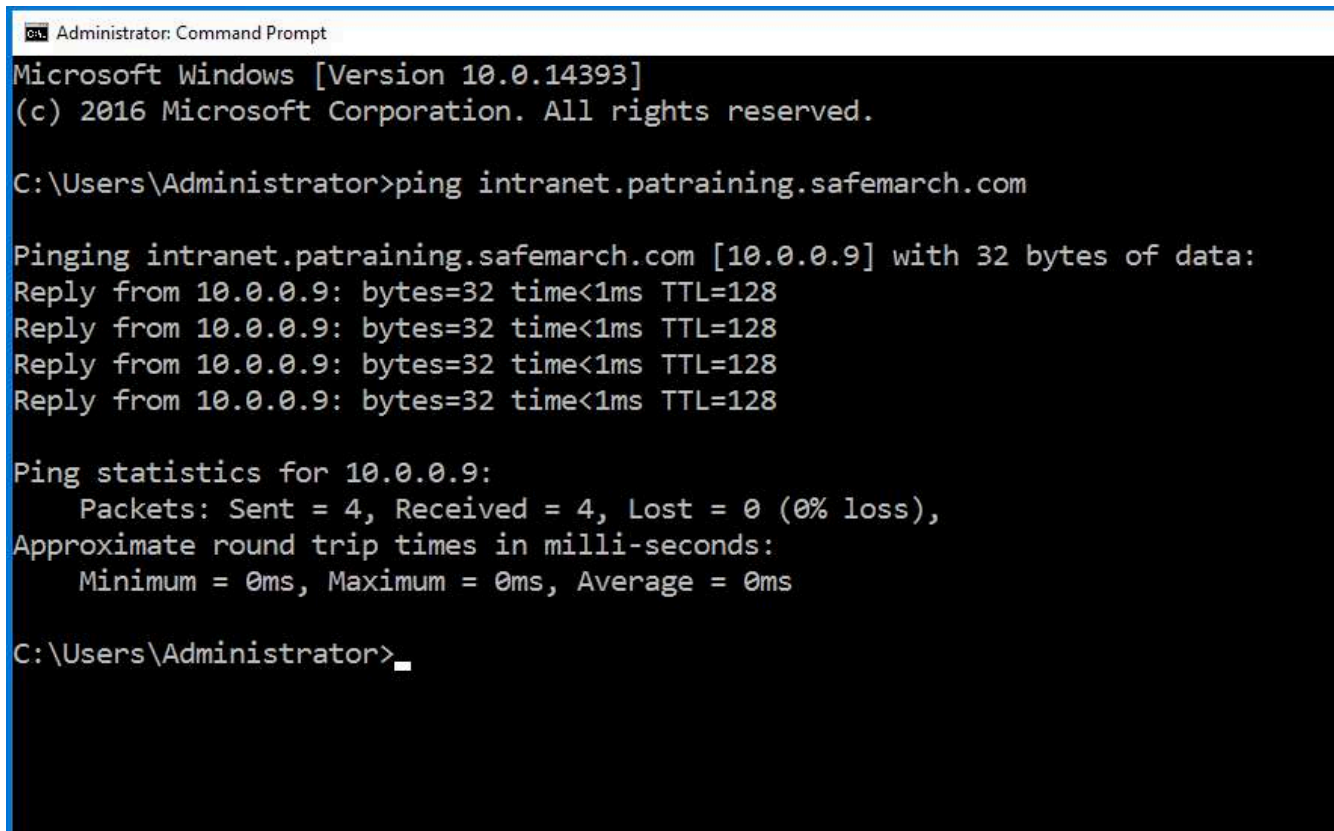
Note: If ICMP Access is disabled in the Application Segment, the internal application will not respond to ping requests or be reachable through ICMP.

2. Enable ICMP Access in the ZPA Admin Portal

- Navigate to **Policies > Access Control > Private Applications > App Segments**
- Locate the Application Segment named **Intranet** and click **Edit**.
- In the Edit Application Segment window, Click **Next** to navigate to the **ICMP Access** field.
- Set ICMP Access to **Enabled**.
- Click **Save** to apply the change.



3. Test ICMP Connectivity:
 - a. Return to the Corp: Client PC.
 - b. Reopen Command Prompt and enter `ping intranet.pattraining.safemarch.com`



```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\Administrator>ping intranet.pattraining.safemarch.com

Pinging intranet.pattraining.safemarch.com [10.0.0.9] with 32 bytes of data:
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128
Reply from 10.0.0.9: bytes=32 time<1ms TTL=128

Ping statistics for 10.0.0.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\Users\Administrator>
```

Expected Result: The internal application should now respond to ping requests, confirming that ICMP traffic is allowed

Note: Expected Result: The internal application should now respond to ping requests, confirming that ICMP traffic is allowed. The ICMP Access option should only be enabled if ICMP connectivity is required for troubleshooting. Avoid using it for general application testing.

Note: The ICMP Access option should only be enabled if ICMP connectivity is required for troubleshooting. Avoid using it for general application testing

Lab 11: Troubleshoot Access to an Internal Application for a Specific User

Scenario: The Zscaler platform administrator has configured the internal application `intranet.pattraining.safemarch.com` using **Zscaler Private Access (ZPA)** to secure user access to company resources. **Joe**, a user **cannot access the application** due to **missing the ZPA service entitlement**. In this lab, you will **troubleshoot and resolve the access** issue by granting the necessary ZPA service entitlement to Joe.

Objectives: By the end of this lab you will:

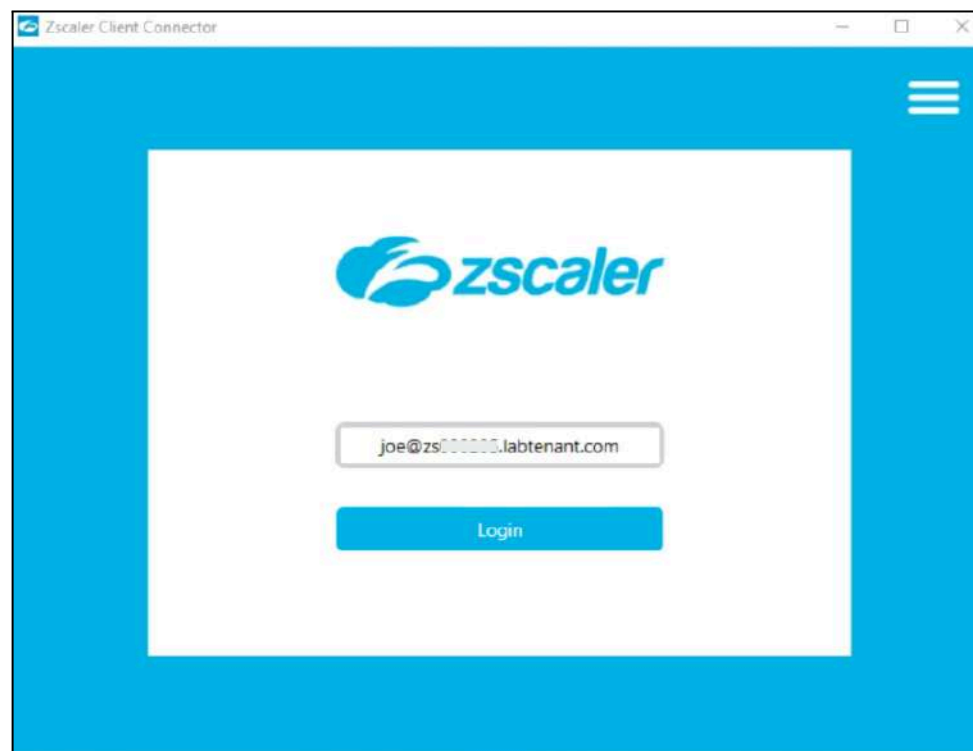
- **Identify** the root cause of **Joe's access issue** using **ZPA** administrative tools.
- **Analyze** the **configuration** and **perform** key **troubleshooting** steps to diagnose the issue.
- **Assign** the **missing ZPA service entitlement** to eliminate access restrictions.
- **Evaluate** the **solution** and confirm that Joe can successfully access the internal application.

Approximate Time: 25 minutes

Task 11.1: Identify the Root Cause of the Internal Application Access Issue

In this task, you will determine why Joe is unable to access the internal application by analyzing his permissions and service entitlements.

1. Simulate the Issue:
 - a. Log in to Zscaler Client Connector on the **Corp: Client PC**:
 - i. Username: **joe@<Student FQDN>** (Refer to Lab 2 for Joe's credentials).
 - b. Open a browser and attempt to access the internal application:
 - i. `http://intranet.patrainig.safemarch.com`



Note: Joe cannot access the application.

2. Troubleshoot User Entitlements:

- a. From the **Experience Center** navigate to **Administration > Identity > Authentication Service> User Management> Users**.
- b. Locate **Joe** and click **Edit**.
- c. Navigate to the **Entitlement** tab and check for assigned services:

Identity

Zidentity

IDP Configuration

User Management

Users

User Groups

Departments

Attributes

User Risk Scores

Domains

Passwords & Authentication

Connectors

Internet & SaaS

Private Access

AnalyticsAdministrationPoliciesInfrastructureLogs

Users

+ Add UserImport CSV

Name Search...

No.	Name	Login ID	Status	Actions
☐	1 Babu	babu@zs000205.labtenant.com	Active	
☐	2 Bev	bev@zs000205.labtenant.com	Active	
☐	3 HVAC	hvac@zs000205.labtenant.com	Active	
☐	4 Instructor	instructor@zs000205.labtenant.com	Active	
☐	5 Joe	joe@zs000205.labtenant.com	Active	
☐	6 OTP Handler	handler5057@zs000205.zslogin.net	Active	
☐	7 Student Admin	student@zs000205.labtenant.com	Active	
☐	8 Yuvaraj	yuvaraj@zs000205.labtenant.com	Active	
☐	9 zsclocloudlabautomation-101 Admin	admin@zs000205.zslogin.net	Active	

Showing 9 of 9

Edit User

General

Entitlements

Security Settings

Additional Attributes

ADMINISTRATIVE ENTITLEMENTS

Name	Cloud Name	Org Name	Roles
 Zscaler Digital Experience	zsccloud.net	zsccloudlabautomation-101	ZDX Read-only Admin
 Zscaler Internet Access	zsccloud.net	zsccloudlabautomation-101	Executive Insights App

SERVICE ENTITLEMENTS

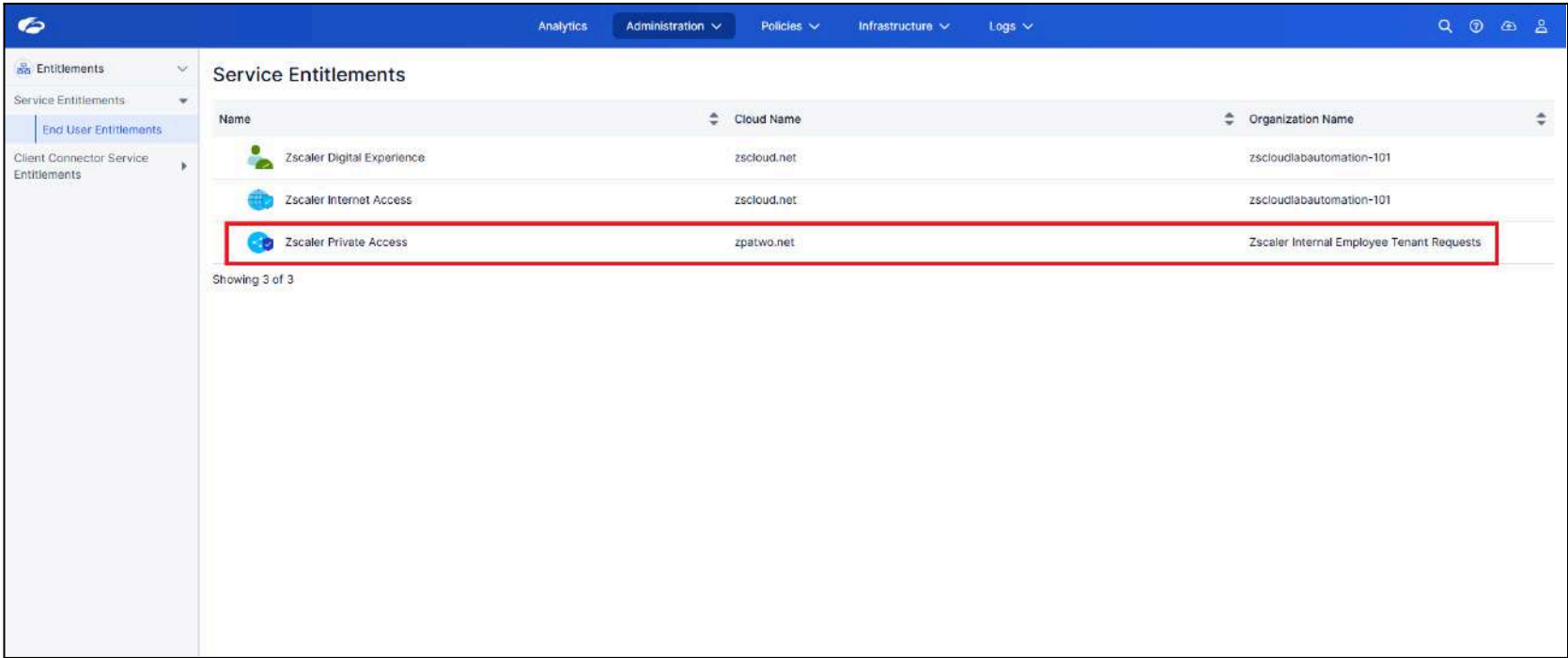
Name	Cloud Name	Org Name
 Zscaler Internet Access	zsccloud.net	zsccloudlabautomation-101

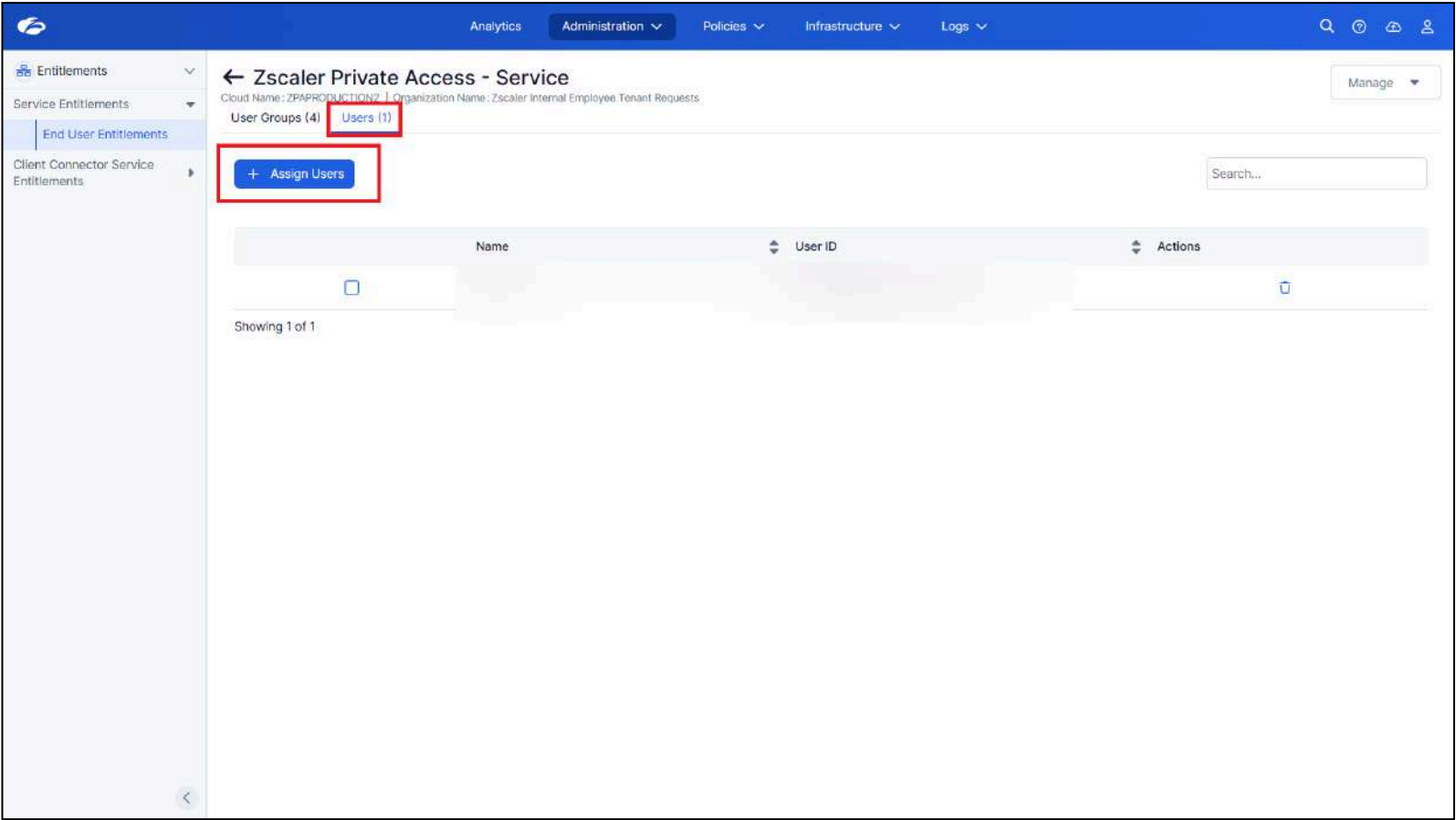
Note: As configured in Lab 2, Joe already has ZIA via the Helpdesk L1 group. ZPA is missing and needs to be assigned in order to resolve Joe’s access issue.”

Task 11.2: Resolve the Internal Application Access Issue

In this task, you will assign the necessary **ZPA service entitlement** to Joe, allowing him to access the intranet application.

- 1. Assign Zscaler Private Access (ZPA) Service Entitlement:
 - a. Navigate to **Administration > Entitlements > Service Entitlements > End User Entitlements**
 - b. Select the **Zscaler Private Access** service from the list.
 - c. Navigate to the **Users** tab and click Assign Users.
 - d. Select the user **Joe** from the list.
 - e. Click **Next**, then click **Assign** to confirm





Entitlements

Service Entitlements

End User Entitlements

Client Connector Service Entitlements

Assign Users

Select Users

Summary

Select Users

Zscaler Private Access - Services | Cloud Name: ZPAPRODUCTION2 | Organization Name: Zscaler Internal Employee Tenant Requests

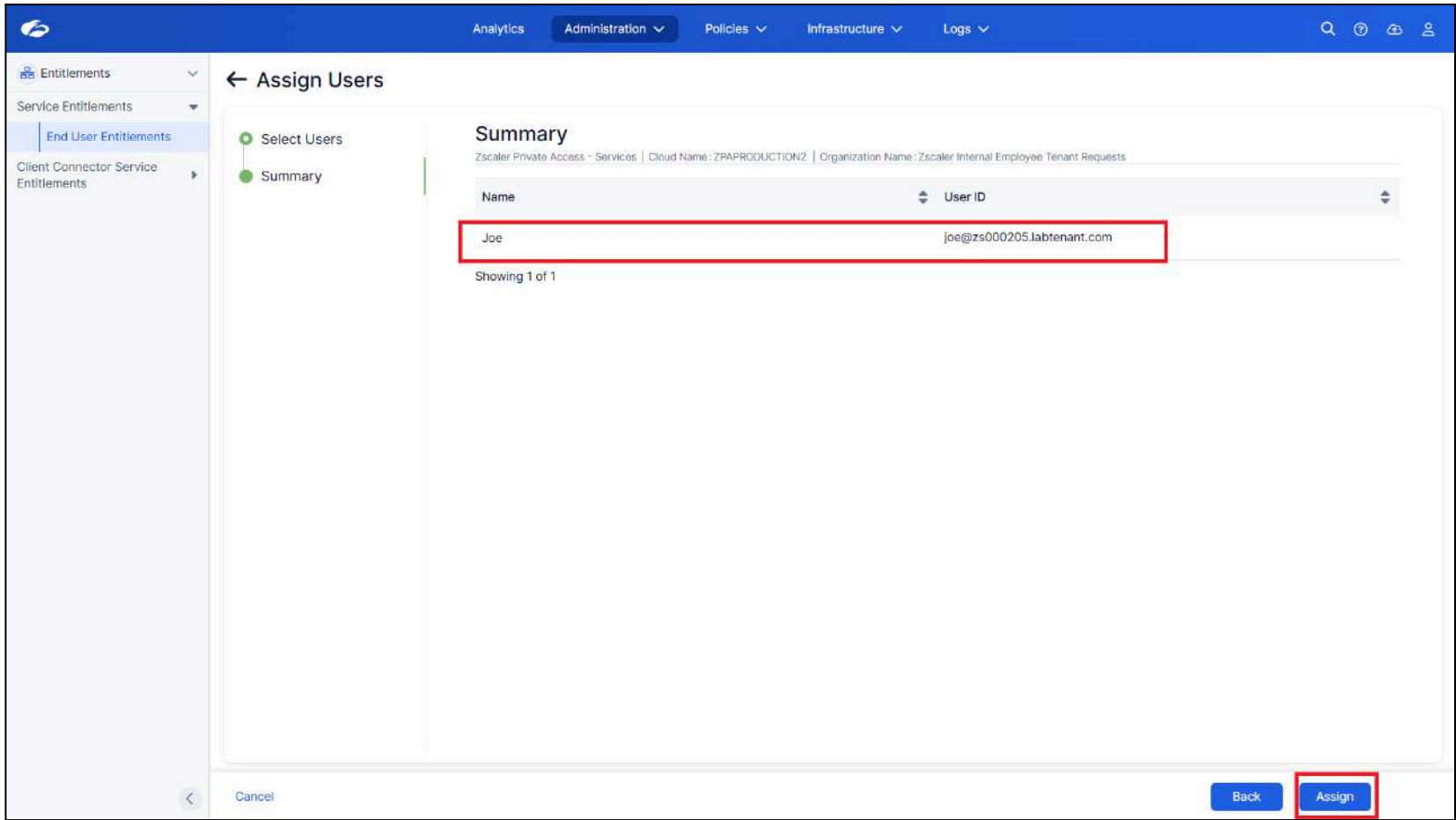
Search...

	Name	User ID
☐	Bev	bev@zs000205.labtenant.com
☐	HVAC	hvac@zs000205.labtenant.com
☐	Instructor	instructor@zs000205.labtenant.com
☒	Joe	joe@zs000205.labtenant.com
☐	OTP Handler	handler5057@zs000205.zslogin.net
☐	Student Admin	student@zs000205.labtenant.com
☐	Yuvaraj	yuvaraj@zs000205.labtenant.com
☐	zscloudiabautomation-101 Admin	admin@zs000205.zslogin.net

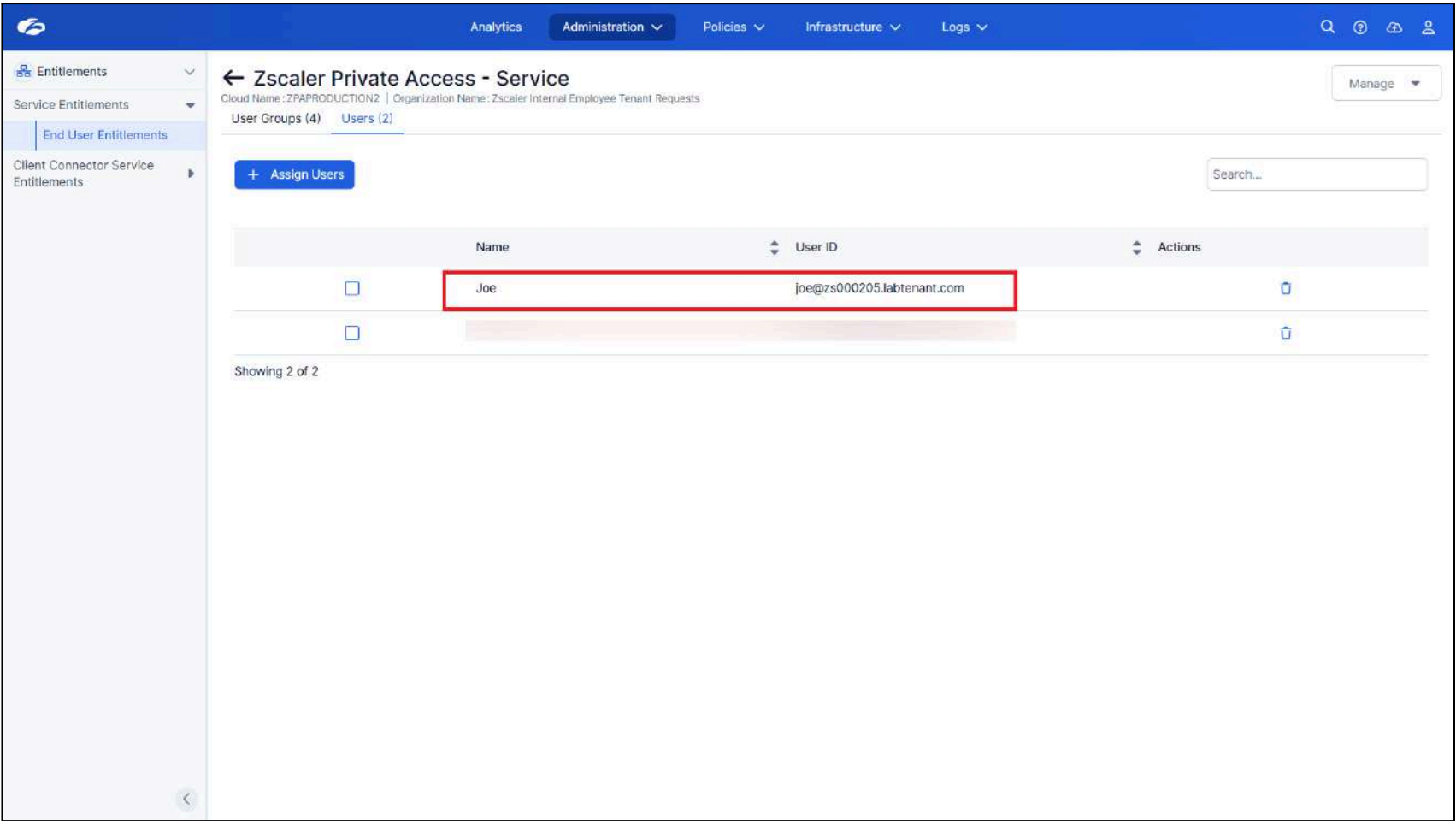
Showing 8 of 8

Cancel

Next



- f. Verify the Assignment:
 - i. Confirm that **Joe** has now been assigned the **ZPA service entitlement** via the **HelpDesk L1 User Group**.



- 2. Test Access After Troubleshooting
 - a. Open **Zscaler Client Connector** and **Update the policy**.
 - i. Confirm that **Private Access** is now visible.
- 3. Launch a web browser and navigate to: `http://intranet.pattraining.safemarch.com`



Lab 12: Traffic Forwarding through SIPA

Scenario: Safemarch Corporation hosts critical cloud applications that require **strict Source IP whitelisting** for access. **Zscaler Internet Access (ZIA)** uses dynamic egress IP addresses, which can create issues when accessing these applications. To meet these requirements, a Zscaler administrator must **configure** the **Source IP Anchoring (SIPA)** feature to **ensure consistent, whitelisted source IPs for accessing public applications** while leveraging ZPA's secure connectivity.

Objectives: By the end of this lab, you will:

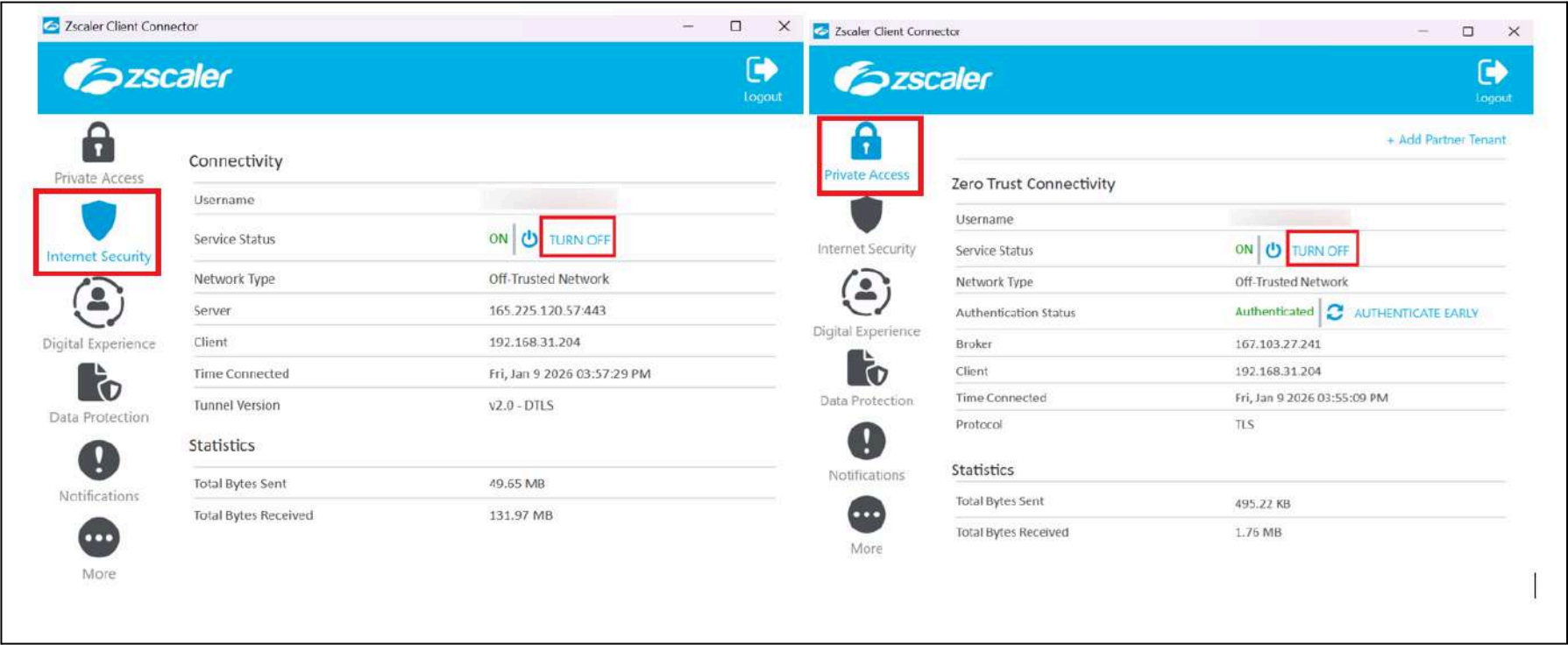
- **Configure** a **Source IP Anchoring application segment** to provide consistent egress IPs for public applications.
- **Define and apply ZPA and ZIA forwarding rules** to enable SIPA for traffic management.
- **Validate** the **configuration** by testing traffic routing through the SIPA-enabled application.

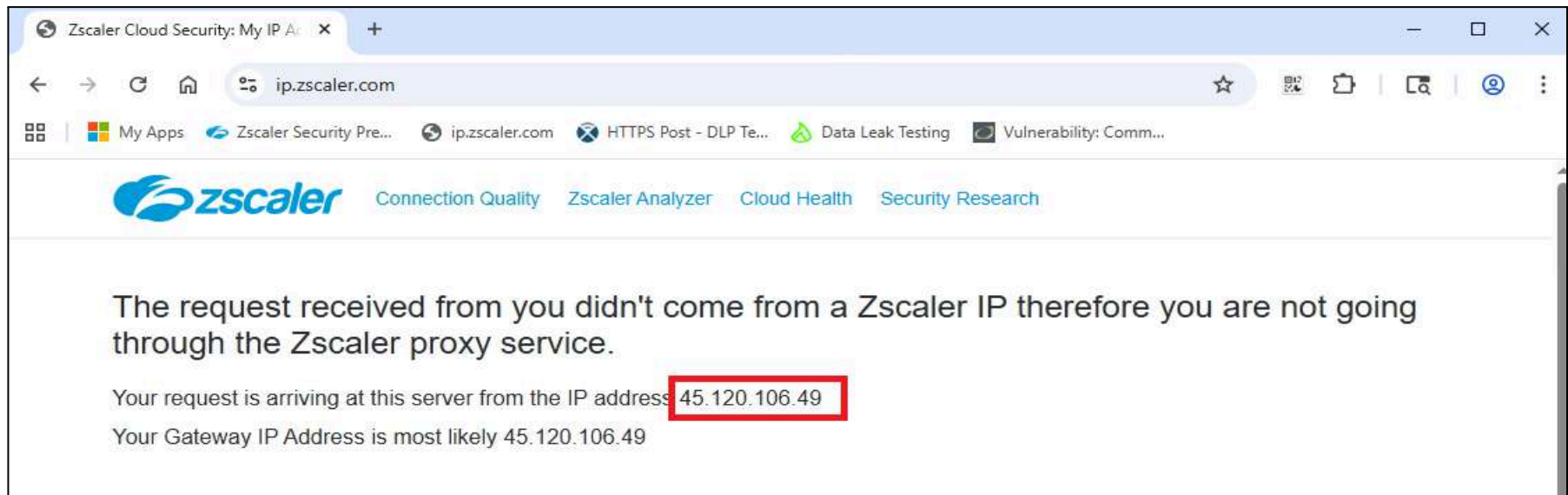
Approximate Time: 30 minutes

Task 12.1: Configure Source IP Anchoring (SIPA) in ZPA

In this task, you will configure the SIPA feature in **ZPA** to enable a fixed, whitelisted egress IP for public applications.

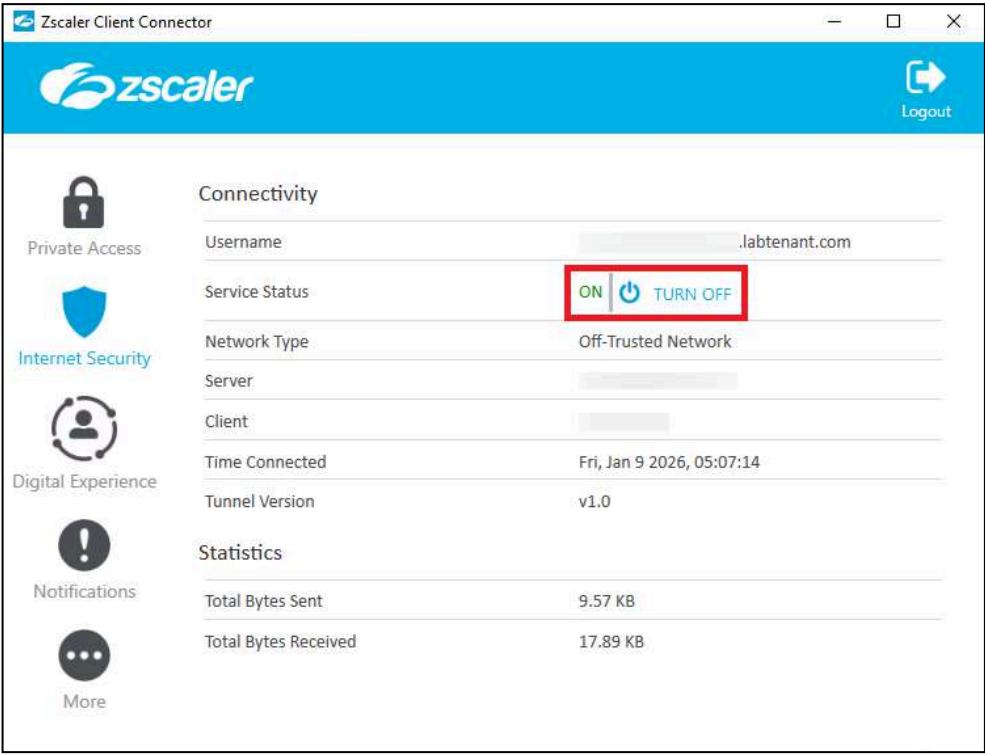
- 1. Verify Current Egress IPs
 - a. Identify your native public IP (without ZCC).
 - i. Open **Zscaler Client Connector (ZCC)** on the **Corp: Client PC** and disable **ZIA** and **ZPA**.
 - ii. Open a browser and navigate to: `ip.zscaler.com`





Note: The IP address displayed reflects your device's native public IP without Zscaler services active.

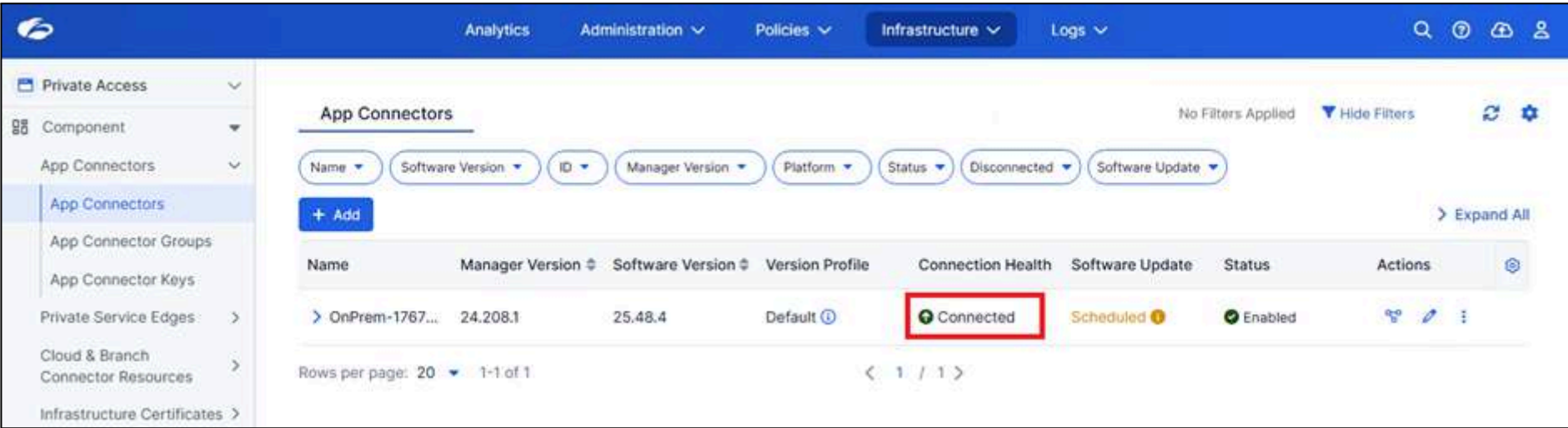
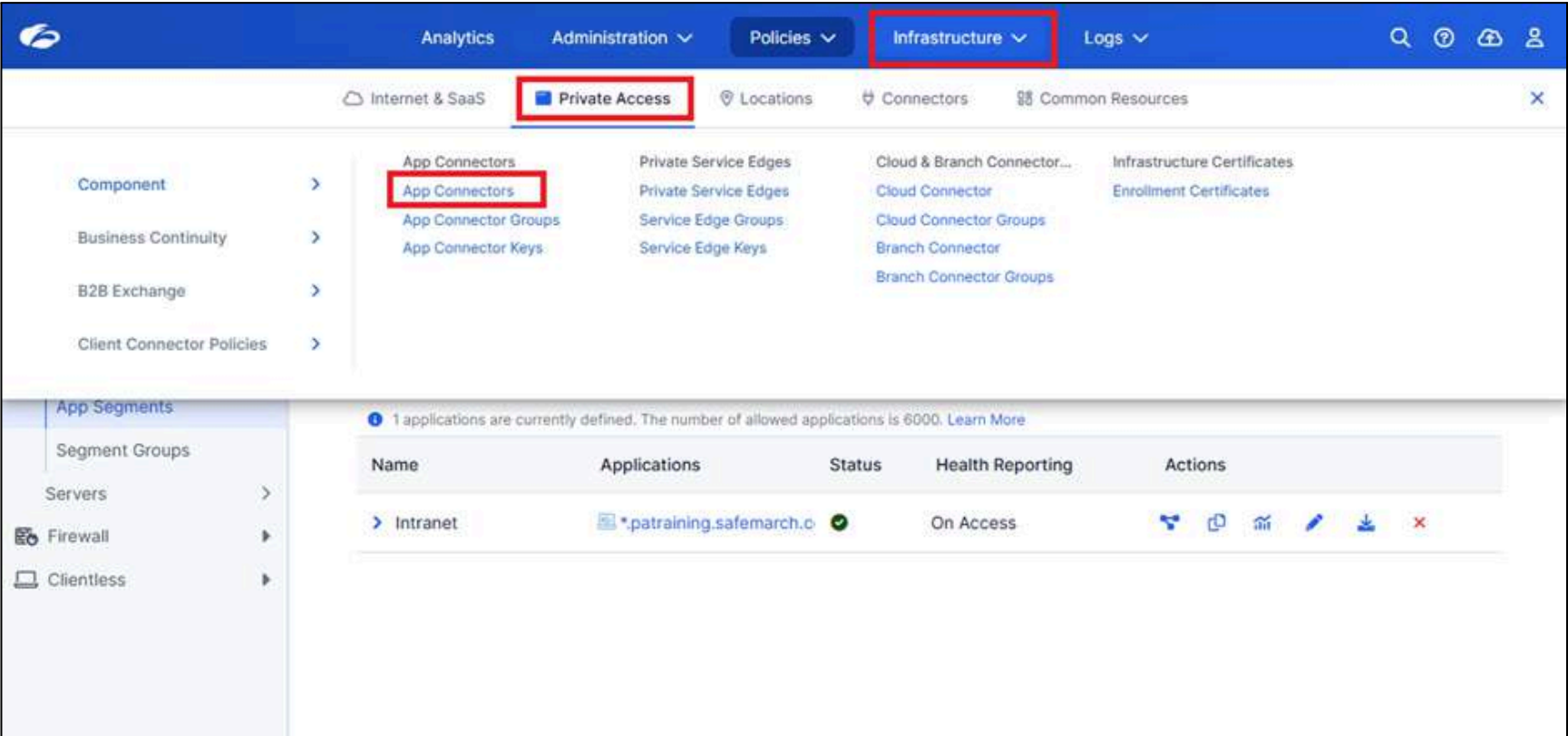
- b. Identify your public IP after enabling Zscaler services:
 - i. Re-enable **ZIA** and **ZPA** in **ZCC**.
 - ii. Open a browser and navigate to: `ip.zscaler.com`



Note: The displayed IP address represents the ZIA egress IP, which changes dynamically and is typical of Zscaler's proxy-based architecture. However, public applications often require a consistent, predictable IP for security purposes. Enabling SIPA solves this issue, as configured in the next steps.

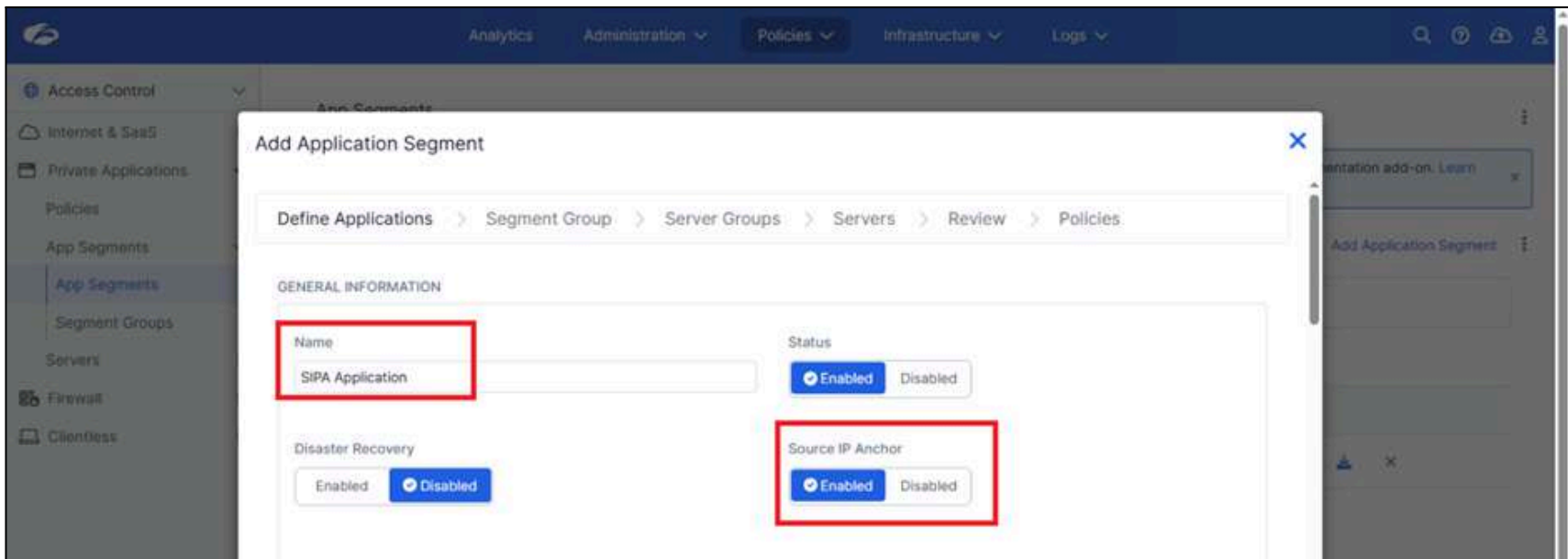
2. Configure SIPA in ZPA

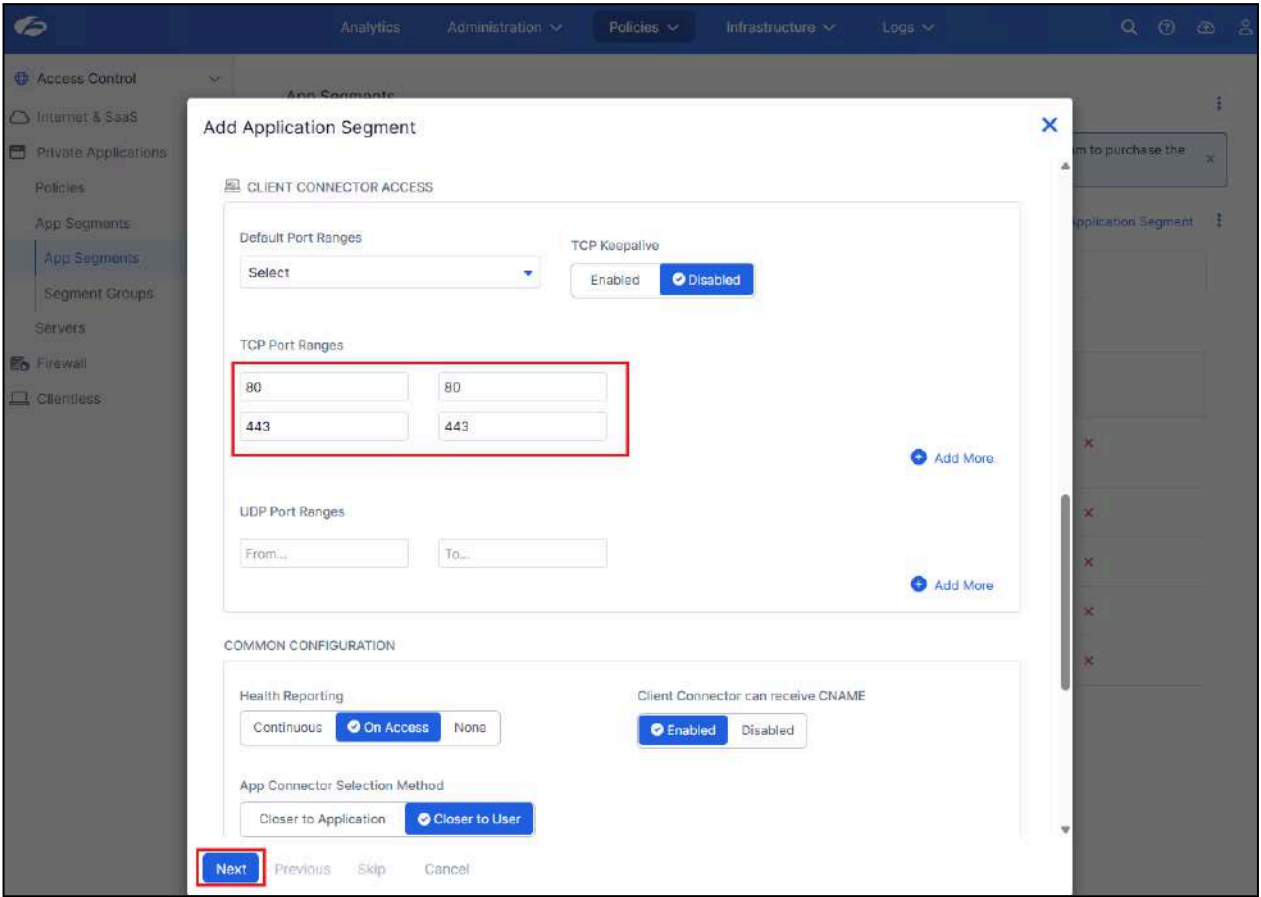
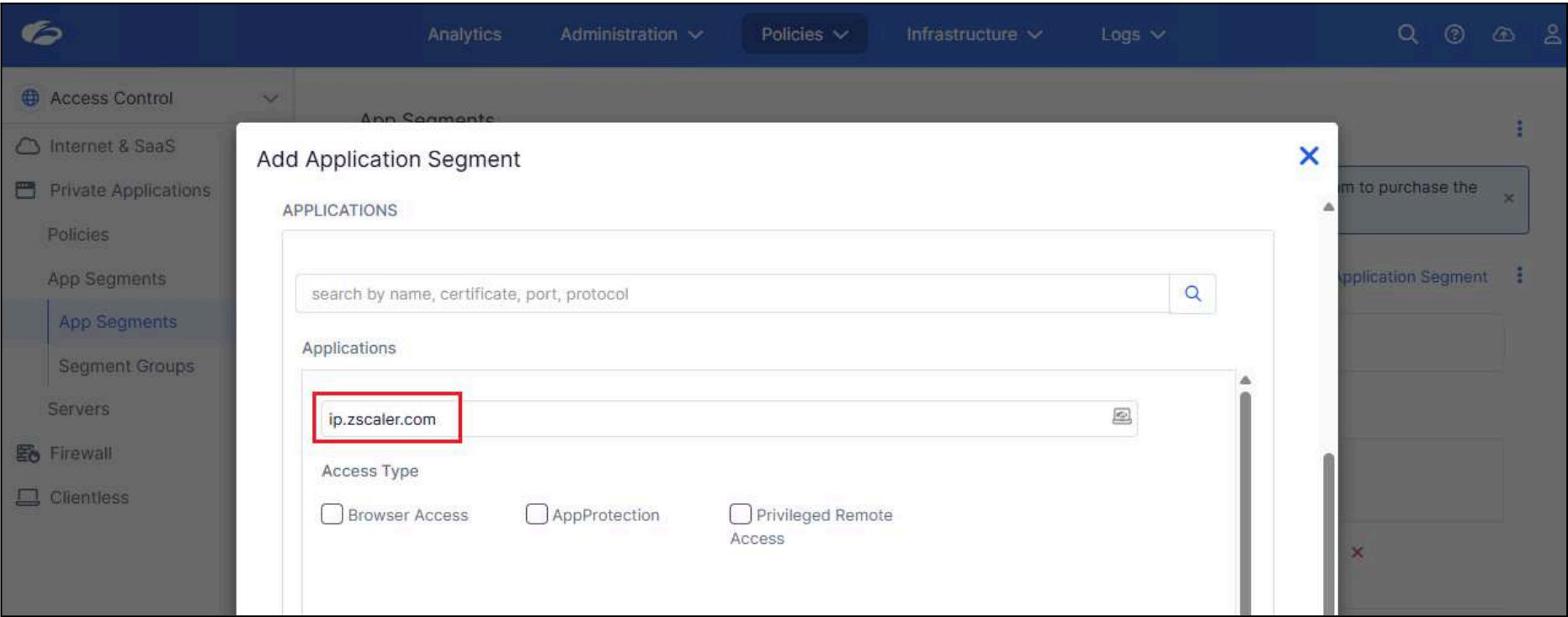
- a. Navigate to the **ZPA Admin Portal** via the **Experience Center**:
 - i. Go to **Infrastructure > Private Access > App Connectors**.
 - ii. Verify that an **App Connector** is deployed and in the **connected** state.



- b. Add a new Application Segment:
- i. Navigate to **Policies > Private Applications > App Segments**.
 - ii. Click Add **Application Segment** and configure as follows:

- Name: **SIPA Application**
- Source IP Anchor: **Enabled**
- Application: `ip.zscaler.com`
- TCP Port Ranges: **80 (HTTP)** and **443 (HTTPS)**.
- Click **Next**.





c. Configure the Segment Group

- i. Name: **SIPA Segment**
- ii. Description: **Optional**
- iii. Status : **Enabled**
- iv. Click **Next**.

The screenshot shows the 'Add Application Segment' dialog in the Zscaler console. The dialog has a breadcrumb trail: Define Applications > Segment Group > Server Groups > Servers > Review > Policies. Below the breadcrumb is a 'Select Segment Group' dropdown and an 'Add Segment Group' button. The 'Name' field is set to 'SIPA Segment' and is highlighted with a red box. The 'Description' field is set to 'SIPA'. The 'Status' is set to 'Enabled' (indicated by a blue circle with a checkmark) and is also highlighted with a red box. At the bottom, the 'Next' button is highlighted with a red box, along with 'Previous', 'Skip', and 'Cancel' buttons.

d. Configure the Server Group:

- i. Name: **SIPA Server**
- ii. Select the App Connector Group: **OnPrem**.
- iii. Click **Next**.

e. Click **Save** to complete the configuration.

Access Control

Internet & SaaS

Private Applications

Policies

App Segments

App Segments

Segment Groups

Servers

Firewall

Clientless

Analytics

Administration

Policies

Infrastructure

Logs

Search

Help

Cloud

User

Add Application Segment

X

Select Server Group

Add Server Group

Name

SIPA Server

Description

SIPA

Status

Enabled

Disabled

Dynamic Server Discovery

On

Off

Connector Type

App Connector

Extranet

App Connector Groups

OnPrem

Next

Previous

Skip

Cancel

Access Control

Internet & SaaS

Private Applications

Policies

App Segments

App Segments

Segment Groups

Servers

Firewall

Clientless

Analytics

Administration

Policies

Infrastructure

Logs

Search

Help

Cloud

User

Add Application Segment

X

Define Applications

Segment Group

Server Groups

Review

Policies

Application Segment

SIPA Application

Application Segment Status

Enabled

Segment Group

SIPA Segment

Segment Group Status

Enabled

Server Group

SIPA Server

Server Group Status

Enabled

Servers

Connector Type

App Connector

App Connector Groups

OnPrem

Review all of the information before clicking Save

Save

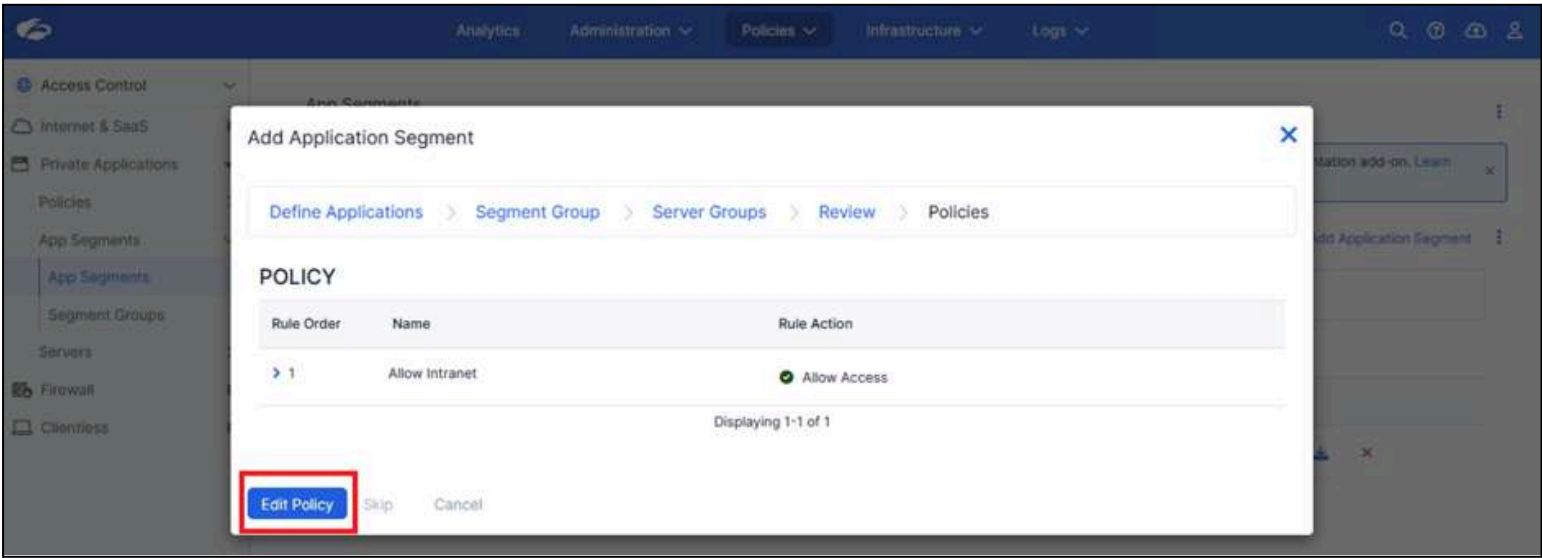
Previous

Skip

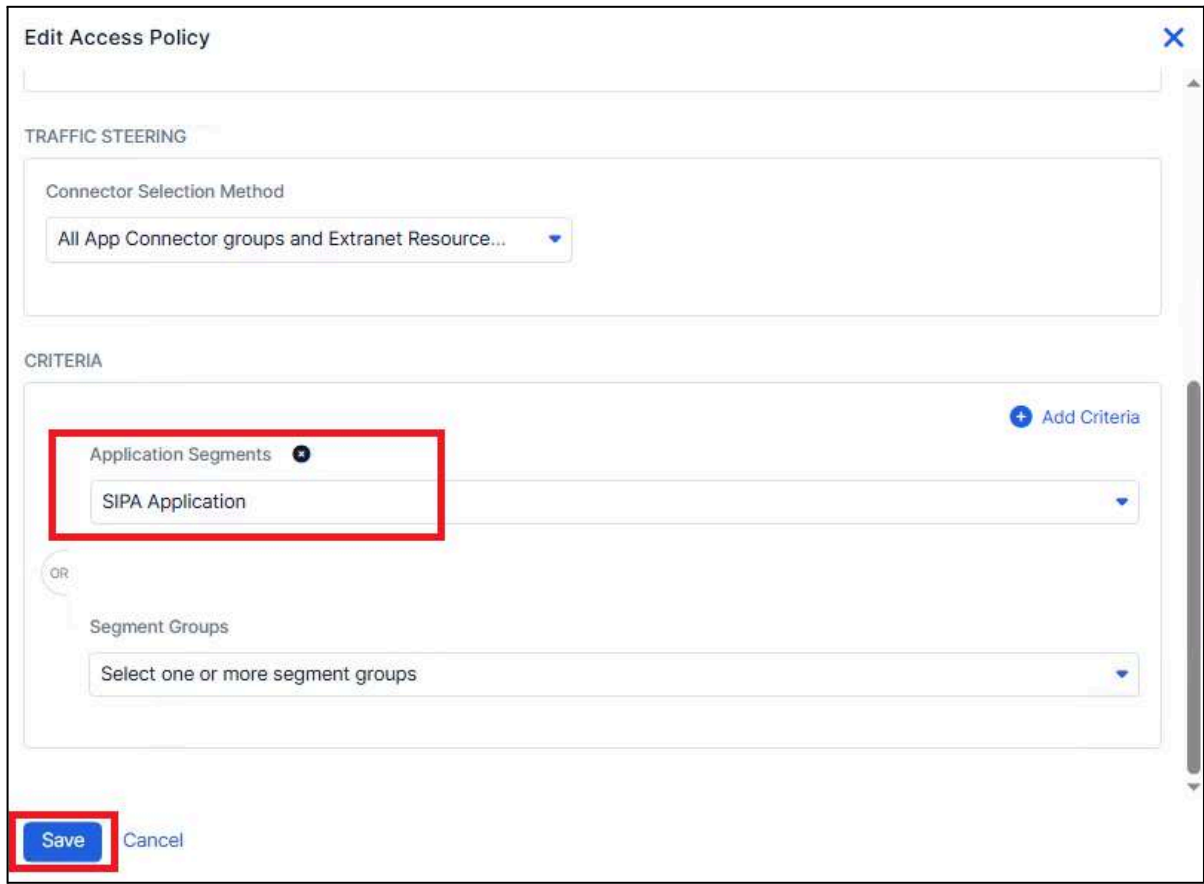
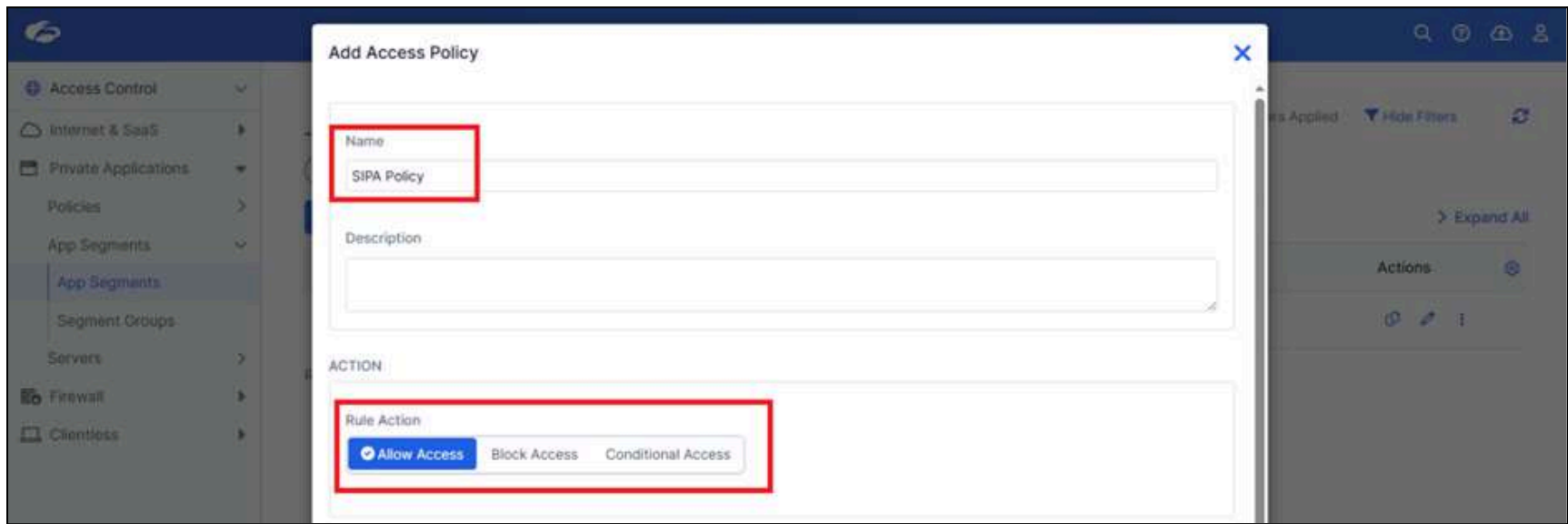
Cancel

3. Configure and Save Access Policies:

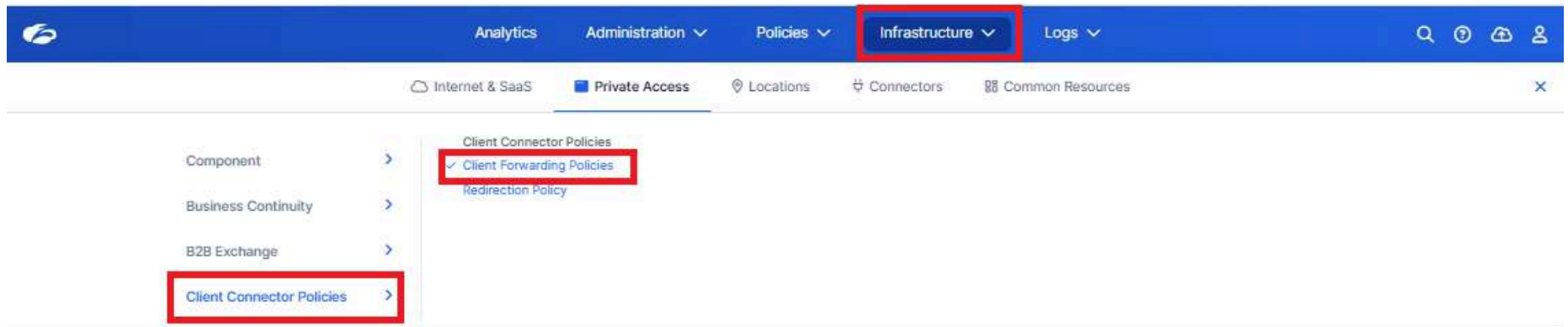
- a. After saving the **SIPA Application Segment**, you will be redirected to the **Access Policy** page. Alternatively, you can navigate to **Policies > Private Applications > Access Policy**.



- b. Add a new policy rule:
 - i. Name: **SIPA Policy**
 - ii. Action: **Allow Access**
 - iii. Define criteria:
 - Select the **applications mapped to the SIPA application segment and segment groups**.
- c. Click **Save**.



4. Configure Client Forwarding Policies:



a. Rule 1: Bypass ZPA for Specific Segment Groups

- i. Navigate to **Infrastructure > Private Access > Client Connector Policies > Client Forwarding Policies**.

Add a Client forwarding policy

- Name: **SIPA Bypass**
 - Rule Action: **Bypass ZPA**.
 - Define the criteria:
 - Applications: Map the **application segments and segment groups**.
 - Client Types: Select **Client Connector**.
 - Click **Save**.

Edit Client Forwarding Policy

Name

SIPA Bypass

Description

ACTION

Rule Action

Forward to ZPA

Only Forward Allowed Applications

Bypass ZPA

CRITERIA

Application Segments

Select one or more application segments

OR

Segment Groups

SIPA Segment

AND

Client Types

Client Connector

Save

Cancel

b. Rule 2: Forward Traffic to ZPA

i. Configure **SIPA Forward**:

- Rule Action: **Forward to ZPA**.
- Define the criteria:
 - Applications: Map the **application segments and segment groups**.
 - Client Types: Select **ZIA Service Edge**.

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- Click **Save**.

Edit Client Forwarding Policy

Name

SIPA Forward

Description

ACTION

Rule Action

Forward to ZPA

Only Forward Allowed Applications

Bypass ZPA

CRITERIA

Application Segments

Select one or more application segments

OR

Segment Groups

SIPA Segment

AND

Client Types

ZIA Service Edge

Add Criteria

Save

Cancel

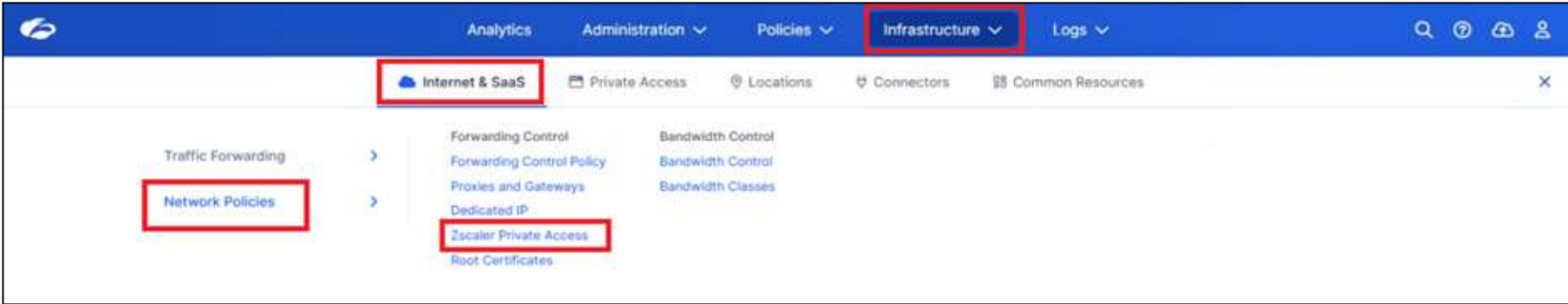
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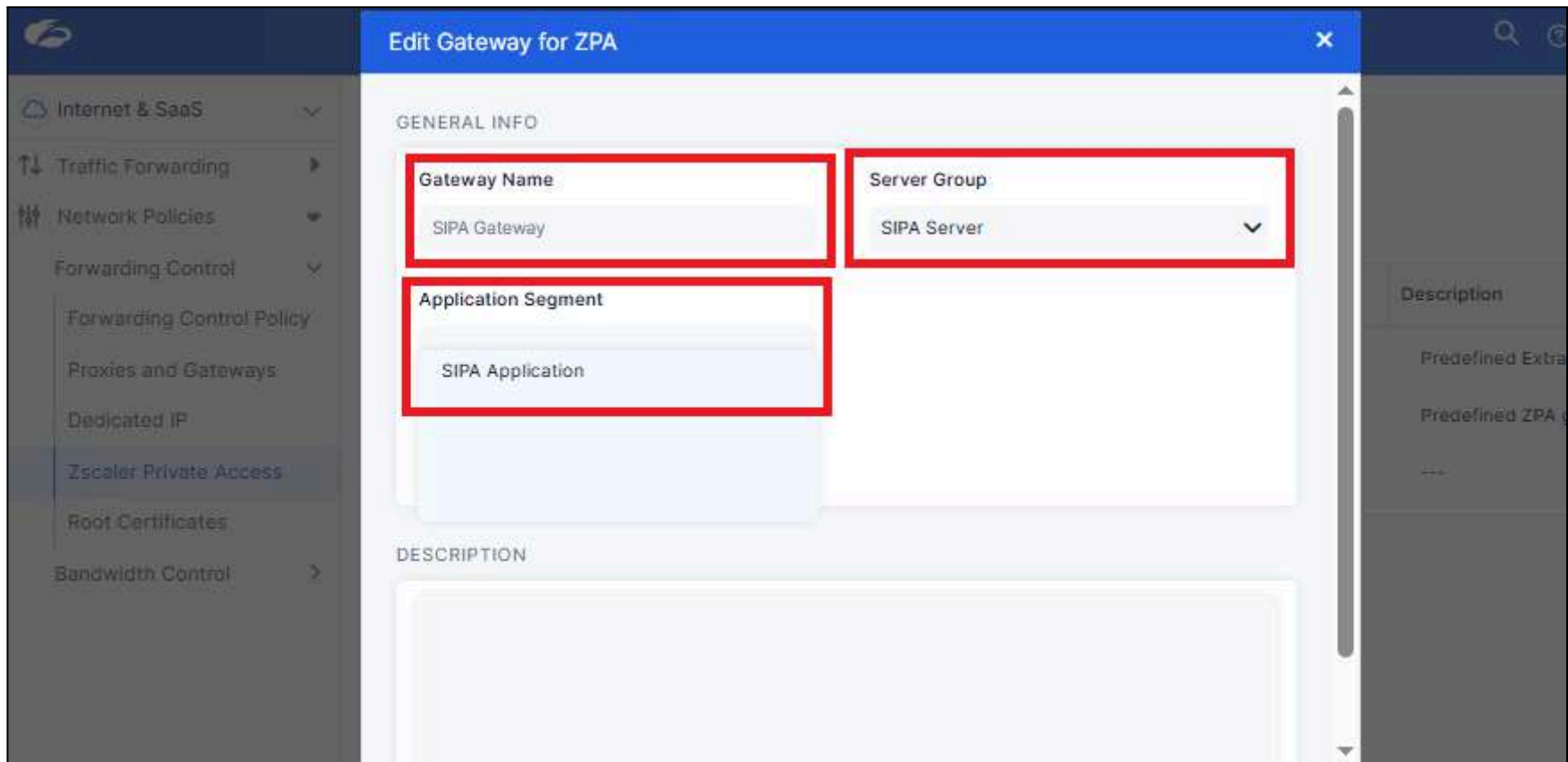
Task 12.2: Configure ZIA Forwarding for Source IP Anchoring

In this task, you will configure forwarding behavior in **Zscaler Internet Access (ZIA)** to ensure **SIPA-enabled applications** are **routed through ZPA** with consistent source IPs.

- 1. Navigate to **Infrastructure > Internet and SAAS > Network Policies > Zscaler Private Access** in the **Experience Center**.

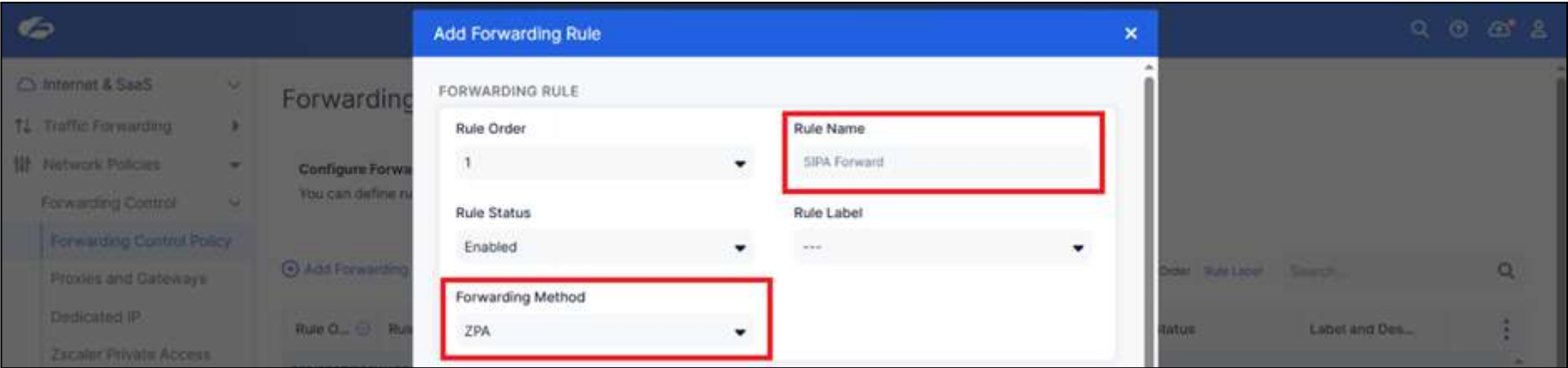


- 2. Define the ZPA Gateway:
 - a. Click on Add Gateway for ZPA and Provide a **name** (e.g., "SIPA Gateway").
 - b. Select the **server group** and **application segment**, which should be automatically fetched based on the ZPA configuration from Task 13.1.



3. Configure Forwarding Control Policy:

- a. On the same page, move to the **Forwarding Control Policy** tab.
- b. Define the following:
 - i. **Rule order**
 - ii. **Rule name : SIPA Forward**
 - iii. Ensure **rule status** is **enabled**.
 - iv. Under the **Forwarding Method**, select: **ZPA**.



4. Set Criteria for Forwarding:
- a. Navigate to the Criteria tab.
 - b. Click **Destination** and define the **ZPA application segment (e.g., "SIPA application")**.
 - c. Set **Forward to ZPA Gateway** action to the **SIPA Gateway** defined earlier.

Add Forwarding Rule

Forwarding Method

ZPA

CRITERIA

General

Applicatio...

Source

Destination

ZPA Application Segment

SIPA Application

ACTION

Forward to ZPA Gateway

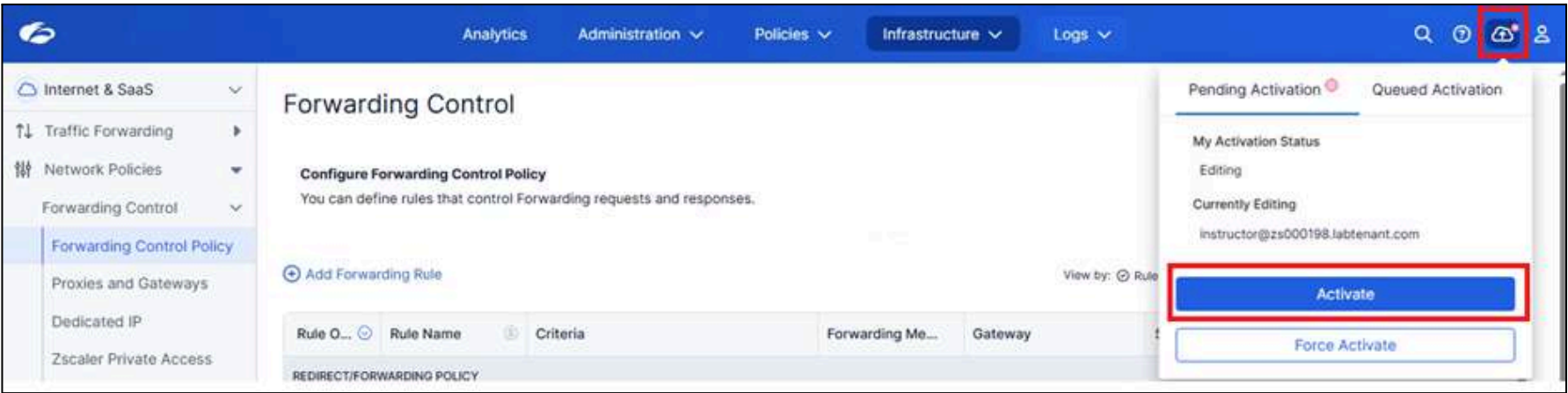
SIPA Gateway

DESCRIPTION

Save

Cancel

d. Click **Save** and then **Activate** the changes.



Task 12.3: Verify Source IP Anchoring (SIPA) Functionality

In this task, you will **verify** that the **SIPA configuration** is functioning correctly by confirming traffic routing through ZPA and observing the effect of modifying the application segment.

1. Perform Initial Verification:

- a. Open a browser on the **Corp: Client PC** and navigate to: `ip.zscaler.com`
- b. Observe the displayed public IP address:
 - a. This IP confirms that traffic is routed from **ZIA** to **ZPA** to the **App Connector** as defined in the **SIPA** rule.

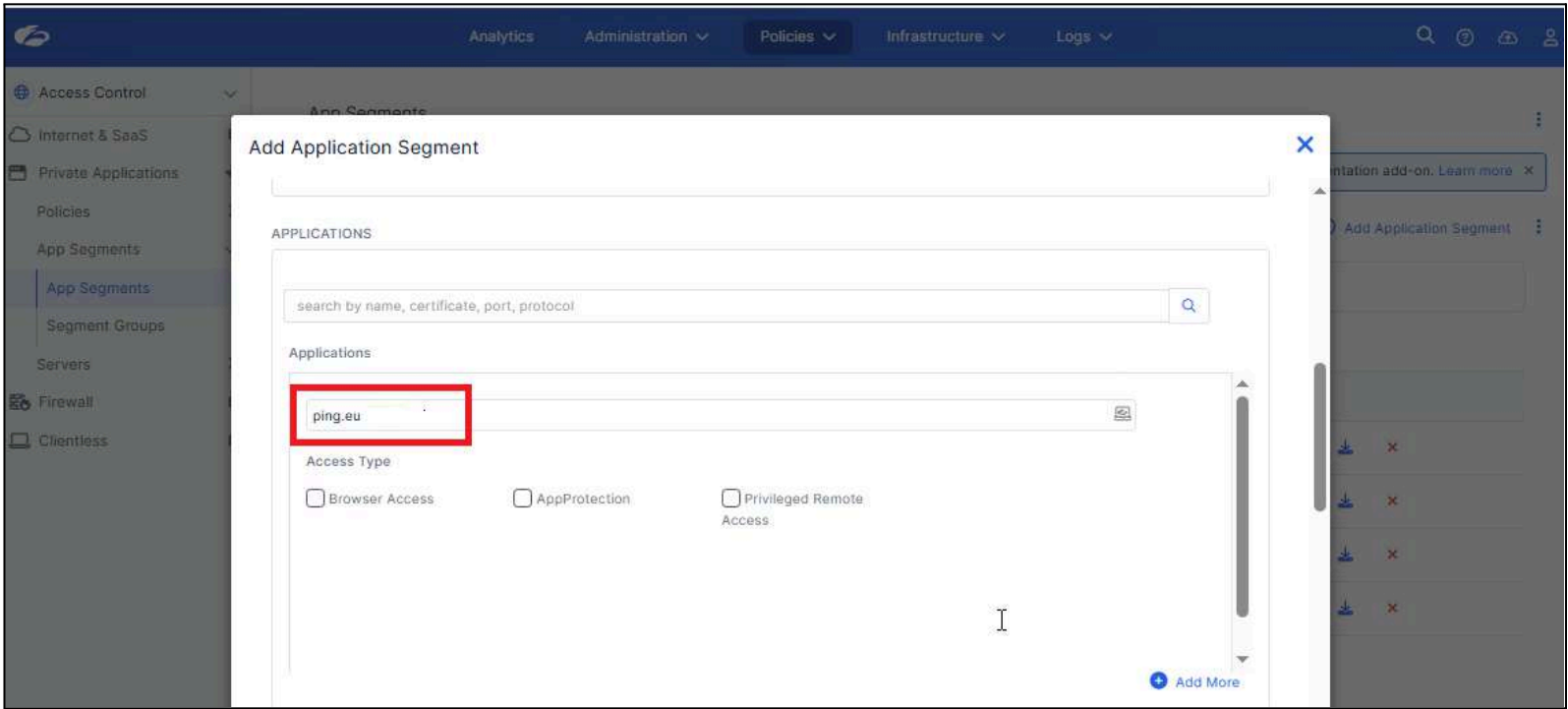
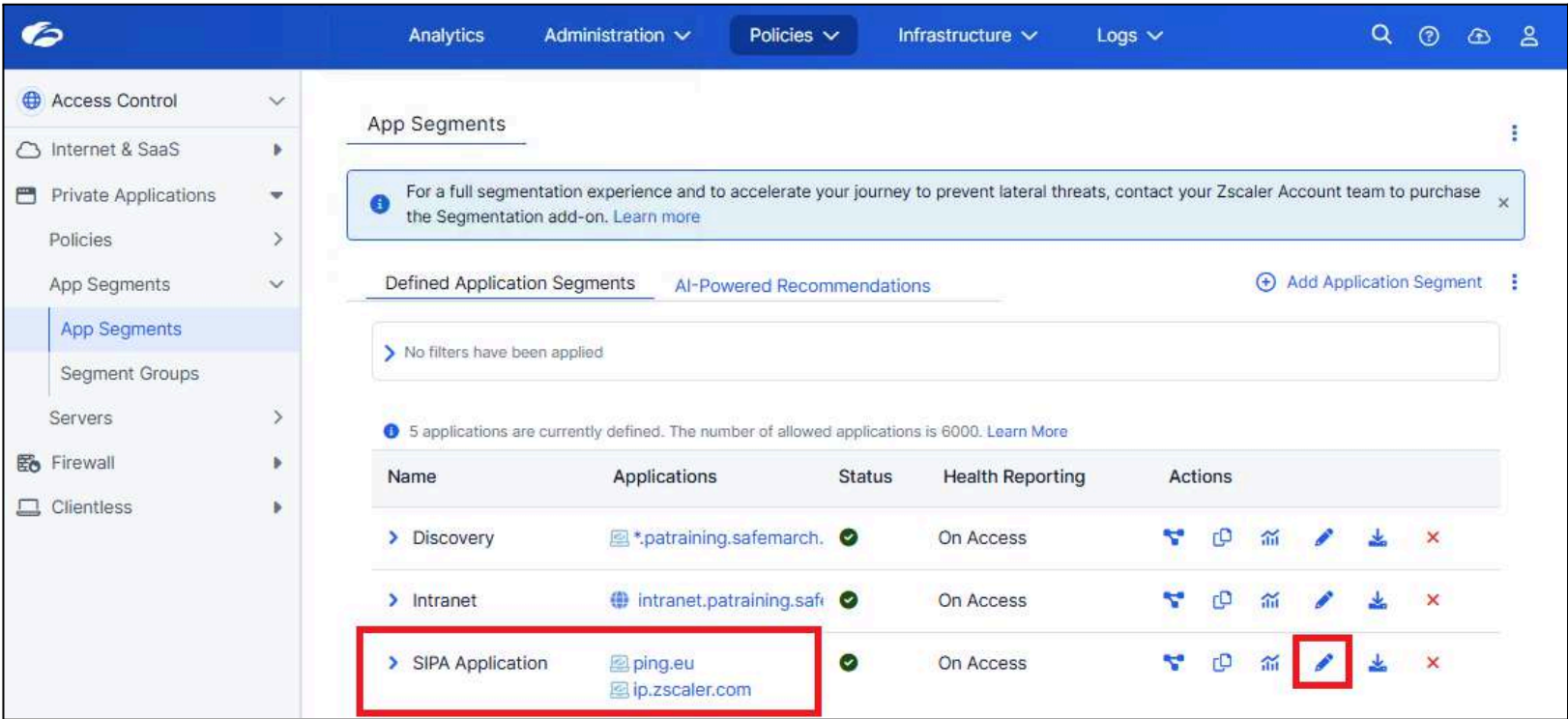


Note: The displayed SIPA IP is distinct from the default ZIA Egress IP (Step 1 of Task 13.1). This demonstrates ZPA's capability to provide a fixed source IP for private applications, crucial for whitelisting security.

2. Dynamic Testing: (Modifying Application Segment):

- a. Navigate to the **ZPA Admin Portal** via the **Experience Center**:
 - i. **Policies > Private Application > App Segments.**
- b. Edit the **SIPA Application Segment** as follows:
 - i. **Remove** `ip.zscaler.com` from the **Application** tab.

ii. Add a new domain (e.g., ping.eu) instead.



3. Test Dynamic Configuration:

- a. Open a browser and navigate to: `ping.eu`
- b. Observe the new IP address displayed by `ping.eu`:

Note: The traffic for the newly defined application is routed through ZPA-SIPA, showing the updated ZPA egress IP.



Lab 13: Access an Internal Application using Remote Desktop Protocol (RDP) [Optional]

Scenario: An internal user reports that they cannot access the internal application intranet.pattraining.safemarch.com. As the Zscaler administrator, troubleshooting begins by confirming application connectivity through **ICMP Access**. After verifying basic connectivity, you attempt to connect to the application using **Remote Desktop Protocol (RDP)**, which **requires TCP port 3389**. Upon reviewing the application segment configuration, you discover that port **3389 is not included in the defined TCP port ranges**, preventing RDP traffic from being forwarded via **ZPA**. To resolve this issue, you will update the application segment to include port **3389** and confirm secure access using RDP.

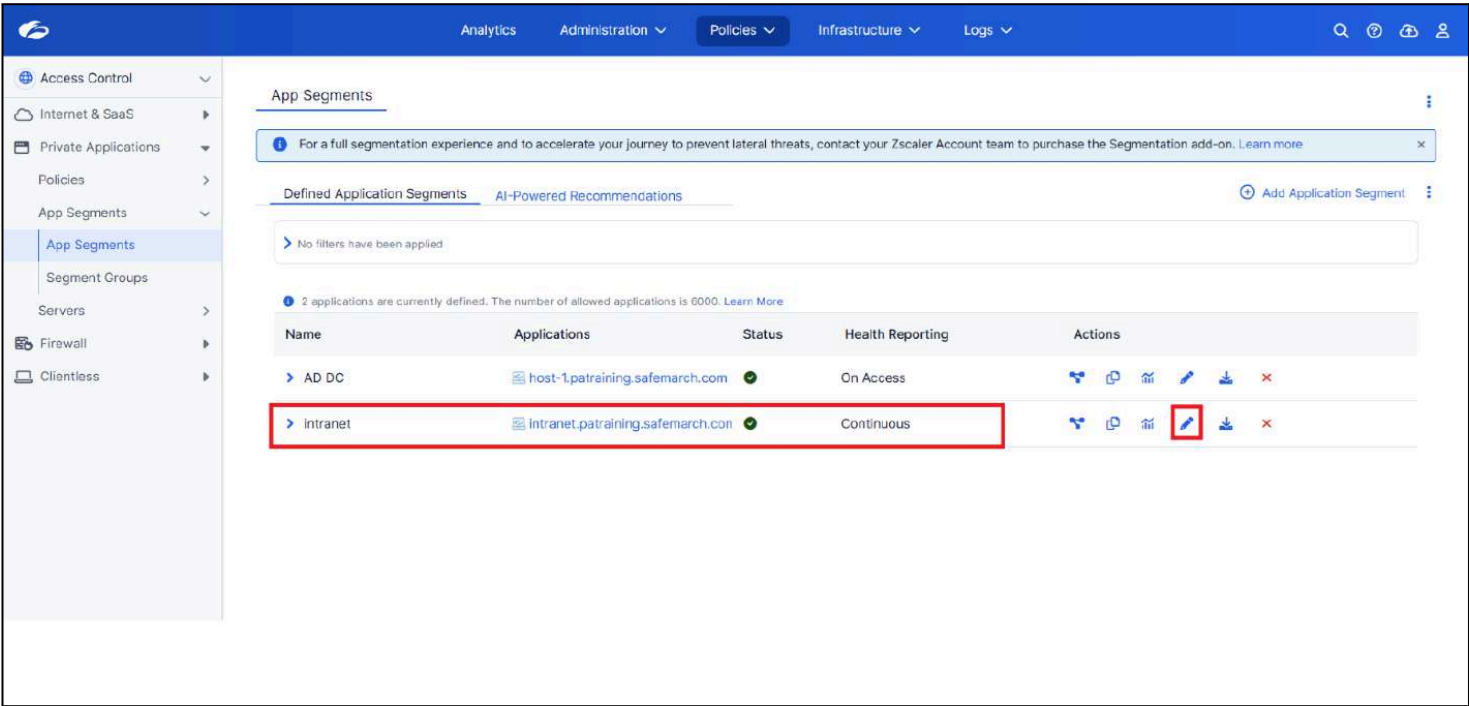
Objectives: By the end of this lab you will

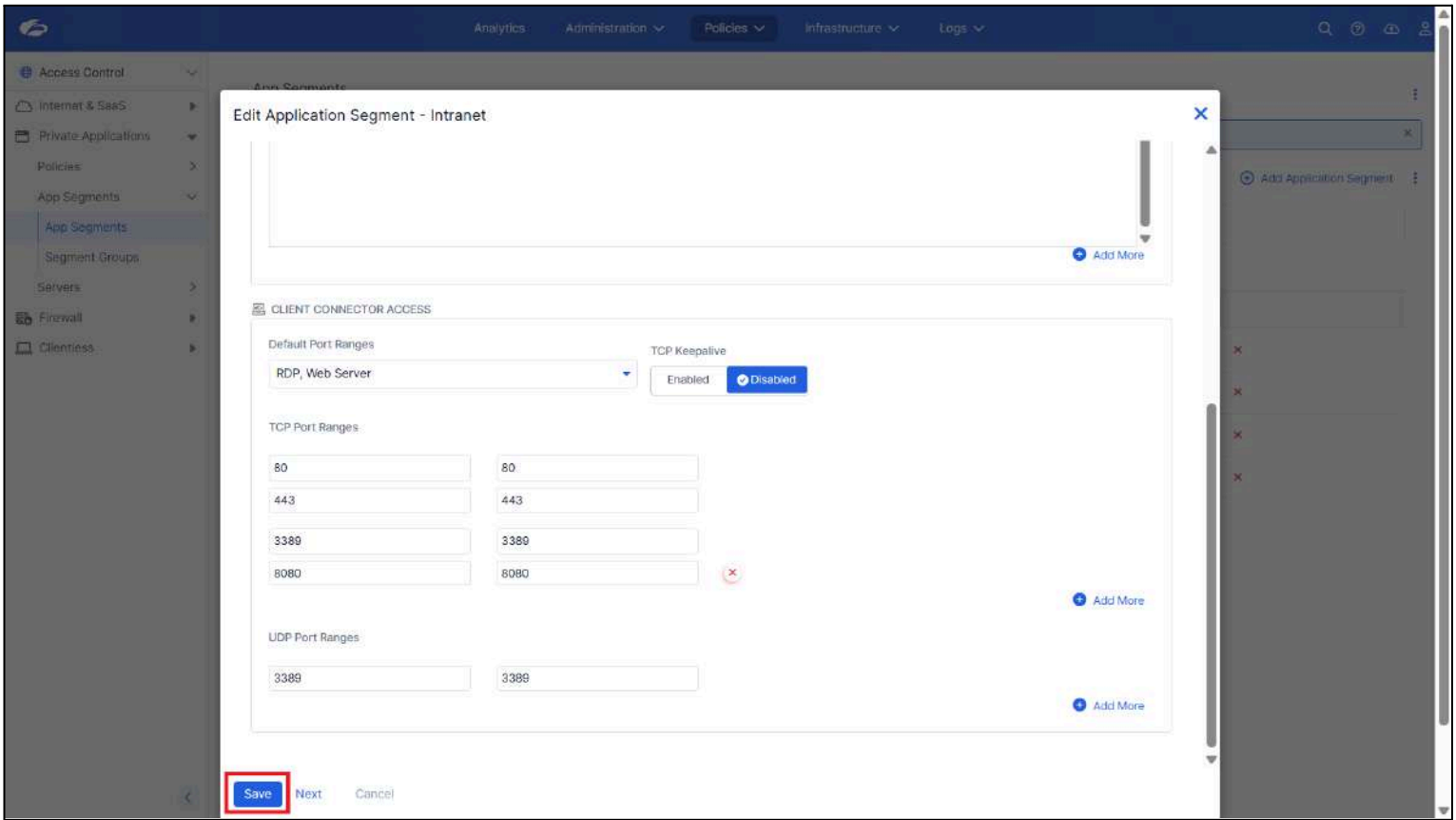
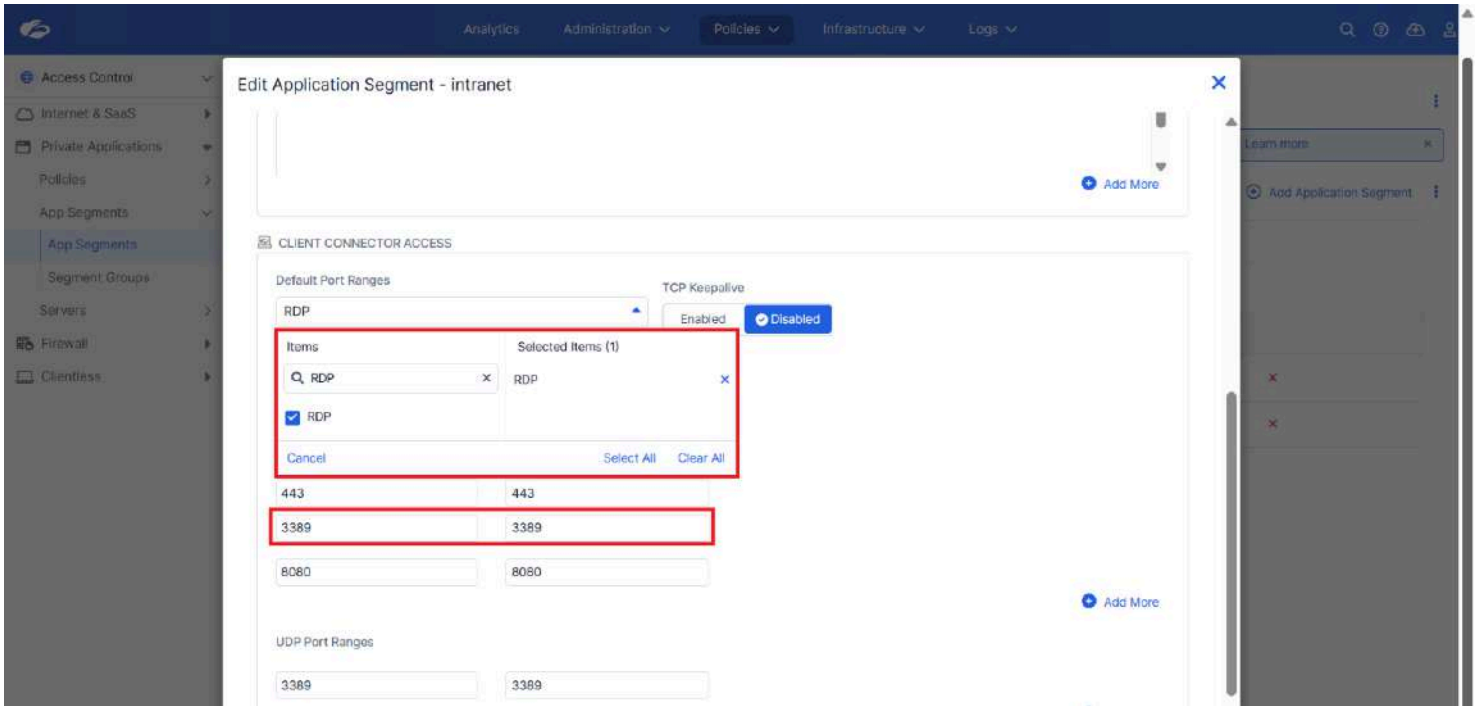
- Configure the **Intranet Application Segment** in ZPA to allow **RDP Access** on port **3389** for troubleshooting purposes, securely connecting through ZPA and
- Verify access details in the logs.

Approximate Time: 10 minutes

Task 13.1:Verify Access to External Application Using RDP

- 1. Enable RDP Port in the Application Segment:
 - a. On the Corp: Client PC, navigate to:
 - i. **Policies > Access Control > Private Applications > App Segment** in the ZPA Admin Portal.
 - b. Find the **Intranet App Segment** and click **Edit**.
 - c. In the Edit Application Segment window:
 - i. Scroll to the port configuration section.
 - ii. Add **RDP Port (3389)** to the TCP Port Ranges.
 - d. Click **Save**.





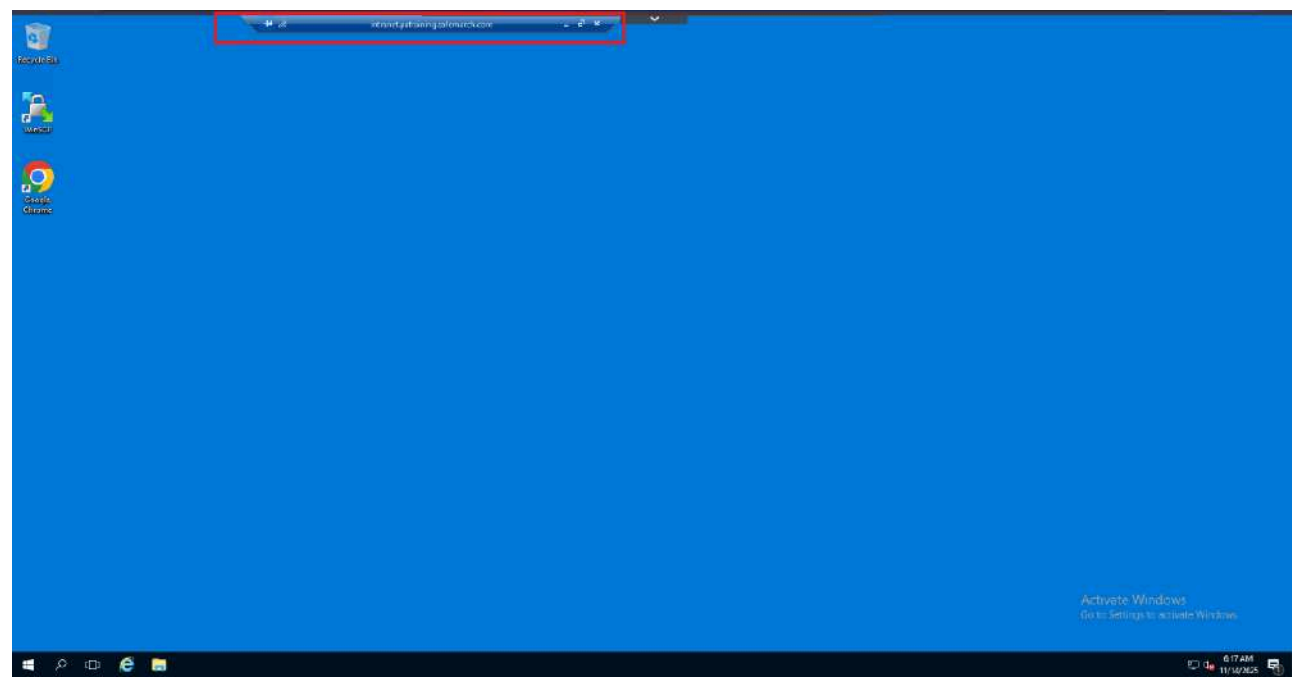
2. Connect Using Remote Desktop Connection
 - a. On the Corp: Client PC, open the **Remote Desktop Connection** application.



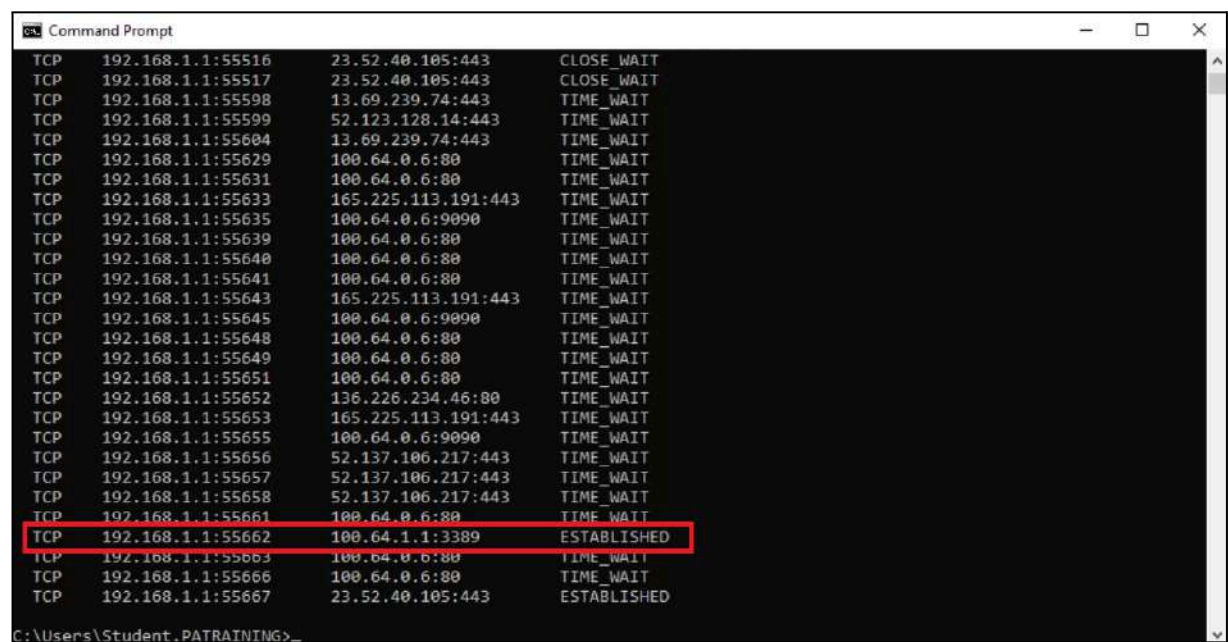
- b. Input the following credentials:
 - i. Computer: intranet.pattraining.safemarch.com
 - ii. Username: **student**
 - iii. Password: **Admin-123!**



- c. Confirm that the internal application is accessed using the Remote Desktop Application.

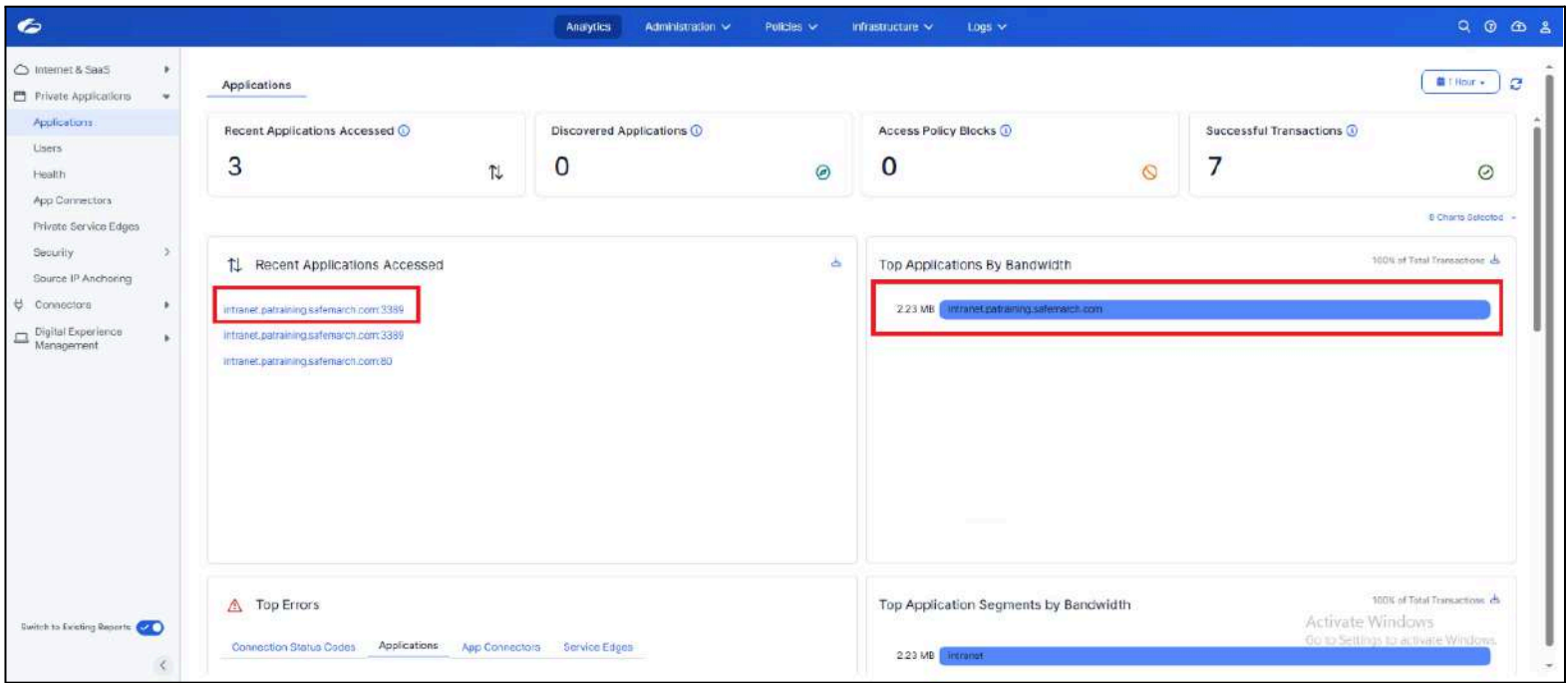


- 3. Verify the RDP Connection and Logs:
 - a. Open the command prompt and type: `netstat -fn`
 - i. Confirm that port **3389** listed as an **Established** connection.

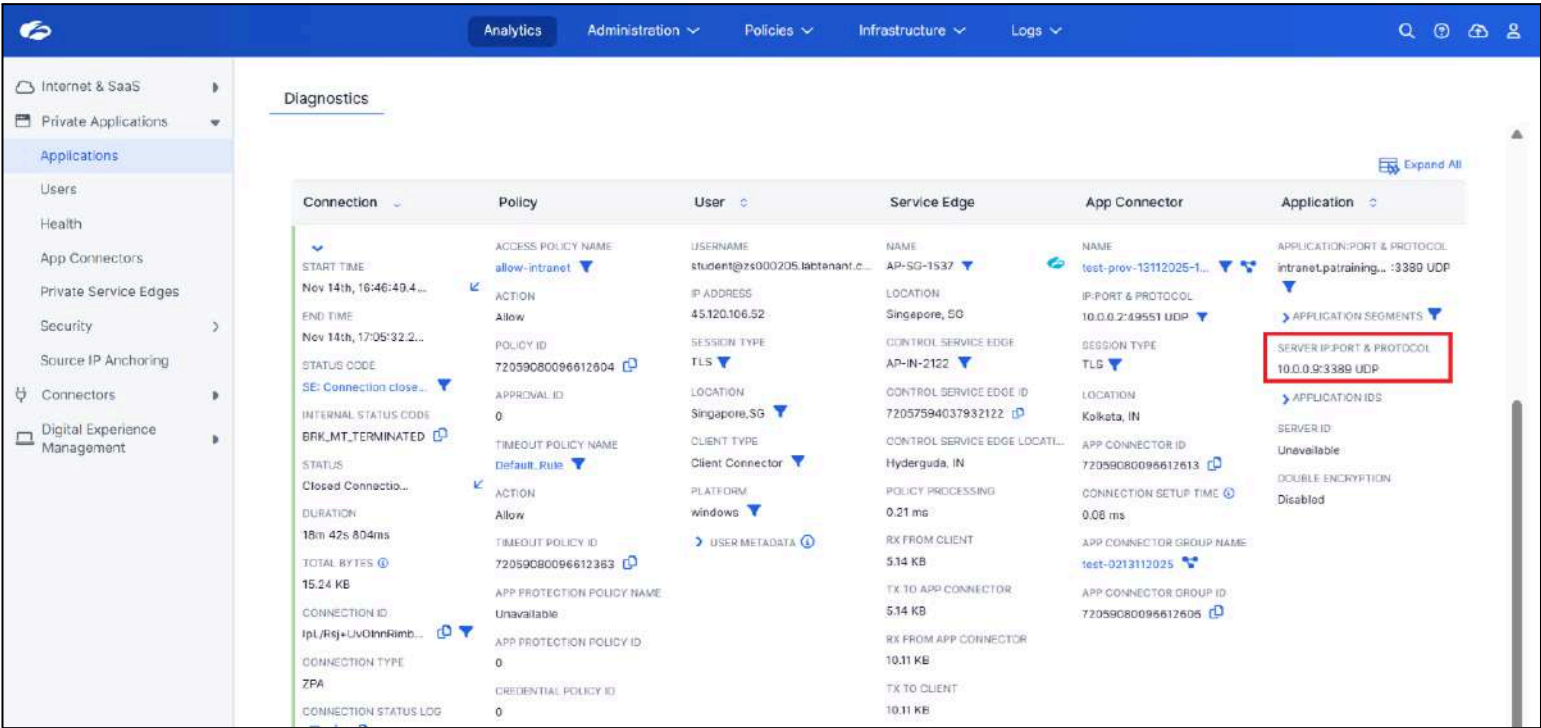


- b. Navigate to **Analytics > Private Applications** and in the **Recently Application Accessed** tab.
 - c. Find the entry for `intranet.pattraining.safemarch.com:3389`.

d. Click the entry to view its diagnostics.



Note: You may need to toggle the setting "Switch to existing reports" to view private application logs.



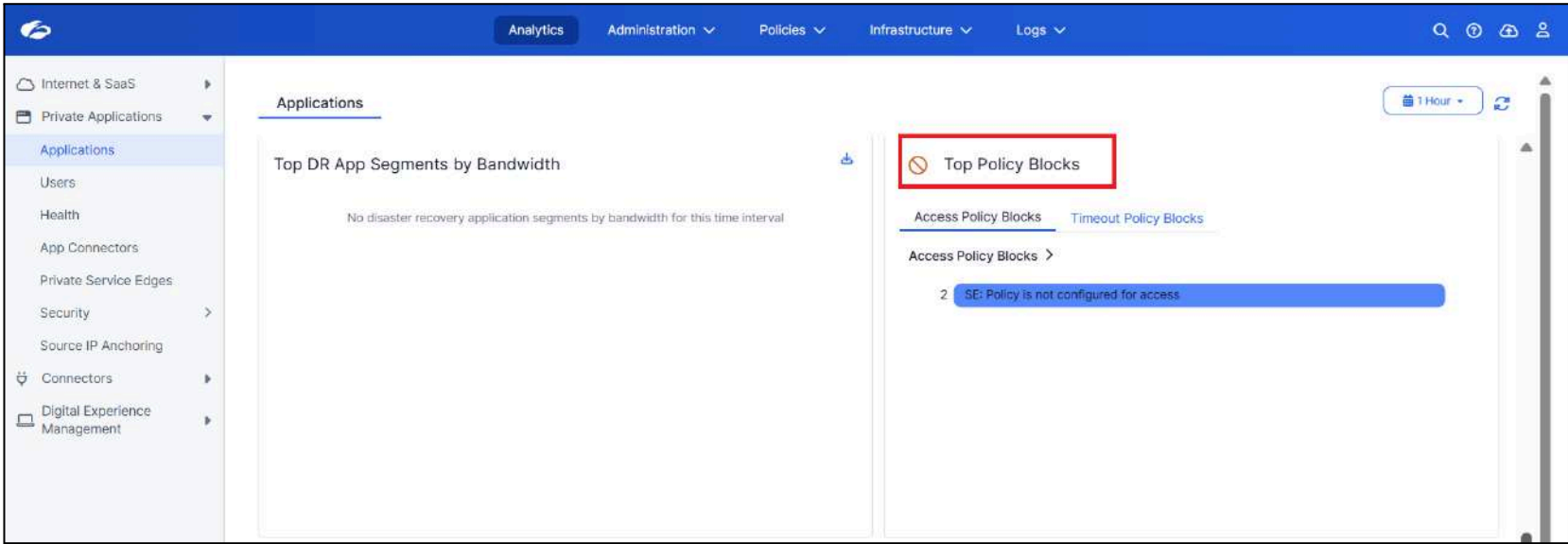
Note: Using Zscaler Private Access (ZPA), you can securely access a Windows server via Remote Desktop from any internet-connected device running the Zscaler Client Connector. Once the administrator is granted the appropriate entitlement, they can also access internal applications hosted in the data center.

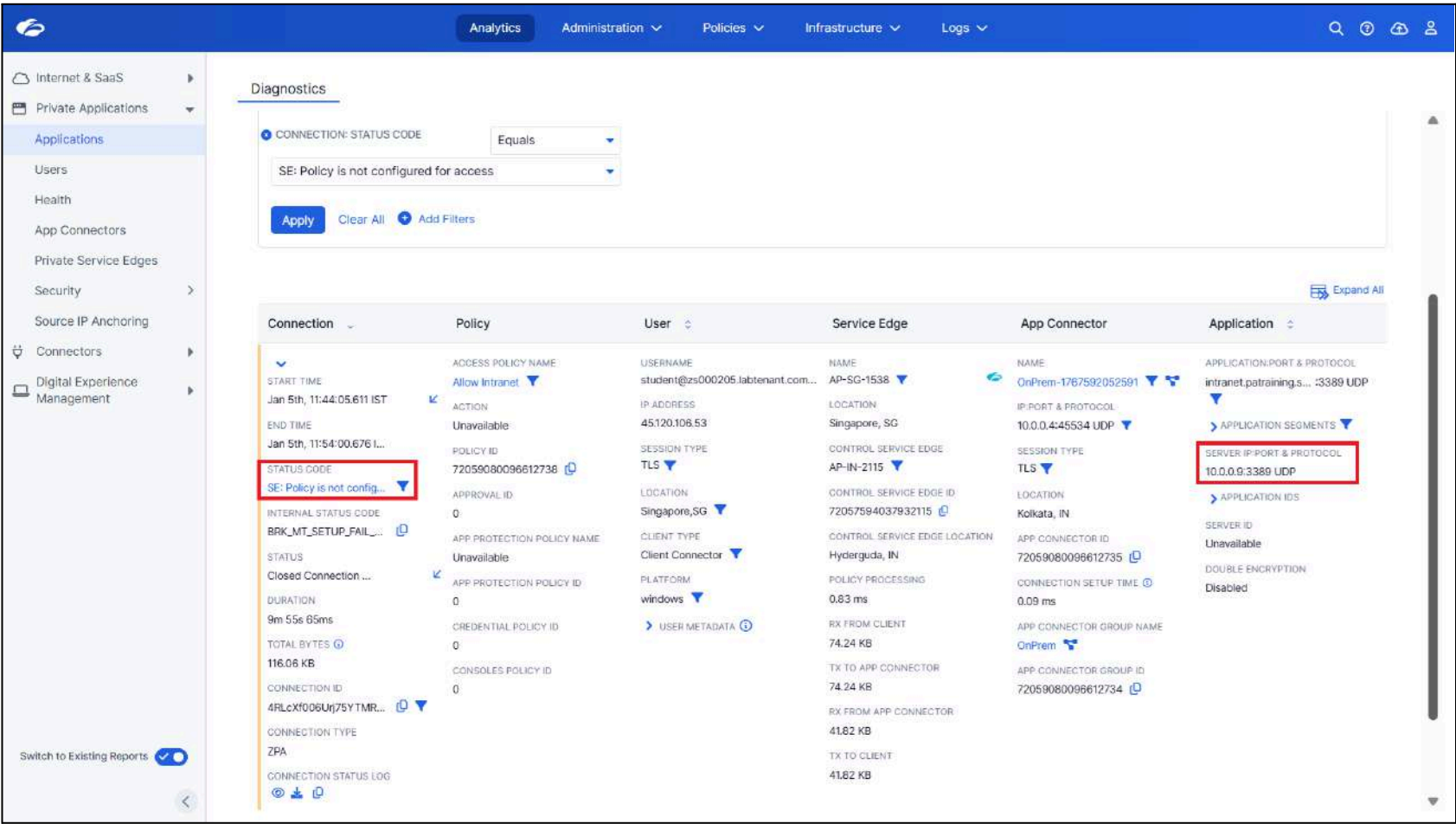
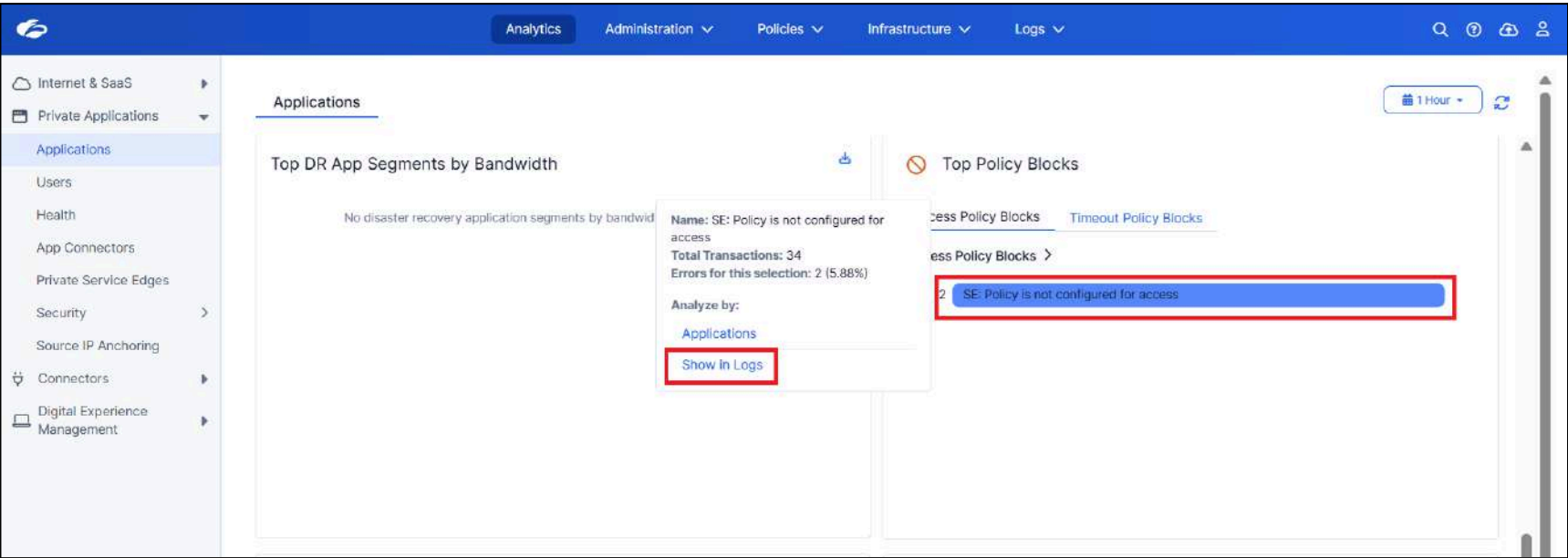
4. (Optional) Troubleshooting Steps:
- a. If issues occur while establishing the RDP connection:
 - i. Edit the **Intranet Application Segment** in the ZPA Admin Portal:
 - Remove **RDP ports (TCP 3389, UDP 3389)** from the **TCP and UDP Port Ranges**.
 - ii. Attempt to reconnect using the **Remote Desktop Connection App**:
 - Enter the same credentials and connection details as before.

Note: The RDP connection fails because port 3389 is no longer defined in the application segment, and ZPA cannot forward the traffic.

5. Validate Diagnostics and Access Policy Blocks:
- a. Navigate to **Analytics > Private Applications** and scroll down to view the **Top Policy Blocks**.
 - b. Click **Access Policy Blocks**, then navigate to the **Show in Logs** option.
 - c. In the **Diagnostics** page:
 - i. Expand the log entry for the access attempt.
 - ii. Review the following details:
 - **Status Code**
 - **Server IP & Protocol**

Note: The Access Policy Blocks section tracks and indicates application requests blocked by access policy rules. As the RDP port (3389) is not defined in the App Segment, the user will be unable to access the internal application, generating an error visible in the diagnostics logs.





Solutions Demo Center Labs

Note: While our hands-on, read-write labs are designed to enhance experiential learning, certain labs remain in read-only mode (SDC) due to specific technical dependencies such as report generation timelines (which may take up to 24 hours or more) and restricted student access to some of the integrated services.

These SDC labs are intentionally designed to expose learners to key features and real-world dashboards. They are especially useful in scenarios where full access may not be feasible. These labs provide valuable insights into **policy enforcement**, **analytics**, and **governance workflows** that are critical to understanding Zscaler's data protection capabilities.

Scenario:

For these following lab exercises, consider a scenario where your organization has deployed **Zscaler Digital Experience** to proactively monitor network and device performance for all users. You are expected to proactively monitor trends and incidents impacting multiple users and to take corrective actions before potential issues escalate. Your team is also responsible for troubleshooting reports of poor user experience identifying the root cause.

1. To Access the **Solutions Demo Center (SDC)**:
 - a. Navigate to <https://console.zscaler.com>
 - b. Login using the **SDC credentials** provided in your lab access information:
 - i. Username: <**ZIA_Admin_User_ID**>
 - ii. Password: <**Session_Password**>

Note: For each lab, you may choose to follow the step-by-step instructions or challenge yourself by reviewing the problem statement and independently identifying the root cause before proceeding. This approach will help you build troubleshooting and critical thinking skills.

Lab 14: Navigate the Incident, Self-Service, Application Overview Dashboards

Scenario: As the Zscaler administrator, you know that your organization relies on **Zscaler Digital Experience (ZDX)** to monitor and improve user experiences across endpoints, networks, and applications. Your role is to analyze signals provided by ZDX dashboards—such as **Incident**, **Self-Service**, **Application Overview**, and **User Overview**—and use this data to identify the root cause of user experience issues. By understanding and interpreting key performance trends and metrics, you will make informed decisions to enhance the **digital experience** across the organization.

Objective: By the end of this lab, you will:

- **Navigate and analyze ZDX's dashboards**, including **Performance Overview**, **Incidents**, **Self-Service**, **Application Overview**, and **User Overview**.
- **Analyze root cause indicators** for **application**, **network**, and **user** issues
- **Troubleshoot key performance problems** and **plan corrective actions**.

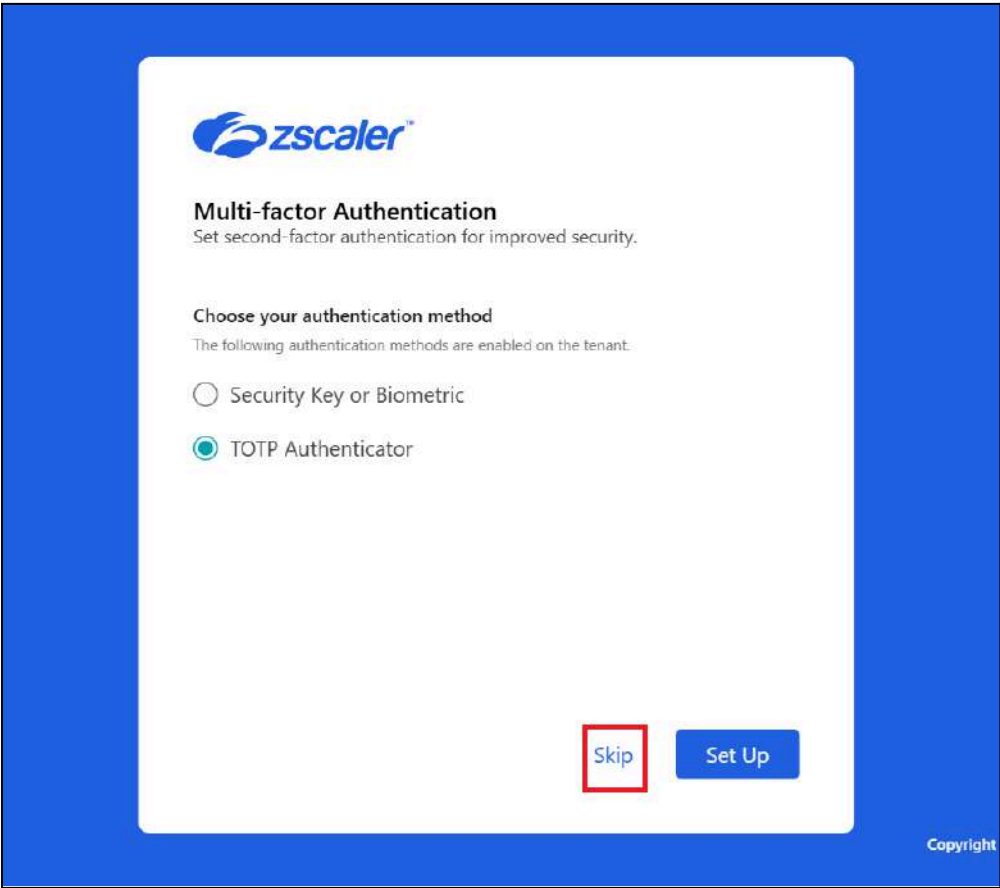
Approximate Time: 20 minutes

Task 14.1: View ZDX Performance Dashboard

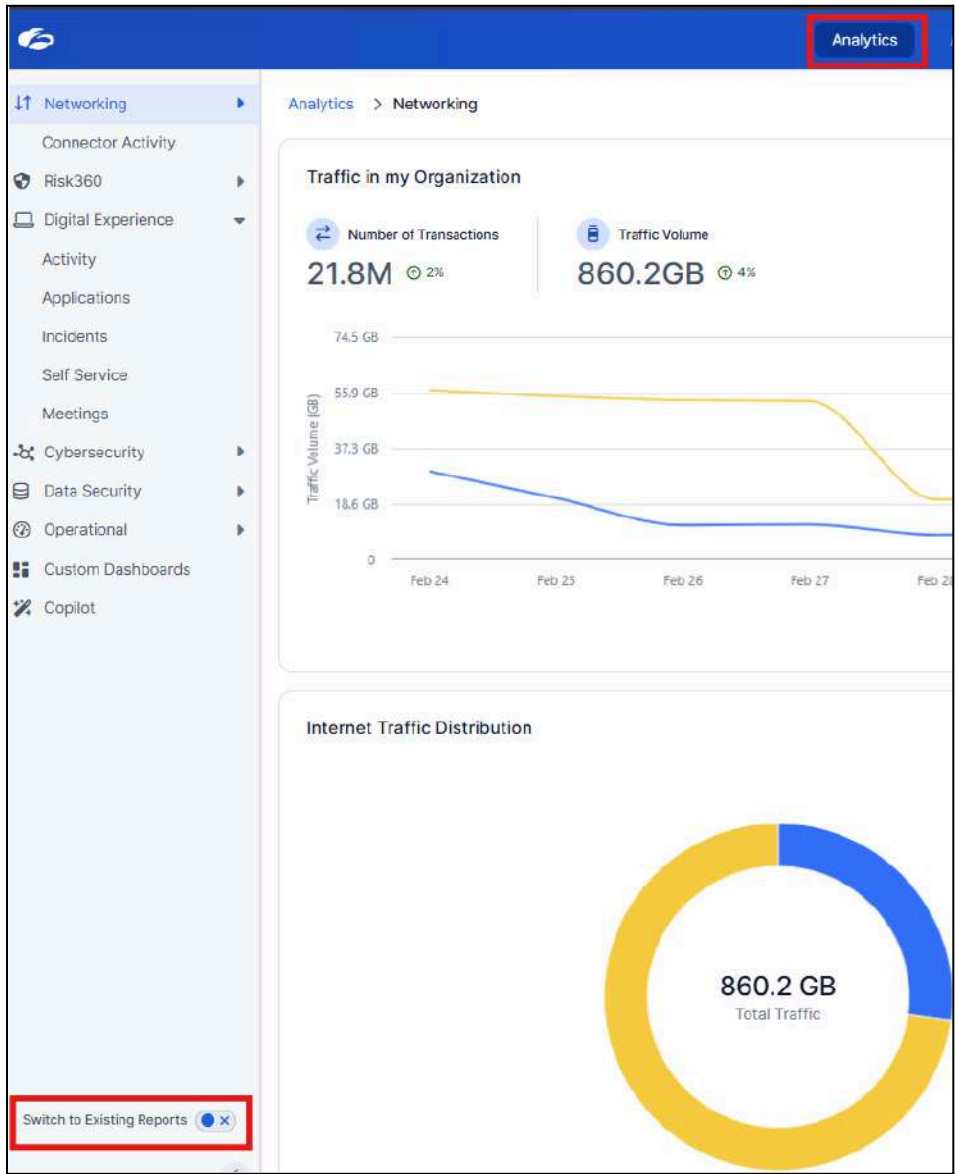
In this task, you will take the role of a Zscaler administrator , and explore the ZDX Performance Dashboard, including **Most Impacted Applications, Regions by ZDX Score, ZDX Score.**

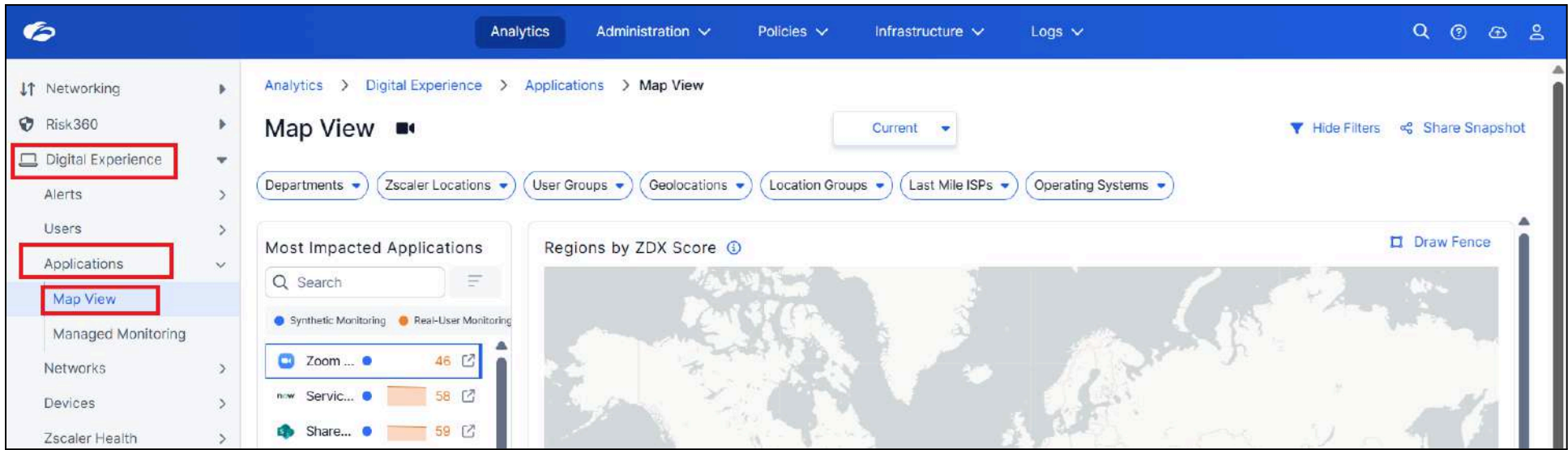
1. Login to the Experience Center:
 - a. Navigate to `https://console.zscaler.com` and authenticate with the student account:
 - i. Username: **SDC Admin Username (present in your lab access information)**
 - ii. Password: **SDC Password**

Note: You will be taken to **Multi-factor Authentication** while logging in to your SDC account, click on **Skip** and move to the next step.

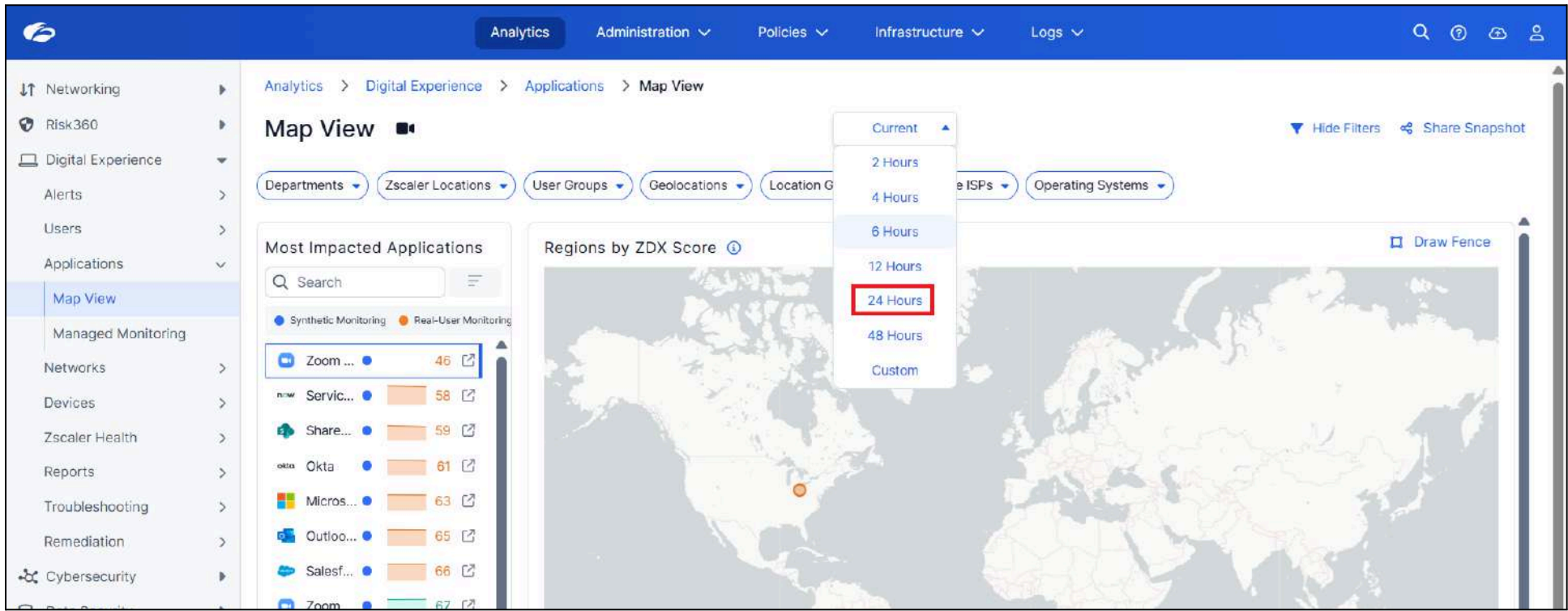


- 2. Go to **Analytics** then click on “**Switch to Existing Reports**”
- 3. Then, navigate to **Analytics > Digital Experience > Applications > Mapview**
- 4. OR, in the **Search** option type **Map View**.





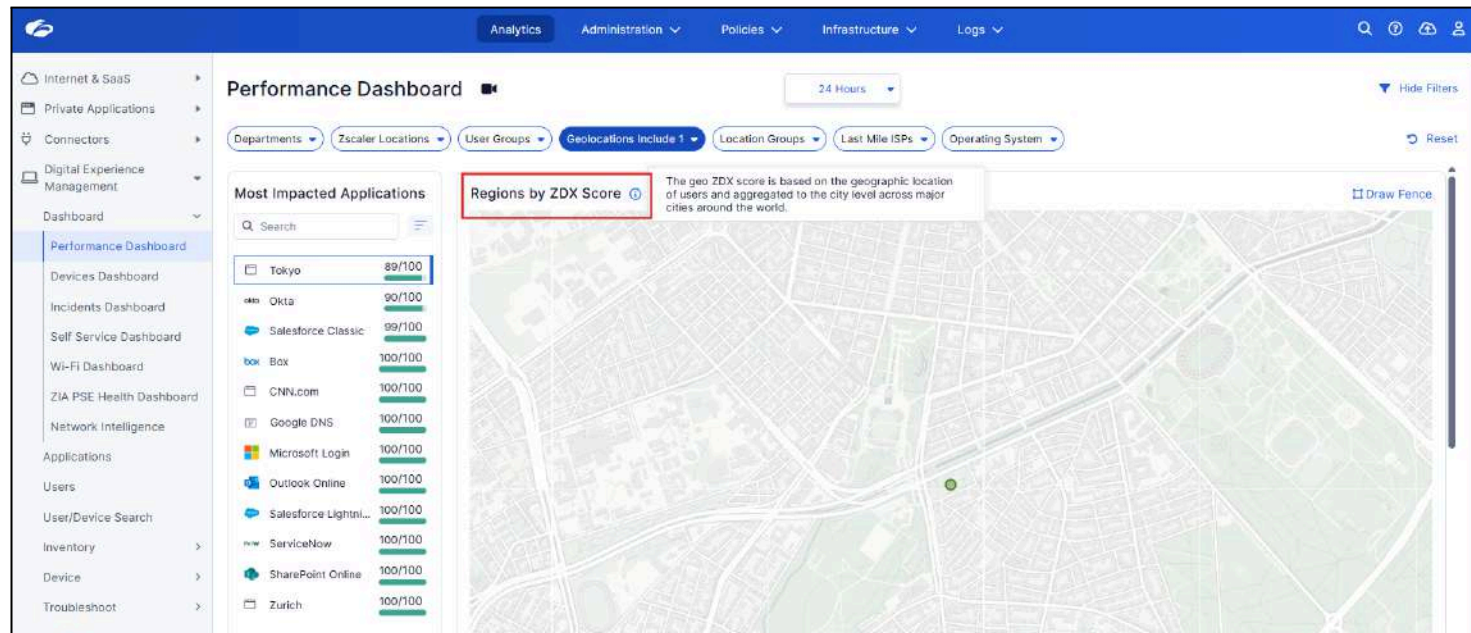
a. At the top of the **Performance Dashboard**, select **24 Hours** from the dropdown list to set the reporting interval.



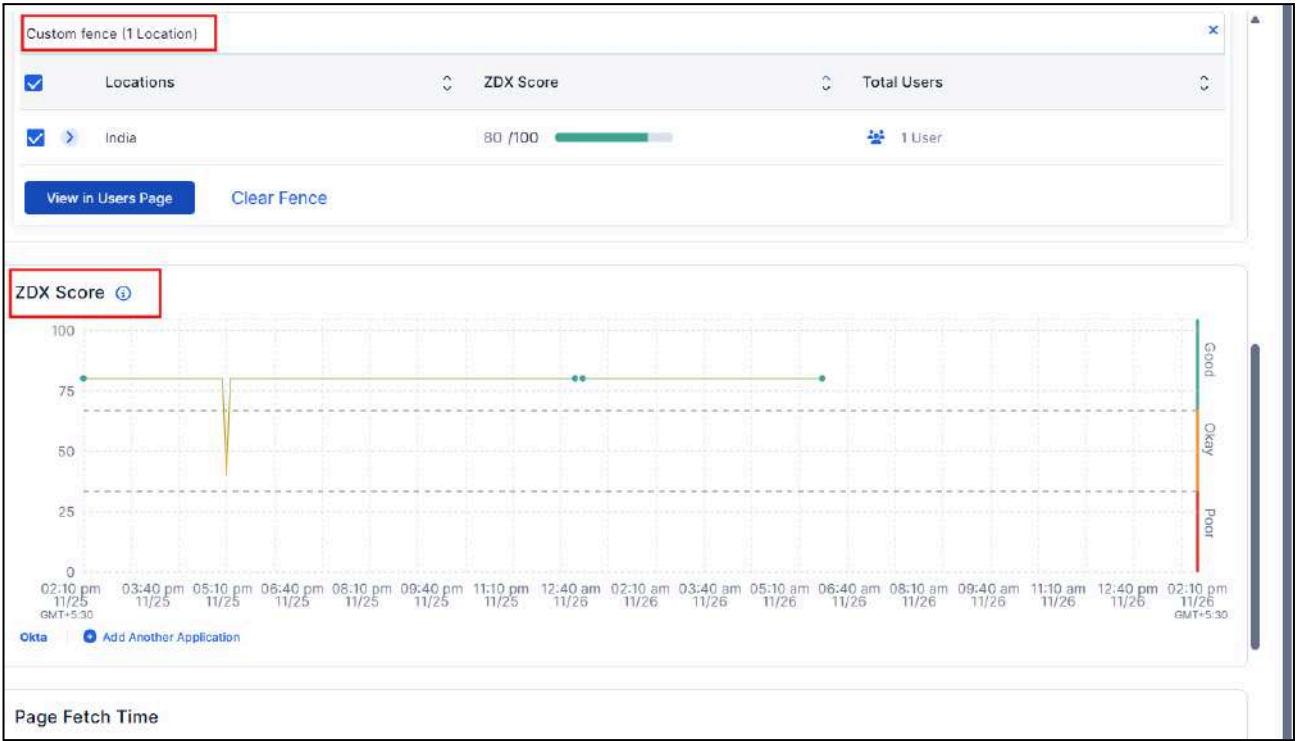
5. Explore Key Dashboard Features:
- a. Review the **Most Impacted Applications**:
 - i. Scroll through the list and click on an application (e.g., “ZPA Application”) to drill into its performance metrics.

b. Analyze geographic performance:

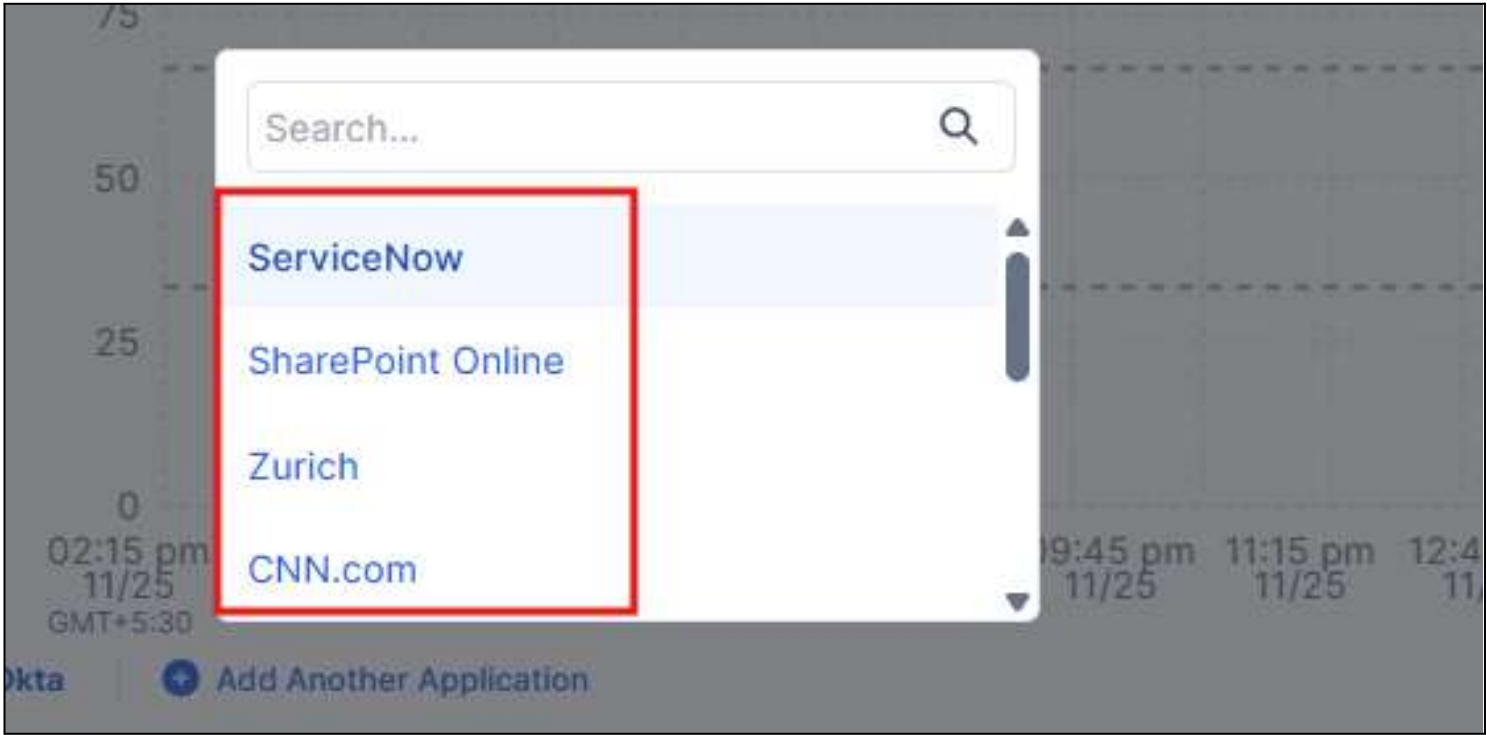
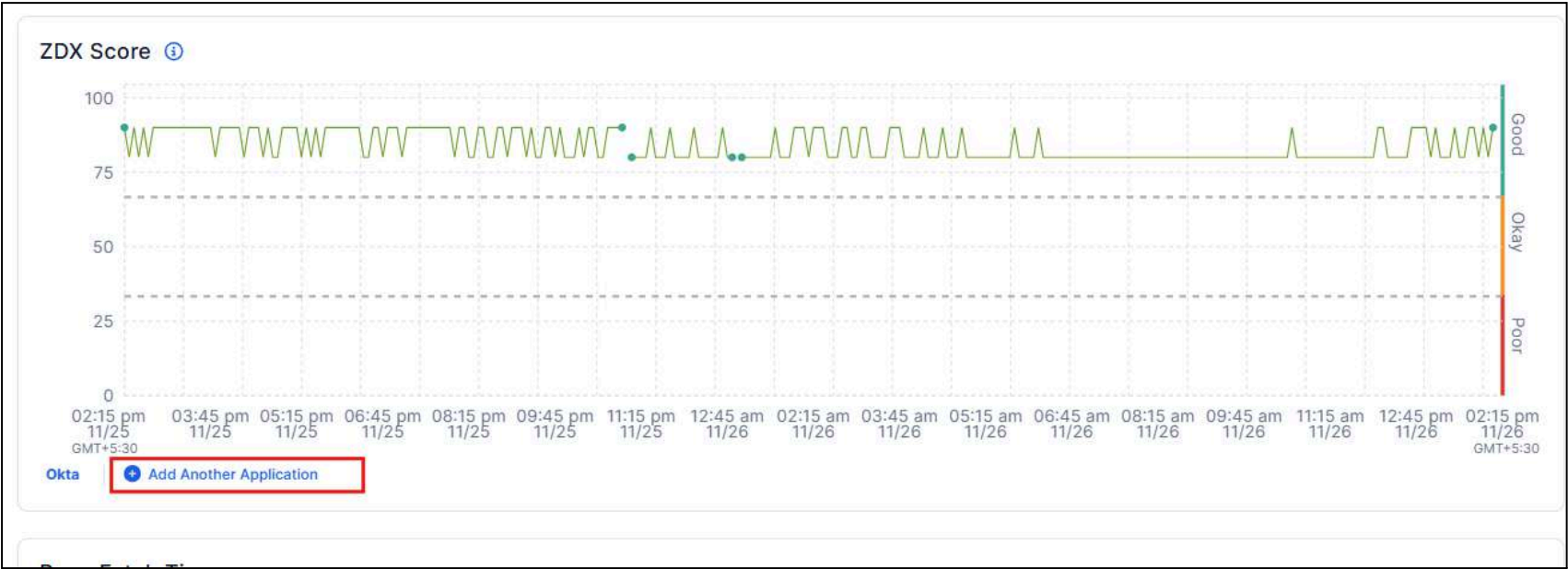
- i. In the **Regions by ZDX Score** map:
- ii. Click a location to view the **ZDX Score** and the number of total users in that region.
- iii. Use the **+/-** controls to zoom in or out.
- iv. Click **Draw Fence**, use your mouse to select a geographic area.
- v. Click **Filter Selection**, and review the fenced region's performance metrics.
- vi. Click on any location to view the **ZDX Score** and the **total users**.
- vii. Click **Reset** (top-right corner of the map) to clear applied settings, if needed.

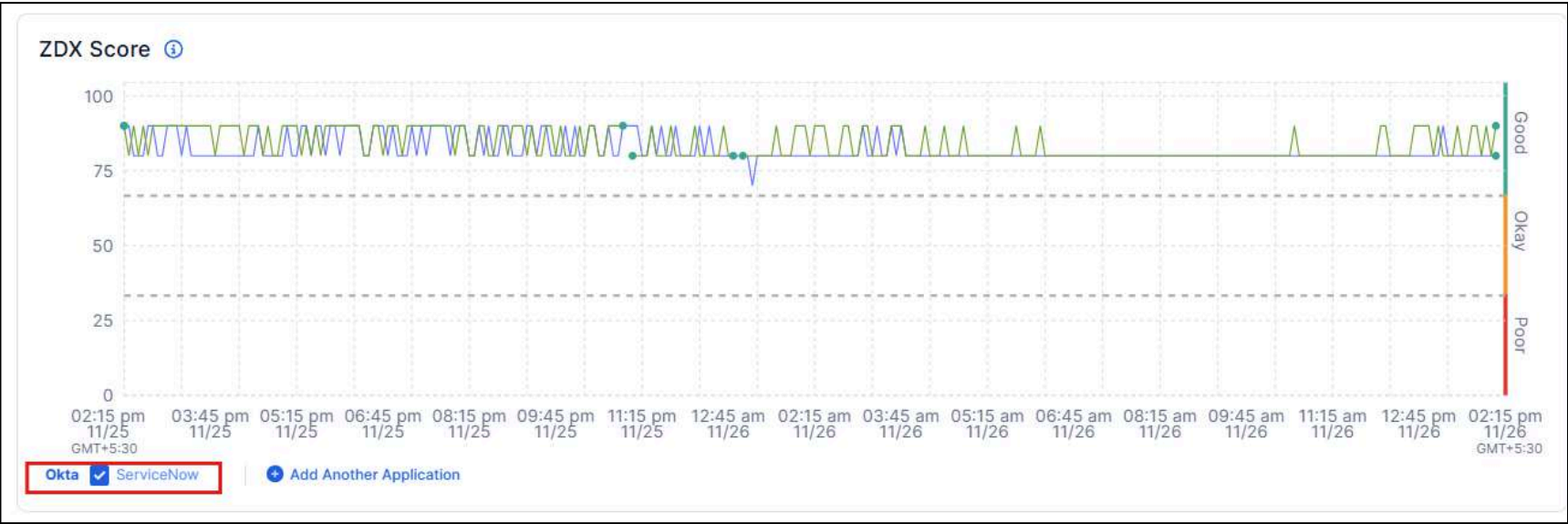




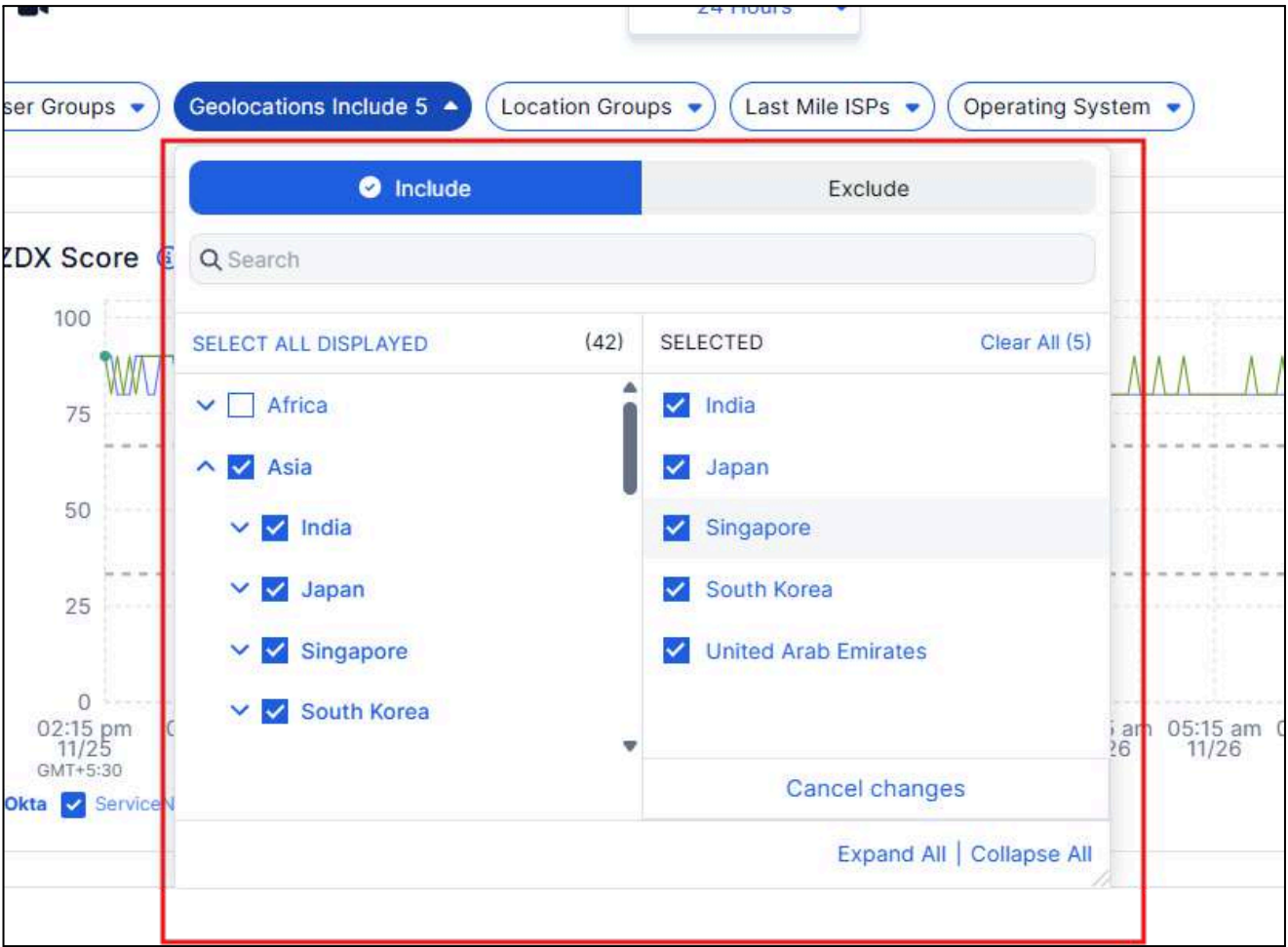


6. Review ZDX Score Details for Selected Applications:
- a. Scroll down to view the **ZDX Score** for the selected application:
 - b. Click on **Add Another Application** at the bottom to compare the ZDX scores of additional applications .

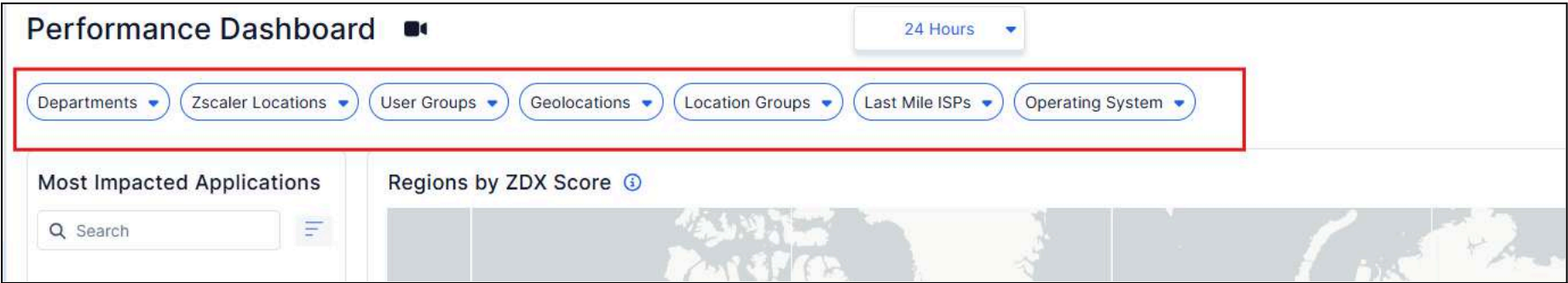


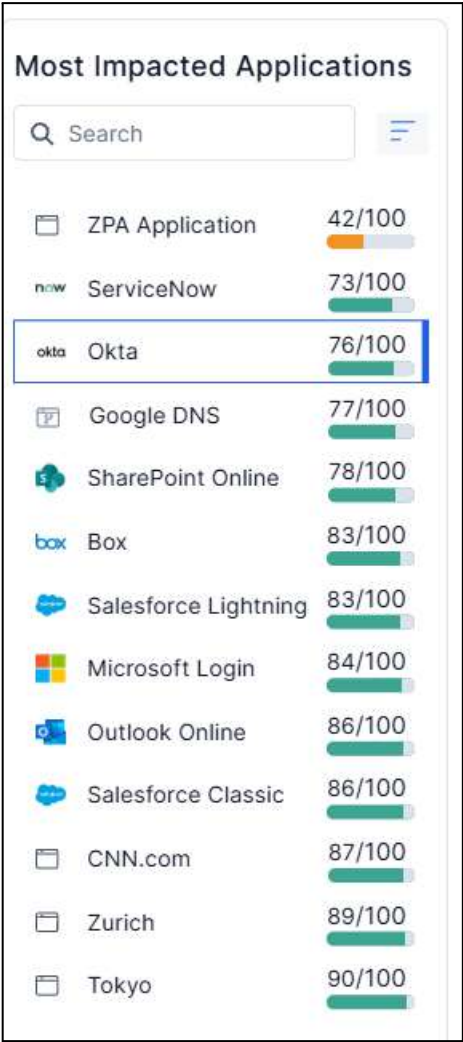


- 7. Filter by Geolocations:
 - a. At the top of the ZDX Dashboard, click **Geolocations**, select a location from the dropdown, and click **Apply**.
 - b. View the **Performance Overview** reports for the chosen location.
 - c. Click **Reset** to clear the filter.

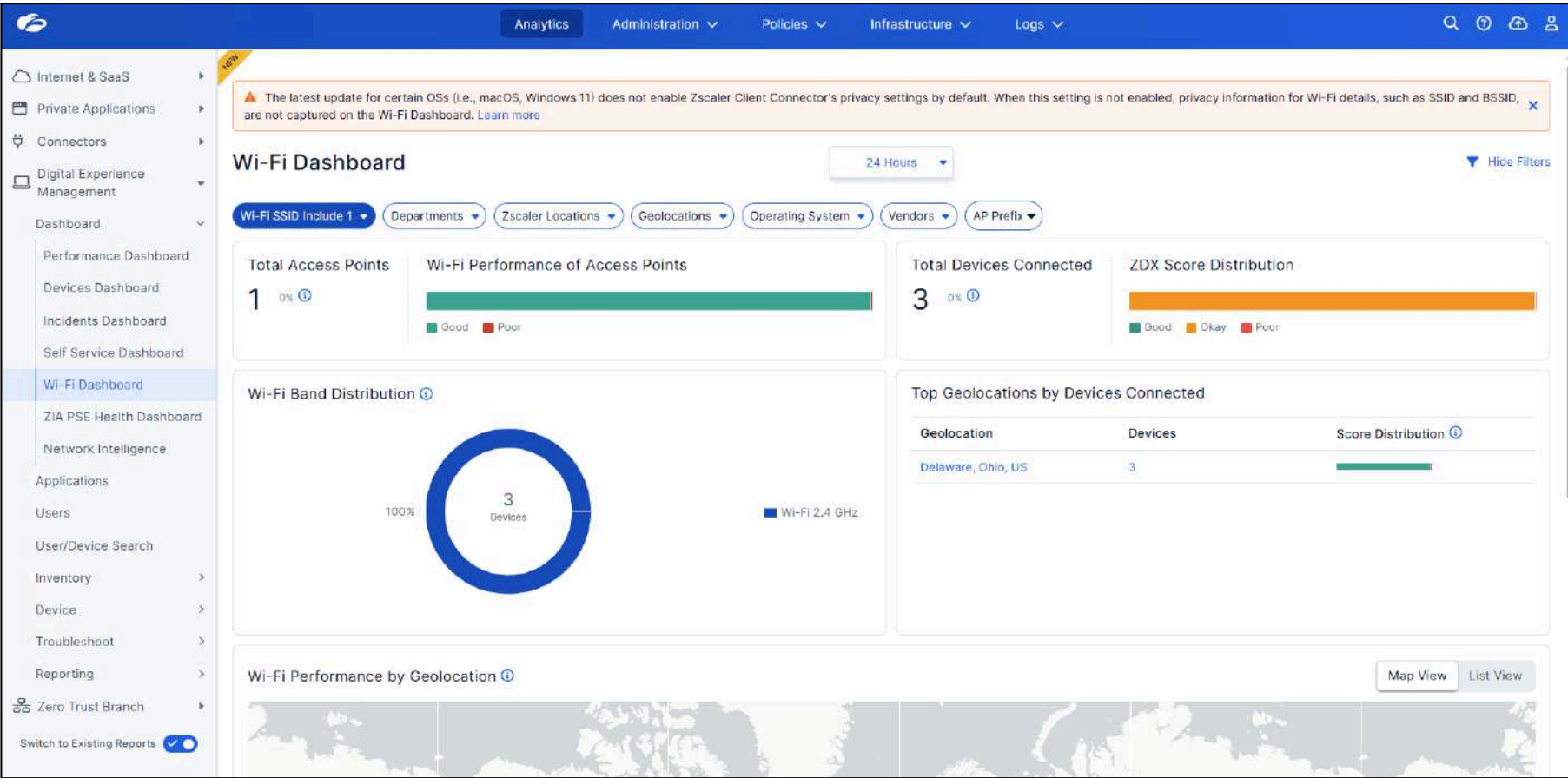


8. Explore Filtering Options:
- a. At the **Most Impacted Application** section, explore multiple filtering options: **Departments, Zscaler Locations, User groups, Location groups, Last Mile ISPs, Operating systems.**
 - b. **Select filters, apply them,** and **review** the results.
 - c. View the **ZDX Score** on the basis the filters applied.





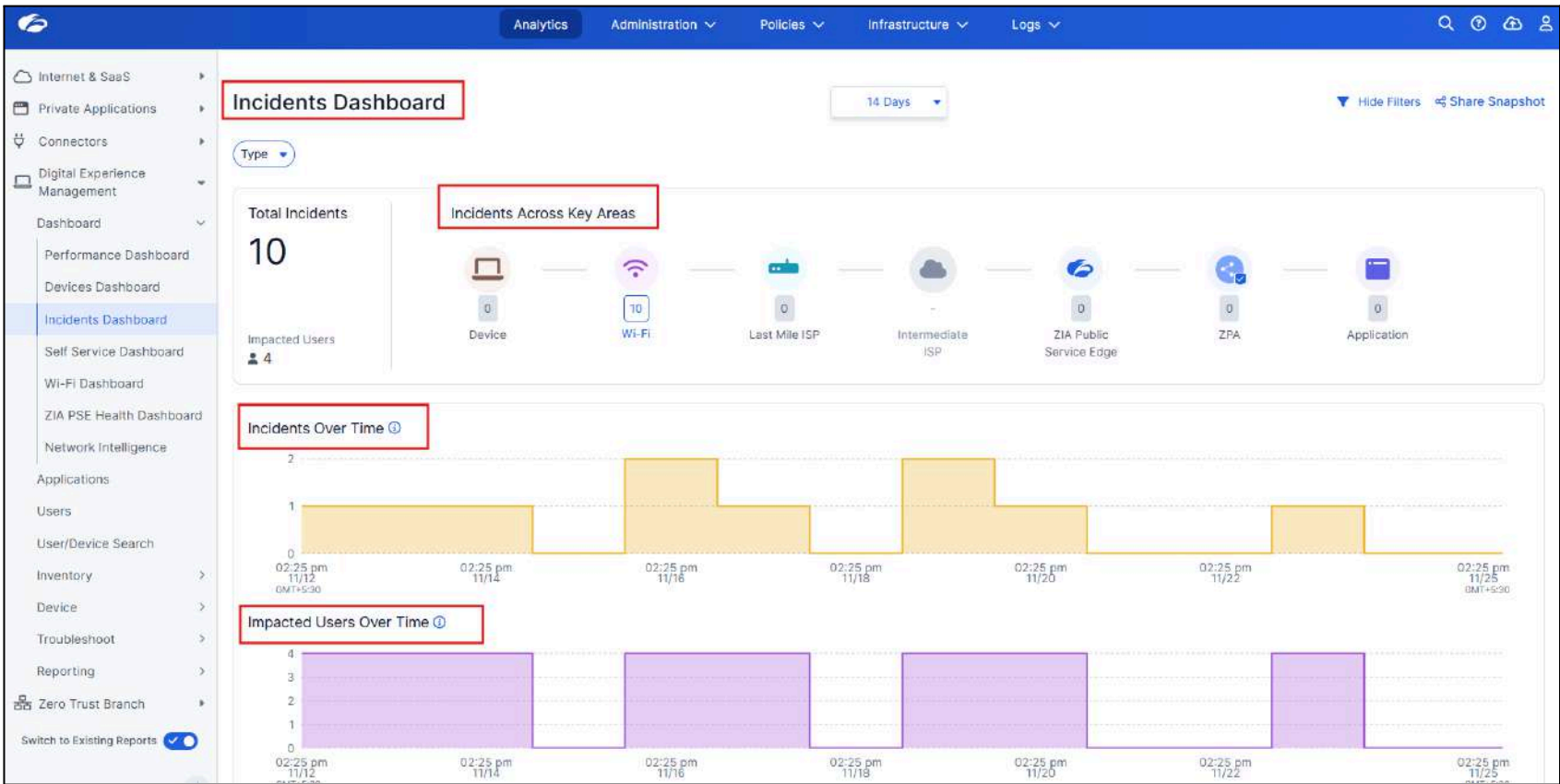
9. Review the Wi-Fi Dashboard:
- a. Navigate to the **Wi-Fi Dashboard** using the navigation bar.
 - b. Review the following Wi-Fi parameters:
 - i. **Total Access Points**
 - ii. **Wi-Fi Performance of Access Points**
 - iii. **Total Devices Connected**



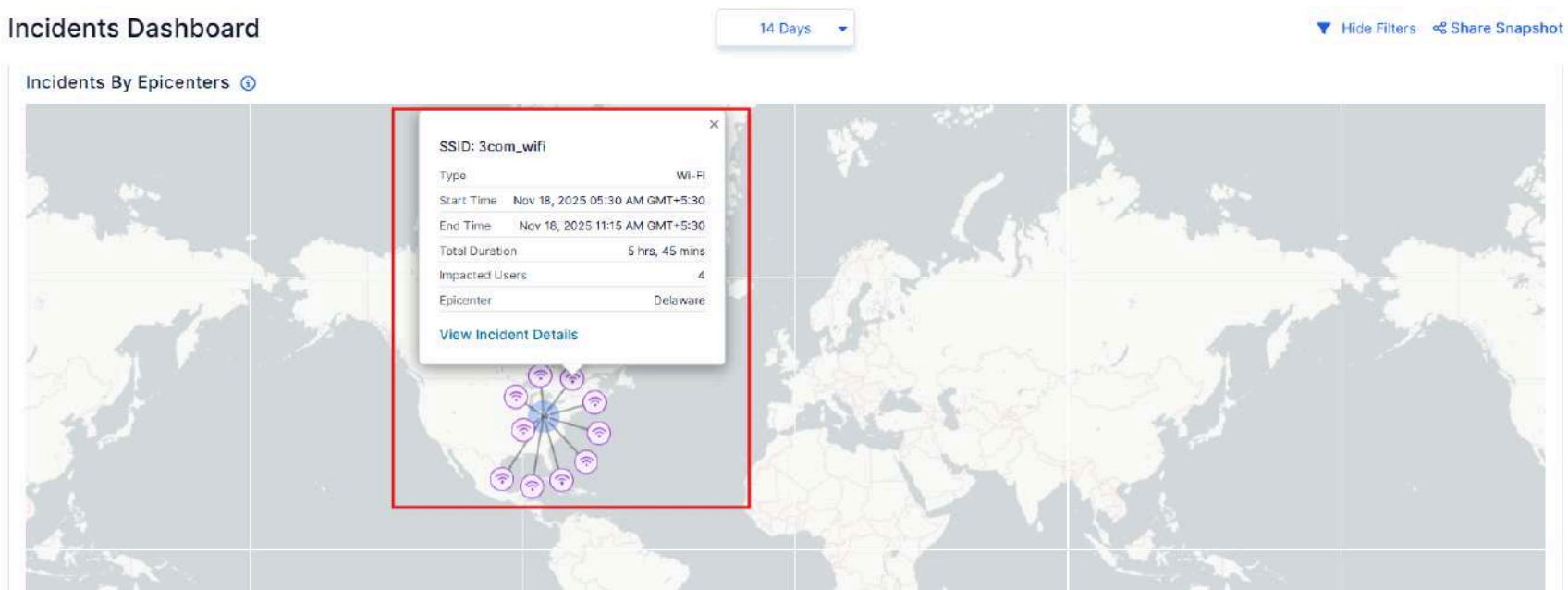
Task 14.2: View Incidents and Self Service Dashboards

In this task, you will analyze incident trends and use self-service insights to identify key issues affecting user application performance.

- 1. Explore the Incidents Dashboard:
 - a. On the **left navigation panel**, go to the **Incident Dashboard** or in the **Search** option type **Incident Dashboard**.
 - b. Review the information for the various incident areas.



c. Scroll down to the **Incidents by Epicenters** map, click on a marker and select one of the incidents.



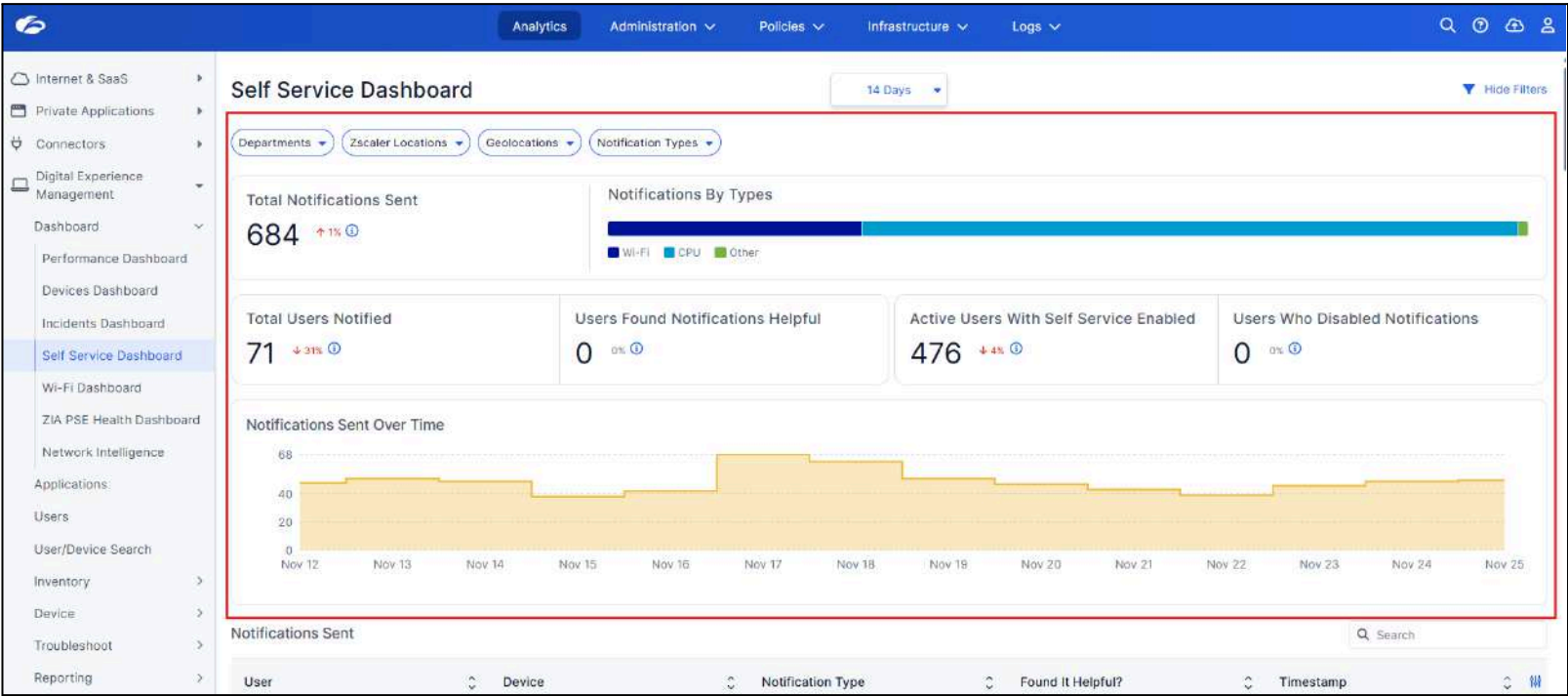
Note: While viewing the Incidents By Epicenters dashboard, if you see the message **“Try again later. If this persists, contact Zscaler Support.”** refresh the page and try again.

d. Click the **View incidents Details** link and review the collected metrics.

2. Explore the Self Service Dashboard:

a. Navigate to **Dashboard > Self Service Dashboard**.

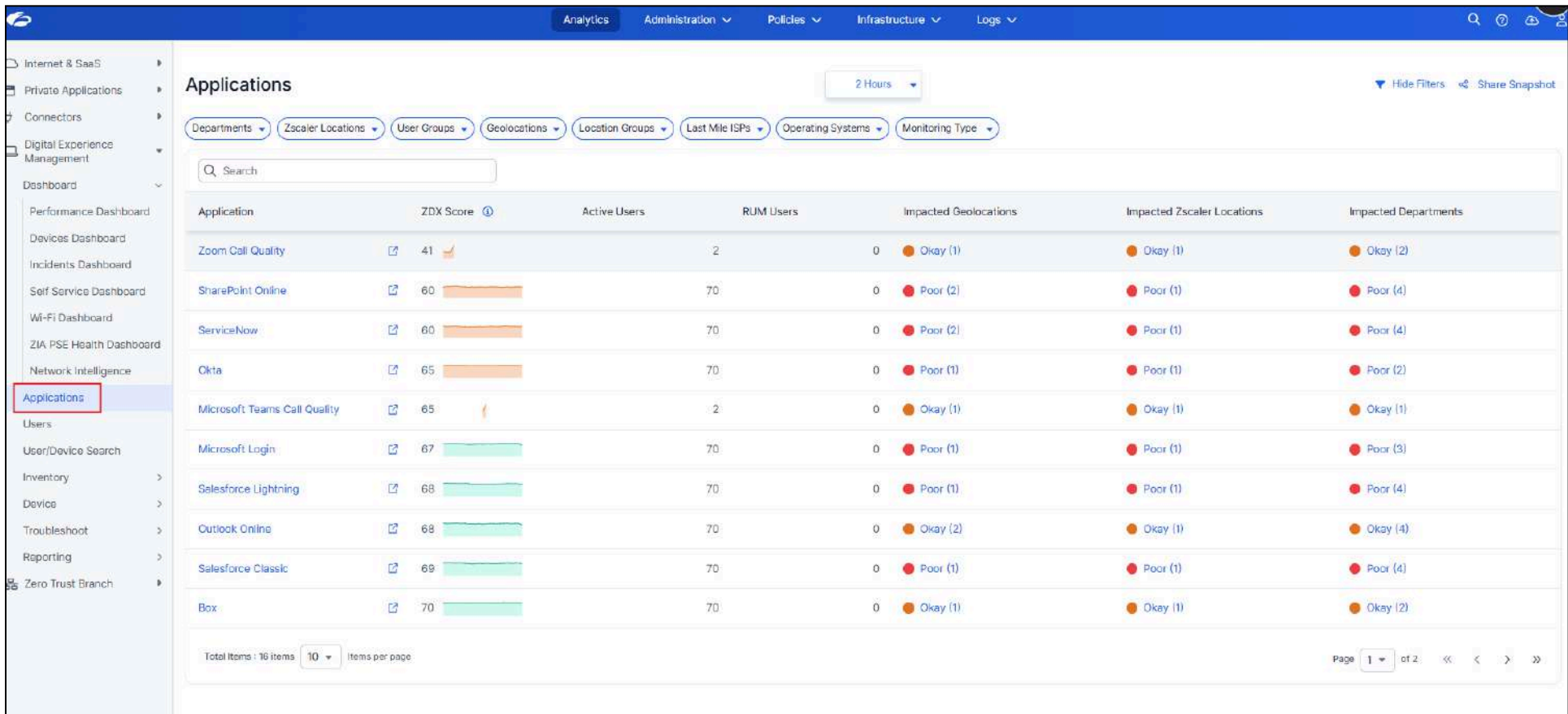
b. Explore the summary graphs about notifications in the selected timeframe.



Task 14.3: View Application Overview Dashboard

In this task, you will investigate predefined and custom application metrics to identify problem patterns and potential root causes for user or device issues.

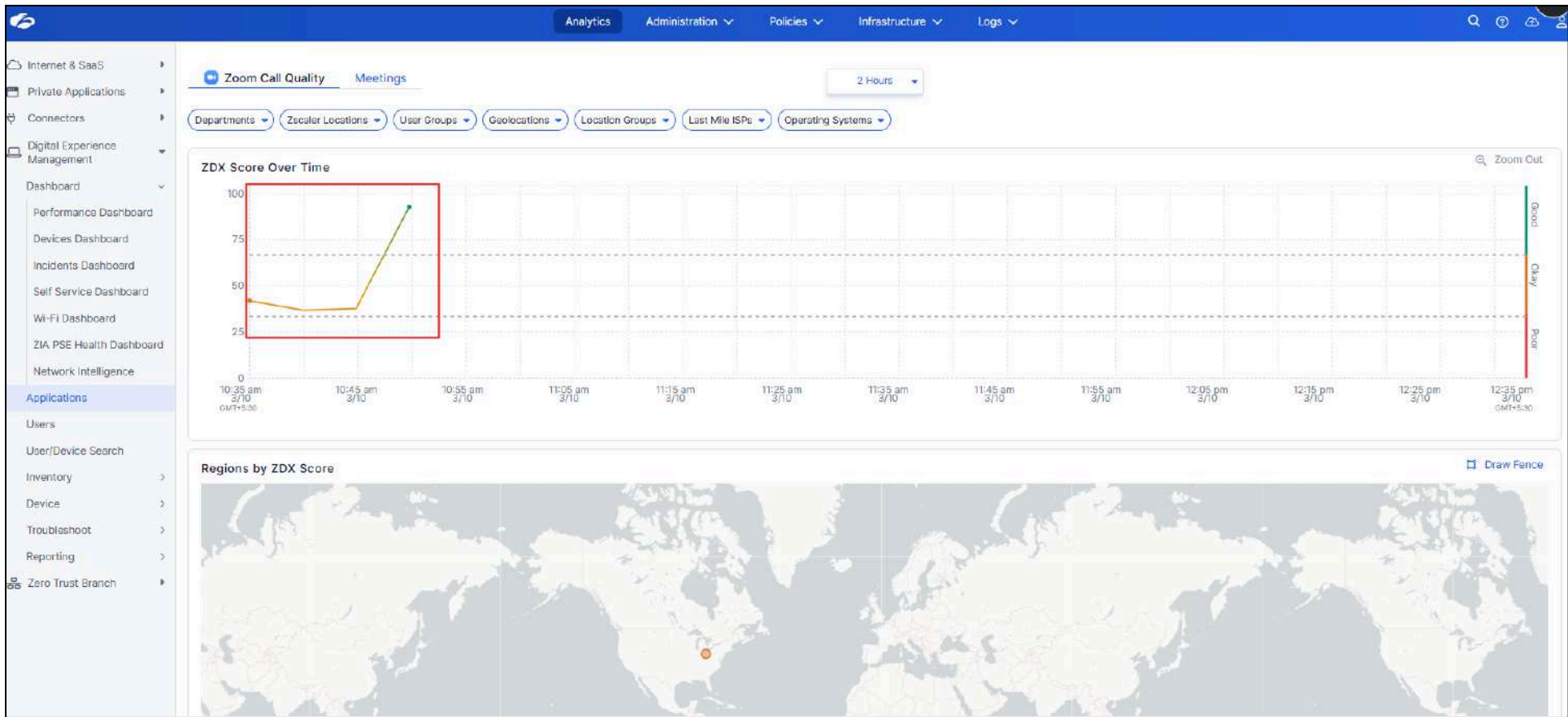
- 1. Navigate to Applications Overview:
 - a. In the left **navigation panel**, go to **Applications** or in the **Search** option type **Applications**.



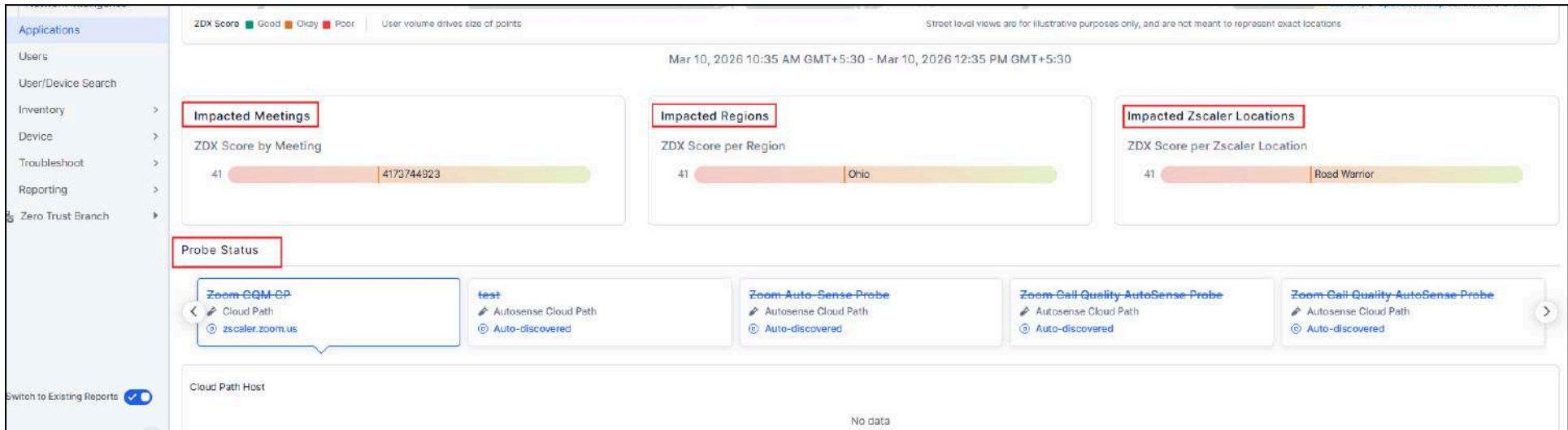
- 2. Identify the Lowest Score:
 - a. Sort the applications table based on **ZDX Score Trend**.
 - b. Click the **application** with the **lowest ZDX Score**.

Note: For different applications you will have different dashboard views, for e.g in UCAAS applications you will not be able to view the Page Fetch Time tab.

- 3. Scope a Time Period:
 - a. In the **ZDX Score Over Time** graph, drag your mouse to zoom in on a time period with a significant change.



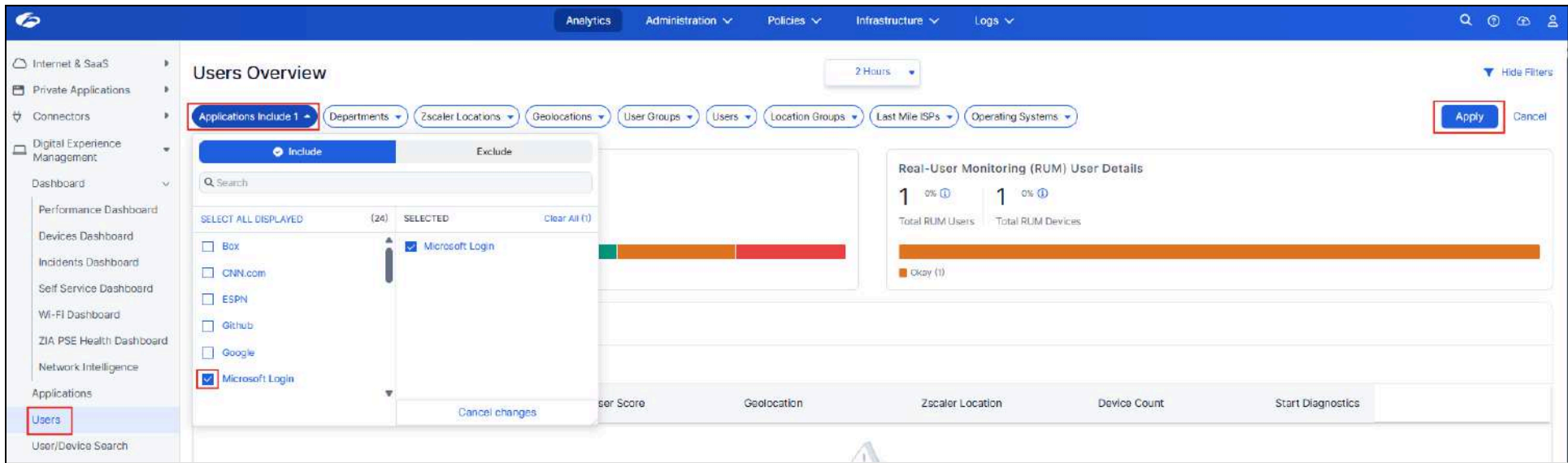
- b. Click **Zoom out**.
- 4. View Impact Dimensions:
 - a. Scroll down to view the impacted **Meetings**, **Regions**, and **Zscaler Locations**.
 - b. View information about the application's configured **Probe Status**.



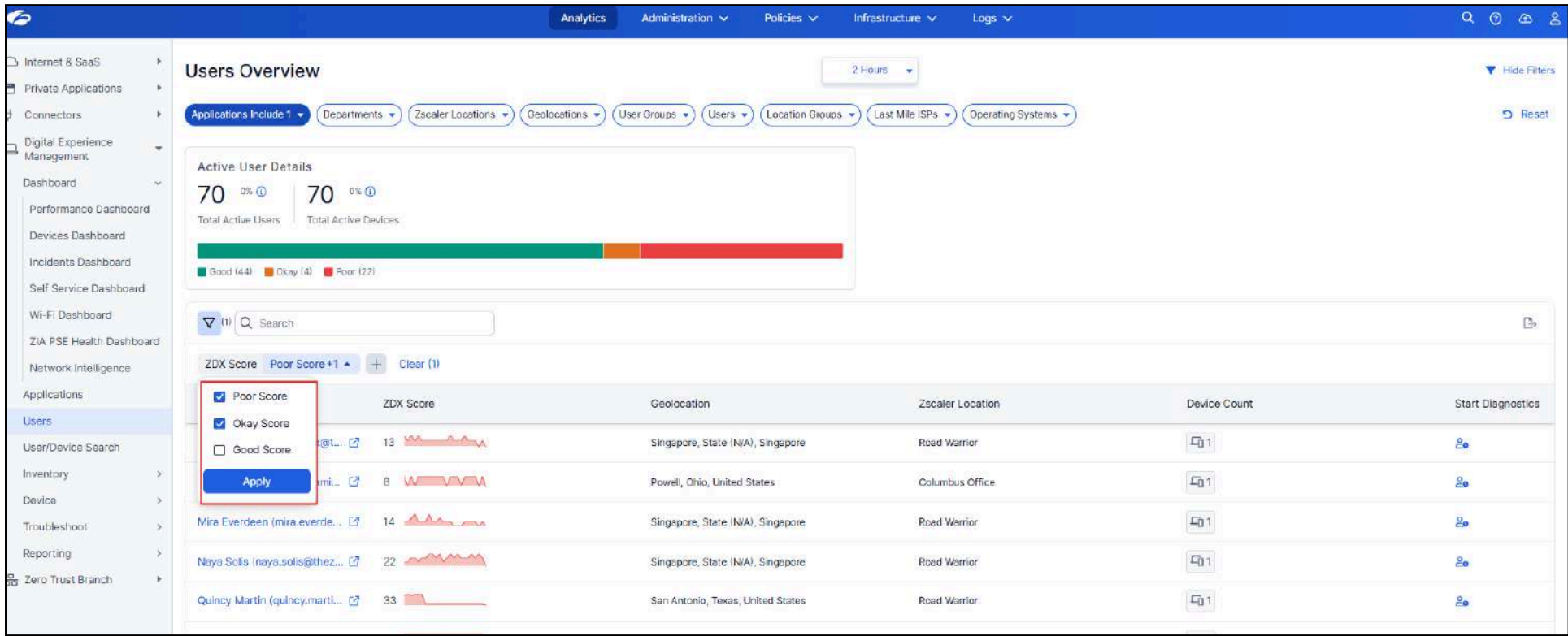
Task 14.4: View User Overview Dashboard

In this task, you will analyze user-specific metrics and explore reports to troubleshoot and improve user experiences.

- 1. Navigate the Users Overview Dashboard
 - a. From the navigation bar, click **Users**.
- 2. Scope by Application:
 - a. Select an application from the dropdown list, and then click **Apply**.

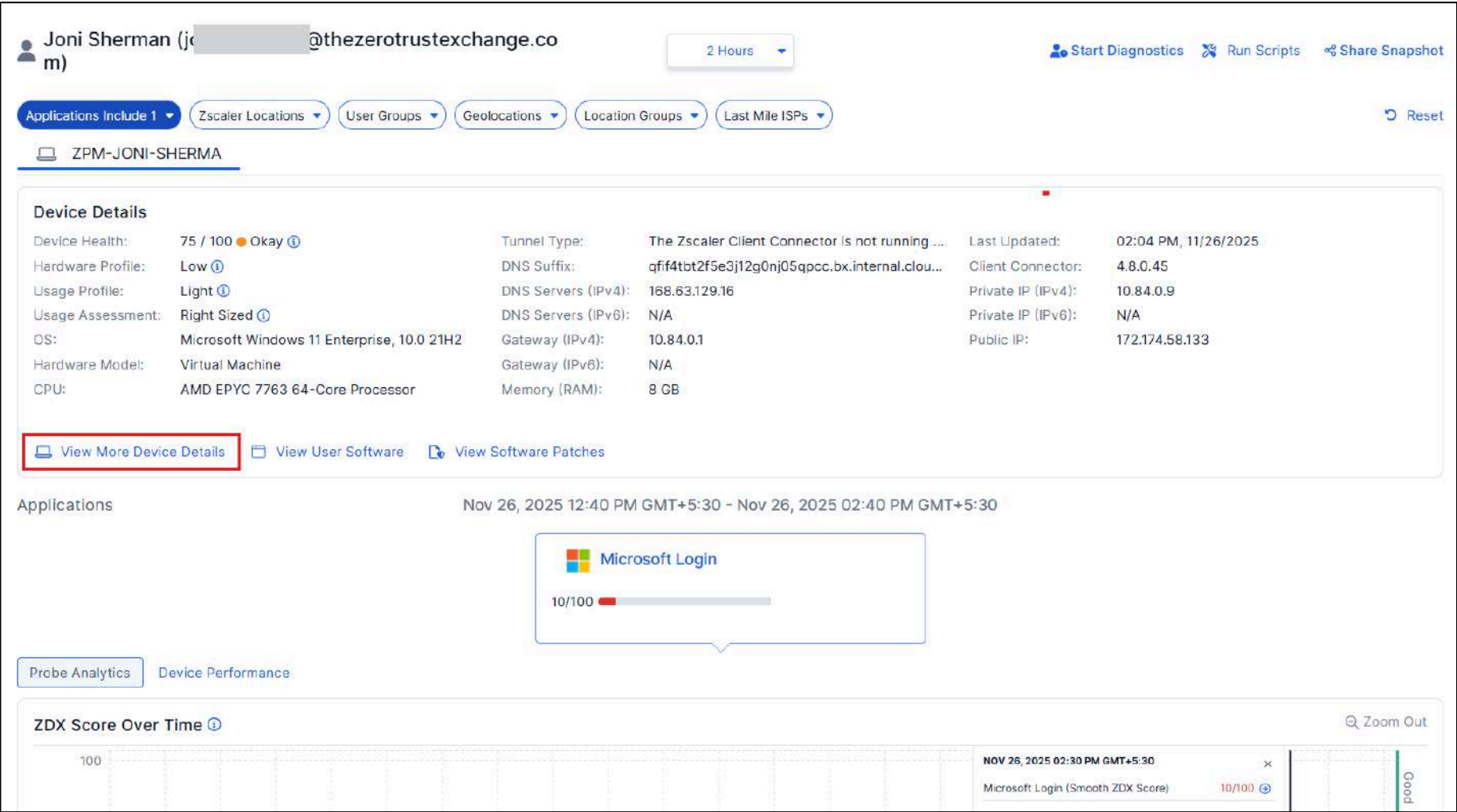


- 3. Find Impacted Users:
 - a. You can apply filters in the **ZDX Score drop down (Poor, Okay, and Good)** to filter the users' table.
 - b. Select a user with either **Poor** or **Okay** performance for deeper analysis.



4. Review User/Device Context:

- a. Review information such as **OS type** , **IP address information**, and other device-specific metrics.
- b. Click **View** of any users and click **View More Device Details**.



c. Analyze details under the **Hardware**, **Network** and **Software** tabs. Then click **Done**.

5. Review Application Performance for this User:

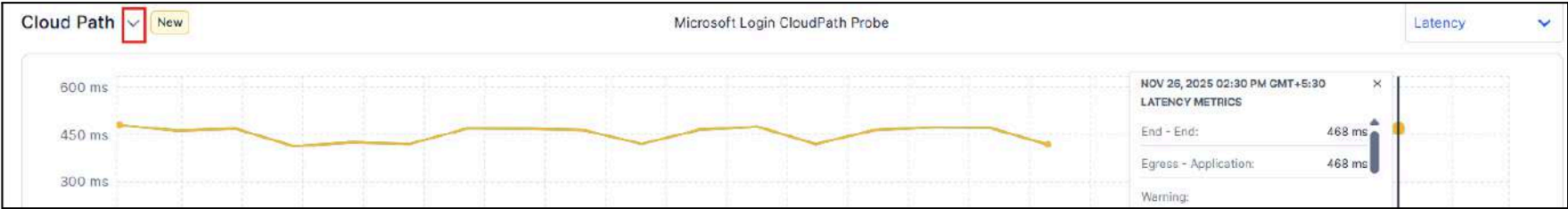
a. View the selected application's **ZDX Score Over Time** graph for that user.



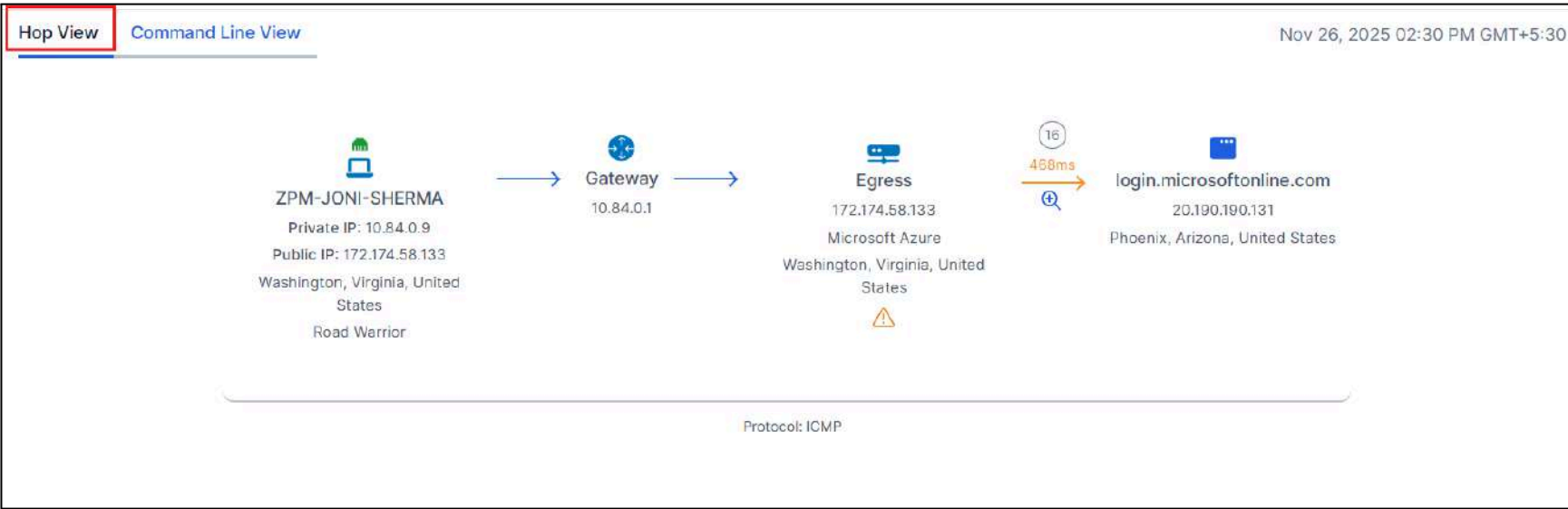
6. Review Web Probe Metrics:
- a. Scroll down to the **Web Probe Metrics** graphs.
 - b. Click to select/unselect web probe graphs as needed.
 - c. Click anywhere on a graph to get more details about spikes in **Page Fetch Time**, **Server Response Time**, etc.



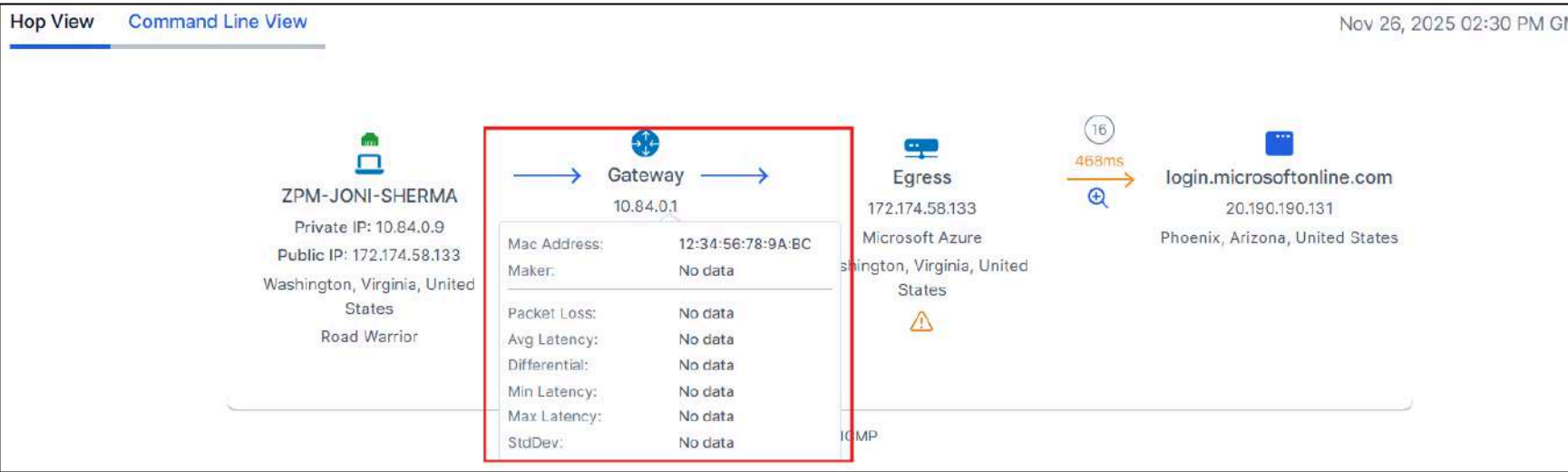
7. Review Cloud Path Metrics:
- a. Scroll down to the **Cloud Path Probe Metrics** graphs.
 - b. View the **Cloud Path** graph for **Latency** and **Packet Loss**.
 - c. Select/unselect any of the **Cloud Path legs**, as needed.



d. Scroll down to the **Hop View** and expand the view for the leg with the **highest differential latency** (shown in **Orange**).



e. **Mouse-over** a hop to get more information.



f. Click the **Command Line View** and identify hops with high packet loss and/or latency.

Hop View

Command Line View

Nov 26, 2025 02:30 PM GMT+5:30

	IP Address	Hop Direction	Service Provl...	Region	Geo	ASN	Assignee	Packet Loss	Packets Faile...	Differential	Averag
1	10.84.0.9	Client	-	Washington, Virginia	United States	-	-	-	-	-	-
2	10.84.0.1	↓	-	-	-	-	-	100%	10/10	-	-
3	172.174.58.133 ⚠	Egress	Microsoft Azure	Washington, Virginia	United States	8075	Microsoft Azure	-	-	-	-
4 - 19	No Response (16)	↓	-	-	-	-	-	100%	10/10	-	-
20	20.190.190.131	Application	Microsoft Azure	Phoenix, Arizona	United States	8075	Microsoft Azure	0%	0/10	468	468

NOTE: ZDX probes collect application performance data. Predefined applications include built-in probes by default, while custom applications require you to configure probes manually. Probes provide both baseline and real-time visibility into where a problem exists before user impact escalates. Troubleshooting in ZDX is probe-driven: probes generate the performance data that allows administrators to pinpoint the exact hop or layer causing degradation, rather than relying on user complaints or assumptions. Understanding how probes work is essential for correctly interpreting ZDX metrics, trends, and alerts, and for performing effective root-cause analysis.

Lab 15: Troubleshoot Poor Wi-Fi

Scenario: As the Zscaler administrator, you are responsible for resolving a poor user experience issue reported by a specific employee. The user has experienced **frequent disconnections, slow internet speeds, and difficulty completing work tasks**, which has negatively impacted their productivity. **Initial insights suggest that the issue may stem from poor Wi-Fi performance, but the details remain unclear.** Using **Zscaler Digital Experience (ZDX)**, your role is to analyze performance trends to pinpoint the root cause. You will troubleshoot the issue and restore the user's digital experience.

Objectives: By the end of this lab you will:

- **Filter** ZDX dashboards to isolate **user-specific device and Wi-Fi performance** data during the incident window.
- **Use** the **ZDX Y-engine Analyze Score feature** to find the factors that contribute to ZDX score degradation.
- **Compare Cloud Path data** and diagnose a degradation of ZDX score to pinpoint root causes.
- **Analyze Device Events** associated with ZDX score changes to identify device events that impact user experience.

Approximate Time: 15 minutes

User: **Curtis Hardwick**

Device: **PC-Curtis-Wi-Fi**

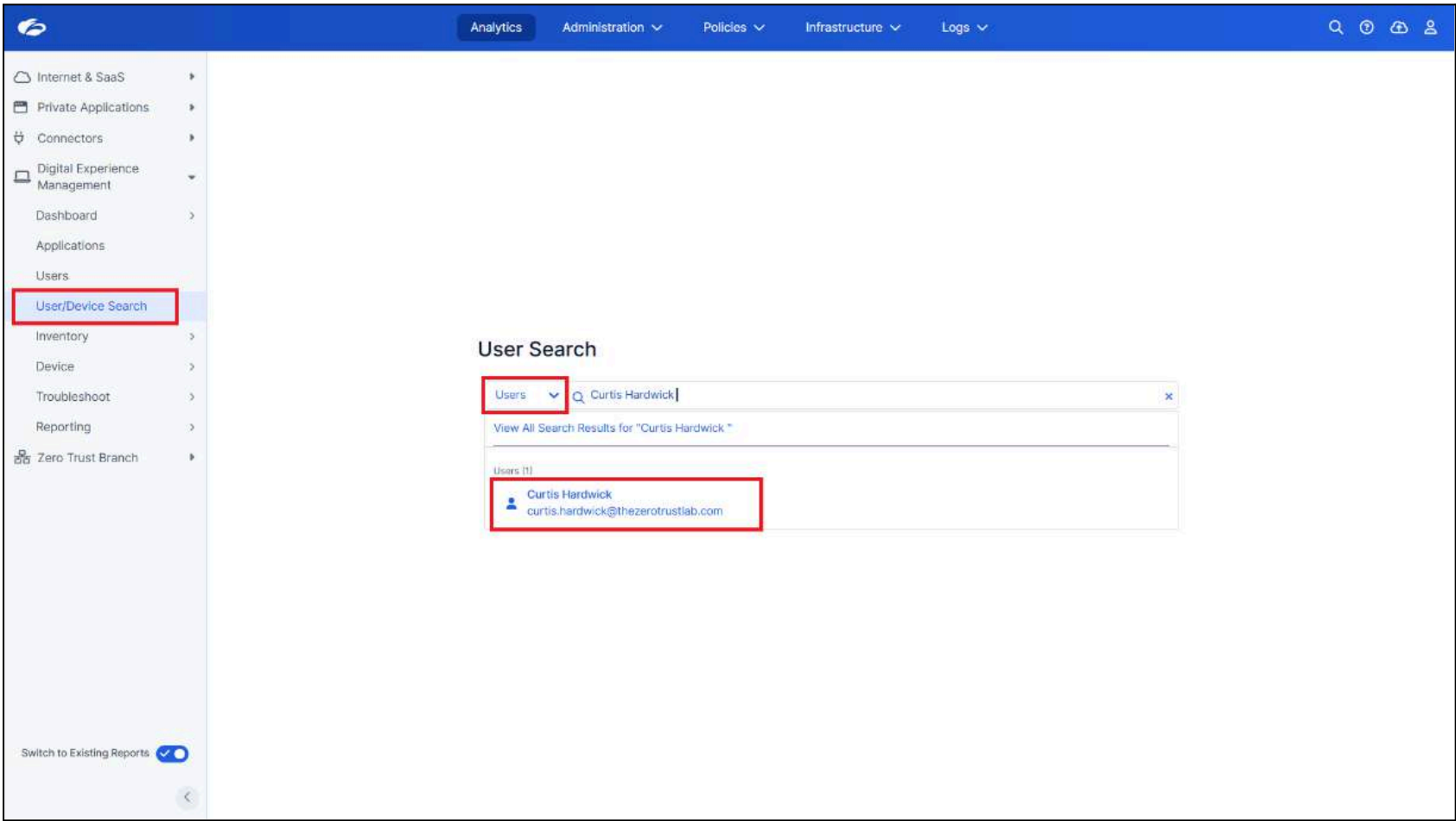
Time Range: **Last 48 Hours**

Highlights: **Poor Wi-Fi Signal, SSID Changing, Y-Engine Comparison**

Task 15.1: View User ZDX Score for an Application

In this task, you will search to see ZDX score data for a user for an application covering the time period that they have been having issues. To view the user ZDX score for an application, follow these steps:

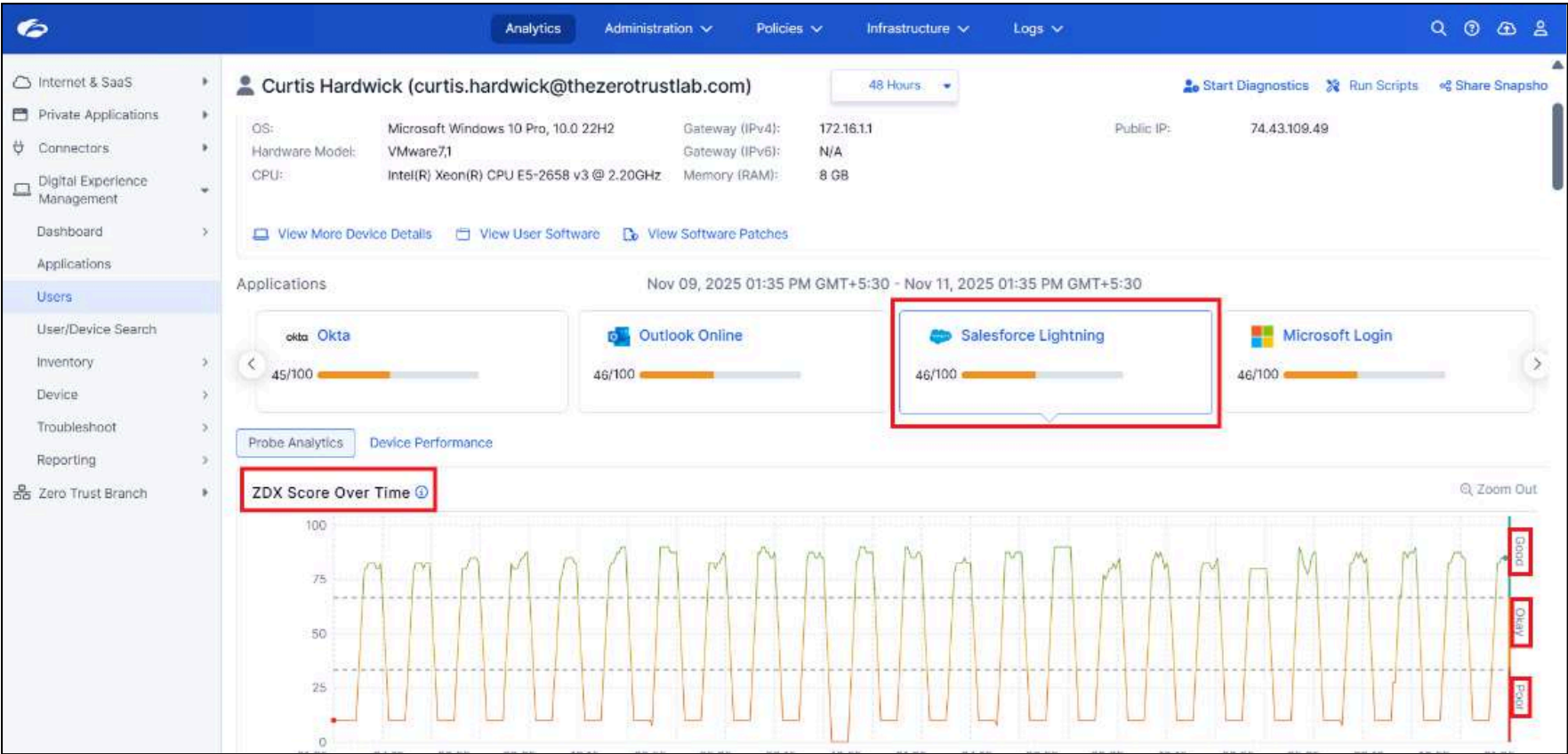
1. Access User / Device Search for the Affected User:
 - a. From the **Experience Center**, go to the **Search** option and type **User/Device Search (Analytics > Digital experience Management > User /Device Search)**
 - b. In the dropdown, select **User Search**.
 - c. In the search box, type: **Curtis Hardwick**
 - d. In the search results, click **Curtis Hardwick** to open the user page.



- 2. You will be taken to the **Users** page of Curtis Hardwick.
- 3. Note that you will need to **Reset** the filter you applied in preceding task 14.4, before viewing the user dashboard.



- 4. Filter by Time Range and Application:
 - a. View the ZDX score for an application (e.g., **Salesforce Lightning**) for the period of the incident.
 - b. Select the time-period for the incident (e.g., **48 Hours**).
 - c. Select an application (**Salesforce Lightning**).
 - d. View the **ZDX Score Over Time** graph.



e. Observe that the ZDX score is fluctuating often between **Good** and **Poor**.

Note: You may have to select custom options from the reporting timeframe to see meaningful data.

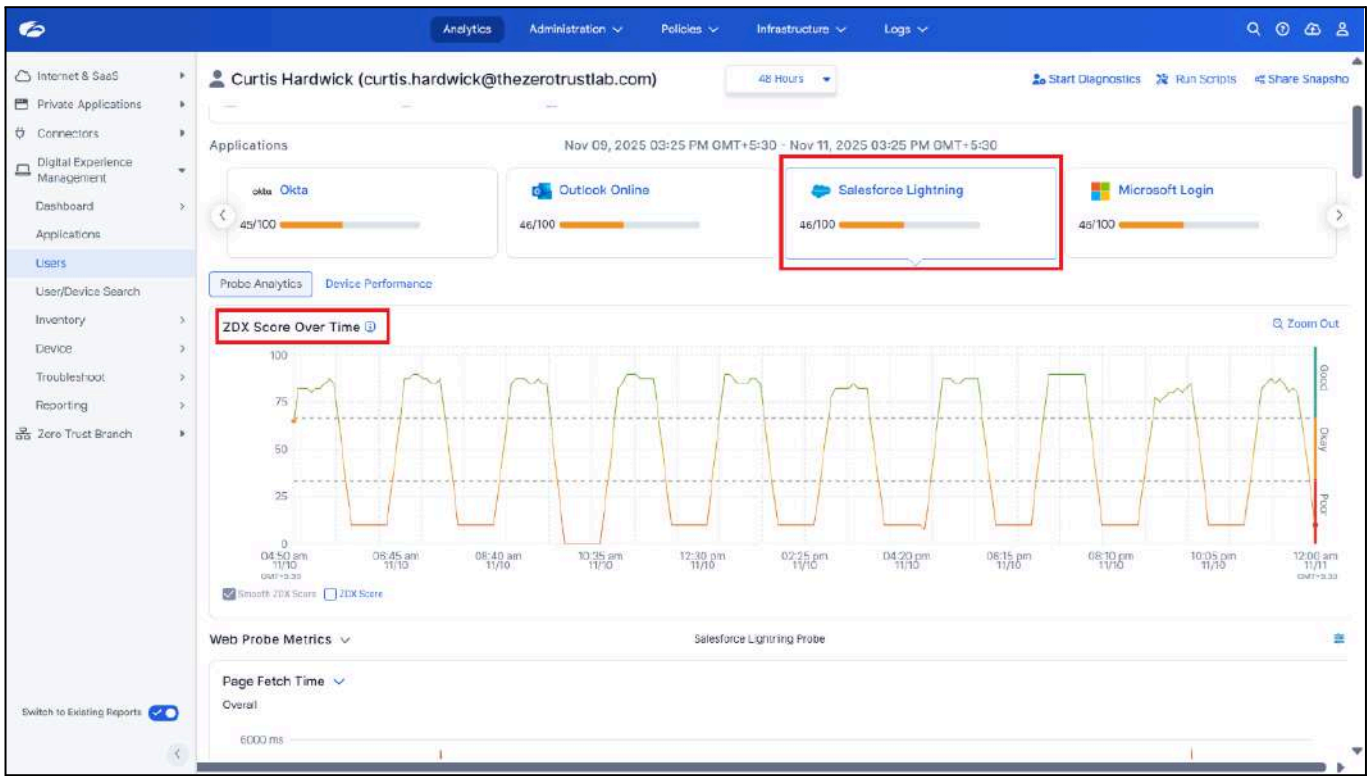
Task 15.2: Analyze Poor ZDX Score Interval

In this task, you will zoom into a period where the ZDX score is Poor, and use the Analyze Score capability of the ZDX Y-Engine to identify key factors that contributed to the degraded score.

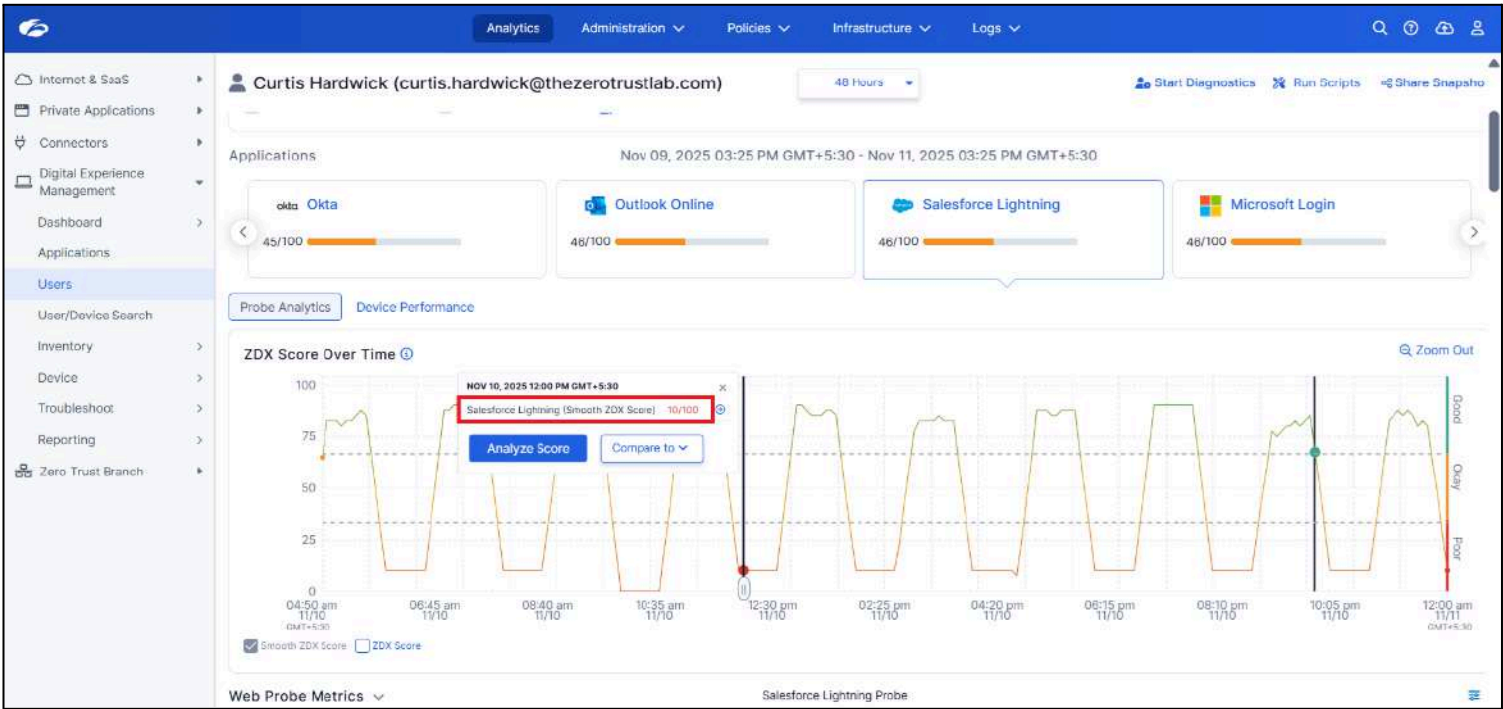
Note: Your results will differ from any screenshots provided in the lab guide. Traffic in the lab is simulated, so the exact values and indicators will vary based on the time range and sample you select. Feel free to repeat these steps with different Poor score points to explore multiple scenarios.

- 1. Zoom into a poor score interval:
 - a. In the **ZDX Score Over Time** graph, locate a time period where the ZDX Score is **Poor**.

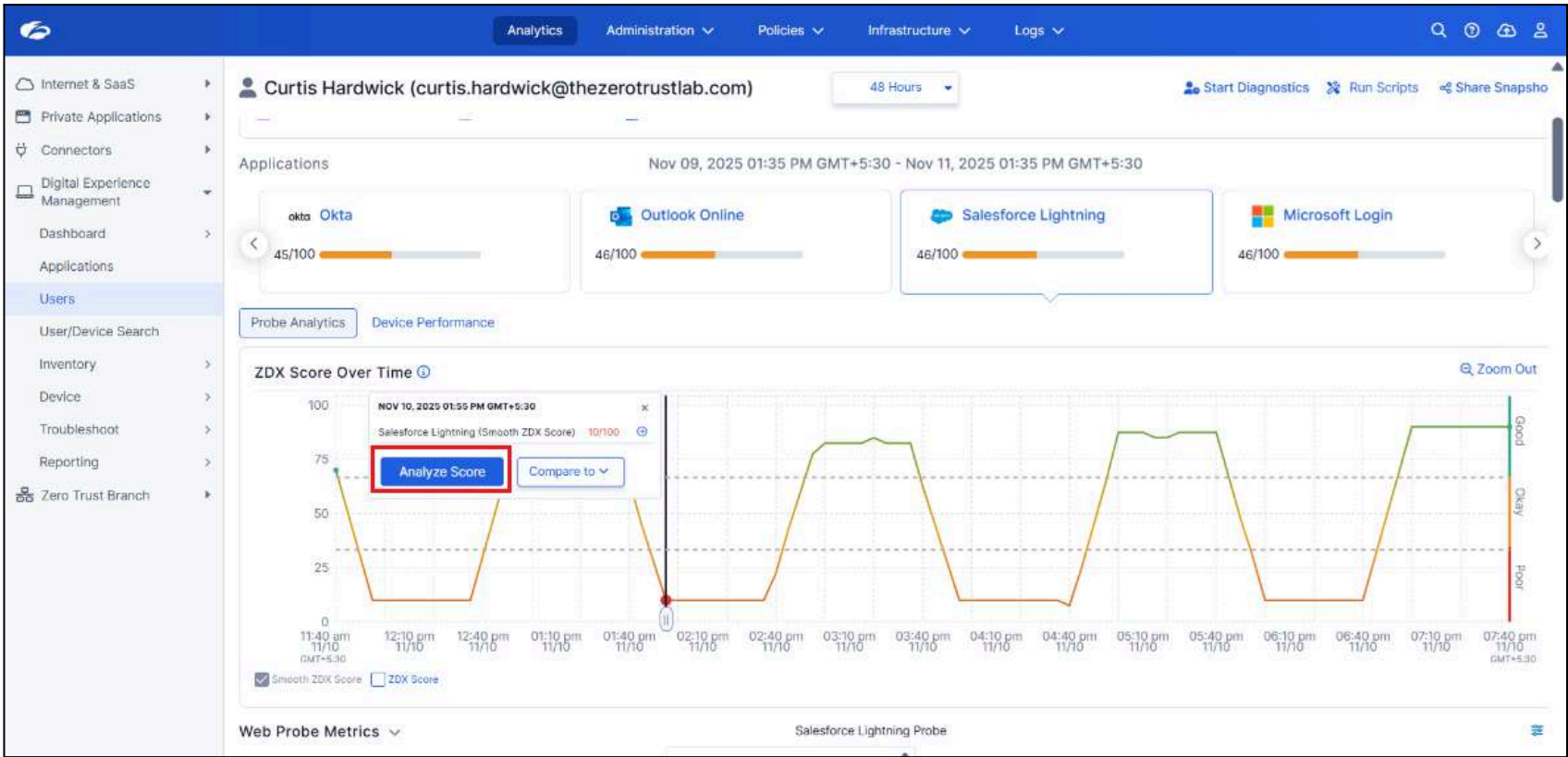
Note: For best results ensure that the selection includes time **before and after** the Poor score period.



- 2. Run Y-Engine Analyze Score:
 - a. Click to set the cursor on a **point on the graph with a Poor score**.
 - b. View the **ZDX Score** displayed for that sample.



- c. Click **Analyze Score**. Analysis will take a few seconds to complete.



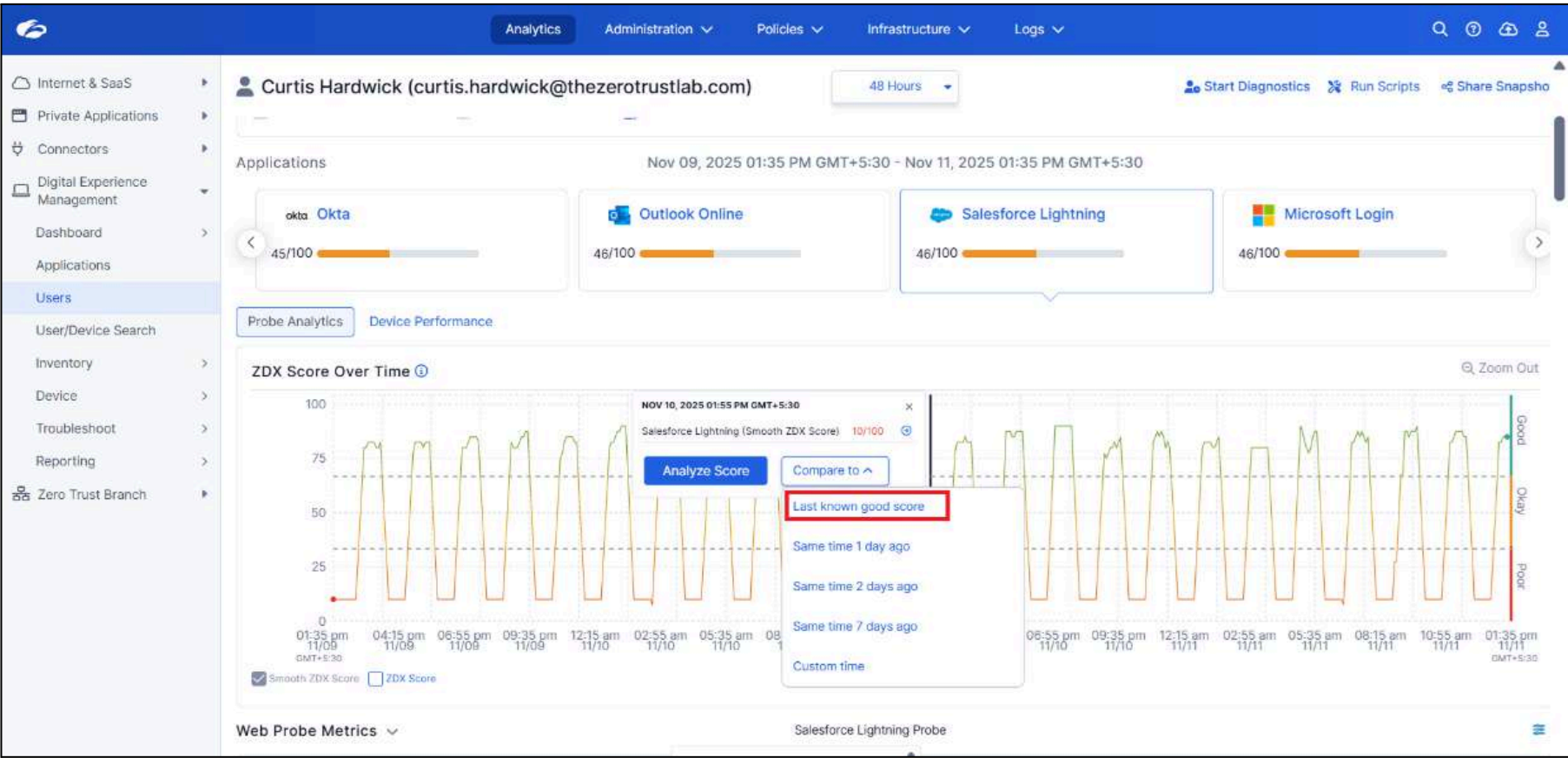
d. View the analysis results table titled the following factors that might have impacted the ZDX score.

Note: In this example, the score of **10/100** most likely came from a Low Wi-Fi Signal, and the Suboptimal Wi-Fi degraded user performance after connecting to the 3Com_Wifi SSID.

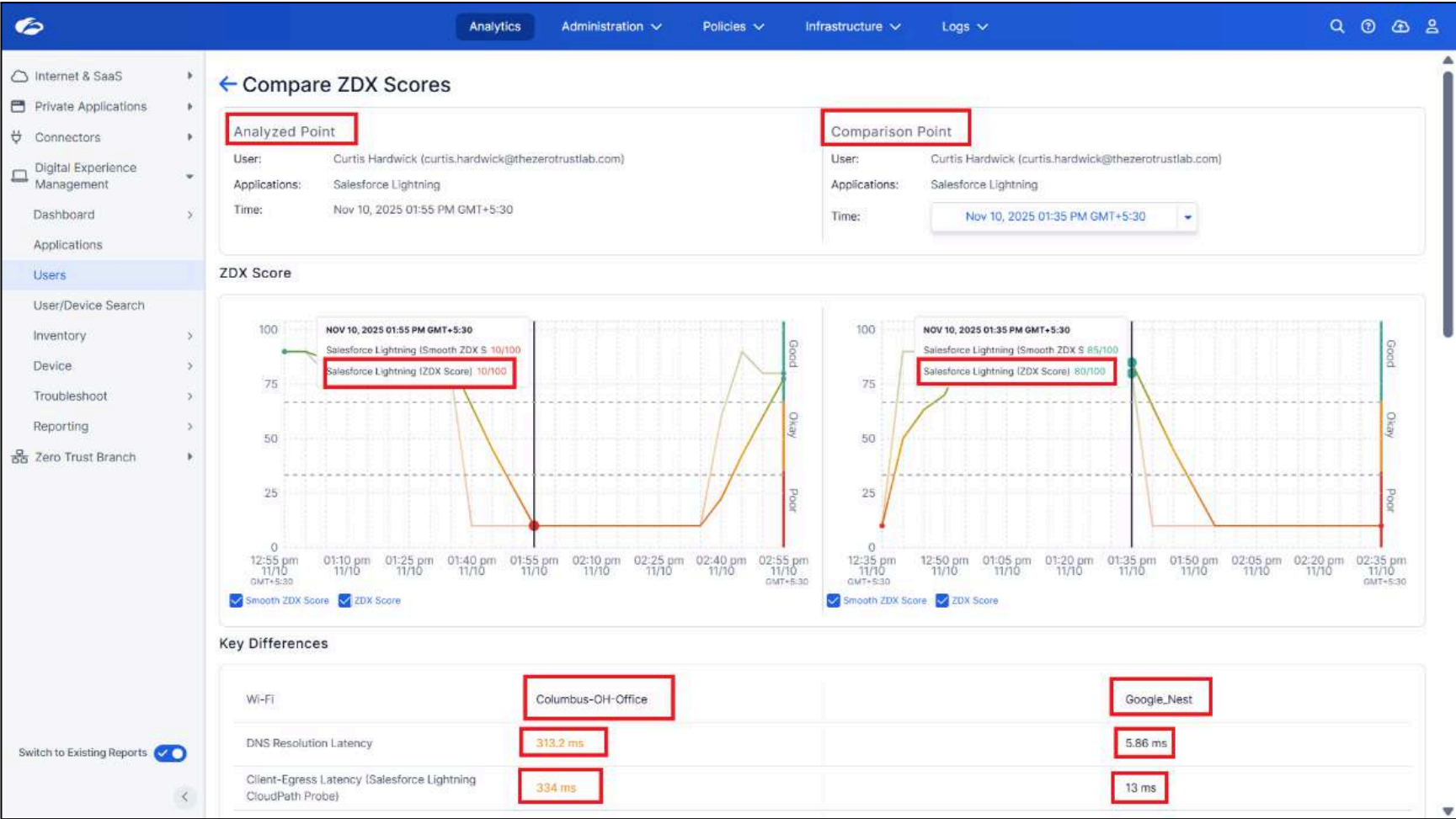
Task 15.3: Compare Poor ZDX Score to Last Known Good Score

In this task, you will compare a Poor ZDX score sample with the last known Good score sample using the ZDX comparison tool.

- 1. Compare Poor vs Last Known Good ZDX Score:
 - a. Click **Compare** to for the selected point on the ZDX Score Over Time graph.
 - b. Select **Last known good score**.



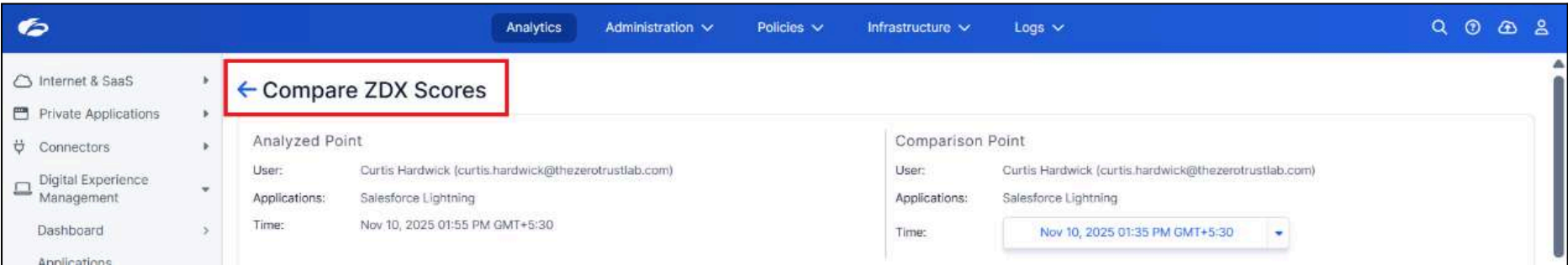
- c. Review the comparison panel that appears, focusing on the differences between:
 - i. **ZDX Score** for the Analyzed Point vs the Comparison Point.
 - ii. **Wi-Fi identifier**
 - iii. **Client-Egress Latency**



Task 15.4: Isolate High Latency Hop

In this task, you will use the Cloud Path Hop View to pinpoint where high latency occurs along the path between the user's device and the application.

1. Return to the user report and select the application:
 - a. Select the **Back** option next to **Compare ZDX Scores** to return to the user report
 - b. Confirm you are still viewing the affected user : **Curtis Hardwick**
 - c. Select the application **Salesforce Lightning**.

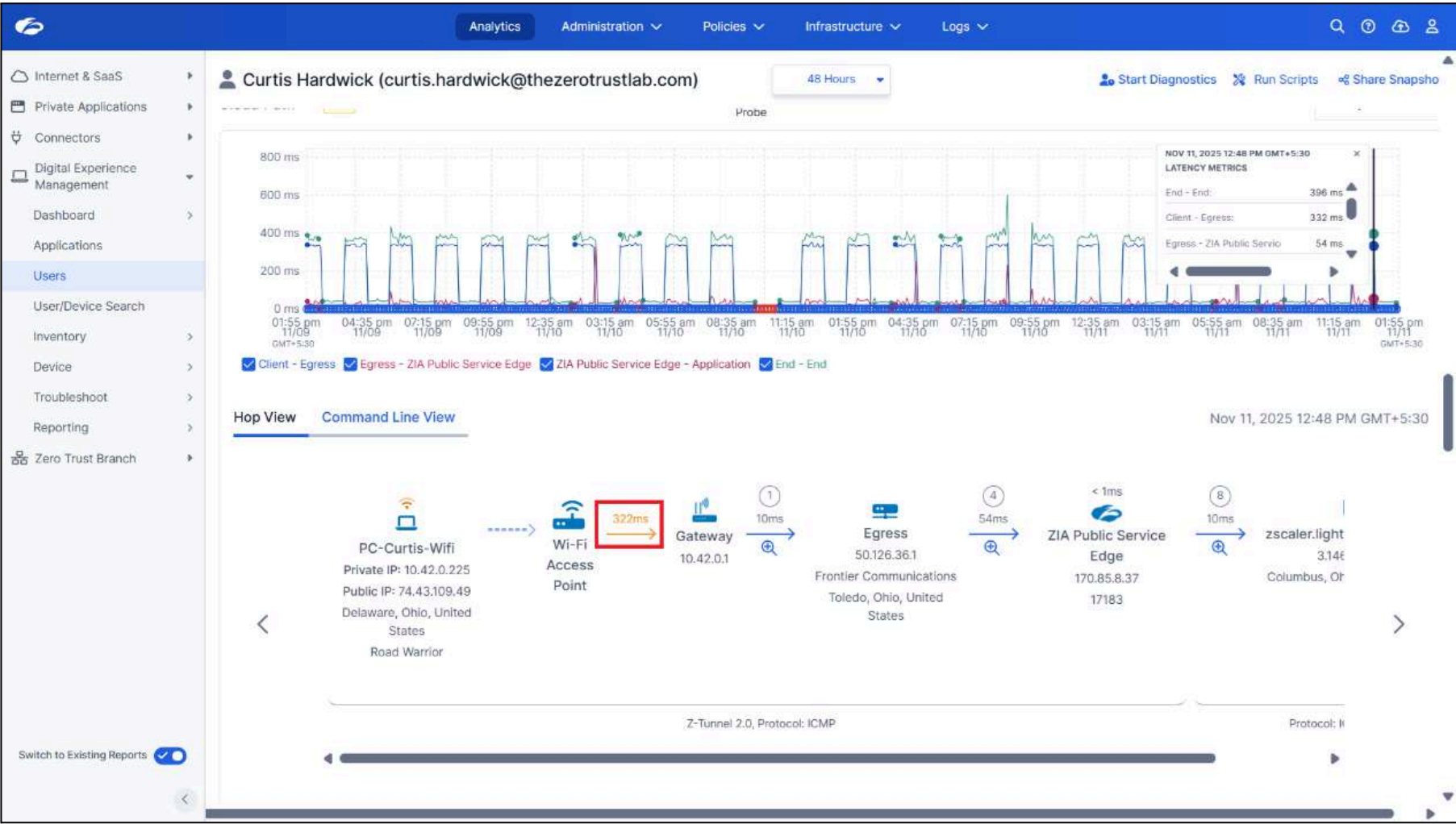


2. Focus on a Poor ZDX Score interval:
 - a. In the **ZDX Score Over Time** graph, zoom in again on a time period where the ZDX Score is **Poor**..
3. Review Cloud Path Latency:
 - a. Scroll down to view the **Cloud Path** graph and latency values at the selected data point.

Note: Results will vary, but you most likely see low latency at the beginning of the selected time period for when the ZDX score was Good, and it will likely change to **high latency** when the score drops to Poor.

4. Examine Hop View to find the Problematic Hop:
 - a. Below the Cloud Path graph, locate the **Hop View** visualization.
 - i. Hop View displays each detected hop between the client and the server for the selected application and sample, along with the latency measured at each hop.
 - b. Examine the hops and identify which hop is contributing the most to **end-to-end latency**.

Note: If the network connection was lost there will be no data available to report on, and the Hop View will not be able to be displayed. Pick a different point in time where data is showing in the graph to analyze.



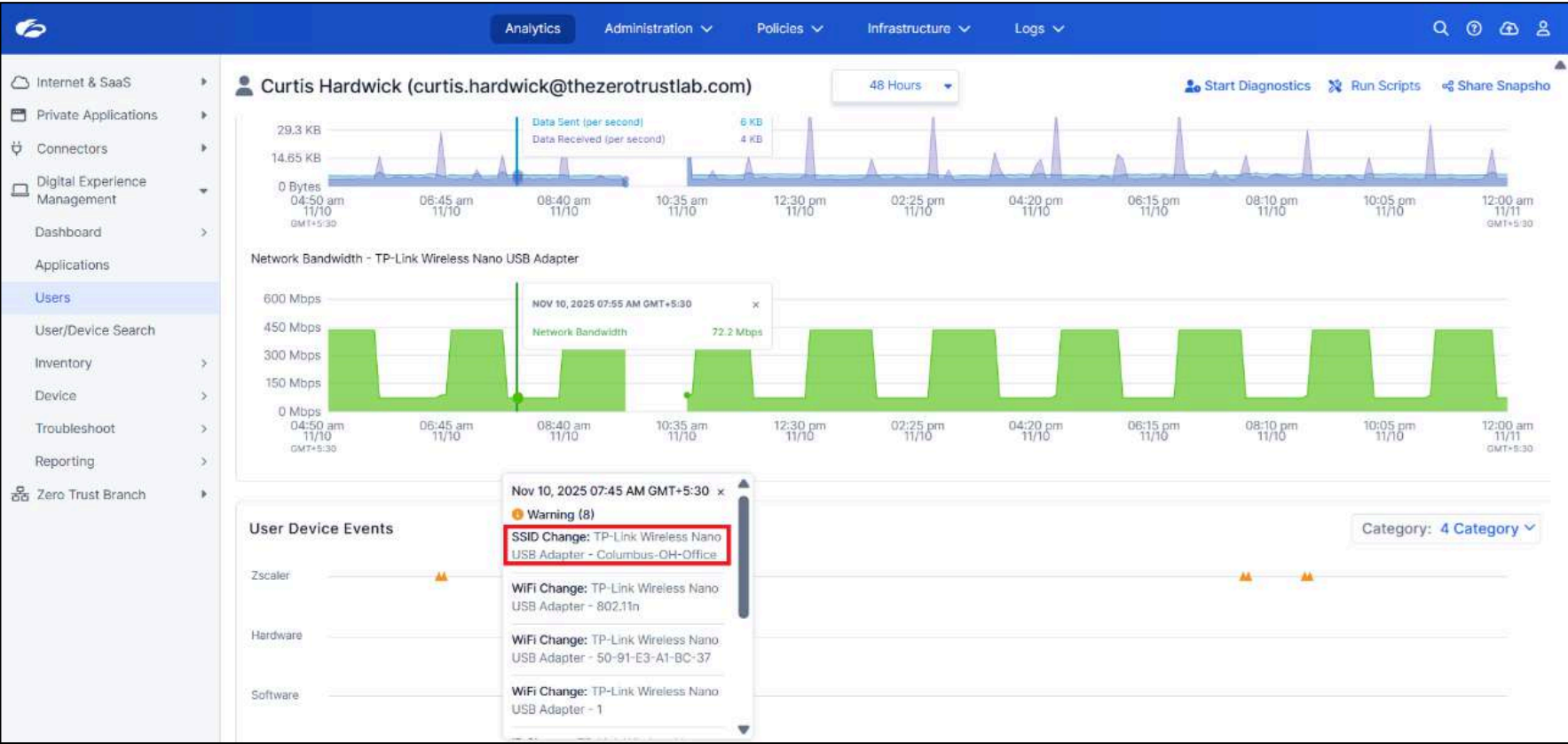
Note: In this example, the Hop View isolates the slow end-to-end latency to the first hop between the user's device and their local gateway.

Task 15.5: View User Device Events Associated with ZDX Score Changes

In this task, you will examine User Device Events and correlate them with ZDX score changes.

1. Access User Device Events timeline:
 - a. From the same user view and time window you used in the previous tasks, **scroll down** to the bottom of the page to locate the **User Device Events** timelines.
2. Correlate device events with the Poor score transition:
 - a. In the **Network** timeline, locate the point in time where the **ZDX Score transitioned to Poor** (based on the ZDX Score Over Time graph you analyzed earlier).
 - b. Mouse over the dot on the Network timeline at that corresponding time.
 - i. A pop-up will appear showing details for the Network Interface data recorded at that sample.
3. Review the details in the pop-up and scroll through the **Network Interface** fields.
 - a. Identify what changed at that time by looking at values recorded in the **New Value** field.

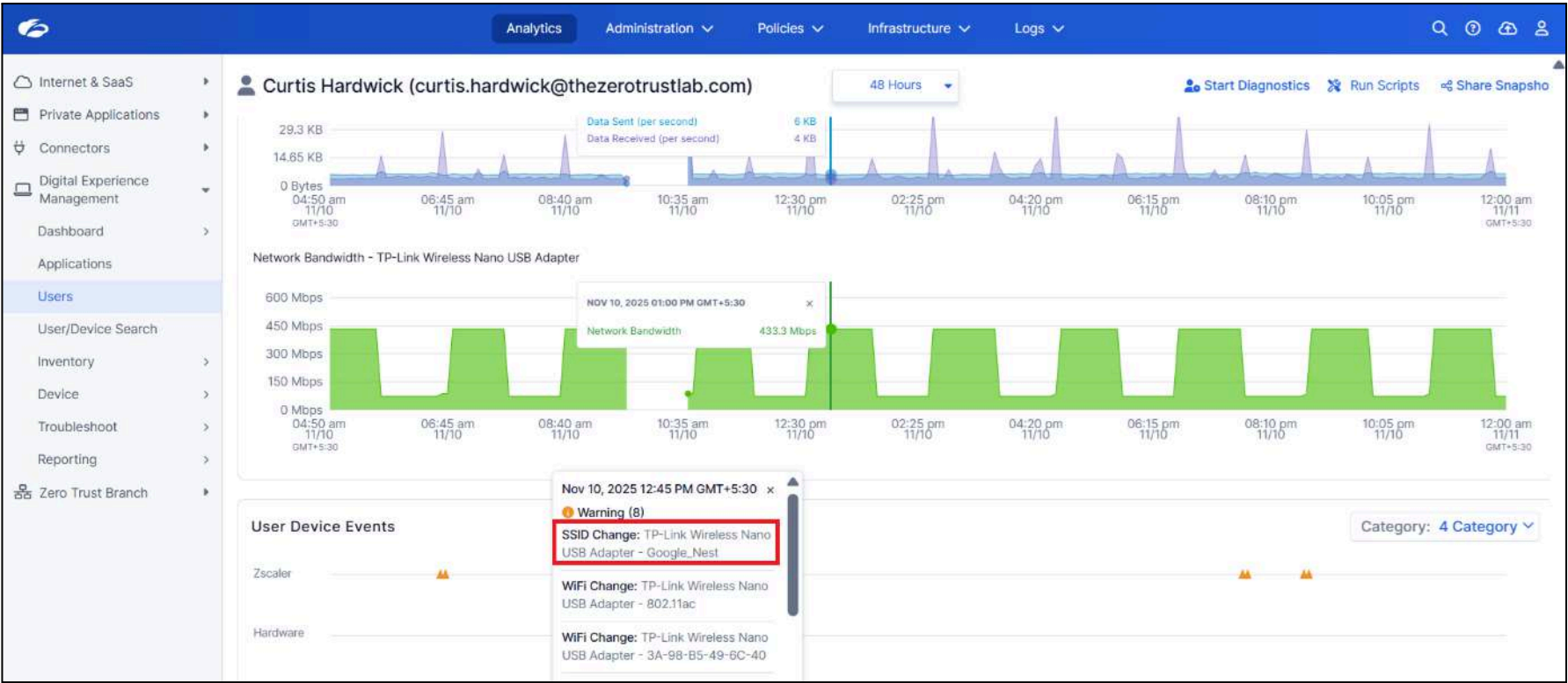
Note: In this example, the **SSID Change** from the **Google_Nest** to the **Columbus-OH-Office** SSID is associated with the **Poor** score transition.



4. Correlate Device Events with the Recovery Back to Good:
- a. In the same **Network** timeline, find the dot that corresponds to when the **ZDX Score transitioned back to Good**.
 - b. Mouse over that dot to view the Network Interface data for the recovery point.

Note: In the example scenario, switching back to the Google_Nest **SSID** is associated with a Good ZDX score, indicating that the user's experience improved when they returned to the original Wi-Fi network.

5. Interpret the impact of device events:
- a. Identify which **Wi-Fi or network events** (such as SSID changes, disconnects, or signal changes) occurred at the same time as score degradation or recovery.
 - b. Use this correlation to support your root-cause conclusion (e.g., **“SSID changes and poor Wi-Fi signal on PC-Curtis-Wi-Fi caused the user’s poor experience”**).



Lab 16: Troubleshoot Slow DNS Response

Scenario: A user, **Linda Lucas**, reports to the service desk that she is experiencing a consistently poor experience when accessing SaaS applications. **Zoom calls work fine once connected, but anything web-based is very slow.** Initial investigation suggests that the issue may be related to **slow DNS response times**, which increase page load latency and degrade the user's digital experience.

Objectives: By the end of this lab you will:

- **Use ZDX Copilot** to identify likely root causes of poor user experience.
- **View a user's overall ZDX score over time** for the incident period.
- **Analyze Page Fetch Time** and related **Web Probe Metrics** to identify latency patterns
- **Confirm** primary **root cause** using the **ZDX Y-Engine**.

Approximate Time: 15 minutes

User: **Linda Lucas**

Device: **HOST-2**

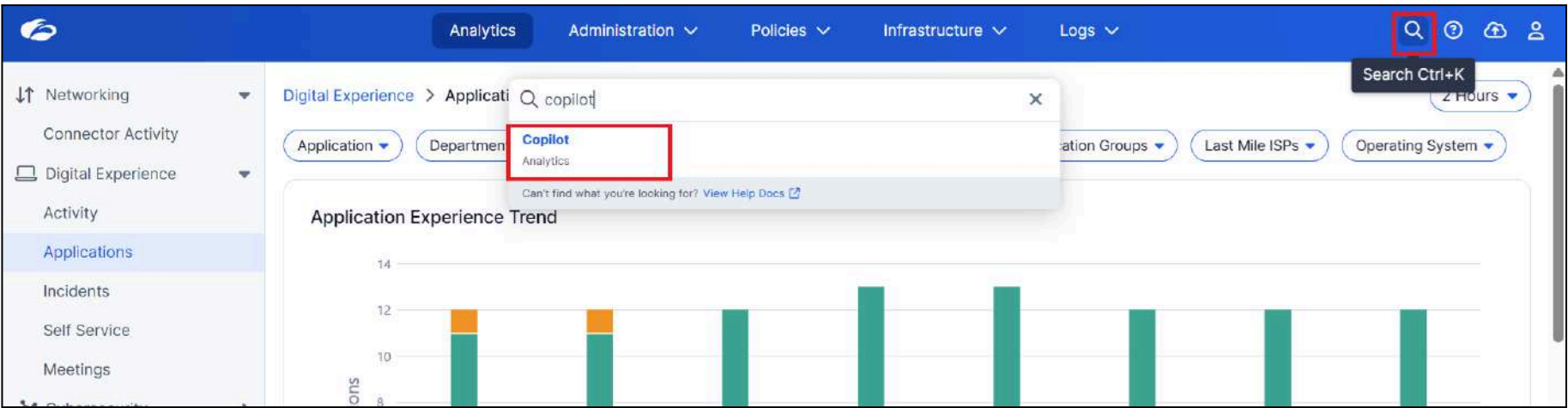
Time Range: **Last 48 Hours**

Highlights: **Poor App Experience, Slow DNS Response Times.**

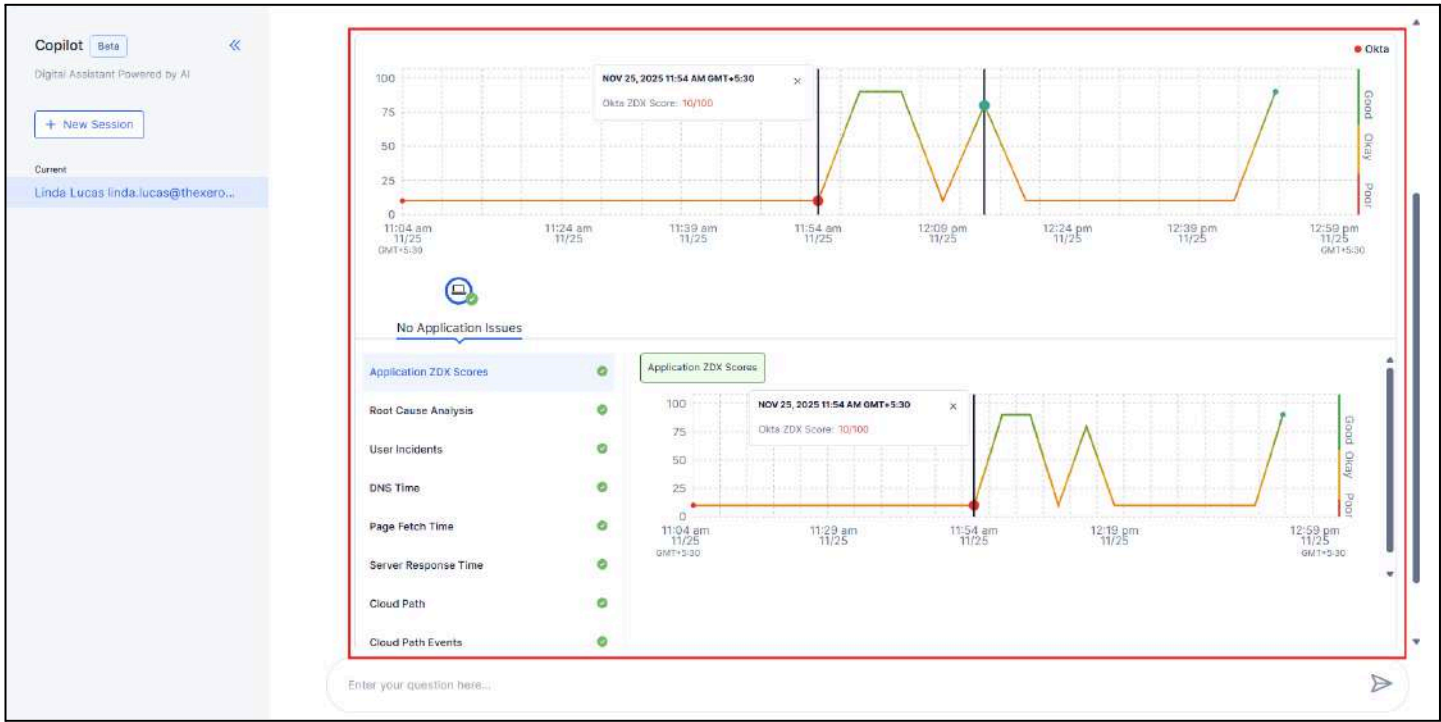
Task 16.1: Use ZDX Copilot for Root Cause Analysis

In this task you will use **ZDX Copilot** to understand why Linda is experiencing poor performance.

- 1. From the **Experience Center**, navigate to **Analytics > Copilot**.
 - a. Or, Use the **Search** option to navigate to **Copilot**.



- b. In the Copilot chat, enter the following prompt and click Send: ***"Linda Lucas linda.lucas@thexerotrustedlab.com reports to the service desk that she is having a consistently poor experience when accessing SaaS apps like okta. analyze the reason behind"***.
 - c. Review the response from ZDX Copilot, focusing on:
 - i. The **reason** it identifies Linda's poor SaaS experience.
 - ii. Any mention of **DNS, latency, or Page Fetch Time** as contributing factors.

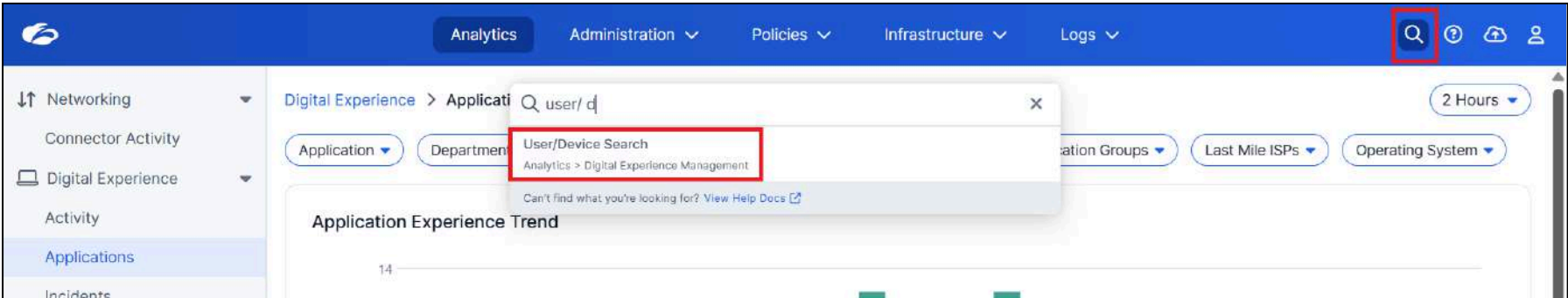


Note: In the next tasks, you will perform a **manual investigation** using ZDX dashboards to validate Copilot's findings and understand how ZDX derives that conclusion.

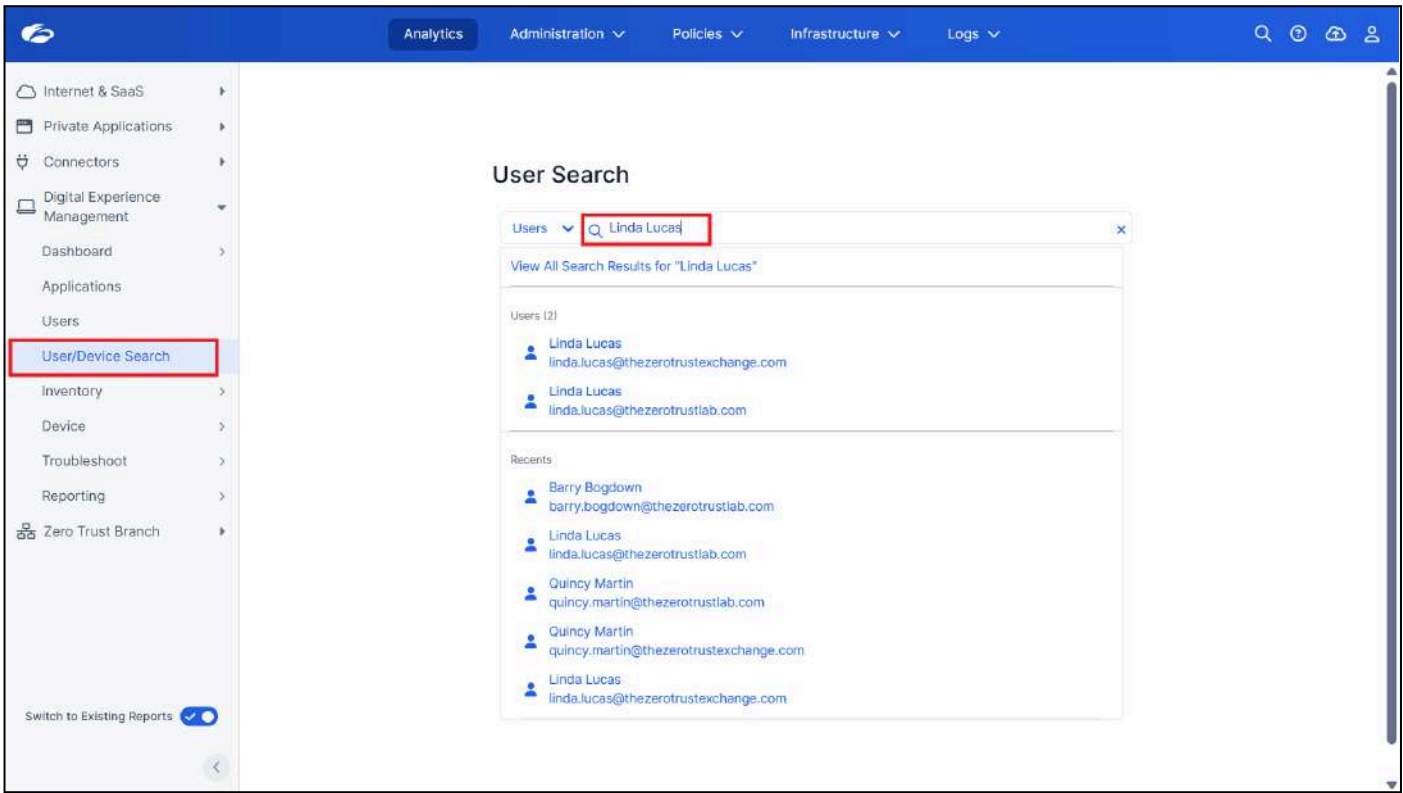
Task 16.2: View User ZDX Score Over Time

In this task, you will view Linda's overall ZDX score for all monitored applications over the period she has been experiencing issues.

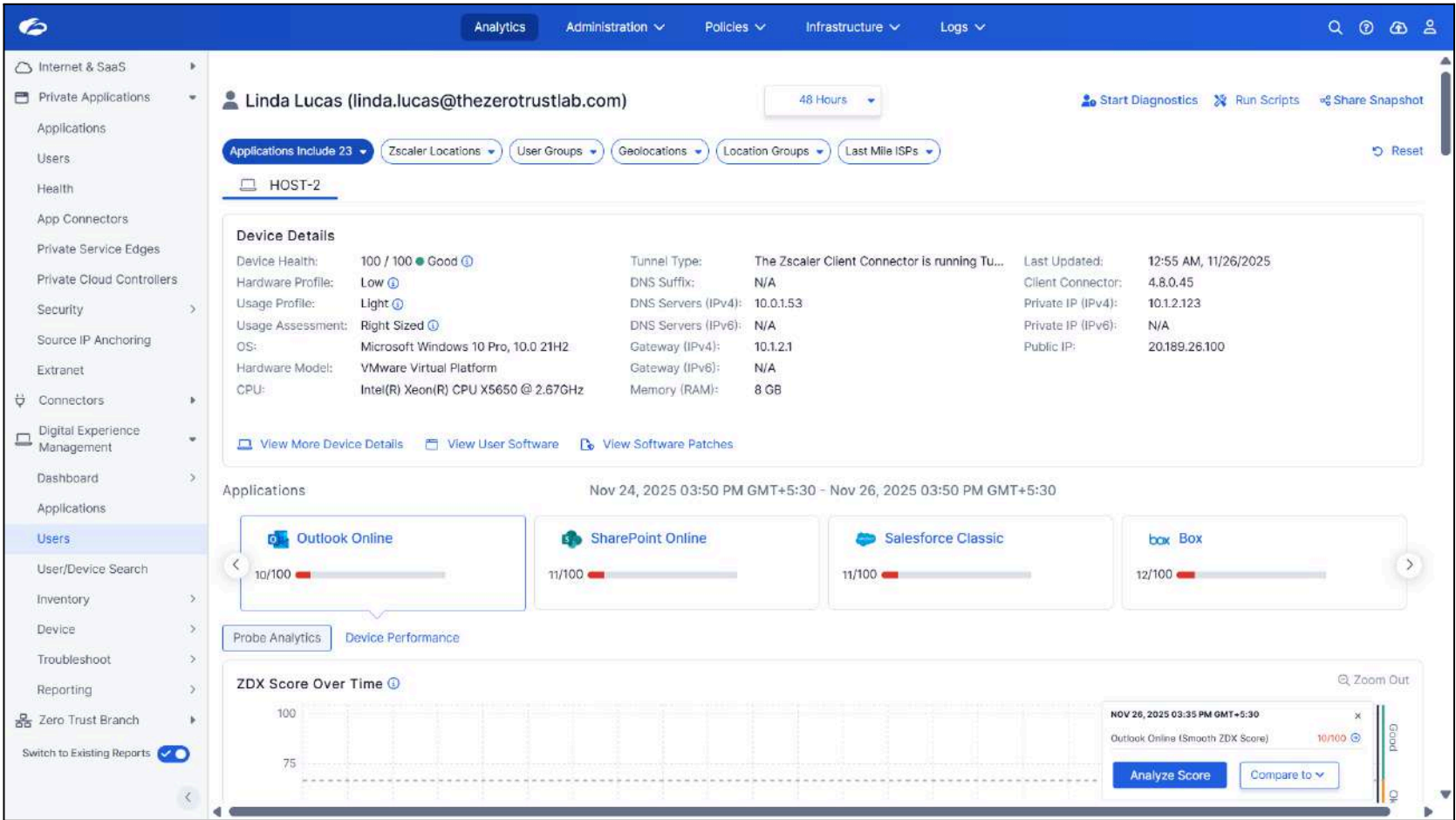
- 1. Navigate to the Analytics > Digital Experience Management > User / Device Search and search for:
 - a. User: **Linda Lucas**.



- b. Device: **HOST-2**.



- 2. View the ZDX score for the period of the incident.
 - a. Select the time period for the incident (e.g., **48 Hours**).
 - b. View the **ZDX Score Over Time** graph.



- 3. Observe the following example behavior:
 - a. The ZDX score is consistently Poor over the entire interval.
 - b. The score for every application is poor (13/100 or lower).

Note: You can also access the user view by clicking Users in the left navigation, searching for Linda Lucas, and then opening the user's detail page.

Task 16.3: Analyze High Page Fetch Time

In this task, you will analyze the web probe metrics for the user to look for patterns that can help to isolate the cause of the user's poor ZDX score.

1. From **Linda's user/device report**, scroll down to the **Web Probe Metrics** section.
2. Review the overall range of **Page Fetch Time values** over the selected period.
 - a. In this example, values may range from approximately 1000 ms to 3000 ms
 - i. indicating high latency for page loads.
3. Click on a specific point in the **Page Fetch Time** graph to analyze a single sample.
4. Compare:
 - a. The **Page Fetch Time** at this point (for example, **1204 ms**).
 - b. The **PFT Baseline** (for example, **506 ms**).

Note: *In this example, a Page Fetch Time of 1204 ms is significantly higher than the PFT Baseline of 506 ms, indicating unusually high latency per page request.*

5. Examine the breakdown of Page Fetch Time at the selected point:
 - a. **Server Response Time**
 - b. **DNS Resolve Time.**
6. In this example, note:
 - i. High **fluctuations in the DNS Resolve Time.**
 - ii. **DNS Resolve Time** accounts for most of the overall Page Fetch Time latency.

Note: *This suggests that slow or unstable DNS is a primary contributor to Linda's poor experience.*

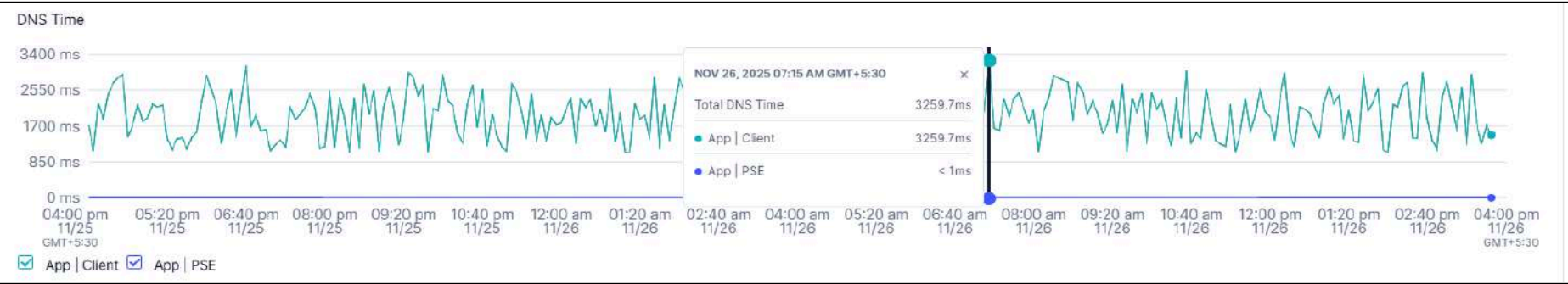
 Linda Lucas (linda.lucas@thezerotrustexchange.com)

Nov 13, 2025 02:55 PM GMT+5:30 - Nov 15, 2025 02:55 PM GMT+5:30

Start Diagnostics

Run Scripts

Share Snapshot



Task 16.4: Use ZDX Root Cause Analysis to Determine Root Cause

In this task, you will use the **ZDX Y-Engine** to **confirm** that slow DNS is the main cause of Linda's poor ZDX score and high Page Fetch Time.

1. Scroll back up to the **ZDX Score Over Time** graph for Linda's device.
2. Click **Analyze Score**.
3. Review the analysis panel showing the factors that might have impacted the ZDX score during the time window.
4. Identify the **top contributing factor(s)**.

Note: In this example, **High DNS** is identified as the most likely **root cause** of poor performance, with a Confidence Level of 100%.

5. Confirm that this aligns with your findings from Task 16.3:
 - a. **DNS Resolve Time** accounted for the majority of the Page Fetch Time.
 - b. DNS latency showed significant variability over time.
6. View the User Devices section to identify the DNS server being used by **Host-2** : 10.0.1.53

Note: This confirms that **slow DNS response from the configured DNS server** is the primary cause of Linda's poor experience when accessing SaaS applications, matching both **manual analysis** and **ZDX Copilot** findings.

